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Anharmonic Phonon-Phonon Interactions

Project Notes



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Theory

1.1. Force Constants

1.1.1. Potential-energy expansion

For theoretical description of lattice vibrations, the crystal potential energy in the atomic displacements is expanded from equilibrium [1, 2],

$$E(\{u_i\}_{i=1}^{n_{\text{dof}}}) = E_0 + \sum_i^{n_{\text{dof}}} \frac{\partial E}{\partial u_i} u_i + \frac{1}{2} \sum_{ij}^{n_{\text{dof}}} \frac{\partial^2 E}{\partial u_i \partial u_j} u_i u_j + \frac{1}{6} \sum_{ijk}^{n_{\text{dof}}} \frac{\partial^3 E}{\partial u_i \partial u_j \partial u_k} u_i u_j u_k + \dots, \quad (1.1)$$

where $n_{\text{dof}} = 3n_{\text{super}} = 3N_{\text{Cell}}n_{\text{prim}}$ is the total number of degrees of freedom (see Section A.3 for the notation). At equilibrium, the first-order term vanishes. The second-order coefficients define the harmonic, or second-order, interatomic force constants (IFC2), the third-order coefficients define the cubic, or third-order, force constants (IFC3). The harmonic approximation retains Eq. (1.1) to second order, and is sufficient for phonon dispersion and most equilibrium thermodynamic quantities [3, 4]. Anharmonic lattice-dynamics methods include the cubic and potentially higher-order terms. These introduce phonon–phonon scattering processes and hence the finite linewidth, frequency shift, and thermal transport properties of crystalline solids [5, 6].

1.1.2. Harmonic force constants (IFC2)

Introducing the primitive-cell degree-of-freedom (DOF) index $\mu = (\kappa, \alpha)$, which bundles the basis atom κ and the Cartesian direction α into a single label, the harmonic force constants are defined as

$$\Phi(0\mu, l'\mu') = \frac{\partial^2 E}{\partial u_{0\mu} \partial u_{l'\mu'}}, \quad (1.2)$$

where $u_{l\mu}$ is the displacement of DOF μ in cell l . Within a perfect crystal, translational symmetry lets us fix the first cell index to the home cell ($l=0$) without loss of generality, so that $\Phi(0\mu, l'\mu')$ depends only on the relative cell separation. Grouping the three Cartesian directions for a fixed pair of atoms (κ, κ') yields a 3×3 block whose rows and columns correspond to force and displacement components, respectively.

The IFC2 quantify how a small displacement of DOF μ' in cell l' changes the force on DOF μ in the home cell. Linearising the Newtonian equations of motion around equilibrium then gives the force–displacement relation

$$F(0\mu) = - \sum_{l'\mu'} \Phi(0\mu, l'\mu') u_{l'\mu'}, \quad (1.3)$$

which is crucial to both harmonic lattice dynamics and the finite-displacement schemes used to compute the IFC2 discussed in Section 1.1.4.

By construction, the IFC2 tensor obeys three exact relations that are useful both for reducing storage and for enforcing physical consistency when force constants are obtained numerically. The first is permutation symmetry: since partial derivatives commute, the tensor is symmetric under exchange of its two legs,

$$\Phi(0\mu, l'\mu') = \Phi(l'\mu', 0\mu), \quad (1.4)$$

which, combined with translational invariance, can equivalently be written as

$$\Phi(0\mu, l'\mu') = \Phi(0\mu', (-l')\mu). \quad (1.5)$$

This form is commonly used in lattice-dynamical treatments of force constants [1, 2, 7].

The second symmetry is translational invariance: a rigid translation of the whole crystal in any Cartesian direction α' cannot generate a restoring force, which leads to the acoustic sum rule

$$\sum_{l'\kappa'} \Phi_{\alpha\alpha'}(0\kappa, l'\kappa') = 0 \quad \forall \kappa, \alpha, \alpha', \quad (1.6)$$

where the condition must hold for each triple $(\kappa, \alpha, \alpha')$ separately [1, 7, 8]. This relation guarantees that the three acoustic branches have zero frequency at the Γ point, and in practice its enforcement is critical for numerically computed IFC2, as even small residual violations produce spurious acoustic gaps in the dispersion. The third is space-group symmetry: under each operation $\{S|v\}$ of the crystal point group, the IFC2 transform as a second-rank Cartesian tensor on the index pair and simultaneously map one atom–cell pair onto another. The resulting linear relations among the entries of Φ typically reduce the number of independent parameters by one or two orders of magnitude, and are exploited systematically by software packages such as phonopy [9] and symfc [10], and are discussed in-depth in Fu et al. [11].

Phonon frequencies and polarisation vectors are obtained as eigenpairs of the mass-weighted Fourier transform of Φ , the dynamical matrix (see Section A.3 for the two phase conventions used in this work). See the standard literature, e.g. Born et al. [1], Maradudin [2], and Dove [3].

1.1.3. Anharmonic force constants (IFC3)

The cubic, or third-order, interatomic force constants are the generalisation of Eq. (1.2) to the next order of the Taylor expansion Eq. (1.1). In the composite-index notation, they read

$$\Phi^{(3)}(0\mu, l'\mu', l''\mu'') = \frac{\partial^3 E}{\partial u_{0\mu} \partial u_{l'\mu'} \partial u_{l''\mu''}}, \quad (1.7)$$

and, as before, translational symmetry lets us fix the first cell index to the home cell. The IFC3 tensor captures the leading anharmonic corrections to lattice dynamics. It sets the amplitude of three-phonon scattering processes, determines the finite linewidth and temperature-dependent frequency shift of phonon modes, and provides the dominant resistive mechanism for lattice thermal conductivity in most crystalline solids [5, 6].

The symmetry properties of IFC3 parallel those of IFC2, but with a richer permutation structure due to the three legs of the tensor. Commutativity of partial derivatives now gives full S_3 symmetry, so that $\Phi^{(3)}(0\mu, l'\mu', l''\mu'')$ is invariant under all six simultaneous permutations of its three cell-DOF pairs:

$$\Phi^{(3)}(0\mu, l'\mu', l''\mu'') = \Phi^{(3)}(l'\mu', 0\mu, l''\mu'') = \Phi^{(3)}(l''\mu'', l'\mu', 0\mu) = \dots \quad (1.8)$$

[6]. Translational invariance again generates acoustic sum rules, now one per leg: summing $\Phi^{(3)}$ over the cell-atom labels of any one leg must vanish for all choices of the remaining indices,

$$\sum_{l''\mu''} \Phi_{\alpha\alpha'\alpha''}^{(3)}(0\kappa, l'\kappa', l''\mu'') = 0 \quad \forall \kappa, \kappa', l', \alpha, \alpha', \alpha'', \quad (1.9)$$

and analogously under summation over the first or second leg [6]. As for the harmonic case, enforcing these sum rules on numerically extracted IFC3 is essential for a consistent phonon self-energy and, hence, for accurate thermal-transport predictions [12]. Space-group symmetry acts as a third-rank Cartesian tensor on the index triple while simultaneously permuting the atom-cell labels, and typically reduces the independent-entry count by one to two orders of magnitude [10, 11, 13].

Despite these reductions, the storage requirements for the full IFC3 grow quickly with system size. After fixing one cell index by translational symmetry, the number of independent entries scales as $\mathcal{O}(n_{\text{prim}} n_{\text{dof}}^2) = \mathcal{O}(N_{\text{Cell}}^2 n_{\text{prim}}^3)$, which rapidly becomes prohibitive for large supercells. This scaling, together with the even more unfavourable cost of the four-tensor contraction that enters the phonon-phonon self-energy, motivates the compression schemes developed in Section 1.1.5.

1.1.4. Computation of force constants

Several established methods exist for obtaining interatomic force constants from first-principles calculations. We summarise the main approaches below. The choice among them involves trade-offs between computational cost, accessible interaction range, the order of anharmonicity that can be treated, and availability of software.

Finite displacements

The most widely used approach constructs a supercell, displaces atoms one at a time along symmetry-inequivalent directions, performs a self-consistent DFT calculation for each displaced structure, and reconstructs the force-constant tensor by inverting the force–displacement relation Eq. (1.3). The method was introduced by Parlinski et al. [14] and is implemented in codes such as phonopy [9], PHON [15], and Phonon [14]. Space-group symmetry reduces the number of independent displacements and hence the number of required DFT calculations, making the approach practical for cells of moderate size.

For cubic force constants, the analogous procedure requires displacing two atoms simultaneously. The number of required supercell calculations therefore grows quadratically with the number of atoms in the primitive cell [12, 13]. Esfarjani et al. [13] introduced a general method that extracts harmonic and anharmonic FCs to arbitrary neighbour shell from a single set of displacements by solving a linear least-squares problem with symmetry constraints. Widely used implementations for the IFC3 include `thirdorder.py` (distributed with ShengBTE) [12] and `phono3py` [16]. The latter also provides the infrastructure for computing phonon linewidths, thermal conductivity, and spectral functions from the extracted IFC3.

Supercell convergence and aliasing. The choice of supercell determines the real-space range over which the force constants can be resolved before periodic wrap-around introduces aliasing. For a supercell of linear dimension N along a lattice direction, the calculated force constant at separation r is the periodically folded sum

$$\Phi_{\text{per}}^{(N)}(r) = \sum_{p \in \mathbb{Z}} \Phi(r + pN). \quad (1.10)$$

In a $2 \times 2 \times 2$ supercell, the resolvable relative translations are limited to those representable within the finite periodic cell – contributions from more distant cells are folded back onto these separations. A larger supercell therefore allows force constants to be resolved over a wider real-space range before aliasing occurs [3, Chs. 2, 6].

This aliasing directly affects the phonon dispersion. At wavevectors $\mathbf{q}$ that are not commensurate with the supercell mesh, the Fourier-interpolated dynamical matrix inherits errors from the folded-back tails of Φ . In practice, convergence must therefore be checked for the target observable—phonon frequencies, thermal conductivity, or self-energy—by increasing the supercell size until the aliased contributions are negligible [9, 14].

For the IFC3, the aliasing issue is compounded by the fact that three cell indices are involved: each pair in the triplet $(0\kappa, l'\kappa', l''\kappa'')$ must lie within the interaction range resolvable by the supercell. This makes third-order properties more demanding in supercell size than their harmonic counterparts [12, 13].

Density-functional perturbation theory (DFPT)

Density functional perturbation theory avoids explicit supercell calculations by computing the linear response of the electron density to a phonon perturbation at an arbitrary wave-vector $\mathbf{q}$ within the primitive cell. The method was introduced by Baroni et al. [17] and Gonze [18], with a comprehensive review by Baroni et al. [19]. Second-order force constants at any $\mathbf{q}$ are obtained directly from the self-consistent solution of the Sternheimer equations, and real-space IFC2 are obtained by Fourier interpolation of the dynamical matrix computed on a regular mesh of $\mathbf{q}$ -points [20].

For cubic force constants, the $2n+1$ theorem allows $\Phi^{(3)}$ to be extracted from the first-order wavefunction and density responses already computed at the IFC2 stage, without requiring second-order responses [21, 22]. Deinzer et al. [23] applied this to compute phonon linewidths in Ge and Si, and Paulatto et al. [24] generalised it to arbitrary triplets $(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3)$ with $\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 = \mathbf{G}$, enabling the direct evaluation of three-phonon scattering matrix elements throughout the Brillouin zone without real-space supercells. Paulatto et al. [25] later extended the approach to compute phonon spectral functions with strong anharmonicity.

DFPT-based codes differ substantially in what they support at the IFC2 and IFC3 levels. For harmonic IFCs, Quantum ESPRESSO [26, 27] (via `ph.x`) supports norm-conserving (NC), ultrasoft (USPP), and projector-augmented-wave (PAW) pseudopotentials, with LDA, GGA, and (in recent versions) hybrid functionals. ABINIT [28] similarly supports both NC and PAW datasets, non-magnetic, collinear, and non-collinear spin, as well as DFT+ U . VASP [29] provides a more rudimentary DFPT implementation (IBRION = 7, 8) that is restricted to displacements commensurate with the supercell ($\mathbf{q}=0$) and to LDA/GGA functionals.

For cubic force constants, the picture is more restrictive. In Quantum ESPRESSO, $\Phi^{(3)}$ is available through the D3Q and `thermal2` packages [24, 30]. These codes support only norm-conserving pseudopotentials and only LDA or GGA exchange–correlation functionals. Neither USPP, PAW, meta-GGA, hybrid functionals, nor DFT+ U are currently implemented at third order. The same limitation applies to all third-order energy derivatives in Quantum ESPRESSO, including Raman cross-sections. In ABINIT, a framework for third-order derivatives of the total energy (3DTE) exists and is used in literature for non-linear optical responses and Raman tensors [31, 32]. The extraction of the full three-phonon IFC3 tensor on a $\mathbf{q}$ -grid analogous to D3Q, however, is not available and, in practice, anharmonic lattice dynamics in ABINIT is usually performed via the supercell-based a-TDEP module [33] rather than through pure DFPT. VASP does not currently implement third-order DFPT, and cubic force constants with VASP are typically obtained through finite-displacement calculations with, e.g., `phono3py` [34]. Hence, when computing IFC3 with DFPT, one is effectively restricted to a LDA/GGA exchange correlation with norm-conserving pseudopotentials, via Quantum ESPRESSO + D3Q.

Compressive sensing and machine-learning approaches

Compressive-sensing lattice dynamics (CSLD) casts the force-constant extraction as an ℓ_1 -regularised regression on DFT forces computed for a small number of randomly displaced configurations [35, 36]. The sparsity prior selects the minimal set of non-negligible FC entries, potentially capturing long-range couplings that would require prohibitively large supercells in the standard finite-displacement scheme. The hiPhive package [37] implements this idea and uses crystal-symmetry reduction and advanced regression algorithms to extract force constants up to arbitrary order from semi-random displacement configurations. These regression-based approaches are particularly attractive when the full enumeration of finite displacements becomes prohibitively expensive for large systems.

1.1.5. FC3 Compression

Storing the full FC3 tensor requires $\mathcal{O}(n_{\text{prim}} n_{\text{dof}}^2) = \mathcal{O}(N_{\text{Cell}}^2 n_{\text{prim}}^3)$ memory, which is prohibitive for large-scale simulations. Beyond storage, the dense four-tensor contraction entering the phonon-phonon self-energy scales as $\mathcal{O}(n_{\text{dof}}^4)$ per frequency point and dominates the cost of the quantum-transport simulation. It is hence desirable to efficiently approximate the FC3 interactions.

Effectively every published anharmonic phonon NEGF implementation avoids the problem by dropping self-energy and FC3 entries at large atomic separations [38–43]. The standard choice is a real-space pair-distance cut-off: keep only those triplets $(0\kappa, l'\kappa', l''\kappa'')$ whose largest pairwise distance lies below some r_c . The same idea is used in the BTE workflows of ShengBTE and its successors, which restrict FC3 to a few nearest-neighbour shells [12].

The drawback is that the approximation error is uncontrolled: there is no variational bound on what is discarded, and the truncation removes genuine non-local contributions that can matter quantitatively in the phonon self-energy. In anharmonic NEGF this is already visible in silicon. Guo et al. use first-principles FC3 in a quantum-transport treatment of cross-plane heat flow in Si thin films, where they compare first- and third-nearest-neighbour FC3 cutoffs [38]. Using the same FC3 data in ShengBTE, they report bulk Si thermal conductivities at 300 K of 120.69, 136.46, and 147.13 W m⁻¹K⁻¹ when FC3 was truncated at the first, second, and third neighbour shell, respectively, so that a first-shell cut-off underestimates the converged bulk value by nearly 20%. For bulk silicon, including up to 4th nearest neighbour interactions gives converged results [12, 44].

For three-dimensional covalent crystals, the cut-off problem is usually less severe than in two-dimensional materials. McGaughey et al. report heat conductivities $\kappa_\ell(300\text{ K}) = 150, 152, 155, 151$, and 149 W m⁻¹K⁻¹ for cubic cut-offs of 2, 3, 4, 5, and 6 neighbour shells, respectively in bulk Si, indicating that once a few shells are retained the residual cut-off dependence is already small [45]. Jiang et al. reached the same qualitative conclusion by contrasting graphene and silicene against bulk Si: whereas the 2D systems require inclusion of up to 14th order NN shells, bulk Si shows “no convergence issue” even for

very small neighbour cut-offs [46].

That behaviour is, however, not universal across 3D materials. Qin et al. showed that bulk SnSe still exhibits strong interactions at distances around 6.15 Å, and that these coincide with a sharp change in the converged κ_ℓ once the pair cut-off exceeds roughly 6 Å [44]. They also show that for the 2D materials studied, the observables in BTE diverge for increasing q -grid when the cut-off is too small.

We therefore seek a representation of the FC3 tensor that

- provides a controllable approximation error,
- achieves significant compression of the stored data, and
- turns the dense $\mathcal{O}(n_{\text{dof}}^4)$ self-energy contraction into a sum of low-order contractions whose cost scales polynomially in a parameter R .

A large body of work has developed such representations in similar settings. In quantum chemistry, tensor hypercontraction (THC) [47, 48] and density fitting [49] factorise the four-index electron-repulsion integral tensor into products of matrices. In many-body physics, tensor networks (matrix product states, tensor trains) [50, 51] compress high-dimensional objects at controlled error. For the present setting, lattice dynamics, tensor decomposition has recently proved effective: Luo et al. used a truncated SVD on the electron-phonon matrix in Wannier basis and recovered high-accuracy transport, spin, and superconducting observables from 1–2% of the singular values [52]. The same group then proposed the permanent CP (PCP) ansatz for the n -IFC tensors and reported 10^3 – 10^4 compression factors across Si, HgTe, MgO, TiNiSn and monoclinic ZrO₂ while retaining > 98% accuracy on κ_ℓ for three- and four-phonon scattering [53]. Compressive-sensing lattice dynamics (CSLD) is a method applied to a related problem at the fitting stage: an ℓ_1 -regularised regression on DFT forces selects a sparse set of non-negligible FC entries directly [35, 36]. Projector-based approaches, implemented in the symfc software, construct a complete orthonormal basis for FCs compatible with all crystal symmetries [10], while group-theoretical reductions enumerate the independent FC3 entries from the space group [11]. In the following, we take the fully-populated FC3 tensor as given (from finite displacements, DFPT, or CSLD) and compress its representation. However, one can likely also fit a compressed tensor from DFT data with an adaption of CSLD or similar methods without ever obtaining a full FC3 tensor.

Kolda et al. [54] organise tensor compression into two principal families: the Tucker decomposition, which writes a tensor as a core tensor multiplied by a factor matrix along each mode (generalising principal-component analysis), and the CP (CANDECOMP/PARAFAC) decomposition, which writes it as a sum of rank-one outer products (generalising the matrix SVD in a different direction). Applied to a tensor with the FC3 index pattern and combined with the possible symmetry constraints on the factor matrices, these two families yield five canonical ansätze, ordered here from least to most tensor-structure-aware:

1. Truncated matricization SVD, i.e. the classical matrix SVD applied to an matricization of the tensor. fully algebraic and Eckart–Young optimal in the chosen matricization, but separating only one leg from the other two.

2. Truncated HOSVD (Tucker), with independent factor matrices per mode and a small dense core.
3. CP, obtained by restricting the Tucker core to be diagonal, with three independent rank- R factor matrices and no symmetry constraint.
4. INDSCAL [55], obtained from CP by setting the factor matrices on the two internal legs equal, exploiting the S_2 symmetry $\Phi_{\mu j k} = \Phi_{\mu k j}$.
5. Symmetric CP, equivalent to the Waring decomposition of a homogeneous cubic polynomial [56–58], obtained from CP by setting all three factor matrices equal (using the $p(\mu)$ lift of a primitive-cell DOF to its supercell counterpart in cell zero), exploiting the full S_3 symmetry.

Entries 1 and 2 are non-iterative. Entries 3–5 are NP-hard in the worst case [59] and use iterative optimisation, with the known caveats of rank-degeneracy and non-closedness of rank- R sets [60, 61]. Each added symmetry constraint from entry 2 onwards reduces the parameter count at matched rank while shrinking the set of representable tensors. As the transport-residual study of Section 2.2 shows, the most-constrained ansatz is not necessarily the least accurate per parameter — the transport observable converges faster than the Frobenius residual the fitters minimise.

Truncated matricization SVD

Stack the $3n_{\text{prim}}$ slices $M_\mu = \Phi_{\mu, :, :} \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$ into a single matrix

$$M_{\text{stack}} = \begin{pmatrix} M_1 \\ \vdots \\ M_{3n_{\text{prim}}} \end{pmatrix} \in \mathbb{R}^{3n_{\text{prim}} n_{\text{dof}} \times n_{\text{dof}}}. \quad (1.11)$$

A truncated matrix SVD at rank R gives

$$\tilde{\Phi}_{\mu j k}^{\text{mSVD}} = \sum_{r=1}^R F_r(\mu, j) h_r(k), \quad (1.12)$$

with $F_r(\mu, j) = \sigma_r [\mathbf{u}_r]_{(\mu-1)n_{\text{dof}}+j}$ and $h_r(k) = [\mathbf{v}_r]_k$, where $\sigma_r, \mathbf{u}_r, \mathbf{v}_r$ are the R leading singular values, left singular vectors, and right singular vectors of M_{stack} . This is equivalent to a Tucker1 decomposition with compression on mode 3 only [54, §4].

Computation. A single matrix SVD of M_{stack} , computed by standard dense LAPACK routines or, for large n_{dof} , by randomised linear algebra algorithms [62]. The Eckart–Young–Mirsky theorem [63, 64] guarantees that, at the chosen $(\mu, j)|k$ bipartition, the truncation is globally optimal among all rank- R approximations in the Frobenius norm. This makes it the natural baseline against which the structurally richer decompositions of Section 1.1.5 must be compared. The ansatz does not exploit the S_2 symmetry on (j, k) : modes 1 and 2 are folded into a single flat index, so parameters are wasted on storing redundant information and the reconstruction is not exactly symmetric in (j, k) unless restored in post-processing.

Parameter count.

$$N_{\text{param}}^{\text{mSVD}}(R) = R((3n_{\text{prim}} + 1)n_{\text{dof}} + 1) \sim \mathcal{O}(R n_{\text{prim}} n_{\text{dof}}) = \mathcal{O}(R N_{\text{Cell}} n_{\text{prim}}^2), \quad (1.13)$$

one factor of n_{prim} worse than the CP-family ansätze below.

Self-energy kernel. The separable structure factorises the ring contraction entering the phonon-phonon self-energy into two independent bilinear forms. The $\mathbf{q}$ -convolution is an FFT over the R^2 rank pairs, replacing the dense n_{dof}^4 contraction by $R^2 n_{\text{dof}}$ -scaling accumulations.

Truncated HOSVD (Tucker decomposition)

The Tucker ansatz writes FC3 as a multilinear contraction of a small core tensor with one factor matrix per mode [54, 65]:

$$\tilde{\Phi}_{\mu j k}^{\text{HOSVD}} = \sum_{p=1}^{R_1} \sum_{q=1}^{R_2} \sum_{r=1}^{R_2} G_{pqr} A_{\mu p} B_{jq} B_{kr}, \quad (1.14)$$

with $A \in \mathbb{R}^{3n_{\text{prim}} \times R_1}$, $B \in \mathbb{R}^{n_{\text{dof}} \times R_2}$ columnwise orthonormal, and a core $G \in \mathbb{R}^{R_1 \times R_2 \times R_2}$. To preserve the S_2 symmetry of the two internal legs, the factor matrices on modes 2 and 3 can be chosen identical ($B = C$) and the core is constrained to satisfy $G_{pqr} = G_{prq}$.

Computation. The truncated higher-order SVD (HOSVD) of De Lathauwer et al. [66] produces A and B in closed form as the R_1 and R_2 leading left singular vectors of the mode-1 and mode-2 unfoldings of Φ , followed by the core contraction $G = \Phi \times_1 A^\top \times_2 B^\top \times_3 B^\top$ (which is automatically S_2 -symmetric when Φ is). This is quasi-optimal: the truncation error is bounded by a factor $\sqrt{3}$ above the best rank- (R_1, R_2, R_2) approximation error [50, 67]. The fit can optionally be refined by a few steps of higher-order orthogonal iteration (HOOI), an alternating least squares (ALS)-type loop [68]. The special case $R_1 = 3n_{\text{prim}}$, $A = I$ (compression of the two internal legs only) is the Tucker1 baseline and is obtained by a single SVD of the corresponding unfolding.

Parameter count.

$$N_{\text{param}}^{\text{HOSVD}}(R_1, R_2) = 3n_{\text{prim}} R_1 + n_{\text{dof}} R_2 + R_1 \frac{R_2(R_2+1)}{2} \sim \mathcal{O}(R_2 n_{\text{dof}} + R_1 R_2^2), \quad (1.15)$$

where the S_2 constraint halves the nominal core cost $R_1 R_2^2$ to $R_1 R_2(R_2 + 1)/2$. The R_2^2 -scaling core term makes Tucker the most parameter-heavy of the five ansätze when R_2 is not small compared to $n_{\text{dof}}^{1/2}$. for strongly anharmonic systems where the effective rank is moderate this is mitigated by allowing $R_1 \ll R_2$.

Self-energy kernel. The self-energy contraction factorises into three mode-wise contractions against A and B , leaving the $R_1 \times R_2 \times R_2$ core as a small dense sub-problem. The total cost scales as $R_2^2(R_1 + n_{\text{dof}})$ per frequency point rather than n_{dof}^4 .

Canonical Polyadic decomposition (CP)

Dropping the off-diagonal entries of the Tucker core yields the CP approximation, also known as CANDECOMP and PARAFAC [55, 69, 70], a sum of rank-one outer products with three independent factor vectors per term:

$$\tilde{\Phi}_{\mu j k}^{\text{CP}} = \sum_{r=1}^R \lambda_r a_r(\mu) b_r(j) c_r(k). \quad (1.16)$$

No symmetry between the modes is imposed. For a tensor that is S_2 - or S_3 -symmetric, the fit does not necessarily converge to a symmetric solution [71]. Uniqueness can be shown under certain conditions [72, 73] and is generically guaranteed for the FC3 tensor of interest here [74].

Computation. The standard algorithm is alternating least squares (CP-ALS), which cycles through the three modes solving, at each step, a linear least-squares problem for one factor matrix while the other two are fixed [54, Sec. 3.4]. Convergence is only to a stationary point and can be slow. Variants include damped Gauss–Newton (dGN), PMF3 [75], line-search extrapolation (ELS) [76], and simultaneous-diagonalisation approaches [77] that recast CP as a generalised eigenproblem when $R \leq \min(3n_{\text{prim}}, n_{\text{dof}})$.

Parameter count.

$$N_{\text{param}}^{\text{CP}}(R) = R(3n_{\text{prim}} + 2n_{\text{dof}} + 1) \sim \mathcal{O}(R n_{\text{dof}}) = \mathcal{O}(R N_{\text{Cell}} n_{\text{prim}}). \quad (1.17)$$

INDSCAL (CP with internal-leg symmetry)

Individual-differences-in-scaling (INDSCAL) [54, 55] is the specialisation of CP in which the factor matrices on two modes are constrained equal, tailored to tensors whose slices $\Phi_{\mu, :, :}$ are symmetric:

$$\tilde{\Phi}_{\mu j k}^{\text{INDSCAL}} = \sum_{r=1}^R d_r(\mu) v_r(j) v_r(k), \quad (1.18)$$

with $d_r \in \mathbb{R}^{3n_{\text{prim}}}$ and $v_r \in \mathbb{R}^{n_{\text{dof}}}$. This imposes the S_2 symmetry $\Phi_{\mu j k} = \Phi_{\mu k j}$. Typical and maximal ranks for partially symmetric tensors were characterised by Ten Berge et al. [71]: for the shapes relevant here, they coincide with or are strictly smaller than the ranks of generic nonsymmetric tensors of the same shape, so the symmetry constraint is not costly in expressive power.

Computation. Carroll et al. [55] originally proposed running unconstrained CP-ALS with the two internal factors B and C updated separately, relying on symmetry of the data to drive $B \rightarrow C$. Ten Berge et al. [71, 78] showed that this convergence is generic but not guaranteed. Modern implementations impose equality explicitly via a symmetric ALS or via a Gauss–Newton method [54, 79]. A cheap closed-form alternative performs a matrix

SVD of Φ reshaped to $3n_{\text{prim}} \times n_{\text{dof}}^2$ followed by an eigendecomposition of each symmetric slice of the resulting right factors, and keeps the R terms with largest amplitude. This is not jointly optimal (the total rank needed to match a given error is typically larger than that of a jointly-fitted INDSCAL) but is useful as an initialisation.

Parameter count.

$$N_{\text{param}}^{\text{INDSCAL}}(R) = R(3n_{\text{prim}} + n_{\text{dof}}) \sim \mathcal{O}(R n_{\text{dof}}) = \mathcal{O}(R N_{\text{Cell}} n_{\text{prim}}), \quad (1.19)$$

smaller than CP by one n_{dof} -length factor per term (a constant ~ 2 , not a factor of n_{dof}), from merging $b_r \equiv c_r$.

Self-energy kernel. Identical in scaling to CP, $\mathcal{O}(R^2 n_{\text{dof}})$ per frequency, but with the additional property that the reconstruction is exactly S_2 -symmetric in (j, k) , so ASR-sensitive observables receive only the quadratic-in-error contribution (Section 2.2).

Symmetric CP (Waring) decomposition

The symmetric CP decomposition imposes equality of all three factor matrices, $A = B = C$, on the fully S_3 -symmetrised FC3 tensor obtained via the supercell lift $p : \{1, \dots, 3n_{\text{prim}}\} \rightarrow \{1, \dots, n_{\text{dof}}\}$. The Waring parametrisation reads:

$$\tilde{\Phi}_{\mu j k}^{\text{Waring}} = \sum_{r=1}^R \lambda_r v_r(p(\mu)) v_r(j) v_r(k), \quad (1.20)$$

with $v_r \in \mathbb{R}^{n_{\text{dof}}}$ and $\lambda_r \in \mathbb{R}$. This is the tensor form of the classical Waring problem: expressing a homogeneous cubic polynomial as a sum of R cubes of linear forms [56–58]. For generic third-order symmetric tensors on n_{dof} indices, the Waring rank equals $\lceil \binom{n_{\text{dof}}+2}{3} / n_{\text{dof}} \rceil$ by the Alexander–Hirschowitz theorem [80] (which is a non-trivial statement and proven by Alexander and Hirschowitz in a set of 5 papers of over 100 pages). An alternative “permanent-CP” parametrisation, introduced by Luo et al. [53] as PCP, uses three distinct vectors per term explicitly symmetrised over S_3 ,

$$\tilde{\Phi}_{\mu j k}^{\text{PCP}} = \sum_{\xi=1}^R \frac{\lambda_{\xi}}{6} \sum_{\sigma \in S_3} u_{\sigma_1}^{\xi}(p(\mu)) u_{\sigma_2}^{\xi}(j) u_{\sigma_3}^{\xi}(k), \quad (1.21)$$

with $u_1^{\xi}, u_2^{\xi}, u_3^{\xi} \in \mathbb{R}^{n_{\text{dof}}}$. PCP fits the same class of symmetric tensors as Waring but with more capacity per term (at threefold parameter cost) and a smoother optimisation landscape.

Computation. Algorithms fall into two categories. Algebraic methods reconstruct the decomposition from moment and flattening matrices via generalised eigenproblems and are exact up to numerical precision when the rank is below the generic bound [56, 58]. Numerical methods optimise $\|\Phi - \tilde{\Phi}\|_F^2$ directly: the shifted symmetric higher-order power

method (SS-HOPM) [81] and the subspace power method [82] extract components one at a time with provable convergence for low-rank tensors; all-at-once gradient and Gauss–Newton methods [79] fit all R components simultaneously. Degeneracy and diverging-norm instabilities can appear because the set of symmetric tensors of rank $\leq R$ is not closed in general [60, 61]. Remedies include norm penalties and second-order line searches.

Parameter count.

$$N_{\text{param}}^{\text{Waring}}(R) = R(n_{\text{dof}} + 1) \sim \mathcal{O}(R n_{\text{dof}}) = \mathcal{O}(R N_{\text{Cell}} n_{\text{prim}}), \quad (1.22)$$

$$N_{\text{param}}^{\text{PCP}}(R) = R(3 n_{\text{dof}} + 1) \sim \mathcal{O}(R n_{\text{dof}}) = \mathcal{O}(R N_{\text{Cell}} n_{\text{prim}}). \quad (1.23)$$

Self-energy kernel. Identical in scaling to CP and INDSCAL, $\mathcal{O}(R^2 n_{\text{dof}})$ per frequency, but with the additional property that the reconstruction is exactly S_3 -symmetric in $(p(\mu), j, k)$, so ASR-sensitive observables receive only the quadratic-in-error contribution (Section 2.2).

Further approximations

The five ansätze of Section 1.1.5 do not exhaust the useful choices. Further approximations appear in the broader literature and can either be combined with them or replace them in specific regimes.

Cross / CUR / skeleton approximations. Rather than fitting a compressed form to a precomputed Φ , one can build the approximation from a small number of sampled fibers of the tensor and interpolate the rest:

$$\tilde{\Phi}_{\mu j k}^{\text{CUR}} = \sum_{p, q, r} \Phi_{\mu, J_q, K_r} \Phi_{I_p, J_q, K_r}^{-1} \Phi_{I_p, j, k}, \quad (1.24)$$

where I, J, K are index subsets of size $\sim R$ selected to maximise the volume of the intersection subtensor, and $\Phi_{I, J, K}$ is inverted in the appropriate pseudoinverse sense. This bypasses the construction of the full FC3 altogether, which is normally the computational bottleneck (finite-displacement DFT). Rank- R tensor cross [83, 84] accesses $\mathcal{O}(R^2 n_{\text{dof}})$ entries. Adaptive cross approximation [85] and CUR-style subset selection [86, 87] give analogous matrix-level guarantees, and fiber-sampling Tucker (FSTD) of Caiafa et al. [88] yields an explicit Tucker form from $\mathcal{O}(R^{N-1})$ fibers. The tensor-CURT extension of Song et al. [89] provides the first relative-error bounds in this setting. For FC3 the main challenge is that cross methods have no intrinsic notion of ASR or S_3 symmetry. A hybrid that samples in the symmetric CP parameter space would combine the two sets of guarantees.

Tensor train and hierarchical Tucker. The tensor-train (TT) format [50] writes an N -index tensor as a contraction of N three-index cores with bond ranks $(R_1, \dots, R_{N-1})$. For third-order tensors this collapses to

$$\tilde{\Phi}_{\mu j k}^{\text{TT}} = \sum_{\alpha_1, \alpha_2} G_1(\mu, \alpha_1) G_2(\alpha_1, j, \alpha_2) G_3(\alpha_2, k), \quad (1.25)$$

which is structurally a Tucker1 SVD of the $(\mu)|(j, k)$ unfolding followed by a Tucker1 SVD of the $(\alpha_1 j)|(k)$ matricisation of the residual. It does not offer independent advantages beyond Section 1.1.5. TT and the hierarchical Tucker (HT) format of Hackbusch [67] become first-class methods at order ≥ 4 and are the natural representation for FC4 (four-phonon) tensors, where CP and Tucker lose efficiency. Lee et al. [90] recently used Matrix Product States (= TT) to solve the Peierls–Boltzmann equation in silicon.

Block-term decomposition (BTD). De Lathauwer [91] and De Lathauwer et al. [92] generalise both CP and Tucker by writing the tensor as a sum of low-multilinear-rank Tucker blocks,

$$\tilde{\Phi}_{\mu j k}^{\text{BTD}} = \sum_{r=1}^R (\mathcal{G}_r \times_1 A_r \times_2 B_r \times_3 C_r)_{\mu j k}, \quad (1.26)$$

with each $\mathcal{G}_r$ of small core shape (e.g. the rank- $(L, L, 1)$ BTD has $\mathcal{G}_r$ of shape $L \times L \times 1$). CP is the rank- $(1, 1, 1)$ special case; Tucker is the single-block special case. For FC3 this structurally expresses block patterns like per-atom-species or per-shell contributions in a single decomposition, and inherits the uniqueness and ALS algorithms of the parent families.

Constrained / symmetry-adapted decompositions (CANDELINC). CANDELINC [93] is CP with factor matrices constrained to user-chosen linear subspaces,

$$A = \Phi_A \hat{A}, \quad B = \Phi_B \hat{B}, \quad C = \Phi_C \hat{C}, \quad (1.27)$$

where Φ_A, Φ_B, Φ_C are columnwise-orthonormal basis matrices. It reduces to an ordinary CP fit in the reduced coordinates $(\hat{A}, \hat{B}, \hat{C})$ applied to the projected tensor $\hat{\Phi} = \Phi \times_1 \Phi_A^\top \times_2 \Phi_B^\top \times_3 \Phi_C^\top$. For FC3 this is the natural framework to couple any of the five ansätze with a space-group-compatible basis: choose Φ_A, Φ_B, Φ_C to span the ASR-respecting, symmetry-invariant subspace (symfc [10], Fu–Marianetti [11], CSLD [35, 36]), then fit INDSCAL or symmetric CP on the reduced tensor.

Sparse / structured CP. Imposing sparsity or smoothness priors on the CP factor vectors yields decompositions whose rank- R components are localised on a few atoms or DOFs. As an example, one can penalise the loss with

$$L(A, B, C) = \frac{1}{2} \|\Phi - (A, B, C)\|_F^2 + \lambda_1 \sum_r (\|a_r\|_1 + \|b_r\|_1 + \|c_r\|_1), \quad (1.28)$$

optionally replacing $\|\cdot\|_1$ by group-sparse or total-variation terms [94, 95], and are solved by ALS or AO-ADMM.

1.2. Phonon NEGF

We apply the non-equilibrium Green's function (NEGF) formalism to phonon quantum transport, taking as starting point the energy expansion in atomic displacements introduced in Eq. (1.1). Truncated at second order, the Hamiltonian generates ballistic (harmonic) transport. The third-order contribution gives the leading anharmonic phonon-phonon scattering, treated perturbatively in the self-consistent Born approximation (SCBA). The phonon NEGF algorithm is in close analogy with the electronic case [96–100], with two major differences: the natural energy variable is ω^2 because the lattice equation of motion is second-order in time, and the particles are bosons whose vacuum-fluctuation contributions persist at $T = 0$. The phonon-NEGF strategy was developed by Mingo and Yang for ballistic transport [101, 102], extended to anharmonicity by Mingo [43] and Wang et al. [103], and by [40, 104]. An explicit derivation of the SCBA self-energy at the diagrammatic level is given by Guo et al. [38].

1.2.1. Hamiltonian and lattice-dynamical setup

Let $u_{l\kappa\alpha}$ denote the Cartesian displacement of atom κ in unit cell l , and let m_κ be its mass. In mass-weighted coordinates $\tilde{u}_{l\kappa\alpha} = \sqrt{m_\kappa} u_{l\kappa\alpha}$, the lattice Hamiltonian truncated at third order is

$$\hat{H} = \frac{1}{2} \sum_{\mu} \tilde{p}_{\mu}^2 + \frac{1}{2} \sum_{\mu\nu} D_{\mu\nu} \tilde{u}_{\mu} \tilde{u}_{\nu} + \frac{1}{3!} \sum_{\mu\nu\lambda} \Phi_{\mu\nu\lambda} \tilde{u}_{\mu} \tilde{u}_{\nu} \tilde{u}_{\lambda} + \mathcal{O}(\tilde{u}^4), \quad (1.29)$$

where the multi-index $\mu = (l, \kappa, \alpha)$ collects unit-cell, sublattice, and Cartesian labels. The mass-weighted second- and third-order force constants are

$$D_{\mu\nu} = \frac{1}{\sqrt{m_\kappa m_{\kappa'}}} \Phi_{\mu\nu}^{(2)}, \quad \Phi_{\mu\nu\lambda} = \frac{1}{\sqrt{m_\kappa m_{\kappa'} m_{\kappa''}}} \Phi_{\mu\nu\lambda}^{(3)}, \quad (1.30)$$

with $\Phi_{\mu\nu}^{(2)} = \partial^2 V / \partial u_{\mu} \partial u_{\nu}$ and analogously for $\Phi^{(3)}$. The acoustic sum rules

$$\sum_{l\kappa} \Phi_{l\kappa\alpha, \mu_2 \dots \mu_n}^{(n)} = 0 \quad (n = 2, 3, \dots), \quad (1.31)$$

plus translational and permutation symmetry are imposed by the definition of the FCs.

Quantizing in the harmonic basis,

$$\tilde{u}_{\mu}(t) = \sum_s \sqrt{\frac{\hbar}{2\omega_s}} e_{\mu s} [a_s e^{-i\omega_s t} + a_s^{\dagger} e^{i\omega_s t}], \quad D e_s = \omega_s^2 e_s, \quad (1.32)$$

we obtain ω^2 as the spectral variable of phonon Green's functions.

1.2.2. Contour-ordered Green's function and Dyson equation

Following the conventions of Wang et al. [100], we define the contour-ordered phonon Green's function on the Keldysh contour $\mathcal{C}$:

$$G_{\mu\nu}(\tau, \tau') = -\frac{i}{\hbar} \langle T_{\mathcal{C}} \tilde{u}_{\mu}(\tau) \tilde{u}_{\nu}(\tau') \rangle. \quad (1.33)$$

Projecting τ onto the upper/lower branches yields the lesser, greater, retarded, advanced, and time-ordered components $G^{\lessgtr}$, $G^{R/A}$, G^t , with the standard relations

$$G^R(t, t') = \theta(t - t') [G^>(t, t') - G^<(t, t')], \quad G^A = (G^R)^{\dagger}, \quad (1.34)$$

$$G^R - G^A = G^> - G^<. \quad G^t = G^< + G^R \quad (1.35)$$

At equilibrium, the fluctuation–dissipation theorem can be written as [100, Eq. 24]

$$G^<(\omega) = 2i n_{\text{BE}}(\omega, T) \text{Im } G^R(\omega), \quad G^>(\omega) = 2i [1 + n_{\text{BE}}(\omega, T)] \text{Im } G^R(\omega). \quad (1.36)$$

Out of equilibrium, Eq. (1.36) does not hold and one must solve the Keldysh equation

$$G^{\lessgtr}(\omega) = G^R(\omega) \Sigma^{\lessgtr}(\omega) G^A(\omega). \quad (1.37)$$

The retarded Green's function can be computed from the Dyson equation

$$[G^R(\omega^2)]^{-1} = (\omega^2 + i\eta) \mathbb{I} - D - \Sigma^R(\omega^2), \quad (1.38)$$

with D the dynamical matrix, $\eta > 0$ a numerical broadening matched to the frequency grid, and the total retarded self-energy decomposed as

$$\Sigma^R(\omega^2) = \Sigma_L^R(\omega^2) + \Sigma_R^R(\omega^2) + \Sigma_s^R(\omega^2). \quad (1.39)$$

The lead self-energies $\Sigma_{L,R}^R$ are due to the semi-infinite harmonic contacts. Σ_s^R collects all many-body scattering and equals zero in the ballistic limit. The spectral function is defined as

$$A(\omega^2) = i[G^R(\omega^2) - G^A(\omega^2)] = -2 \text{Im } G^R(\omega^2), \quad (1.40)$$

with sum rule $\int_0^\infty \frac{d(\omega^2)}{2\pi} [A(\omega^2)]_{\mu\mu} = 1$ and poles only at positive ω^2 .

1.2.3. Transverse periodic NEGF

We consider both finite and transversely periodic geometries, with transport along x and periodicity in (y, z) . Following [38, 105, 106], the transverse wave vector $\mathbf{q}_\perp$ (we use q for phonons, k for electrons) is sampled on a uniform Monkhorst–Pack mesh

$$k_i = \frac{n_i + 0.5}{N_i} - 0.5 + \text{shift}_i, \quad n_i = 0, \dots, N_i - 1, \quad (1.41)$$

with total count N_q chosen to converge the transverse Brillouin-zone integral (e.g. $N_q = 400$ on a 20×20 grid). The Dyson equation becomes $\mathbf{q}_\perp$ -dependent,

$$[G^R(\omega^2, \mathbf{q}_\perp)]^{-1} = (\omega^2 + i\eta)\mathbb{I} - D(\mathbf{q}_\perp) - \Sigma^R(\omega^2, \mathbf{q}_\perp). \quad (1.42)$$

For fixed $\mathbf{q}_\perp = (q_x, q_y)$, the on-site and nearest-neighbour transport-direction blocks are obtained by partial Fourier transform over the q_z mesh:

$$H_{00}(\mathbf{q}_\perp) = \frac{C}{N_{q_z}} \sum_{m=0}^{N_{q_z}-1} D^{(B)}\left(q_x, q_y, \frac{m}{N_{q_z}}\right), \quad (1.43)$$

$$H_{01}(\mathbf{q}_\perp) = \frac{C}{N_{q_z}} \sum_{m=0}^{N_{q_z}-1} D^{(B)}\left(q_x, q_y, \frac{m}{N_{q_z}}\right) e^{-2\pi i m / N_{q_z}}. \quad (1.44)$$

Here $D^{(B)}$ is the bulk mass-weighted dynamical-matrix block and C is the force-constant-to-(THz)² conversion constant of Section A.2 (the sum index m runs over the q_z mesh). At nonzero $\mathbf{q}_\perp$, these blocks are generally complex and may equivalently be written as the 2D Fourier transform of the real-space FC2:

$$H_{0n_z}(\mathbf{q}_\perp) = \sum_{n_x, n_y} \Phi(n_x, n_y, n_z) e^{2\pi i (q_x n_x + q_y n_y)}, \quad (1.45)$$

with hermiticity enforcing $H_{10}(\mathbf{q}_\perp) = H_{01}^\dagger(\mathbf{q}_\perp)$.

1.2.4. Contact self-energy

The contact self-energies $\Sigma_{L,R}^R$ embed the semi-infinite harmonic leads via the surface Green's function g_s , computed iteratively by the renormalization–decimation algorithm of Sancho et al. [107]. The method has quadratic-doubling convergence: each iteration doubles the number of decimated layers, so the iteration error falls as $\|T\|^{2^n}$ with n the iteration index.

Decimation iteration. Given the bulk on-site and coupling blocks H_{00} and H_{01} , initialize

$$\epsilon = \epsilon_s = (\omega^2 + i\eta)\mathbb{I} - H_{00}, \quad \alpha = -H_{10}, \quad \beta = -H_{01}, \quad (1.46)$$

and iterate (within each sweep all four right-hand sides are evaluated with the ϵ from the previous sweep, i.e. the updates are simultaneous, not sequential)

$$\epsilon_s \leftarrow \epsilon_s - \alpha \epsilon^{-1} \beta, \quad (1.47)$$

$$\epsilon \leftarrow \epsilon - \alpha \epsilon^{-1} \beta - \beta \epsilon^{-1} \alpha, \quad (1.48)$$

$$\alpha \leftarrow \alpha \epsilon^{-1} \alpha, \quad (1.49)$$

$$\beta \leftarrow \beta \epsilon^{-1} \beta, \quad (1.50)$$

terminating when $\|\alpha\| + \|\beta\| < \text{tol}$. The surface Green's function is then $g_s = \epsilon_s^{-1}$. For the left lead $g_L = g_s(H_{00}, H_{01})$. For the right lead, the coupling direction is reversed, $g_R = g_s(H_{00}, H_{01}^\dagger)$. The contact self-energies follow from

$$\Sigma_L^R = D_{10} g_L D_{01}, \quad \Sigma_R^R = D_{01} g_R D_{10}, \quad (1.51)$$

with $D_{10} = D_{01}^\dagger$ the device-lead coupling block (the off-diagonal of the full mass-weighted dynamical matrix linking the device edge to the first lead layer; equal to the H_{01} block when device and lead are the same material). and the broadening matrices, which set the rate at which lead ℓ injects phonons into the device, are

$$\Gamma_\ell(\omega^2) = i[\Sigma_\ell^R(\omega^2) - \Sigma_\ell^A(\omega^2)]. \quad (1.52)$$

Convergence rate and concrete error bound. The decimation can be written as a transfer-matrix recursion. Eliminating odd-numbered layers in the bulk lead leads to an effective coupling between layers separated by 2, 4, 8, ... unit cells. In matrix form, writing T_n for the effective transfer operator after n decimation steps,

$$T_{n+1} = T_n^2, \quad T_0 = -\epsilon_0^{-1} H_{01}. \quad (1.53)$$

The relevant decay rates $\lambda_j = e^{ik_j a}$, with k_j the complex band wavevectors at energy ω^2 , are the eigenvalues of the $2N \times 2N$ companion linearisation of the quadratic eigenproblem $(\lambda^2 H_{10} + \lambda(H_{00} - \omega^2 \mathbb{I}) + H_{01})\xi = 0$ (not of the $N \times N$ operator T_0 alone). In a bulk gap, $|\lambda_j| < 1$ strictly for evanescent solutions and $|\lambda_j| \rightarrow \infty$ for those that grow into the lead; only the former survive the iteration. After n steps,

$$\|\alpha_n\|, \|\beta_n\| \sim \rho^{2^n}, \quad \rho := \max_j \{|\lambda_j| : |\lambda_j| < 1\} < 1, \quad (1.54)$$

i.e. double-exponential convergence in n . On a propagating band, $\max_j |\lambda_j| = 1$ exactly, and the convergence is entirely due to $\eta > 0$: writing $|\lambda_j(\omega^2 + i\eta)| = 1 - c_j \eta + \mathcal{O}(\eta^2)$,

$$\|\alpha_n\| \sim e^{-c_j \eta 2^n}, \quad (1.55)$$

which is still exponential in 2^n but with a small effective rate. At van Hove singularities where $v_g \rightarrow 0$, $c_j \rightarrow 0$ and the convergence becomes arbitrarily slow (the effective exponential rate $c_j \eta$ collapses), so many more decimation steps are needed.

Alternative algorithms. Several alternatives exist. Direct eigenvalue methods solve the quadratic eigenvalue problem $(\lambda^2 H_{10} + \lambda(H_{00} - \omega^2 \mathbb{I}) + H_{01})\xi = 0$ to obtain g_s in closed form [108, 109], and have been claimed to be faster than Sancho-Rubio for electron transport [109]. For phonons they have the additional advantage of producing mode-resolved transmission directly.

Ideal-lead boundary conditions. The contacts embed *ideal*, scattering-free reservoirs: the decimation Eqs. (1.47) to (1.50) is built from the bare harmonic lead blocks H_{00}, H_{01} and never sees the device anharmonic self-energy Σ_s . The scattering self-energy enters only the *device* Dyson equation Eq. (1.38); it is not folded back into the open-boundary problem. Feeding Σ_s into the contacts – effectively assuming the semi-infinite reservoir is as anharmonic as the device – makes the right-contact surface eigenproblem intermittently ill-conditioned once Σ_s grows under self-consistency, and was a source of SCBA divergence at finite anharmonicity. Keeping the leads ideal removes it and is the standard atomistic-NEGF boundary condition [40, 110].

1.2.5. Phonon–phonon scattering self-energy

The anharmonic self-energy has been reported in slightly different forms in the literature: the prefactor reported for the same diagram in Wang et al. [99, 103], Mingo [43], and Luisier [40] differ by factors of $\pi/9$ and 4, as analysed by Guo et al. [38] who also provided the diagrammatic derivation. We give a derivation below.

Let $\hat{V}_3(\tau) = \frac{1}{3!} \sum_{\mu\nu\lambda} \Phi_{\mu\nu\lambda} \tilde{u}_\mu(\tau) \tilde{u}_\nu(\tau) \tilde{u}_\lambda(\tau)$ be the cubic interaction in the interaction picture (With respect to the harmonic operators). For the interaction we compute the S-matrix [98, 111]

$$S_{\mathcal{C}} = T_{\mathcal{C}} \exp \left[-\frac{i}{\hbar} \oint_{\mathcal{C}} d\tau \hat{V}_3(\tau) \right]. \quad (1.56)$$

Wick’s theorem (valid for the contour-ordered Green’s function [99]) decomposes contour-ordered products into pair contractions. The full Green’s function is

$$G_{\mu\nu}(\tau, \tau') = -\frac{i}{\hbar} \frac{\langle T_{\mathcal{C}} S_{\mathcal{C}} \tilde{u}_\mu(\tau) \tilde{u}_\nu(\tau') \rangle_0}{\langle S_{\mathcal{C}} \rangle_0}, \quad (1.57)$$

and the linked-cluster theorem cancels the denominator against the disconnected diagrams in the numerator, leaving only connected diagrams in the self-energy.

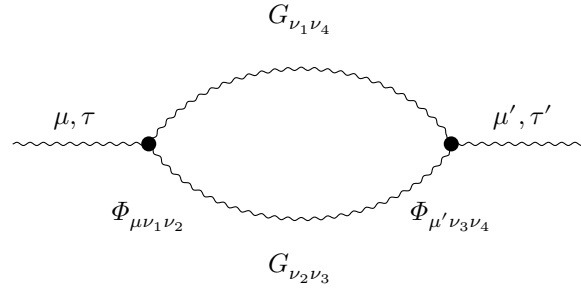


Figure 1.1.: Sunset (two-vertex bubble) Feynman diagram for the leading $(\Phi^{(3)})^2$ phonon–phonon self-energy of Eq. (1.58). Wavy lines denote phonon propagators G , filled circles denote cubic FC3 vertices. The two equivalent internal lines produce the symmetry factor $\frac{1}{2}$.

To order $(\Phi^{(3)})^2$, the proper self-energy is given by the “sunset” diagram (Fig. 1.1), in which two cubic vertices are connected by two internal phonon propagators. The value of the diagram, including all combinatorial factors, is [38, 100]

$$\Sigma_{\mu\mu'}^{(2)}(\tau, \tau') = \frac{i\hbar}{2} \sum_{\nu_1\nu_2\nu_3\nu_4} \Phi_{\mu\nu_1\nu_2} G_{\nu_1\nu_4}(\tau, \tau') G_{\nu_2\nu_3}(\tau, \tau') \Phi_{\mu'\nu_3\nu_4}. \quad (1.58)$$

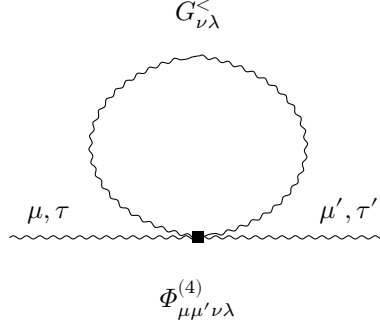


Figure 1.2.: Hartree-Fock (one-loop) tadpole self-energy from a single quartic FC4 vertex. The filled square denotes $\Phi_{\mu\mu'\nu\lambda}^{(4)}$. The two internal lines contract into the closed loop $G_{\nu\lambda}^{<}(\tau, \tau)$. The diagram is real and frequency-independent and acts as a temperature-dependent renormalisation $D \mapsto D + \Sigma^{(4),\text{HF}}$.

A second diagram, lower order in derivatives but one loop, survives: the FC4 Hartree-Fock “loop” tadpole $\Sigma_{\mu\mu'}^{(4),\text{HF}} = \frac{1}{2} \sum_{\nu\lambda} \Phi_{\mu\mu'\nu\lambda}^{(4)} \langle u_\nu u_\lambda \rangle = \frac{i\hbar}{2} \sum_{\nu\lambda} \Phi_{\mu\mu'\nu\lambda}^{(4)} G_{\nu\lambda}^{<}(\tau, \tau)$ from a single quartic vertex (Fig. 1.2). The loop self-energy is real and frequency-independent. It therefore produces no phonon linewidth, but shifts the phonon dispersion, $D \mapsto D + \Sigma^{(4),\text{HF}}$, providing the leading temperature-dependent mode renormalisation of self-consistent phonon theory [100, 112]. In a transport-focused SCBA targeting the imaginary part of Σ_s , the loop can be absorbed into a renormalised harmonic spectrum computed independently (e.g. via SCP or SSCHA [25, 112]) and is not re-included diagram-by-diagram in the iteration.

The factor 1/2 above corresponds to the fully-symmetric FC3 $\Phi_{\mu\nu\lambda}$ defined by $\Phi_{\mu\nu\lambda} = \partial^3 V / (\partial u_\mu \partial u_\nu \partial u_\lambda)$, divided by 3! in Eq. (1.29). Mingo [43] obtains an extra factor of $\pi/9$ which Guo et al. [38] attribute to a different convention in the symmetrization of the cubic vertex.

Applying the Langreth analytic continuation rules [98, 113] to the contour product of two G ’s in Eq. (1.58) yields real-time lesser/greater components,

$$\Sigma_{\mu\mu'}^{>}(\omega) = \frac{i\hbar}{2} \sum_{\nu_1\nu_2\nu_3\nu_4} \Phi_{\mu\nu_1\nu_2} \left[\int \frac{d\omega'}{2\pi} G_{\nu_1\nu_4}^{>}(\omega') G_{\nu_2\nu_3}^{>}(\omega - \omega') \right] \Phi_{\mu'\nu_3\nu_4}, \quad (1.59)$$

which is the form derived in detail in [38, 100]. The convolution enforces the energy-conservation $\delta(\omega - \omega' - \omega'')$ implicit in the contour integral. Substituting equilibrium

FDT Eq. (1.36) reproduces Fermi's-golden-rule expressions for three-phonon scattering rates: in mode basis,

$$\begin{aligned} \text{Im } \Sigma_s^R(\omega_s, \mathbf{q}_\perp) = & -\frac{\pi\hbar}{8\omega_s(\mathbf{q}_\perp)} \sum_{s_1 s_2} \sum_{\mathbf{q}_\perp'} |V_3(s\mathbf{q}_\perp; s_1\mathbf{q}_\perp'; s_2, \mathbf{q}_\perp - \mathbf{q}_\perp')|^2 \\ & \times \left\{ [1 + n_1 + n_2] \delta(\omega_s - \omega_{s_1} - \omega_{s_2}) + 2[n_2 - n_1] \delta(\omega_s - \omega_{s_1} + \omega_{s_2}) \right\}, \end{aligned} \quad (1.60)$$

with $n_i = n_{\text{BE}}(\omega_{s_i}, T)$, exhibiting the splitting/recombination ($1 \rightarrow 2$) channel and the absorption ($2 \rightarrow 1$) channel. This expression coincides with the Boltzmann result of Klemens [114] in the relaxation-time approximation.

Retarded self-energy

By construction Σ^R is causal in time, so its frequency components satisfy a Kramers–Kronig relation. Starting from $\Sigma^R = \frac{1}{2}(\Sigma^t - \Sigma^{\bar{t}}) + \frac{1}{2}(\Sigma^> - \Sigma^<)$ and using the spectral identity for $\Sigma^{t, \bar{t}}$, one obtains [38, 100]

$$\Sigma^R(\omega) = \frac{1}{2}[\Sigma^>(\omega) - \Sigma^<(\omega)] + i\mathcal{P} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{\Sigma^>(\omega') - \Sigma^<(\omega')}{\omega - \omega'}. \quad (1.61)$$

The antihermitian (dissipative) first term carries inverse-lifetime information,

$$\tau_s^{-1}(\mathbf{q}_\perp) \equiv -\frac{1}{\omega_s(\mathbf{q}_\perp)} \text{Im } \langle s | \Sigma^R(\omega_s, \mathbf{q}_\perp) | s \rangle, \quad (1.62)$$

while the dispersive second term encodes the anharmonic frequency renormalization $\Delta\omega_s \approx \text{Re } \Sigma^R/(2\omega_s)$ (mode hardening or softening). Numerical evaluation of the principal value is expensive, so in practice the dispersive part is often dropped:

$$\Sigma^R(\omega) \approx \frac{1}{2}[\Sigma^>(\omega) - \Sigma^<(\omega)]. \quad (1.63)$$

Half rule versus the causal transform. Dropping the dispersive term is harmless at finite η and on short devices, where $\text{Re } \Sigma^R$ is a small renormalisation and the lead broadening keeps the propagator well conditioned. At $\eta \rightarrow 0$ on longer devices it is not. The half rule Eq. (1.63) keeps only the anti-Hermitian part, so its Σ^R is no longer the causal Kramers–Kronig partner of the bubble $\Sigma^{\geq}$; the propagator $G^< = G^R \Sigma^< G^A$ built from it is then not the causal resolvent, and although the *lead-summed* balance of Eq. (1.77) still holds (the Keldysh route makes $D(\omega) \equiv 0$), the *per-interface* currents Eq. (1.74) become non-uniform and can turn negative, so the $\eta = 0$ fixed point is lost. Restoring the full transform Eq. (1.61) — evaluated by FFT through the Hilbert transform of the $\Sigma^{\geq}$ data — recovers a causal Σ^R , physical per-interface currents, and a stable conserving fixed point. The half rule is therefore the finite- η default, while the FFT/Kramers–Kronig form is mandatory for the $\eta \rightarrow 0$ limit of Section 1.2.7.

Storage sign convention. The production solver stores *occupation-positive* propagators, $-iG^{\lessgtr}(\omega) \gtrless 0$ for $\omega > 0$, the same sign as the lead injection $\Sigma_\ell^{\lessgtr} = +i n_\ell(\pm 1)\Gamma_\ell$ (Section A.2). The bubble Eq. (1.59) is quadratic in G and therefore returns a *textbook-signed* self-energy, $-i\Sigma^{\lessgtr} \gtrless 0$ — the opposite sign to what the Keldysh feedback Eq. (1.37) expects in this convention. The implementation accordingly negates $\Sigma^{\lessgtr}$ on output, while the dissipative retarded part Eq. (1.63) is formed from the *raw* (un-negated) values so that the damping $\Gamma_\Sigma = i(\Sigma^R - \Sigma^A) \gtrless 0$ keeps its sign. With the wrong sign the feedback is anti-dissipative and the SCBA diverges at any coupling; the single-shot positivity gate `sigma_convention` (Chapter F) verifies $-i\Sigma^{\lessgtr} \gtrless 0$ and catches exactly this error, which the sign-blind energy balance Eq. (1.89) cannot.

Detailed balance

The $\Sigma^{\lessgtr}$ of Eq. (1.59) satisfy detailed balance at equilibrium,

$$\Sigma^>(\omega) = e^{\beta\hbar\omega} \Sigma^<(\omega), \quad (1.64)$$

which together with the Keldysh equation Eq. (1.37) guarantees that $G^{\lessgtr}$ also satisfy detailed balance, recovering Eq. (1.36) when Σ_L, Σ_R correspond to a single temperature. The presence of two leads at different temperatures breaks detailed balance and drives a steady-state heat current.

Higher-order corrections

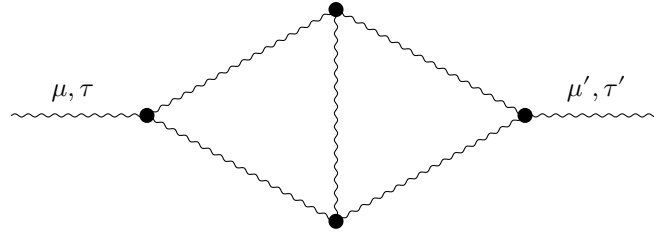


Figure 1.3.: Two-loop $\Phi^{(3)}$ vertex correction to the phonon self-energy ($\mathcal{O}((\Phi^{(3)})^4)$). Four cubic vertices form a 1PI “diamond” subgraph.

Equation (1.59) captures the leading three-phonon process. Higher orders include [38, 100]: (i) the four-phonon tadpole from $\Phi^{(4)}$, $\Sigma_{\mu\mu'}^{(4),\text{HF}} = \frac{i\hbar}{2} \sum_{\nu\lambda} \Phi_{\mu\mu'\nu\lambda}^{(4)} G_{\nu\lambda}^<(\tau, \tau)$, which is real, frequency-independent, and renormalizes $D \mapsto D + \Sigma^{(4),\text{HF}}$, contributing the temperature-dependent thermal expansion and frequency shifts [100] (Fig. 1.2), (ii) the two-loop $\Phi^{(3)}$ vertex correction, relevant at high T where $\hbar\omega_{\text{max}}/k_B T \lesssim 1$ and three-phonon processes saturate (Fig. 1.3) and (iii) crossed and ladder graphs (Fig. 1.4) that are needed to maintain Φ -derivability.

The Φ -derivability framework of Baym and Kadanoff [115, 116] states that Σ must be the functional derivative of a generating functional $\Phi[G]$ for the conservation laws

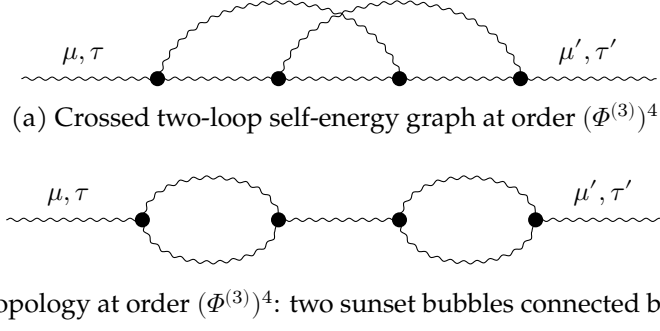


Figure 1.4.: Non-trivial two-loop self-energy topologies at order $(\Phi^{(3)})^4$ that lie beyond bubble-SCBA.

(energy, particle number) to be satisfied identically. The sunset diagram of Eq. (1.58) is Φ -derivable when the same G enters both internal lines, but only asymptotically as the SCBA iteration converges [104]. This is e.g. addressed by the LOA-based scheme of Lee et al. [104] which is exactly conserving at every iteration.

Transverse periodic case

For a 3D system with periodicity transverse to transport, the self-energy acquires a $\mathbf{q}_\perp$ argument and an internal sum over transverse momenta [38]:

$$\begin{aligned} \Sigma_s^{\geq}{}_{xx'}(\omega, \mathbf{q}_\perp) &= \frac{i\hbar}{2} \sum_{y_1 y_2 y_3 y_4} \frac{1}{N} \sum_{\mathbf{q}_\perp'} \tilde{\Phi}_{xy_1 y_2}(\mathbf{q}_\perp', \mathbf{q}_\perp - \mathbf{q}_\perp') \tilde{\Phi}_{x' y_3 y_4}(\mathbf{q}_\perp' - \mathbf{q}_\perp, -\mathbf{q}_\perp') \\ &\times \int \frac{d\omega'}{2\pi} G_{y_1 y_4}^{\geq}(\omega', \mathbf{q}_\perp') G_{y_2 y_3}^{\geq}(\omega - \omega', \mathbf{q}_\perp - \mathbf{q}_\perp'), \end{aligned} \quad (1.65)$$

where x, y denote transport-direction DOF indices, i.e. $x = (l_x, \kappa, \alpha)$ with l_x enumerating transport-direction slabs/supercells (here (κ, α) is the intra-slab DOF, distinct from the global index $\mu = (l, \kappa, \alpha)$ of Eq. (1.29)). The partially Fourier-transformed FC3 entering this expression is

$$\tilde{\Phi}_{xy_1 y_2}(\mathbf{q}_\perp, \mathbf{q}_\perp') = \tilde{\Phi}_{\mu\nu_1\nu_2}^{lm_1 m_2}(\mathbf{q}_\perp, \mathbf{q}_\perp') = \sum_{\substack{n_1, n_2 \\ n_{ix} = m_{ix}}} \Phi_{\mu\nu_1\nu_2}^{ln_1 n_2} e^{-i\mathbf{q}_\perp \cdot (n_1 - m_1)} e^{-i\mathbf{q}_\perp' \cdot (n_2 - m_2)}, \quad (1.66)$$

i.e. the sum runs over transverse supercells at fixed transport-direction cell. Unlike the harmonic FC2 object which depends on a single transverse wavevector, the cubic tensor depends on two: the $\mathbf{q}_\perp'$ sum couples all transverse wavevectors, so evaluating $\Sigma_s(\mathbf{q}_\perp)$ requires Green's functions at every $\mathbf{q}_\perp'$, destroying the embarrassing parallelism of the ballistic problem. The arithmetic complexity scales as $\mathcal{O}(N_q^2 N_\omega^2 N_{\text{DOF}}^4)$ (the N_{DOF}^4 , rather than the naive N_{DOF}^6 of the four-index ring, already assumes the FC3 block-locality of Section 1.3.4) before any low-rank or stochastic acceleration. This is the dominant cost of the anharmonic simulation, as reported in [38, 40, 104].

1.2.6. Observables

We collect the observables computable from $G^{R,A,\lessgtr}$ in both the ballistic and anharmonic regimes.

Spectral function and local density of states

The spectral function and LDOS are

$$A(\omega^2) = i[G^R - G^A], \quad \rho_\mu(\omega^2) = \frac{1}{\pi} A_{\mu\mu}(\omega^2), \quad (1.67)$$

related to the standard ω -resolved density of states by $\rho(\omega) = 2\omega \rho(\omega^2)$. In the transverse periodic case $\rho_\mu(\omega) = (2\omega/N_q) \sum_{\mathbf{q}_\perp} A_{\mu\mu}(\omega^2, \mathbf{q}_\perp)/\pi$.

Ballistic transmission and conductance

In the ballistic limit ($\Sigma_s = 0$) the transmission function is given by the Caroli formula [96, 101, 103],

$$\mathcal{T}(\omega^2, \mathbf{q}_\perp) = \text{Tr}[\Gamma_L G^R \Gamma_R G^A], \quad (1.68)$$

with the transverse-averaged transmission

$$\bar{\mathcal{T}}(\omega^2) = \frac{1}{N_q} \sum_{\mathbf{q}_\perp} \mathcal{T}(\omega^2, \mathbf{q}_\perp). \quad (1.69)$$

The thermal conductance per unit area is the Landauer integral

$$G(T) = \frac{1}{A_c} \int_0^\infty \frac{d\omega}{2\pi} \bar{\mathcal{T}}(\omega) \hbar\omega \frac{\partial n_{\text{BE}}}{\partial T}, \quad (1.70)$$

with A_c the cross-sectional area and

$$\frac{\partial n_{\text{BE}}}{\partial T} = \frac{\hbar\omega}{k_B T^2} \frac{e^{\hbar\omega/k_B T}}{(e^{\hbar\omega/k_B T} - 1)^2}. \quad (1.71)$$

The integrand vanishes at both endpoints, so a trapezoidal rule on the sampled frequency grid suffices.

Mode-resolved transmission

Diagonalizing Γ_ℓ at fixed $(\omega, \mathbf{q}_\perp)$ yields incoming and outgoing mode bases of the leads. Writing $\Gamma_L = U_L \Lambda_L U_L^\dagger$ and similarly for R , the transmission matrix element from incoming mode $m \in L$ to outgoing mode $m' \in R$ is

$$t_{m'm}(\omega, \mathbf{q}_\perp) = \sqrt{\Lambda_R^{m'}} [U_R^\dagger G^R U_L]_{m'm} \sqrt{\Lambda_L^m}, \quad (1.72)$$

with $\sum_{m'm} |t_{m'm}|^2 = \mathcal{T}$. This is the phonon analogue of the Fisher–Lee relation [117]. For phonons the mode-matching formulation [109] provides an alternative route to the same quantity through the eigenproblem of the bulk transfer matrix.

Meir–Wingreen heat current

In the anharmonic regime the Caroli formula no longer holds: there is no $\mathcal{T}(\omega)$ that captures both elastic and inelastic processes. The generalization is the phonon Meir–Wingreen formula [97, 99, 103],

$$I_L = \int_0^\infty \frac{d\omega}{4\pi} \hbar\omega \text{Tr}[\Sigma_L^<(\omega) G^>(\omega) - \Sigma_L^>(\omega) G^<(\omega)]. \quad (1.73)$$

Equation (1.73) is exact for any many-body Σ_s , provided $G^\lessgtr$ are obtained from the full Keldysh equation Eq. (1.37). In the ballistic limit, the Keldysh equation gives $G^< = G^R(\Sigma_L^< + \Sigma_R^<)G^A$, and substitution into Eq. (1.73) reproduces Eq. (1.68) weighted by $\hbar\omega[n_L - n_R]$, recovering Landauer.

Local heat current and conservation

The bond-resolved heat current of Hardy [118] can provide atomic insight. Between two transport-direction slabs i and $j = i + 1$,

$$I_{i,i+1} = - \int_0^\infty \frac{d\omega}{2\pi} \hbar\omega \text{Re Tr}[H_{i,i+1} G_{i+1,i}^<(\omega)], \quad (1.74)$$

with $H_{i,i+1}$ the inter-slab coupling block. In a converged SCBA solution the steady-state continuity equation enforces $I_{i,i+1}$ independent of i throughout the device, providing physically meaningful convergence test:

$$\delta I = \max_i I_{i,i+1} - \min_i I_{i,i+1} \quad (1.75)$$

measures the violation of energy conservation by approximations in Σ_s (incomplete SCBA iteration, Hilbert-transform truncation, finite frequency grid). A convergence criterion for SCBA can be $\delta I/\bar{I} \lesssim \varepsilon$ [38, 40] for appropriate ε (e.g. 10^{-3} in [38]).

Heat conservation at the level of Eq. (1.58) is guaranteed by Φ -derivability [116] only at full self-consistency. At any finite iteration count, $\delta I/\bar{I}$ is non-zero. The Hilbert-transform neglect of Eq. (1.63) introduces an additional $\mathcal{O}(\Delta\omega/\omega)$ conservation error.

The conservation identity. The statement can be made exact at the level of Eq. (1.73). Writing the $\hbar\omega$ -weighted energy current *into* the device from each terminal $a \in \{L, R, s\}$ (the two leads and the scattering “terminal”),

$$J_a = \int_0^\infty \frac{d\omega}{4\pi} \hbar\omega \text{Tr}[\Sigma_a^<(\omega) G^>(\omega) - \Sigma_a^>(\omega) G^<(\omega)], \quad \Sigma_{\text{tot}} = \Sigma_L + \Sigma_R + \Sigma_s, \quad (1.76)$$

steady-state energy balance is $J_L + J_R + J_s = 0$. Summing Eq. (1.76) over a and inserting the Keldysh form Eq. (1.37) $G^\lessgtr = G^R \Sigma_{\text{tot}}^\lessgtr G^A$ gives the identity

$$J_L + J_R + J_s = \int_0^\infty \frac{d\omega}{4\pi} \hbar\omega D(\omega), \quad D(\omega) = \text{Tr}[\Sigma_{\text{tot}}^<(G^R \Sigma_{\text{tot}}^> G^A - G^A \Sigma_{\text{tot}}^> G^R)], \quad (1.77)$$

where $D(\omega) \equiv \text{Tr}[\Sigma_{\text{tot}}^< G^> - \Sigma_{\text{tot}}^> G^<]$ has been rewritten as the *ordering commutator* on the right using the cyclic property of the trace. Energy conservation is thus the pointwise statement $D(\omega) \equiv 0$, which holds through either of two independent mechanisms. (i) *Keldysh route* ($\eta \rightarrow 0$). The half rule Eq. (1.63) satisfies $\Sigma^R - \Sigma^A = \Sigma^> - \Sigma^<$ (any Hermitian Kramers–Kronig part of Eq. (1.61) cancels in the difference), and the regulator term of Eq. (1.38), being $\mathcal{O}(\eta)$, drops from $[G^R]^{-1} - [G^A]^{-1}$; hence $G^> - G^< = G^R - G^A$, the two resolvent orderings in Eq. (1.77) coincide, and $D \equiv 0$ for arbitrary data. (ii) *Reciprocity route* (any η). If every matrix is complex-symmetric, $M = M^T$ — the reciprocity of a phonon system whose force constants obey Eq. (1.31) — the transpose identity $\text{Tr } M = \text{Tr } M^T$ maps one trace ordering in Eq. (1.77) into the other and $D \equiv 0$ identically. At finite η and broken reciprocity neither route applies and $D \neq 0$.

Total symmetry of the vertex and Φ -derivability. Vanishing of J_s alone is more restrictive. It requires that the scattering self-energy be the genuine functional derivative $\Sigma_s = \delta\Phi/\delta G$ of a Luttinger–Ward functional, evaluated with the *same* G on every internal line, and that the cubic vertex be totally symmetric under the permutation group S_3 , $\Phi_{\mu\nu\lambda} = \Phi_{\sigma(\mu)\sigma(\nu)\sigma(\lambda)}$ for all $\sigma \in S_3$. Only then do the $\hbar\omega$ -weighted triads of the sunset bubble Eq. (1.59) telescope pairwise to give $J_s = P_{\text{in}} - P_{\text{out}} = 0$, the energy-current conservation of the SCBA established in [99, 119] (the explicit S_3 telescoping is the present derivation). This symmetry is automatic for a vertex defined as a third derivative of the potential (Eq. (1.29)) [40]: a vertex that is not totally symmetric is not the third derivative of any potential, the theory is then not Φ -derivable [116], and $J_s \neq 0$ even at self-consistency. At an SCBA fixed point with an exactly S_3 vertex the lead currents balance, $J_L = -J_R$, at any η ; away from the fixed point (a mixed iterate, $\Sigma_s \neq \delta\Phi/\delta G[G]$) $J_s \neq 0$ and the slab imbalance Eq. (1.74) is finite. The imbalance is therefore a *convergence* diagnostic, not a defect of the formulation.

The broadening η as a ghost reservoir. The residual finite- η term in Eq. (1.77) has a transparent origin. The total anti-Hermitian broadening of the propagator,

$$i[(G^R)^{-\dagger} - (G^R)^{-1}] = \Gamma_L + \Gamma_R + \Gamma_s + \Gamma_\eta, \quad \Gamma_a = i(\Sigma_a^R - \Sigma_a^A), \quad (1.78)$$

shows that the numerical regulator of Eq. (1.38) adds a damping channel $\Gamma_\eta = \mathcal{O}(\eta)$ with *no* companion fluctuation term $\Sigma_\eta^<$. A conserving (Kadanoff–Baym) theory requires every dissipative self-energy to carry a matching $\Sigma^<$ fixed by fluctuation–dissipation balance, so η acts as a “ghost reservoir” at undefined occupation. In equilibrium, or whenever G is reciprocal, its in- and out-flows cancel and Eq. (1.77) vanishes; under a temperature bias the nonequilibrium $G^<$ is no longer complex-symmetric, the bubble inherits the asymmetry, and the product $\eta \times \text{asym}$ survives in $D(\omega)$ as a small spurious current. Consequently the slab/lead imbalance does not reach zero at finite η under bias but floors at the value of Eq. (1.77); the robust convergence gates are the self-energy residual and the pairing-exact bubble balance $P_{\text{in}} = P_{\text{out}}$, while transport ratios should be extrapolated to $\eta \rightarrow 0$. Chapter F verifies each step of this analysis numerically and quantifies the floor on production devices.

1.2.7. SCBA iterations

Equations (1.37), (1.38), (1.63) and (1.65) together define a nonlinear fixed-point problem $\Sigma_s = \mathcal{F}[G[\Sigma_s]]$, solved iteratively in the SCBA. Convergence is two-fold: numerical (consecutive iterates close in some norm) and physical (current conservation, Eq. (1.74)). Mixing schemes are necessary because pure fixed-point iteration of Eq. (1.65) is often divergent or oscillatory at finite anharmonicity [40, 103].

Linear mixing

The simplest scheme,

$$\Sigma^{(n+1)} = (1 - \alpha) \Sigma^{(n)} + \alpha \Sigma^{\text{new}}, \quad (1.79)$$

with damping parameter $\alpha \in (0, 1]$. Linear mixing converges if the fixed-point Jacobian $J = \partial \mathcal{F} / \partial \Sigma_s$ has spectral radius $\sigma((1 - \alpha)I + \alpha J) < 1$, i.e. $\alpha < 2 / (1 + |\sigma(J)|_{\max})$. For weak anharmonicity ($|\sigma(J)| < 1$) any $\alpha \leq 1$ works; for strong anharmonicity the spectrum of J extends close to -1 in some directions and small $\alpha \sim 0.1\text{--}0.3$ is required, with linear convergence rate $\max_\lambda |1 - \alpha + \alpha \lambda|$ set by the eigenvalue of J nearest $+1$ (which tends to 1 as the iteration approaches marginal stability).

Anderson mixing

Anderson acceleration [120] constructs a quasi-Newton update from a history of residuals,

$$\mathbf{x}_{k+1} = (\mathbf{x}_k + \alpha \mathbf{f}_k) - (\Delta \mathbf{X} + \alpha \Delta \mathbf{F}) \boldsymbol{\gamma}_k, \quad (1.80)$$

with $\mathbf{f}_k = G(\mathbf{x}_k) - \mathbf{x}_k$ the residual, $\Delta \mathbf{X}, \Delta \mathbf{F}$ the matrices of consecutive iterate and residual differences, and $\boldsymbol{\gamma}_k$ solving the regularized least-squares problem

$$(\Delta \mathbf{F}^H \Delta \mathbf{F} + \lambda \mathbb{I}) \boldsymbol{\gamma} = \Delta \mathbf{F}^H \mathbf{f}_k, \quad \lambda = 10^{-8} \text{Tr}(\Delta \mathbf{F}^H \Delta \mathbf{F}). \quad (1.81)$$

For linear problems Anderson mixing is essentially equivalent to GMRES applied to the residual equation [121, 122], with provable local q -linear convergence for depth-1 Anderson when the underlying iteration is a contraction [123, 124]. Local convergence in nonlinear electronic-structure problems is well established under mild Lipschitz assumptions on the fixed-point map [122, 123]. In practice, Anderson mixing dramatically accelerates convergence when the fixed-point equation is well-conditioned, with rates approaching that of Newton's method. However, storing the history is often prohibitive, so that Anderson mixing becomes infeasible for large structures.

Extrapolation and Newton root-finders for unstable fixed points

Linear and Anderson mixing both presuppose that the fixed point is *stable* under iteration, i.e. that the map Jacobian $G' \equiv \partial \mathcal{F} / \partial \Sigma_s$ is contractive, $|\sigma(G')| < 1$. As the numerical broadening is taken to zero (Section 1.2.7) this fails for sharp, weakly damped resonances: a few eigenvalues of G' acquire modulus ≥ 1 and no contraction-based scheme converges.

Three progressively stronger tools target this regime, all MPI-distributed over the energy grid; their global reductions (the residual inner products and norms, the RRE/RPM Gram solves $Z^H G' Z$, and the JFNK trust-region cap) are carried by the collective primitives in `quatrex.core.mpi_linalg` (`global_dot`, `global_norm`, `global_gram`, `trust_cap`), so the serial linear algebra written below is evaluated across the distributed energy grid.

Reduced-rank extrapolation (RRE). RRE [125] forms the affine combination $\Sigma = \sum_i \gamma_i \Sigma^{(i)}$, $\sum_i \gamma_i = 1$, of a short window of iterates that minimises the residual norm $\|\sum_i \gamma_i \mathbf{f}_i\|$ (a minimal-residual vector extrapolation; distinct from minimal-polynomial extrapolation, MPE). Restarted every few iterates, it recovers a mildly unstable fixed point from the algebra of the residual history rather than by contraction, at the cost of one small real Gram solve per cycle and no extra map evaluations.

Recursive projection method (RPM). When the unstable subspace is low-dimensional, the RPM of Shroff et al. [126] applies Newton on that subspace and Picard on its contractive complement,

$$\Sigma^{(n+1)} = \mathcal{F}[\Sigma^{(n)}] + Z(\mathbb{I}_k - H)^{-1} H Z^H \mathbf{f}^{(n)}, \quad \mathbf{f}^{(n)} = \mathcal{F}[\Sigma^{(n)}] - \Sigma^{(n)}, \quad (1.82)$$

where the columns of Z span the k -dimensional unstable invariant subspace and $H = Z^H G' Z$ is the restricted Jacobian. Both are obtained without extra map evaluations – Z from the dominant singular vectors of a sliding buffer of increments (a dynamic-mode decomposition) and H from the projected secants $Z^H \Delta \mathcal{F} = H Z^H \Delta \Sigma$. The correction multiplies $\mathbf{f}$ and so vanishes identically at the fixed point; the safeguarded modal step $\phi(\lambda) = \lambda/(1 - \lambda)$ is applied only on directions with $|\lambda(H)| > 1$, the rest being left to Picard.

Jacobian-free Newton–Krylov (JFNK). When the unstable subspace is larger, or its eigenvalues very large, we instead solve the Newton system for the residual $R(\Sigma) = \mathcal{F}[\Sigma] - \Sigma$,

$$(G' - \mathbb{I}) \delta = -R, \quad (1.83)$$

by GMRES [127], which requires only Jacobian–vector products. These are formed matrix-free by a finite difference of the map,

$$(G' - \mathbb{I}) \mathbf{v} = \frac{\mathcal{F}[\Sigma + \epsilon \mathbf{v}] - \mathcal{F}[\Sigma]}{\epsilon} - \mathbf{v}, \quad (1.84)$$

at one self-energy evaluation per Krylov vector. Because the self-energy map conjugates $-\Sigma^R$ through the Kramers–Kronig relation Eq. (1.61), $G^A = (G^R)^H$, and the bubble leg structure – G' is real-linear but not complex-analytic, so the Krylov solve is run in the real embedding $[\text{Re } \Sigma, \text{Im } \Sigma]$. The inner solve is truncated inexactly with an Eisenstat–Walker forcing term [128] and globalised by a trust-region cap on $\|\delta\|$, following Brown et al. [129] and Kelley [130]. A Levenberg–Marquardt / pseudo-transient shift $G' - \mathbb{I} \rightarrow G' - \mathbb{I} + \mu \mathbb{I}$ is inappropriate here: the Newton Jacobian of Eq. (1.83) is indefinite (contractive modes contribute eigenvalues in $[-2, 0)$, unstable modes positive eigenvalues), so a positive shift drives a negative eigenvalue through zero; the trust region is the only safe globalisation.

The $\eta \rightarrow 0$ limit and grid-consistent regularisation

The broadening η in Eq. (1.38) is a numerical device; the conserving transport limit is $\eta \rightarrow 0$. Its difficulty is a resolvent-sensitivity effect. Differentiating the Dyson equation gives $dG^R = G^R(d\Sigma^R)G^R$, so near a mode of frequency Ω and linewidth Γ the map gain scales as

$$\left| \frac{\partial G^R}{\partial \Sigma^R} \right| \sim |G^R|^2 \sim \frac{1}{(2\Omega\Gamma)^2}, \quad |\lambda(G')| \sim \frac{\Phi^2}{\max(\Gamma, c\Delta\omega)^2}, \quad (1.85)$$

the floor $c\Delta\omega$ arising because a Lorentzian narrower than the grid samples as an undamped pole. The iteration is therefore marginal or unstable wherever a heat-carrying mode is sharper than the grid, $\Gamma \lesssim \Delta\omega$ – the regime the root-finders above are built for. We pair them with two regularisations that vanish as $\Delta\omega \rightarrow 0$ and preserve the Φ -derivable conservation laws.

Smooth spectral window. Above the harmonic band top and inside any spectral gap the device Green's functions have no support, yet the convolution Eq. (1.65) leaks a small self-energy there which, at $\eta \rightarrow 0$, is divided by an undamped resolvent. Masking those bins with a hard window is itself harmful: a step discontinuity in Σ has a Kramers–Kronig partner $\sim \ln|\omega - \omega_{\text{edge}}|$ that rings (Gibbs), manufacturing a marginal mode at the mask edge. We instead apply a *smooth* multiplicative window $w(\omega) = w_{\text{supp}}(\omega) w_{\text{occ}}(\omega) \in [0, 1]$: a raised-cosine ramp of width a few $\Delta\omega$ on the distance to the harmonic support (computed once from the band structure $D(k) = D_0 + D_1 e^{ik} + D_1^H e^{-ik}$, independent of G), times a logistic in the Bose occupation that freezes thermally inactive spectator modes. The window multiplies the input Green's-function legs, the output $\Sigma^\lessgtr$, and the Kramers–Kronig input; because the same per-frequency factor multiplies both sides of the energy-balance identity Eq. (1.89), the equality $P_{\text{in}} = P_{\text{out}}$ is preserved and conservation is left intact.

Infrared occupancy taper. The Bose factor $n(\omega) \simeq k_B T / \hbar\omega$ diverges as $\omega \rightarrow 0$; sampled at the first grid bin it injects a spurious $\sim 1/\Delta\omega$ linewidth, whereas the acoustic sum rule forces $\Gamma \sim \omega^2 \rightarrow 0$. We restore the correct onset with the taper

$$t(\omega) = \frac{\omega^2}{\omega^2 + \omega_{\text{reg}}^2}, \quad \omega_{\text{reg}} = C\Delta\omega, \quad (1.86)$$

applied identically to both contact occupations so that every $G^<$ leg inherits it and the balance is preserved; as $\Delta\omega \rightarrow 0$, $\omega_{\text{reg}} \rightarrow 0$ and $t \rightarrow 1$, and the physical conductance is the $\omega_{\text{reg}} \rightarrow 0$ extrapolant. This taper is a *pragmatic* regulator: it is a cutoff, removed by the $\omega_{\text{reg}} \rightarrow 0$ extrapolation, and because it multiplies the occupations it slightly suppresses the genuine low-frequency transport. That suppression is small — the heat current is $\hbar\omega$ -weighted, so the infrared region carries little conductance — and, as Section 1.2.7 shows, the bare untapered bubble is itself exactly conserving; the taper is therefore a stabiliser of the marginal $\eta = 0$ iteration, not a correction to the physics.

Frequency-proportional regulator and the sub-grid soft-mode floor. The constant broadening $(\omega^2 + i\eta)\mathbb{I}$ of Eq. (1.38) is a convenient idealisation; the production solver instead uses a *frequency-proportional* regulator

$$[G^R]^{-1} = (\omega^2 + 2i\eta|\omega|)\mathbb{I} - D - \Sigma^R, \quad (1.87)$$

i.e. ω^2 shifted by $2i\eta|\omega|$ rather than by $i\eta$ or $(\omega + i\eta)^2$. The $(\omega + i\eta)^2$ form carries a spurious $-\eta^2$ band-edge shift that makes modes with $\omega \lesssim \eta$ artificially evanescent and suppresses the acoustic plateau; the linear-in- $|\omega|$ damping avoids this and vanishes as $\omega \rightarrow 0$, so it does not pollute the infrared. The price is that the acoustic *soft* modes ($D \rightarrow 0$ as $\omega \rightarrow 0$) are then left unregularised at $\eta \rightarrow 0$ ($G^R \sim 1/\omega^2$) and the marginal SCBA diverges. We restore conditioning with a DC-concentrated soft-mode broadening floor added to the *device* system matrix only (never to the ideal-lead OBC),

$$\Delta z^2(\omega) = i \Gamma_{\text{floor}} \frac{\omega_c^2}{\omega^2 + \omega_c^2}, \quad \Gamma_{\text{floor}} = (c_{\text{floor}} \Delta\omega)^2, \quad \omega_c = 2\Delta\omega, \quad (1.88)$$

a Lorentzian low-pass that damps only the lowest, grid-unresolved (and essentially heat-free) bins while leaving the resolved low- ω physics and the plateau untouched. Since $\Gamma_{\text{floor}} \propto (\Delta\omega)^2 \rightarrow 0$ under grid refinement, it is grid-consistent and removed in the same limit as the taper Eq. (1.86). The floor is *annealed* during the SCBA: c_{floor} ramps from a larger start to its target over the first iterations (the config knobs `eta_ir_floor_cells` $\rightarrow$ `eta_ir_floor_final_cells` over `eta_ir_floor_ramp_iterations`), so the soft modes are damped while Σ develops and the broadening is then relaxed toward zero at the converged fixed point. This floor-plus-anneal — not the spectral window or the occupancy taper alone — is what makes the $\eta = 0$ fixed point of soft systems such as the d5a nanowire reachable (Section 2.8).

Linewidth feasibility criterion. Whether $\eta = 0$ is reachable on a given grid can be decided a priori, without any SCBA, by evaluating the golden-rule linewidth $\Gamma_{\text{anh}}(\omega)$ (Eq. (1.60)) directly from $\Phi^{(3)}$ and the harmonic eigenvectors and comparing it to $\Delta\omega$: by Eq. (1.85) the fixed point is well posed only where every heat-carrying mode satisfies $\Gamma_{\text{anh}} \gtrsim \Delta\omega$.

Energy-balance diagnostic. Beyond the lead imbalance $\delta I/\bar{I}$, full Φ -derivable self-consistency implies the per-iteration identity

$$P_{\text{in}} = \sum_{\omega} \hbar\omega \text{Tr}[\Sigma^<(\omega)G^>(\omega)] = \sum_{\omega} \hbar\omega \text{Tr}[\Sigma^>(\omega)G^<(\omega)] = P_{\text{out}}, \quad (1.89)$$

the net power exchanged by the scattering self-energy. Its residual isolates vertex and convolution errors from contact errors and is the most stringent of our convergence checks.

Infrared limit of the heat current

The taper Eq. (1.86) regularises the iteration, not the physics; the infrared behaviour itself is benign, and no exact bubble-level subtraction is required. The Bose factor $n(\omega) = 1/(e^{\beta\hbar\omega} - 1)$ enters three places — the contact injection $\Sigma_\ell^\lessgtr = i\Gamma_\ell[n_\ell + (1 \pm 1)/2]$, the bubble legs $G^\lessgtr(\omega') \sim n(\omega')A(\omega')$ of Eq. (1.59), and the Meir–Wingreen integral Eq. (1.73) — through its Laurent (Bernoulli) expansion

$$n(\omega, T) = \frac{k_B T}{\hbar\omega} - \frac{1}{2} + \frac{\hbar\omega}{12k_B T} - \frac{(\hbar\omega)^3}{720(k_B T)^3} + \dots, \quad (1.90)$$

whose singular part is the c-number pole $k_B T / \hbar\omega$.

The pole is physical only in the observable. In the heat current the $1/\omega$ is cancelled by the explicit $\hbar\omega$ weight. The constant $-1/2$ is temperature-independent and drops from any lead difference, $\hbar\omega[n_L - n_R] = k_B(T_L - T_R) + \mathcal{O}(\omega^2)$, so the spectral density has the finite infrared limit

$$j(\omega) = \frac{\hbar\omega}{2\pi} \mathcal{J}(\omega) [n_L(\omega) - n_R(\omega)] \xrightarrow{\omega \rightarrow 0} \frac{k_B \Delta T}{2\pi} \mathcal{J}(0), \quad \mathcal{J}(0) = N_{\text{ac}}, \quad (1.91)$$

with N_{ac} the number of acoustic channels, each contributing the quantised thermal conductance $g_0 = \pi^2 k_B^2 T / 3h$ [131–133] (Chapter G for the derivation). This plateau is the physical infrared target; a regulator that multiplies the occupations by ω^2 , as the taper Eq. (1.86) does, removes a finite, quantised piece of it (Fig. 2.16).

There is no pole to subtract in the bubble. It is tempting to remove the $1/\omega$ from the scattering self-energy by the singularity-subtraction device of boundary-integral numerics [134–136]. Applied to the bubble legs this is both unnecessary and harmful. The acoustic spectral weight vanishes linearly, $A(\omega') \rightarrow A'(0)\omega'$, so the equilibrium lesser function $G_{\text{eq}}^\lessgtr = -i n A$ stays finite despite $n(\omega') \sim 1/\omega'$. Out of equilibrium the bosonic fold $G^\lessgtr(-\omega) = G^\gtr(\omega)^T$ together with the bounded spectral function forces the $1/\omega$ parts of $G^\lessgtr$ and $G^\gtr$ to be equal and opposite, hence both zero; and the contact broadening $\Gamma_\ell(\omega)$, being the bosonic (odd) anti-Hermitian part of Σ_ℓ^R , vanishes as $\omega \rightarrow 0$, so $\Gamma_\ell n_\ell$ is finite. The device bubble therefore carries no net infrared pole. Subtracting one regardless feeds the bubble a modified G , so that $\Sigma \neq \mathcal{F}[G]$ and the Φ -derivable balance Eq. (1.89) is broken. The accurate and conserving choice near $\omega = 0$ is the *bare* bubble; the taper is retained only as a numerical stabiliser of the marginal $\eta = 0$ iteration (Section 1.2.7), its small effect on the conductance removed by the $\omega_{\text{reg}} \rightarrow 0$ extrapolation rather than by any analytic subtraction. In a single nanotube the long-wavelength three-phonon linewidths scale per polarisation ($\Gamma_{\text{LA}} \sim |k|^{-1/2}$, $\Gamma_{\text{flexural}} \sim |k|^0$, $\Gamma_{\text{twist}} \sim |k|^{1/2}$) [137–139], so the converged anharmonic plateau need not equal the ballistic $\mathcal{J}(0) = N_{\text{ac}}$.

Convergence diagnostics

We monitor three quantities across iterations: (i) the relative Frobenius norm $\|\Sigma_s^{(n+1)} - \Sigma_s^{(n)}\| / \|\Sigma_s^{(n)}\|$, (ii) the conductance $G(T)$ from Eq. (1.70) or, in the inelastic case, the

Meir–Wingreen current, and (iii) the slab-resolved current imbalance $\delta I/\bar{I}$. Convergence is declared when (i) and (iii) fall below prescribed tolerances simultaneously. Under a temperature bias at finite η , however, (iii) does not reach zero but floors at the ghost-reservoir term of Eq. (1.77) (Chapter F); the primary gates are then (i) and the pairing-exact bubble balance $P_{\text{in}} = P_{\text{out}}$, with (iii) reserved as an η -budget diagnostic.

Lowest-order approximation.

An attractive alternative to direct SCBA is the lowest-order approximation (LOA), introduced for inelastic quantum transport by Mera et al. [140] and applied to anharmonic phonon transport by Lee et al. [104] and Lee et al. [141], implemented through Padé resummation. LOA collects only the conserving (Φ -derivable) diagrams at each order in $\Phi^{(3)}$, ensuring exact current conservation at every iteration, then accelerates the resulting series by Padé approximants. Compared to SCBA, LOA achieves comparable accuracy at $\sim 2\times$ lower cost for thermal transport in 3D nanostructures [104].

1.2.8. Decay properties and Truncation errors

Truncations at every step (real-space FC cutoff, finite device length, finite ω grid, finite Sancho-Rubio iteration count) introduce errors with quantifiable scaling. We summarize, from first principles, the decay rates of each of the involved quantities.

Real-space decay of harmonic force constants

The asymptotic behaviour of $\Phi^{(2)}(R)$ at large $|R|$ is controlled by the analytic structure of the dielectric response of the underlying electronic system. Cancès et al. [142] prove the following within the reduced Hartree–Fock (or random-phase approximation) model:

Insulators at zero electronic temperature. The dynamical matrix $D(\mathbf{q}_\perp)$ is singular at $\mathbf{q}_\perp = 0$, with the non-analytic dipole–dipole structure given in [143]:

$$D_{\kappa\alpha,\kappa'\beta}^{\text{LR}}(\mathbf{q}_\perp \rightarrow 0) \sim \frac{4\pi}{\Omega_{\text{cell}}} \frac{(\mathbf{q}_\perp \cdot \mathbf{Z}_\kappa^*)_\alpha (\mathbf{q}_\perp \cdot \mathbf{Z}_{\kappa'}^*)_\beta}{\mathbf{q}_\perp \cdot \epsilon_\infty \cdot \mathbf{q}_\perp}, \quad (1.92)$$

where $\mathbf{Z}_\kappa^*$ is the Born effective charge tensor of atom κ and ϵ_∞ the optical dielectric tensor. Inverse Fourier transform of this $\mathbf{q}_\perp \rightarrow 0$ singularity gives an algebraic real-space tail to Φ .

Materials at finite electronic temperature (metals). Cancès et al. [142] prove that the dynamical matrix is analytic in $\mathbf{q}_\perp$, and $\Phi^{(2)}(R)$ decays exponentially:

$$|\Phi^{(2)}(R)| \leq C e^{-|R|/\xi_e}, \quad \xi_e \sim \frac{\hbar v_F}{k_B T_{\text{el}}}. \quad (1.93)$$

Concrete truncation error. Truncating at distance R_c introduces an error in the dynamical matrix of order

$$\|D(\mathbf{q}_\perp) - D^{R_c}(\mathbf{q}_\perp)\| \lesssim C \frac{e^{-R_c/\xi_e}}{1 - e^{-1/\xi_e}} \quad (\text{metals, exponential}), \quad (1.94)$$

or, in polar insulators, $\sim 1/R_c^3$ for the bare truncation, but exponential, $\sim e^{-R_c/\xi_g}$, once the $1/r^3$ Cochran–Cowley tail has been subtracted analytically and added back exactly via Ewald summation [7].

Transverse $\mathbf{q}_\perp$ -mesh convergence

The convergence of observables in N_q depends on the differentiability of the integrand. For the Caroli formula Eq. (1.68), the $\mathbf{q}_\perp$ -integrand is generically smooth and trapezoidal Monkhorst–Pack sampling converges as $\mathcal{O}(N_q^{-2})$ in 2D. At Brillouin-zone boundaries or near van Hove singularities, convergence degrades to $\mathcal{O}(N_q^{-1})$; a Gaussian smearing or tetrahedron method may then be preferable. The self-energy convolution Eq. (1.65) couples $\mathbf{q}_\perp$ and $\mathbf{q}_\perp'$, so the effective integrand has $N_q^{-1/2}$ statistical-noise structure under random sampling, and reducing this toward $\mathcal{O}(N_q^{-1})$ requires variance reduction beyond plain importance sampling, e.g. quasi-Monte-Carlo / low-discrepancy sampling [104].

Convergence rate of Sancho-Rubio decimation revisited

As discussed in Section 1.2.4, the decimation iteration Eqs. (1.47) to (1.50) effectively doubles the number of decimated layers at each step, with iterates obeying $\|\alpha_n\|, \|\beta_n\| \sim \rho^{2^n}$ inside gaps and $e^{-c\eta \cdot 2^n}$ on bands. The practical implication is that *doubling the tolerance roughly costs one extra iteration*; total cost is $\mathcal{O}(N_\omega N_q N_{\text{DOF}}^3 N_{\text{iter}})$ with N_{iter} essentially constant (and small, ~ 10 – 30).

1.3. Computation of the Phonon-Phonon Scattering Self Energy

The phonon-phonon SSE in Eq. (1.65) involves three computational steps that dominate the cost of an SCBA iteration of the anharmonic SCBA equations: a four-index ring contraction over y_1, y_2, y_3, y_4 , an internal sum over the transverse wavevector $\mathbf{q}'_\perp$, and a frequency convolution along ω . The frequency convolution is structurally identical to the polarisation that appears in the GW formalism [144],

$$P_{ij}^{\gtrless}(E) = -i \int dE' G_{ij}^{\gtrless}(E') G_{ji}^{\gtrless}(E' - E), \quad (1.95)$$

and reuses the same FFT-based machinery. The $\mathbf{q}'_\perp$ -sum couples distinct transverse wavevectors through the FC3 vertices $\tilde{\Phi}$ and is unique to the transverse-periodic geometry. The inner four-index contraction has no analogue in GW: in Eq. (1.95), the contraction structure of P is local in the orbital pair (i, j) , whereas the phonon bubble couples $y_1, \dots, y_4$ across the supercell. In what follows we omit the $\gtrless$ superscripts unless the

distinction matters; all derivations apply unchanged to lesser, greater, and (via the fluctuation–dissipation construction of Eq. (1.63)) retarded self-energies.

1.3.1. Frequency convolution and discretisation

The integrand in Eq. (1.65) contains a single frequency convolution $\int \frac{d\omega'}{2\pi} G_1(\omega') G_2(\omega - \omega')$. We turn it into a pointwise product in time through the Fourier transform pair

$$g(\tau; \cdot) = \int \frac{d\omega}{2\pi} G(\omega; \cdot) e^{-i\omega\tau}, \quad G(\omega; \cdot) = \int d\tau g(\tau; \cdot) e^{+i\omega\tau}, \quad (1.96)$$

under which the 1D convolution theorem gives

$$\int \frac{d\omega'}{2\pi} G_1(\omega'; \cdot) G_2(\omega - \omega'; \cdot) = \int d\tau g_1(\tau; \cdot) g_2(\tau; \cdot) e^{+i\omega\tau}. \quad (1.97)$$

Substituting Eq. (1.97) into Eq. (1.65), the bubble takes the τ -domain form

$$\begin{aligned} \Sigma_{xx'}(\omega; \mathbf{q}_\perp) &= \frac{i\hbar}{2} \sum_{y_1 y_2 y_3 y_4} \frac{1}{N} \sum_{\mathbf{q}'_\perp} \tilde{\Phi}_{xy_1 y_2}(\mathbf{q}'_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp) \tilde{\Phi}_{x' y_3 y_4}(\mathbf{q}'_\perp - \mathbf{q}_\perp, -\mathbf{q}'_\perp) \\ &\quad \times \int d\tau g_{y_1 y_4}(\tau; \mathbf{q}'_\perp) g_{y_2 y_3}(\tau; \mathbf{q}_\perp - \mathbf{q}'_\perp) e^{+i\omega\tau}. \end{aligned} \quad (1.98)$$

The ω -integral has been replaced by a τ -integral, the two propagators are now multiplied pointwise, and a final $\tau \rightarrow \omega$ Fourier transform delivers Σ on the original frequency grid.

Discrete frequency grid. Numerically, the SSE is evaluated on a uniform grid $\omega_n = \omega_{\min} + n\Delta\omega$ for $n = 0, \dots, N_\omega - 1$, with broadening $\eta = \eta_{\text{factor}}(\Delta\omega)^2$ to suppress spurious poles at the resolution scale of the grid. The convolution is evaluated by FFT on a zero-padded grid of length

$$N_{\text{fft}} = 2N_\omega - 1, \quad (1.99)$$

which is the smallest length that avoids cyclic wrap-around between the two factors when both have support of width N_ω . Stages 1 and 3 (Section 1.3.7) operate on N_{fft} -long arrays, the SSE on the original grid is recovered by truncation to the first N_ω points after the inverse FFT.

Convolution support: $f_{\max} \geq 2\omega_{\max}/2\pi$. The harmonic Green's functions $G^{\gtrless}(\omega)$ have spectral support bounded by the largest phonon frequency $\omega_{\max}$ of the device, i.e. the spectrum of $H_{00} + H_{01} + H_{01}^\dagger$. The product $g_1(\tau) g_2(\tau)$ in Eq. (1.97) has Fourier support that is the convolution sum of the individual supports, so $\Sigma^{\gtrless}(\omega)$ is nonzero on

$$\omega \in [-2\omega_{\max}, 2\omega_{\max}] \quad (1.100)$$

and only there. The truncated linear convolution Eq. (1.99) is unbiased on that interval *only when the grid covers it*, i.e. when $f_{\max} \geq 2\omega_{\max}/2\pi$. If the grid is shorter, $f_{\max} <$

$2\omega_{\max}/2\pi$, the bubble contributions at $|\omega| \in (f_{\max}, 2\omega_{\max})$ are missing from the inverse FFT and, on a discrete grid, the residual aliases back into the visible window. The bubble then carries large spurious contributions that fake a strong-coupling self-energy and drive the SCBA loop unstable. The required grid size is therefore set by the stiffest mode of the device, not by the thermally populated bands; in particular, an H-terminated wire with $\omega_{\max}/2\pi \sim 64$ THz from the Si-H stretch needs $f_{\max} \gtrsim 130$ THz, well above the ~ 32 THz a bulk-Si device ($\omega_{\max}/2\pi \sim 16$ THz) would require. Chapter E documents the runtime safety net that enforces this condition.

Negative-frequency handling. The bosonic phonon Green's functions live on a real frequency grid centred on $\omega = 0$. Both $\Sigma^<$ and $\Sigma^>$ involve products of $G^{\lessgtr}$ at shifted arguments. Depending on the relative sign of ω and ω' , these give two distinct convolution patterns. Following the convention used in the GW SSE [144], we can split each product into a forward convolution on the positive-frequency support and a cross-correlation against the conjugate on the negative-frequency support. Concretely, defining

$$\text{conv}(f, g)[\omega] = \int d\omega' f(\omega') g(\omega - \omega'), \quad \text{corr}(f, g)[\omega] = \int d\omega' f(\omega') g(\omega' - \omega), \quad (1.101)$$

the bubble splits as

$$\Sigma^{\lessgtr}(\omega) \propto \text{conv}(g^{\lessgtr}, g^{\lessgtr})[\omega] + \text{corr}(g^{\lessgtr}, (g^{\lessgtr})^*)[\omega], \quad (1.102)$$

the second term capturing contributions where one factor's frequency exceeds ω in absolute value. Both pieces are evaluated by FFT on the padded grid Eq. (1.99). The only book-keeping is the sign convention on the frequency axis. The retarded SSE is constructed from $\Sigma^> - \Sigma^<$ via Eq. (1.63), optionally with the Hilbert-transform real-part correction obtained by another FFT-based pass.

Exact bosonic continuation (production). The conjugation form $\text{corr}(g^{\lessgtr}, (g^{\lessgtr})^*)$ above is exact only when the propagators are complex-symmetric (equilibrium, or a reciprocal device). Under a temperature bias $G^{\lessgtr}$ are no longer symmetric, and the negative-frequency (absorption) leg must be continued with the full bosonic relation

$$G_{ij}^<(\mathbf{q}_{\perp}, -\omega) = G_{ji}^>(-\mathbf{q}_{\perp}, \omega), \quad (1.103)$$

i.e. with a spatial *transpose* ($ij \rightarrow ji$) and a transverse-momentum negation ($\mathbf{q}_{\perp} \rightarrow -\mathbf{q}_{\perp}$), not merely a conjugation. The production kernel (`quatrex.phonon.sse_phonon_phonon, _build_fold_plan/_fold_reversed_legs`) folds the $\omega < 0$ contribution onto the one-sided $\omega \geq 0$ grid through Eq. (1.103), so each $\Sigma^{\lessgtr}$ is the sum of the forward ring and the transposed/ $\mathbf{q}_{\perp}$ -reversed ring. This retains the inter-block (off-diagonal) coherence that the equilibrium conjugation shortcut discards and keeps the bubble exact and conserving away from equilibrium (Chapter F).

1.3.2. Inner ring contraction

At each working point $(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp)$, the SSE entry $\Sigma_{xx'}$ contains the four-tensor ring contraction

$$\mathcal{B}_{xx'} = \sum_{y_1 y_2 y_3 y_4} \Phi_{xy_1 y_2} g_{y_1 y_4} g_{y_2 y_3} \Phi_{x' y_3 y_4}, \quad (1.104)$$

where Φ stands for either $\tilde{\Phi}$ at the working $(\mathbf{q}_\perp, \mathbf{q}'_\perp)$ or the real-space FC3, and the indices each range over $n_{\text{dof}} = N_{AO} = N_B N_{BS}$ degrees of freedom. The graph defined by the index structure is a four-cycle: $x \rightarrow y_1 \rightarrow y_4 \rightarrow x'$ via Φ_L and g , and $x' \rightarrow y_3 \rightarrow y_2 \rightarrow x$ via Φ_R and the second g . The two Φ legs and two g legs alternate around the ring.

A ring of four tensors admits three structurally distinct evaluation orders, distinguished by which pair of adjacent tensors is contracted first.

Outside-in. Contract one Φ leg at a time, starting from one corner of the ring. The intermediate $A_{x,c,d} = \sum_e \Phi_{x,c,e} g_{e,d}$ has three free indices and per-entry cost $\mathcal{O}(n_{\text{dof}})$. Building A for all (x, c, d) costs $\mathcal{O}(n_{\text{dof}}^4)$ and uses $\mathcal{O}(n_{\text{dof}}^3)$ memory; iterating around the ring gives two more steps of the same shape, for a total cost $\mathcal{O}(n_{\text{dof}}^4)$.

Balanced tree. Compute the two halves

$$T_1(x)_{y_2 y_4} = \sum_{y_1} \Phi_{xy_1 y_2} g_{y_1 y_4}, \quad T_2(x')_{y_2 y_4} = \sum_{y_3} g_{y_2 y_3} \Phi_{x' y_3 y_4}, \quad (1.105)$$

in parallel, then contract along the shared (y_2, y_4) indices,

$$\mathcal{B}_{xx'} = \sum_{y_2 y_4} T_1(x)_{y_2 y_4} T_2(x')_{y_2 y_4}. \quad (1.106)$$

Each half-tree has the same arithmetic shape as one outside-in step ($\mathcal{O}(n_{\text{dof}}^4)$ to build, $\mathcal{O}(n_{\text{dof}}^3)$ memory per piece). The final pairing has cost $\mathcal{O}(n_{\text{dof}}^4)$. The total is again $\mathcal{O}(n_{\text{dof}}^4)$ but now exposes two independent halves that can be distributed across disjoint resources.

Inside-out. Contract the two propagators first into the four-index intermediate $C_{y_1 y_4 y_2 y_3} = g_{y_1 y_4} g_{y_2 y_3}$, then sandwich with Φ_L, Φ_R . This intermediate has four free spatial indices and $\mathcal{O}(n_{\text{dof}}^4)$ entries; building it costs $\mathcal{O}(n_{\text{dof}}^4)$ and the subsequent contraction with each Φ costs $\mathcal{O}(n_{\text{dof}}^5)$. The intermediate also defeats sparsity exploitation, because the propagator pair is generically denser than either g alone. We discard this order in what follows.

Outside-in (i) and balanced-tree (ii) have identical asymptotic cost

$$\mathcal{C}_{\text{ic}}^{\text{dense}} = \mathcal{O}(n_{\text{dof}}^4), \quad (1.107)$$

which we take as the unit cost throughout. preferable when the two halves can be evaluated on different ranks, which is the case in our distribution scheme; otherwise the two are interchangeable. Outside-in is the natural form for a single-rank evaluation and is the form we use to express the per-block kernel in Section 1.3.6.

1.3.3. Q-space and real-space formulations

We present two evaluation strategies of Eq. (1.98) that differ in how they treat the $\mathbf{q}'_{\perp}$ -sum: a direct evaluation in the Brillouin zone (“q-space” below), or a reformulation in real space via a second application of the convolution theorem.

Q-space formulation. The direct evaluation retains the explicit $\mathbf{q}'_{\perp}$ -loop. At each working point $(\tau, \mathbf{q}_{\perp}, \mathbf{q}'_{\perp})$, the partially Fourier-transformed FC3 $\tilde{\Phi}_{xy_1y_2}(\mathbf{q}'_{\perp}, \mathbf{q}_{\perp} - \mathbf{q}'_{\perp})$ is needed. We assemble it on the fly by streaming through the $N_{\perp}^2$ non-zero real-space FC3 entries with the corresponding phase factors. Because $N_{\perp}^2 \ll n_{\text{dof}}^2$ for realistic FC3 cut-offs, this phase assembly is negligible compared with the subsequent ring contraction. We never materialise the dense $\tilde{\Phi}$ at every $(\mathbf{q}_{\perp}, \mathbf{q}'_{\perp})$. Defining the τ -bubble at fixed transverse $\mathbf{q}_{\perp}$,

$$\begin{aligned} \mathcal{B}_{xx'}(\tau; \mathbf{q}_{\perp}) \equiv & \frac{1}{N} \sum_{\mathbf{q}'_{\perp}} \sum_{y_1 y_2 y_3 y_4} \tilde{\Phi}_{xy_1y_2}(\mathbf{q}'_{\perp}, \mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) g_{y_1 y_4}(\tau; \mathbf{q}'_{\perp}) \\ & \times g_{y_2 y_3}(\tau; \mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) \tilde{\Phi}_{x' y_3 y_4}(\mathbf{q}'_{\perp} - \mathbf{q}_{\perp}, -\mathbf{q}'_{\perp}), \end{aligned} \quad (1.108)$$

the SSE follows from a single $\tau \rightarrow \omega$ Fourier transform. The total cost is

$$\mathcal{W}_q = N_{\tau} N_q^2 \mathcal{C}_{\text{ic}}, \quad (1.109)$$

with $N_{\tau} = N_{\text{fft}}$ the number of frequency grid points, N_q the number of transverse Brillouin-zone samples (e.g. $N_q = 400$ for a 20×20 grid). The factor N_q^2 is the direct consequence of the two transverse momentum arguments of $\tilde{\Phi}$ and is the primary cost increase of the transverse-periodic problem over the purely local one.

Real-space formulation. The N_q^2 factor in Eq. (1.109) can in principle be eliminated by applying the convolution theorem to the $\mathbf{q}'_{\perp}$ -sum and evaluating the resulting expression on a finite real-space lattice of $\mathbf{R}$ values.

Substituting the Fourier expansion Eq. (1.66) of both FC3 factors into Eq. (1.65):

$$\tilde{\Phi}_{xy_1y_2}(\mathbf{q}'_{\perp}, \mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) = \sum_{\boldsymbol{\rho}_1, \boldsymbol{\rho}_2} \Phi_{xy_1y_2}(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) e^{-i\mathbf{q}'_{\perp} \cdot \boldsymbol{\rho}_1} e^{-i(\mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) \cdot \boldsymbol{\rho}_2}, \quad (1.110)$$

$$\tilde{\Phi}_{x' y_3 y_4}(\mathbf{q}'_{\perp} - \mathbf{q}_{\perp}, -\mathbf{q}'_{\perp}) = \sum_{\boldsymbol{\rho}_3, \boldsymbol{\rho}_4} \Phi_{x' y_3 y_4}(\boldsymbol{\rho}_3, \boldsymbol{\rho}_4) e^{-i(\mathbf{q}'_{\perp} - \mathbf{q}_{\perp}) \cdot \boldsymbol{\rho}_3} e^{+i\mathbf{q}'_{\perp} \cdot \boldsymbol{\rho}_4}. \quad (1.111)$$

Collecting the four phase factors gives

$$e^{-i\mathbf{q}'_{\perp} \cdot \boldsymbol{\rho}_1} e^{-i(\mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) \cdot \boldsymbol{\rho}_2} e^{-i(\mathbf{q}'_{\perp} - \mathbf{q}_{\perp}) \cdot \boldsymbol{\rho}_3} e^{+i\mathbf{q}'_{\perp} \cdot \boldsymbol{\rho}_4} = \underbrace{e^{+i\mathbf{q}_{\perp} \cdot (\boldsymbol{\rho}_3 - \boldsymbol{\rho}_2)}}_{\mathbf{q}_{\perp}\text{-dep.}} \cdot \underbrace{e^{-i\mathbf{q}'_{\perp} \cdot \mathbf{S}}}_{\mathbf{q}'_{\perp}\text{-dep.}}, \quad (1.112)$$

with $\mathbf{S} = \boldsymbol{\rho}_1 - \boldsymbol{\rho}_2 + \boldsymbol{\rho}_3 - \boldsymbol{\rho}_4$. The $\mathbf{q}_{\perp}$ -dependence is now a single phase multiplying everything, and the $\mathbf{q}'_{\perp}$ -dependence is in the second factor and in the two g -arguments.

Pulling the $\mathbf{q}_\perp$ -phase outside the $\mathbf{q}'_\perp$ -sum and applying the convolution theorem to the $\mathbf{q}'_\perp$ -sum yields

$$\begin{aligned} \frac{1}{N} \sum_{\mathbf{q}'_\perp} e^{-i\mathbf{q}'_\perp \cdot \mathbf{S}} G_{y_1 y_4}(\omega'; \mathbf{q}'_\perp) G_{y_2 y_3}(\omega - \omega'; \mathbf{q}_\perp - \mathbf{q}'_\perp) \\ = \sum_{\mathbf{R}} g_{y_1 y_4}(\omega'; \mathbf{R} - \mathbf{S}) g_{y_2 y_3}(\omega - \omega'; \mathbf{R}) e^{-i\mathbf{q}_\perp \cdot \mathbf{R}}, \end{aligned} \quad (1.113)$$

where the shift by $\mathbf{S}$ in the first g argument reflects the multiplication by $e^{-i\mathbf{q}'_\perp \cdot \mathbf{S}}$ in $\mathbf{q}'_\perp$ -space. A change of variables $\mathbf{R}' = \mathbf{R} - (\boldsymbol{\rho}_3 - \boldsymbol{\rho}_2)$ combined with the identity

$$\boldsymbol{\rho}_3 - \boldsymbol{\rho}_2 - \mathbf{S} = \boldsymbol{\rho}_4 - \boldsymbol{\rho}_1 \quad (1.114)$$

cancels the $\mathbf{q}_\perp$ -dependent phase against the $e^{-i\mathbf{q}_\perp \cdot \mathbf{R}}$ factor produced by Eq. (1.113), leaving

$$\Sigma_{xx'}^{\geq}(\omega; \mathbf{q}_\perp) = \frac{i\hbar}{2} \sum_{\mathbf{R}} \int d\tau \mathcal{B}_{xx'}(\tau; \mathbf{R}) e^{+i\omega\tau} e^{-i\mathbf{q}_\perp \cdot \mathbf{R}}, \quad (1.115)$$

with the real-space bubble

$$\begin{aligned} \mathcal{B}_{xx'}(\tau; \mathbf{R}) \equiv \sum_{\substack{\boldsymbol{\rho}_1, \dots, \boldsymbol{\rho}_4 \\ y_1, \dots, y_4}} \Phi_{xy_1 y_2}(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) \Phi_{x'y_3 y_4}(\boldsymbol{\rho}_3, \boldsymbol{\rho}_4) \\ \times g_{y_1 y_4}(\tau; \mathbf{R} + \boldsymbol{\rho}_4 - \boldsymbol{\rho}_1) g_{y_2 y_3}(\tau; \mathbf{R} + \boldsymbol{\rho}_3 - \boldsymbol{\rho}_2). \end{aligned} \quad (1.116)$$

The structure of Eq. (1.116) replaces the explicit $\mathbf{q}'_\perp$ -sum by a loop over $N_D \leq N_\perp^4$ FC3-displacement quadruples and an effective lattice $\mathcal{R}_{\text{eff}}$ of size $N_T^{\text{eff}} = (4N_{\text{cut}} + 1)^2$, the latter following from the largest possible $\mathbf{R} + \boldsymbol{\rho}_i - \boldsymbol{\rho}_j$ shift with $|\boldsymbol{\rho}_i| \leq N_{\text{cut}}$. The total cost is

$$\mathcal{W}_R = N_\tau N_D N_T^{\text{eff}} \mathcal{C}_{\text{ic}}. \quad (1.117)$$

Comparing Eqs. (1.109) and (1.117), the real-space approach becomes favourable when $N_D N_T^{\text{eff}} < N_q^2$. With $N_{\text{cut}} = 3$ and $N_\perp = 7$ (a typical FC3 support along each transverse direction), $N_D \sim 7^4 \approx 2400$ and $N_T^{\text{eff}} = 13^2 \approx 170$, giving $N_D N_T^{\text{eff}} \sim 4 \times 10^5$. For a 20×20 transverse grid ($N_q = 400$, $N_q^2 = 1.6 \times 10^5$), the q-space form is already cheaper, and the gap widens for finer Brillouin-zone sampling.

1.3.4. Sparsity layers

Four sparsity patterns reduce the dense ring cost Eq. (1.107) by several orders of magnitude before any further approximation is introduced. These mirror the sparsity exploitations of the GW solver [144, 145] and follow from the same physical assumption – a real-space cut-off on inter-atomic interactions – applied to the harmonic and cubic parts of the lattice Hamiltonian.

We denote by r_{cut} the physical real-space cut-off radius applied to both IFC2 and IFC3 (in Å), by r_c the corresponding bandwidth on the block-tridiagonal representation of $G^{\geq}$ (the number of primitive blocks within r_{cut} along the transport direction), by $N_{\text{c},x}$ the FC3 cut-off in transport-cell units along the transport direction ($N_{\text{c},x} \leq 3$ typically), and by C_a the number of atoms within r_{cut} of any reference atom.

(a) BT-band cut-off on $G^{\geq}$. As in the electronic GW problem, applying r_{cut} to the harmonic dynamical matrix gives $G^{\geq}$ a block-banded structure of width r_c along the transport direction. Block entries $G_{KK'}$ are non-zero only for $|K - K'| \leq r_c$. The total non-zero count of G per frequency, before any intra-block sparsity, is

$$\text{nnz}(G)_{\text{BB}} = \mathcal{O}(N_B r_c N_{BS}^2) = \mathcal{O}(N_{AO} r_c N_{BS}). \quad (1.118)$$

(b) Inter-block FC3 cut-off. The decay of cubic force constants with pair distance (Section 1.1.5) admits the same real-space cut-off r_{cut} on $\Phi^{(3)}$. For transport-cell labels (I, K_1, K_2) , $\tilde{\Phi}_{I,K_1,K_2}^{(3)}$ is non-zero only when

$$|I - K_1|, \quad |I - K_2|, \quad |K_1 - K_2| \leq N_{c,x}. \quad (1.119)$$

The block-pair index

$$\mathcal{P}(I, J) = \{(K_1, K_2) : \Phi_{I,K_1,K_2}^{(3)} \text{ and } \Phi_{J,K_2,K_1}^{(3)} \text{ both present}\}, \quad (1.120)$$

encodes which inner block-pairs contribute to a given output block-pair (I, J) . From Eq. (1.119) together with the BT-band requirement on the output ($|I - J| \leq 1$), the cardinality of $\mathcal{P}(I, J)$ is bounded by $|\mathcal{P}(I, J)| = \mathcal{O}(N_{c,x}^2)$, with the proportionality constant set by the lattice geometry. For a cubic lattice and $N_{c,x} = 3$, $|\mathcal{P}(I, J)| \leq 49$ in the worst case. The partial Fourier transform $\Phi^{(3)} \rightarrow \tilde{\Phi}^{(3)}(\mathbf{q}_{\perp}, \mathbf{q}'_{\perp})$ over the transverse cells preserves block-sparsity along the transport direction unchanged.

(c) Output BT band. Σ inherits the BT structure of the upstream SCBA solver: only (I, J) with $|I - J| \leq 1$ are retained. The number of distinct output block-pairs is therefore $\mathcal{O}(N_B)$, and the bubble decomposes into a finite sum

$$\Sigma_{IJ}(\omega; \mathbf{q}_{\perp}) = \frac{i\hbar}{2} \sum_{(K_1, K_2) \in \mathcal{P}(I, J)} \mathcal{B}_{IJ}^{K_1 K_2}(\omega; \mathbf{q}_{\perp}). \quad (1.121)$$

This output truncation is consistent with the input truncation (a) on G .

(d) Intra-block atom-pair sparsity. Within a single transport cell of size $N_{BS} = 3N_a N_U$ (N_a atoms per primitive cell, N_U primitive cells per transport cell), the same cut-off r_{cut} acts at the atomic level. Each atom couples to only C_a neighbours within r_{cut} , so each row of an intra-block matrix carries $\mathcal{O}(C_a)$ non-zero atom pairs (times $3 \times 3 = 9$ Cartesian-direction entries each, absorbed into the constant). The block-level non-zero counts are therefore

$$\text{nnz}(G_{KK'}^{\geq}) = \mathcal{O}(N_{BS} C_a), \quad \text{nnz}(\Phi_{I,K_1,K_2}^{(3)}) = \mathcal{O}(N_{BS} C_a^2). \quad (1.122)$$

Sparsity propagation through the kernel. The sparsity patterns above are not lost by the half-tree contractions in Eqs. (1.105) and (1.106). They propagate through the pipeline in the same way that they do in sparse-sparse matrix multiplication. We make this precise for the half-tree $T_1(x)_{y_2 y_4} = \sum_{y_1} \Phi_{x, y_1, y_2} g_{y_1, y_4}$.

For fixed transport-cell labels (I, K_1, K_2) within FC3 support, the non-zero entries of $\Phi_{I, K_1, K_2}^{(3)}$ at the atomic level correspond to triples (μ, ν_1, ν_2) where atoms μ, ν_1, ν_2 are mutually within r_{cut} of one another. For each such (μ, ν_1) , g_{ν_1, ν_4} is non-zero only for ν_4 within r_{cut} of ν_1 . The contraction therefore scans $\text{nnz}(\Phi) \cdot C_a = \mathcal{O}(N_{BS} C_a^3)$ entries to produce T_1 at fixed (I, K_1, K_2) , and T_1 itself has support on (I, K_1, K_2) atoms-within-cut-off triples, with non-zero count $\mathcal{O}(N_{BS} C_a^2)$. The second half-tree has the same shape; the final ring-closing contraction over the shared (ν_2, ν_4) indices runs over the joint support, again $\mathcal{O}(N_{BS} C_a^3)$ per working point. The per-pair ring cost reduces from N_{BS}^4 to

$$\mathcal{C}_{\text{ic}}^{\text{sparse}} = \mathcal{O}(N_{BS} C_a^3), \quad (1.123)$$

a saving of $(N_{BS}/C_a)^3 = (N_a/C_a)^3$ over the dense kernel. For Si nanoribbons with $N_a/C_a \sim 25$, this is roughly 1.5×10^4 .

Combining Eq. (1.123) with the q-space cost Eq. (1.109) and the $\mathcal{P}$ -cardinality bound $|\mathcal{P}(I, J)| = \mathcal{O}(N_{c,x}^2)$, the total q-space SSE cost reduces to

$$\mathcal{W}_{\text{phph}} = \mathcal{O}(N_\tau N_q^2 N_B N_{c,x}^2 N_{BS} C_a^3) = \mathcal{O}(N_\tau N_q^2 N_{AO} N_{c,x}^2 C_a^3). \quad (1.124)$$

1.3.5. Separable FC3 approximation

The intra-block sparsity layer (d) reduces the per-pair ring cost from N_{BS}^4 to $N_{BS} C_a^3$, but C_a^3 remains the dominant arithmetic factor at production block sizes. The factor can be reduced further by exploiting low-rank structure in the FC3 itself, replacing Φ by a sum-of-products approximation that factorises the ring into a product of two independent bilinear forms.

Treating $\Phi_{xy_1 y_2}(\rho_1, \rho_2)$ as a matrix with row index (ρ_1, y_1) and column index (ρ_2, y_2) , a rank- R truncated SVD yields

$$\Phi_{xy_1 y_2}(\rho_1, \rho_2) \approx \sum_{r=1}^R f_{y_1, r}^x(\rho_1) h_{y_2, r}(\rho_2), \quad (1.125)$$

with $R \ll N_{BS}$, and analogously for the second FC3 factor with factors f', h' at rank R . Substituting both into Eq. (1.104) gives the factorised ring

$$\mathcal{B}_{xx'} = \sum_{r, s} \underbrace{\left[\sum_{y_1, y_4} f_{y_1, r}^x g_{y_1 y_4} h'_{y_4, s} \right]}_{P_{rs}^x} \cdot \underbrace{\left[\sum_{y_2, y_3} h_{y_2, r} g_{y_2 y_3} f'_{y_3, s} \right]}_{Q_{rs}^{x'}}. \quad (1.126)$$

Naive evaluation of P_{rs}^x as R^2 independent bilinears costs $\mathcal{O}(R^2 n_{\text{dof}}^2)$ per working point. Sharing the matrix-vector products across (r, s) pairs gives a two-stage evaluation:

1. For each s , compute the projected propagator $\mathbf{v}_s^{y_1} = \sum_{y_4} g_{y_1, y_4} h'_{y_4, s}$. With g sparse (nnz $N_{BS} C_a$ per block, banded with bandwidth r_c), each $\mathbf{v}_s$ costs $\mathcal{O}(n_{\text{dof}} r_c C_a)$. Total over s : $\mathcal{O}(R n_{\text{dof}} r_c C_a)$.
2. For each (r, s, x) , compute the inner product $P_{rs}^x = \sum_{y_1} f_{y_1, r}^x \mathbf{v}_s^{y_1}$ at cost $\mathcal{O}(C_a)$ per output (intra-block-sparse f). Total over (r, s, x) : $\mathcal{O}(R^2 n_{\text{dof}} C_a)$.

The combined per-pair cost is

$$\mathcal{O}(R n_{\text{dof}} r_c C_a + R^2 n_{\text{dof}} C_a), \quad (1.127)$$

to be compared with the sparse ring cost $\mathcal{O}(N_{BS} C_a^3)$ of Eq. (1.123) under the assumption that intra-block sparsity is exploitable (cf. the forward note in Section 1.3.4: in the current dense-kernel implementation $C_a = N_{BS}$, and the comparison collapses to the asymptotically equivalent $\mathcal{O}(N_{BS}^3)$ baseline). For $R \ll C_a^2$ the separable form is the cheaper one. Structurally richer decompositions (Tucker, CP, INDSCAL, symmetric CP; Section 1.1.5) plug into the same kernel with different factor matrices and the same two-stage evaluation.

In the transverse-periodic setting, Fourier-transforming each factor in Eq. (1.125) turns the $\mathbf{q}'_{\perp}$ -sum at fixed (r, s) into a convolution evaluated by FFT. The dense $n_{\text{dof}}^2 \times N_q^2$ ring is replaced by R^2 pointwise products of two FFT-convolved factors, each of which inherits the two-stage assembly cost Eq. (1.127).

1.3.6. Algebraic structure: FFT placement and evaluation order

The bubble involves five elementary operations: forward FFT $\mathcal{F}_{\omega}$ along ω , inverse FFT $\mathcal{F}_{\tau}^{-1}$ along τ , contractions $\mathcal{C}_L$ and $\mathcal{C}_R$ with Φ_L and Φ_R , and the pointwise product $\mathcal{M}_{\tau}$ that closes the ring at fixed τ . Because $\mathcal{F}_{\omega}$ acts on the ω -axis while $\mathcal{C}_{L/R}$ acts only on index axes, the two commute,

$$\mathcal{F}_{\omega} \circ \mathcal{C}_{L/R} = \mathcal{C}_{L/R} \circ \mathcal{F}_{\omega}, \quad (1.128)$$

and we may choose the order of operations.

FFT placement. Two orderings are possible.

(i) *FFT-first.* Take the input $G^{\geq}(\omega)$ to the τ -domain once, and perform all FC3 contractions on $g(\tau)$. The FFT input has nnz(G) entries (Eq. (1.118) times intra-block sparsity), each FFT'd independently along ω at cost $\mathcal{O}(N_{\omega} \log N_{\omega})$. Total FFT work:

$$W_{\text{FFT}}^{\text{first}} = \mathcal{O}(N_{\omega} \log N_{\omega} \cdot N_{AO} r_c C_a). \quad (1.129)$$

(ii) *Contract-first.* Perform all index contractions in the ω -domain to produce a four-index intermediate of FC3 shape, then FFT the result. The intermediate has nnz proportional to $\text{nnz}(\Phi^{(3)}) \sim N_{AO} N_{c,x}^2 C_a^2$, each entry needs an FFT along ω at cost $\mathcal{O}(N_{\omega} \log N_{\omega})$. Total FFT work:

$$W_{\text{FFT}}^{\text{after}} = \mathcal{O}(N_{\omega} \log N_{\omega} \cdot N_{AO} N_{c,x}^2 C_a^2). \quad (1.130)$$

The FFT-first ordering is therefore the clearly superior choice.

Per-pair kernel. With the FFT-first ordering, the balanced-tree (or, equivalently for one-rank evaluation, outside-in) contraction at fixed $(I, J, K_1, K_2, K'_1, K'_2) \in \mathcal{P}$ – with $\Phi_L \equiv \Phi_{I, K_1, K_2}^{(3)}$, $\Phi_R \equiv \Phi_{J, K'_2, K'_1}^{(3)}$, and the (generally off-diagonal) link propagators $g_a \equiv g_{K_1 K'_1}(\tau)$, $g_b \equiv g_{K_2 K'_2}(\tau)$ – takes the form

$$A_{w,a,c,d} = \sum_e (\Phi_L)_{a,c,e} (g_b)_{w,e,d}, \quad (1.131)$$

$$B_{w,a,b,d} = \sum_c A_{w,a,c,d} (g_a)_{w,c,b}, \quad (1.132)$$

$$\hat{\Sigma}_{w,a,J} = \sum_{b,d} B_{w,a,b,d} (\Phi_R)_{J,d,b}, \quad (1.133)$$

where w enumerates the padded τ -grid of size $N_{\text{fft}} = 2N_\omega - 1$, $\Phi_L \equiv \Phi_{I, K_1, K_2}^{(3)}$ and $\Phi_R \equiv \Phi_{J, K'_2, K'_1}^{(3)}$ are the FC3 blocks for the working triplet, and $g_a \equiv \mathcal{F}_\omega^{-1}[G_{K_1 \bullet}^{\geq}]$, $g_b \equiv \mathcal{F}_\omega^{-1}[G_{K_2 \bullet}^{\geq}]$ are the FFTs of the relevant blocks of G along the transport direction. The intermediates A, B are stored only on the joint sparse support of their inputs: A on $\text{supp}(\Phi_L) \cap \text{supp}(g_b)$ (which has $\text{nnz} \mathcal{O}(N_{BS} C_a^2)$ per τ at fixed (I, K_1, K_2)), and similarly for B .

An optimisation arises from sharing intermediates across FC3 triplets that share an inner block. For fixed K_1 , all $(I, K_2) \in \mathcal{P}$ produce A -tensors that share the same g_a -input. The $\mathcal{C}_L$ contraction can therefore be batched into a single sparse einsum over the FC3 entries with shared K_1 . Analogous sharing applies across K_2 .

1.3.7. Distribution and parallelisation

The FFT-first pipeline factorises into three computational stages, each of which is local in a different index. Let P_E be the number of ranks distributed along the ω -axis, P_S the number of ranks distributed along the transport-direction blocks I (the spatial decomposition of [145]), and $P = P_E P_S$ the total rank count.

Stages. We can split the computation in three stages,

$$g_{ij}(\tau) = \mathcal{F}_{\omega \rightarrow \tau}[G_{ij}(\omega)] \quad (\text{Stage 1: forward FFT, local in } (i, j)), \quad (1.134)$$

$$\sigma_{xx'}(\tau) = \sum_{(K_1, K_2) \in \mathcal{P}(I, J)} \mathcal{B}_{IJ}^{K_1 K_2}(\tau) \quad (\text{Stage 2: ring contraction, local in } \tau), \quad (1.135)$$

$$\Sigma_{xx'}(\omega) = \frac{i\hbar}{2} \mathcal{F}_{\tau \rightarrow \omega}^{-1}[\sigma_{xx'}(\tau)] \quad (\text{Stage 3: inverse FFT, local in } (I, J)). \quad (1.136)$$

Stage 1 acts independently on each (i, j) entry, Stage 2 acts independently at each τ but couples block-diagonal g -blocks at different (K_1, K_2) , and Stage 3 acts independently on each output (I, J) .

Following the GW solver, we use a block-banded sparse container with two interconvertible distributions:

- **stack**: each rank holds an ω -slice (or τ -slice after FFT) for the full BT band. The non-zero pattern of the BT band is replicated on every rank, the ω -axis is partitioned across P_E .
- **nnz**: each rank holds the full ω -axis for an (i, j) -slice of the BT band. The frequency axis is replicated, the non-zero pattern is partitioned across $P_E P_S = P$.

The `dtranspose` primitive converts between the two via a balanced `Alltoall`. Stage 1 and Stage 3 are local in **nnz**; Stage 2 is local in **stack**. Placing each stage in its preferred distribution gives a four-redistribution pipeline:

- (i) **stack** $\rightarrow$ **nnz** on $G^{\geq}$.
- (ii) **nnz**: FFT $G \rightarrow g(\tau)$.
- (iii) **nnz** $\rightarrow$ **stack** on $g(\tau)$.
- (iv) **stack**: per- τ ring contraction Eqs. (1.131) to (1.133) (Stage 2).
- (v) **stack** $\rightarrow$ **nnz** on $\sigma(\tau)$.
- (vi) **nnz**: IFFT $\sigma \rightarrow \Sigma$ (Stage 3).
- (vii) **nnz** $\rightarrow$ **stack** on Σ for hand-off to the next SCBA iteration.

The four redistributions occur at steps (i), (iii), (v), (vii).

Per-rank cost derivation. Each stage's cost follows from the sparsity bookkeeping of Section 1.3.4 together with the distribution above.

*Stage 1 (FFT, G in **nnz**).* Memory: each rank holds the full ω -axis for its share of the BT-band non-zero blocks. With $\text{nnz}(G)$ per ω given by Eq. (1.118), the per-rank footprint is

$$M_{\text{rank}}^{(1)} = \mathcal{O}\left(\frac{N_\omega \cdot N_{AO} r_c N_{BS}}{P}\right). \quad (1.137)$$

Computation: each non-zero (I, J) block is FFT'd elementwise along ω ; per-rank work is the same expression times $\log N_\omega$:

$$W_{\text{rank}}^{(1)} = \mathcal{O}\left(\frac{N_\omega \log N_\omega \cdot N_{AO} r_c N_{BS}}{P}\right). \quad (1.138)$$

Communication: the **stack** $\rightarrow$ **nnz** `dtranspose` has the same volume as the memory footprint (this is the entire payload that traverses the `dtranspose`).

*Stage 2 (ring contraction, $g(\tau)$ in **stack**).* The arithmetic at each working point is bounded by Eq. (1.107); summed over all working points and distributed across all ranks,

$$W_{\text{rank}}^{(2)} = \mathcal{O}\left(\frac{N_q^2 N_\tau N_{AO} N_{c,x}^2 N_{BS}^3}{P}\right). \quad (1.139)$$

Memory: dominated by the FC3 dictionary, which is partitioned along its first leg I across P_S but not along ω (the FC3 is ω -independent). Each P_S rank therefore holds

$$M_{\text{rank}}^{(2)} = \mathcal{O}\left(\frac{N_{AO} N_{c,x}^2 N_{BS}^2}{P_S}\right). \quad (1.140)$$

The $g(\tau)$ payload itself is in `stack` and adds an $\mathcal{O}(M_{\text{rank}}^{(1)})$ contribution which is sub-leading at production block sizes. Communication: the only Stage 2 communication is the spatial halo exchange discussed below.

Stage 3 (IFFT, $\sigma(\tau)$ in `nnz`). The output Σ inherits the BT band of G : only $|I - J| \leq 1$ blocks are retained, each stored densely. The output entry count per ω is therefore $\mathcal{O}(N_{AO} N_{BS})$ (the BT-band factor r_c of G is replaced by the strict $|I - J| \leq 1$ output band). Memory and computation scale as

$$M_{\text{rank}}^{(3)} = \mathcal{O}\left(\frac{N_{\omega} N_{AO} N_{BS}}{P}\right), \quad W_{\text{rank}}^{(3)} = \mathcal{O}\left(\frac{N_{\omega} \log N_{\omega} N_{AO} N_{BS}}{P}\right). \quad (1.141)$$

Spatial decomposition and halo. When $P_S > 1$ ranks partition the transport-direction blocks, the FC3 cut-off and the G -band imply that an output block $I \in \mathcal{J}_p$ couples only to band links $g_{KK'}$ within a shell of width $N_h = N_{c,x} + r_c$ of $\mathcal{J}_p$ (the pair-index constraint that Φ_{J,K'_2,K'_1} exist often tightens this to the vertex reach $N_{c,x}$). Outputs are assigned by the $\min(I, J)$ rule so the owner can both compute and store Σ_{IJ} (Theorem 1.1); each rank then needs only its assigned range $\mathcal{J}_p$ plus a width- N_h halo fetched from its two immediate neighbours. The halo traffic per (P_E, P_S) rank is

$$V_{\text{halo}}^{\text{rank}} = \mathcal{O}\left(\frac{N_{\tau} N_h N_{BS}^2}{P_E}\right), \quad (1.142)$$

independent of N_B (a surface term), plus $\mathcal{O}(\alpha)$ latency for the $\mathcal{O}(1)$ neighbour messages per iteration.

Aggregate per-rank cost. The dominant cost contributions balance across stages and ranks, giving

$$W_{\text{phph}}^{\text{rank}} = \mathcal{O}\left(\frac{N_q^2 N_{\tau} N_{AO} N_{c,x}^2 N_{BS}^3}{P}\right), \quad (1.143)$$

$$V_{\text{phph}}^{\text{rank}} = \mathcal{O}\left(\frac{N_{\omega} N_{AO} r_c N_{BS}}{P}\right), \quad (1.144)$$

$$M_{\text{phph}}^{\text{rank}} = \mathcal{O}\left(\frac{N_{\omega} N_{AO} r_c N_{BS}}{P}\right) + \mathcal{O}\left(\frac{N_{AO} N_{c,x}^2 N_{BS}^2}{P_S}\right). \quad (1.145)$$

The two terms in Eq. (1.145) reflect the two kinds of data resident on a rank: the ω -distributed G and Σ payloads (split over P) and the ω -independent FC3 dictionary (split over P_S only). Stage 2 dominates the arithmetic; Stages 1 and 3 dominate the communication.

1.3.8. Memory and batching

The peak memory consumer in the per-pair kernel is the intermediate A or B in Eqs. (1.131) and (1.132), both of which are dense N_{BS}^3 -shaped tensors per τ , i.e. $\mathcal{O}(N_{\text{fft}} N_{BS}^3)$ entries across the padded grid. At production block sizes ($N_{BS} \sim 2 \times 10^3$, $N_{\omega} \sim 10^4$), this is several TB per block triplet in complex FP64 – well beyond a single device. Two batching axes keep the resident working set bounded:

- The τ -axis is point-wise within Stage 2 and can be batched without inducing additional communication. Batches of N_b contiguous τ -slices are sized adaptively from the free HBM at runtime, mirroring the GW polarisation kernel’s batching strategy (PCoulombScreening.compute in [145]).
- The output (I, J) -axis is batched analogously, since the contributions from different (I, J) pairs are independent; the per-batch intermediate footprint is then $\mathcal{O}(N_b N_{BS}^3)$ per active (I, J) pair.

The FC3 dictionary, with non-zero count $\mathcal{O}(N_B N_{c,x}^2 N_{BS}^3)$ for block-dense storage, is partitioned along its first leg I across the P_S spatial ranks. With $N_{BS} \sim 2000$, $N_{c,x} = 3$ shells, the per-rank FC3 footprint is hundreds of GB on a single device – the single-device regime is only practical for much smaller transport cells ($N_{BS} \lesssim 300$). The compressed FC3 representations of Eq. (1.125) and Section 1.1.5 replace the per-block N_{BS}^3 count by $R N_{BS}$ at moderate R and integrate into the same kernel and distribution scheme without further modification; together with the GPU-resident dense BLAS that already dominates Stage 2’s arithmetic, they bring per-rank FC3 storage and per-pair arithmetic back to the few-GB / few-TFLOP regime that fits a single HBM device.

1.3.9. Implemented distributed phonon-phonon SSE

The current quatrex production implementation (`quatrex.phonon.sse_phonon_phonon.SigmaPhononPhono`) realises the dense-block kernel of Eq. (1.104) as a *full off-diagonal ring*: every band link $g_{KK'}^{\geq}$ with $|K - K'| \leq r_c$ enters Stage 2 – not merely the diagonal g_{KK} – so the inter-slab coherence that a diagonal- G approximation discards is retained, while the output band stays $|I - J| \leq 1$. It uses no FC3 decomposition (Section 1.4 remains a future optimisation) and follows the FFT-first pipeline of Section 1.3.7 verbatim, distributing over the frequency (P_E , `comm.stack`) and transport-block (P_S , `comm.block`) axes.

FFT-first pipeline. A single SCBA iteration runs the six steps of Section 1.3.7, reusing the GW solver’s dt transpose machinery [145] for the four redistributions:

- State-restoration contract.* The incoming distribution state of every buffer is recorded on entry and restored in a `finally` block, so the routine composes with the surrounding SCBA loop (which hands the Green’s functions over in `nnz`).
- Forward FFT in nnz.* With $G^{\geq}$ in `nnz` (full ω local, non-zeros partitioned over P), the padded forward transform $g^{\geq}(\tau) = \mathcal{F}\{G^{\geq}(\omega)\}$ is a local operation into cached

length- N_τ buffers that share G 's exact non-zero ordering; the FFT work and this storage divide over P .

- (iii) *Redistribute $g(\tau)$ to stack.* Each `comm.stack` rank now owns a τ -slice of the full band, block-addressable; the spatial halo (below) supplies the off-window neighbour blocks.
- (iv) *Ring contraction per τ (Stage 2), in stack.* For each owned output (I, J) the rank sums $\mathcal{P}(I, J)$ of $\Phi g_{K_1 K'_1}(\tau) g_{K_2 K'_2}(\tau) \Phi$ at its τ -slice (the three-einsum `ring_contract` kernel), streaming Φ from the block dictionary.
- (v) *Redistribute $\sigma(\tau)$ to nnz.* A *partition*, not a broadcast: each (I, J) full- τ lands on exactly one rank, which is what prevents a P_E -fold double count.
- (vi) *Inverse FFT in nnz.* $\Sigma^\geq(\omega) = \frac{i\hbar}{2} \mathcal{F}^{-1}\{\sigma^\geq(\tau)\}|_{[0:N_\omega]}$ added into the outputs; the retarded part follows from the bosonic Hilbert transform $\Sigma^R = \frac{1}{2}(\Sigma^> - \Sigma^<) + \frac{i}{2}\mathcal{H}\{\Sigma^> - \Sigma^<\}$, evaluated along the (locally complete) ω axis.

Because the contraction is in τ -space and partitioned over P_E , the full- ω band is never replicated on every rank – the memory bottleneck of the earlier diagonal-gather design.

Spatial halo and addressing. Under $P_S > 1$ the output (I, J) is assigned to the rank owning block $\min(I, J)$; arrow partitioning (Theorem 1.1) guarantees that this rank stores Σ_{IJ} and can write it through the block indexer. The ring needs band blocks $g_{KK'}(\tau)$ that may lie one shell outside the rank's window; those are fetched from the immediate `comm.block` neighbours by a width- N_h ($N_h = N_{c,x} + r_c$) halo. Each rank sends a neighbour exactly the links that neighbour's outputs need and which fall in its own window, and receives the symmetric set; the send/receive lists follow from the global pair index and block ranges, so both ends enumerate them in the same order and the non-overtaking `Isend/Irecv` match per (source, tag) (the bidirectional exchange of `qttools.datastructures.routines.arrow_partition_halo_comm`, reused by the recursive Green's-function solver). The halo blocks are read transparently alongside local blocks during the contraction.

Per-rank cost. The arithmetic is the FFT plus the dense ring summed over the rank's owned outputs and τ -slice, both divided over the full $P = P_E P_S$:

$$W_{\text{phph,impl}}^{\text{rank}} = \mathcal{O}\left(\frac{N_\tau \log N_\tau N_B r_c N_{BS}^2}{P}\right) + \mathcal{O}\left(\frac{N_\tau N_B N_{c,x}^2 N_{BS}^4}{P}\right). \quad (1.146)$$

Communication is the four energy redistributions plus the spatial halo,

$$V_{\text{phph,impl}}^{\text{rank}} = \underbrace{\mathcal{O}\left(\frac{N_\omega N_B r_c N_{BS}^2}{P}\right)}_{4 \text{ dt transpose}} + \underbrace{\mathcal{O}\left(\frac{N_\tau N_h N_{BS}^2}{P_E}\right)}_{\text{halo, Eq. (1.142)}} + \mathcal{O}(\alpha), \quad (1.147)$$

with α the per-message latency of the $\mathcal{O}(1)$ neighbour exchanges. The halo term is independent of N_B (surface, not volume). Memory divides over P for the frequency-resident data, over P_S for the ω -independent FC3 dictionary, plus a transient halo buffer:

$$M_{\text{phph,impl}}^{\text{rank}} = \mathcal{O}\left(\frac{N_\tau N_B r_c N_{BS}^2}{P}\right) + \mathcal{O}\left(\frac{N_B N_{c,x}^2 N_{BS}^3}{P_S}\right) + \mathcal{O}\left(\frac{N_\tau N_h N_{BS}^2}{P_E}\right). \quad (1.148)$$

Every intrinsic-data term thus divides over the full rank count; the only residual overheads are the two *irreducible* couplings – the energy all-to-all (the convolution couples all frequencies) and the spatial halo (the operator is banded) – both $\mathcal{O}(\text{surface})$. This is the communication-optimal point for the finite block-tridiagonal device.

Lemma 1.1 (Arrow partitioning resolves the boundary block). *The distributed container partitions the block-sparse non-zeros arrow-wise (`dsdbcoo.py`): rank r , owning the global element window $[s_r, e_r)$, stores every entry whose row and column both lie at or beyond s_r and at least one of which lies before e_r . Consequently each band block $g_{KK'}$ is stored on, and addressable by, the rank owning block $\min(K, K')$. The boundary block g_{I_0, I_0-1} that a rank cannot index in its own frame (the column lies below its window) is exactly the block its lower neighbour holds in the down-arm of its arrow – so no raw-data access is needed, only the nearest-neighbour halo above.*

Distribution-algorithm comparison. Table 1.1 contrasts the per-rank bounds of the retired diagonal-gather design, the implemented FFT-first $\times$ spatial-halo algorithm, and the q -resolved variants (Section 1.3.3). The implemented algorithm is preferred whenever $N_B/P_S \gg N_h$ (so the halo surface is negligible against the owned volume); a band-replication variant ($P_S = 1$, no halo) is simpler but does not divide the band over the spatial axis and is therefore confined to short devices.

algorithm	work W^{rank}	comm. V^{rank}	memory M^{rank}
diagonal gather (retired)	$\frac{N_\omega \log N_\omega N_B N_{c,x}^2 N_{BS}^4}{P_S}$	$\frac{N_\omega N_B N_{BS}^2}{P}$	$\frac{N_\omega N_B N_{BS}^2}{P} + \frac{N_B N_{c,x}^2 N_{BS}^3}{P_S}$
FFT-first $\times$ halo (<i>impl.</i>)	$\frac{N_\tau N_B N_{c,x}^2 N_{BS}^4}{P}$	$\frac{N_\omega N_B r_c N_{BS}^2}{P} + \frac{N_\tau N_h N_{BS}^2}{P_E}$	$\frac{N_\tau N_B r_c N_{BS}^2}{P} + \frac{N_B N_{c,x}^2 N_{BS}^3}{P_S}$
q -dense vertex	$\frac{N_q^2 N_\tau N_{AO} N_{c,x}^2 N_{BS}^3}{P}$	+ dense $\tilde{\Phi}$: $\mathcal{O}(N_q^2 N_{BS}^3)$	+ $\mathcal{O}(N_q^2 N_{BS}^3)$
q -streamed (real-space)	$\frac{N_q N_T^{\text{eff}} N_\tau \cdots}{P}$	no dense vertex	$\mathcal{O}(N_q N_{BS}^2)$

Table 1.1.: Per-rank cost of the candidate self-energy distribution algorithms. The implemented FFT-first $\times$ spatial-halo scheme divides every intrinsic-data term over $P = P_E P_S$, leaving only the two irreducible $\mathcal{O}(\text{surface})$ couplings (the energy all-to-all and the nearest-neighbour halo). For the transverse-periodic problem the real-space-streamed vertex replaces the dense $\mathcal{O}(N_q^2 N_{BS}^3)$ Fourier vertex once $N_D N_T^{\text{eff}} < N_q^2$ (Section 1.3.3).

Validation. The implementation is pinned bit-for-bit against an independent dense oracle in `tests/quatrex/phonon/test_sse_phonon_phonon.py`: single transport cell, off-diagonal output, and multi-block ($N_B \geq 3$) with genuine off-diagonal vertices, all to $< 10^{-10}$ relative. The full off-diagonal ring is further checked on *real* bulk-silicon multi-cell force constants (`phonon/scripts/verify/phph_real_fc3_parity.py`) and the self-energy is verified *bit-identical across rank counts*: under `comm.stack` ($P_E = 1, 2, 3, 4$) and under `comm.block` $\times$ `comm.stack` factorisations ($2 \times 2, 3 \times 2$ with $N_B \geq 4$, so the boundary outputs exercise the halo), to $\sim 10^{-15}$ (`phph_block_parallel_check.py`). A separate test pins the state-restoration contract.

Composition with the Interaction registry. The implemented bubble is registered as `PhononPhononInteraction`, one of several concrete `Interaction` instances iterated by the SCBA driver (`quatrex.core.scba.SCBA._compute_interactions`). The registry is the architectural seam for adding a coupled electron-phonon SSE: a future `ElectronPhononInteraction` reads from both the electron and phonon subsystems' Green's functions and writes its contribution into the appropriate σ_* buffer, using the same `comm.stack/comm.block` grid and the same `dtranspose` machinery as the existing interactions, with no further driver-side changes required. In particular, the state-restoration contract above guarantees that `PhononPhononInteraction.compute` can be sequenced before or after a future `ElectronPhononInteraction.compute` without spurious distribution-state transitions between them.

1.4. Phonon-Phonon SSE with decomposed FC3

The sparsity layers of Section 1.3.4 reduce the per-pair ring contraction from $\mathcal{O}(N_{BS}^4)$ to $\mathcal{O}(N_{BS} C_a^3)$. The factor C_a^3 remains the dominant cost in Eq. (1.124). We now show that any decomposition of the FC3 in Section 1.1.5 reduces this further to $\mathcal{O}(R N_{BS} r_c C_a + R^2 N_{BS})$ by collapsing the four-cycle of the ring contraction into a single Gram matrix on the propagator. The reduction is independent of which decomposition is used in the sense that all CP-family decompositions (CP, INDSCAL, Waring, PCP) produce the same kernel structure. The symmetry constraints distinguishing them affect only which factor vectors enter the Gram product and how much it can be symmetrised.

We derive this under two assumptions that hold for the SSE of a single SCBA iteration on a single FC3 dataset. First, both vertices of the sunset diagram (Fig. 1.1) are evaluated against the same $\Phi^{(3)}$: $\Phi_L \equiv \Phi_R \equiv \Phi$. Second, the two internal propagator lines are restrictions of the same Green's function $g(\tau)$ to (possibly different) block-row supports.

The section is organised as follows. Section 1.4.1 fixes the notation and the abstract separable form. Section 1.4.2 derives the factorised kernel and identifies the algebraic simplification that distinguishes the symmetric ansätze from plain CP. Section 1.4.3 specialises to each ansatz of Section 1.1.5. Section 1.4.4 traces through the three block-sparsity layers of Section 1.3.4 under decomposition. Section 1.4.5 gives a concrete three-step algorithm and its cost. Section 1.4.6 addresses the partial Fourier transform. Section 1.4.7 reworks the distributed pipeline of Section 1.3.7.

1.4.1. Setup

Every decomposition considered in Section 1.1.5 writes the FC3 vertex as a sum of separable terms. The most general form is

$$\Phi_{xy_1y_2} \approx \sum_{r=1}^R \lambda_r \alpha_r(x) \beta_r(y_1) \gamma_r(y_2), \quad (1.149)$$

with weights $\lambda_r \in \mathbb{R}$ and factor vectors $\alpha_r, \beta_r, \gamma_r \in \mathbb{R}^{n_{\text{dof}}}$. The symmetry constraints are

$$\begin{aligned} \text{CP:} & \quad \text{no constraint,} \\ \text{INDSCAL:} & \quad \beta_r = \gamma_r, \\ \text{Waring:} & \quad \alpha_r = \beta_r = \gamma_r, \\ \text{Tucker:} & \quad \beta, \gamma \text{ taken from a common } R_2\text{-dim basis,} \\ & \quad \alpha \text{ from a separate } R_1\text{-dim basis, plus a small core.} \end{aligned} \quad (1.150)$$

These constraints reduce parameter count (Eqs. (1.13), (1.15), (1.17), (1.19), (1.22) and (1.23)) and enforce permutation symmetry of the reconstructed tensor in (y_1, y_2) (INDSCAL, Waring) or all three legs (Waring, PCP). PCP is treated separately in Section 1.4.3 because its 6-term explicit symmetrisation interacts non-trivially with the kernel.

We collect the factor vectors as columns of three matrices $A, B, C \in \mathbb{R}^{n_{\text{dof}} \times R}$ with $A_{x,r} = \alpha_r(x)$, $B_{y,r} = \beta_r(y)$, $C_{y,r} = \gamma_r(y)$, and the weights as a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_R)$. Under the constraints Eq. (1.150), $B = C$ for INDSCAL and $A = B = C \equiv U$ for Waring. The propagator $g \equiv g(\tau) \in \mathbb{C}^{n_{\text{dof}} \times n_{\text{dof}}}$ is the time-domain Green's function obtained from Stage 1 of Section 1.3.7. We will not yet distinguish lesser, greater, or retarded components, the derivation applies unchanged to all three.

1.4.2. Factorised kernel

The per-pair ring Eq. (1.104) is

$$\mathcal{B}_{xx'} = \sum_{y_1 y_2 y_3 y_4} \Phi(x, y_1, y_2) g(y_1, y_4) g(y_2, y_3) \Phi(x', y_3, y_4). \quad (1.151)$$

Substituting Eq. (1.149) for both copies of Φ , factoring the g -independent parts, and collecting yields

$$\mathcal{B}_{xx'} = \sum_{r,s=1}^R \lambda_r \lambda_s \alpha_r(x) \alpha_s(x') P_{rs} Q_{rs}, \quad (1.152)$$

$$P_{rs} \equiv \beta_r^\top g \gamma_s = (B^\top g C)_{rs}, \quad (1.153)$$

$$Q_{rs} \equiv \gamma_r^\top g \beta_s = (C^\top g B)_{rs}. \quad (1.154)$$

The four-cycle in the original ring collapses to two independent two-cycles. Each two-cycle is a scalar bilinear form on the propagator. The full $N_{BS} \times N_{BS}$ propagator entered

Eq. (1.104) four times through tensor contractions. In Eqs. (1.152) to (1.154), it enters twice as the kernel of a quadratic form.

The matrices P and Q defined in Eqs. (1.153) and (1.154) are not independent. They satisfy the algebraic identity

$$Q = C^\top g B = (B^\top g^\top C)^\top. \quad (1.155)$$

INDSCAL ($B = C$). The two bilinear forms are

$$P_{rs} = \beta_r^\top g \beta_s = (B^\top g B)_{rs} = Q_{rs}, \quad (1.156)$$

identical entry-by-entry. The result holds without any symmetry assumption on g .

Waring ($A = B = C = U$). Same as INDSCAL, with the additional property that the assembly factor α also reduces to U :

$$\mathcal{B}_{xx'}^{\text{Waring}} = \sum_{r,s} \lambda_r \lambda_s U_{x,r} U_{x',s} [(U^\top g U)_{rs}]^2. \quad (1.157)$$

The bilinear form is squared elementwise; the entire kernel is parametrised by a single $n_{\text{dof}} \times R$ matrix U and the weights λ .

For CP without symmetry, $P \neq Q$ in general.

Unified kernel. Under the simplifications above, the bubble Eq. (1.152) can be written as a single GEMM–Hadamard–GEMM sequence:

$$V = g W, \quad (1.158)$$

$$P = W^\top V, \quad (1.159)$$

$$H_{rs} = \lambda_r \lambda_s P_{rs} P_{rs}^*, \quad (1.160)$$

$$\Sigma = A H A^\top, \quad (1.161)$$

where the choice of W and P^* depends on the ansatz:

$$\begin{aligned} \text{Waring: } W &= U, \quad P^* = P, \quad A = U, \\ \text{INDSCAL: } W &= B, \quad P^* = P, \quad A = A_{\text{INDSCAL}}, \\ \text{CP: } W &= (B, C), \quad P^* = Q, \quad A = A_{\text{CP}}. \end{aligned} \quad (1.162)$$

For CP, $W = (B, C)$ denotes that two SpMMs are needed (one against B , one against C) to obtain $V_B = gB$ and $V_C = gC$. INDSCAL and Waring need only one SpMM. This is the dominant cost saving from imposing internal symmetry on the decomposition.

1.4.3. Specialisation to each ansatz

The CP-family ansätze all fit the unified kernel Eqs. (1.158) to (1.161) with the choices Eq. (1.162). Tucker and the matricisation SVD require minor modifications, given below.

CP. $\Phi_{xy_1y_2}^{\text{CP}} = \sum_r \lambda_r a_r(x) b_r(y_1) c_r(y_2)$ with A, B, C independent. The kernel requires both $V_B = gB$ and $V_C = gC$; $P = B^\top V_C$ and (assuming $g = g^\top$) $Q = P^\top$, giving $H_{rs} = \lambda_r \lambda_s P_{rs} P_{sr}$. The reconstructed tensor is generically not S_2 -symmetric in (y_1, y_2) ; this asymmetry survives into Σ unless restored in post-processing.

INDSCAL. $\Phi_{xy_1y_2}^{\text{INDSCAL}} = \sum_r d_r(x) v_r(y_1) v_r(y_2)$ with $B = C \equiv V$ and $A_{\text{INDSCAL}} = D = [d_1 \dots d_R]$. The kernel computes $V_B = gV$, $P = V^\top V_B$ (a single GEMM-like product), and assembles via $\Sigma = D(P \circ P) D^\top$ with the elementwise square $(P \circ P)_{rs} = \lambda_r \lambda_s P_{rs}^2$. The reconstructed tensor is exactly S_2 -symmetric, so the SSE inherits exact $\Sigma_{xx'} = \Sigma_{x'x}$ (requires $g = g^\top$, i.e. equilibrium / reciprocal G) at the algorithmic level.

Waring. $\Phi_{xy_1y_2}^{\text{Waring}} = \sum_r \lambda_r u_r(p(x)) u_r(y_1) u_r(y_2)$, where p is the supercell lift of Section 1.1.5. With $A = B = C = U$, the kernel reduces further: a single matrix U parametrises both the SpMM input and the output assembly. The bubble is symmetric in (r, s) by construction, so P is symmetric ($P = P^\top$, requires $g = g^\top$, i.e. equilibrium / reciprocal G) and may be computed as a SYRK (upper-triangular GEMM) at half the cost of the general INDSCAL GEMM.

PCP. $\Phi_{xy_1y_2}^{\text{PCP}} = \sum_\xi \frac{\lambda_\xi}{6} \sum_{\sigma \in S_3} u_{\sigma_1}^\xi(p(x)) u_{\sigma_2}^\xi(y_1) u_{\sigma_3}^\xi(y_2)$. Each $\sigma \in S_3$ contributes a CP-shaped term, so PCP is algebraically equivalent to CP at rank $6R$ with a specific structure on the $6R$ triplets. The kernel Eqs. (1.158) to (1.161) applies with $R \rightarrow 6R$. For SSE evaluation, PCP is dominated by Waring at matched fitting cost; we include it only for completeness.

Tucker. $\Phi_{xy_1y_2}^{\text{Tucker}} = \sum_{pqr} G_{pqr} A_{xp} B_{y_1q} B_{y_2r}$ with $A \in \mathbb{R}^{n_{\text{dof}} \times R_1}$, $B \in \mathbb{R}^{n_{\text{dof}} \times R_2}$ columnwise orthonormal and a core $G \in \mathbb{R}^{R_1 \times R_2 \times R_2}$ with $G_{pqr} = G_{prq}$. The factorised kernel becomes matrix-valued:

$$V_B = g B \in \mathbb{R}^{n_{\text{dof}} \times R_2}, \quad (1.163)$$

$$M = B^\top V_B \in \mathbb{R}^{R_2 \times R_2}, \quad (1.164)$$

$$H_{pp'} = \sum_{qrq'r'} G_{pqr} G_{p'q'r'} M_{qr'} M_{rq'}, \quad (1.165)$$

$$\Sigma = A H A^\top, \quad (1.166)$$

The bilinear form is a single $R_2 \times R_2$ Gram matrix M (a single SpMM-GEMM); the core contraction Eq. (1.165) is purely propagator-independent and has cost $\mathcal{O}(R_1^2 R_2^3)$ after S_2 symmetrisation. The propagator-side cost is $\mathcal{O}(R_2 N_{BS} r_c C_a + R_2^2 N_{BS})$, identical in form to the CP family with $R \rightarrow R_2$.

1.4.4. Block sparsity

The unified kernel Eqs. (1.158) to (1.161) contracts g against a dense factor matrix W in step Eq. (1.158) and contracts H against A in step Eq. (1.161). Two of the four block-

sparsity layers of Section 1.3.4 survive the decomposition unchanged; one is absorbed into the factor vectors; one becomes the support pattern of P .

(a) BT-band of g — preserved. The cut-off r_c on the dynamical matrix gives $g(\tau)$ a block-banded structure of bandwidth r_c along the transport direction (Eq. (1.118)). The SpMM $V = gW$ is therefore a block-banded matrix times a dense $n_{\text{dof}} \times R$ matrix. The non-zero pattern of V is dictated by g : for any block I on the transport axis, $V_I = \sum_{|J-I| \leq r_c} g_{IJ} W_J$, a band of width $2r_c + 1$ in the input. The non-zero count of V is

$$\text{nnz}(V) = R \cdot \text{nnz}_{\text{cols}}(g) = \mathcal{O}(R N_{BS} N_B) = \mathcal{O}(R n_{\text{dof}}), \quad (1.167)$$

provided W is dense (factor vectors with full support along transport). The cost of the SpMM is $\mathcal{O}(R \cdot \text{nnz}(g)) = \mathcal{O}(R n_{\text{dof}} r_c C_a)$.

(b) Intra-block atom-pair sparsity of g — preserved. Within each $N_{BS} \times N_{BS}$ block, g_{IJ} has nominal density C_a/N_{BS} (with $C_a = N_{BS}$ in the present dense-block kernel; see the forward note in Section 1.3.4). The block-level SpMM $V_I = \sum_J g_{IJ} W_J$ is therefore a sparse–dense block product with cost $\mathcal{O}(R N_{BS} C_a)$ per source block J , summed over the $\mathcal{O}(r_c)$ blocks within bandwidth, giving $\mathcal{O}(R N_{BS} r_c C_a)$ per output block and $\mathcal{O}(R n_{\text{dof}} r_c C_a)$ for the full SpMM.

(c) FC3 inter-block support — absorbed. The FC3 cut-off Eq. (1.119) restricts $\Phi_{I,K_1,K_2}^{(3)}$ to block triples with mutual transport-direction distance $\leq N_{c,x}$. After decomposition, the FC3 cut-off is absorbed into the support of the factor vectors. Specifically, for a CP fit of the full FC3 tensor, each rank-one component $\lambda_r a_r \otimes b_r \otimes c_r$ has full support $\text{supp}(a_r) \times \text{supp}(b_r) \times \text{supp}(c_r)$; all such supports must lie within the FC3 cut-off, so each individual factor vector has support within $\mathcal{O}(N_{c,x} + \text{spread})$ transport blocks. The total non-zero count of W is therefore

$$\text{nnz}(W) = \mathcal{O}(R N_{BS} N_{c,x}) = \mathcal{O}(R n_{\text{dof}} N_{c,x}/N_B), \quad (1.168)$$

substantially smaller than the dense $R n_{\text{dof}}$ when $N_{c,x} \ll N_B$ (true for any transport device longer than the FC3 cut-off).

In practice, an unconstrained CP-ALS fit produces factor vectors without compact support: the rotational gauge of CP allows globally delocalised factors to represent locally supported tensors. To preserve the FC3 cut-off as factor-vector support, one of three strategies is used: (i) initialise CP-ALS from a spatially localised seed (e.g. the slice-wise SVD initialisation of Section 1.1.5); (ii) regularise with an ℓ_1 or group-sparse penalty as in [94, 95]; or (iii) work in the CANDELINC framework of Section 1.1.5 with a basis that respects the FC3 cut-off. All three preserve Eq. (1.168) at matched accuracy; the choice is empirical and beyond the scope of the SSE kernel.

(d) Output BT band — preserved by output projection. The output Σ lives on the same BT band as G , with $|I - J| \leq 1$. The assembly step Eq. (1.161) is therefore restricted to this band. *Crucially*, this restriction is applied at the output, not at the intermediate H : the $R \times R$ matrix H has dense support and does not inherit the band structure of Σ . We compute H once per τ and project to the BT band when constructing Σ .

Effective sparsity of P . The Gram matrix P has entries $P_{rs} = \beta_r^\top g \gamma_s$. Two effects can render P sparse:

- *Disjoint factor supports.* If $\text{supp}(\beta_r) \cap N_{r_c}(\text{supp}(\gamma_s)) = \emptyset$ (where N_{r_c} is the r_c -block neighbourhood induced by g 's bandwidth), then $P_{rs} = 0$ exactly.
- *Approximate orthogonality.* For Waring with fitted u_r that are approximately orthonormal in the metric g , the off-diagonal entries of P are small. This is a refit-dependent property and we do not assume it.

The first effect can be substantial: for FC3 with $N_{c,x} = 3$ and a device of N_B transport blocks, the fraction of P_{rs} pairs with overlapping supports scales as $N_{c,x}/N_B$ — between 10^{-2} and 10^{-1} at production device sizes. Exploiting this sparsity in step Eq. (1.159) converts the dense $R^2 N_{BS}$ GEMM into a sparse–sparse product at cost $\mathcal{O}(R^2 N_{c,x})$, dominated by the SpMM cost.

1.4.5. Computation strategy

The per- τ kernel Eqs. (1.158) to (1.161) maps onto three numerical primitives, all standard in modern linear algebra libraries:

1. *Step Eq. (1.158): batched sparse SpMM.* $V = gW$, where g is block-banded sparse and W is $n_{\text{dof}} \times R$ with FC3-induced support Eq. (1.168). Implemented as a sparse matrix times dense matrix product with R right-hand sides batched into a single SpMM call. Cost: $\mathcal{O}(R n_{\text{dof}} r_c C_a)$ flops, memory traffic $\mathcal{O}(\text{nnz}(g) + R n_{\text{dof}})$. With R batched RHSs, the arithmetic intensity is $\mathcal{O}(R)$ flops/byte, placing the kernel firmly in the compute-bound regime of modern accelerators for $R \gtrsim 32$.
2. *Step Eq. (1.159): small dense GEMM.* $P = W^\top V$, a dense $R \times n_{\text{dof}}$ times $n_{\text{dof}} \times R$ product yielding an $R \times R$ matrix. For Waring, P is symmetric and the product reduces to SYRK. Cost: $\mathcal{O}(R^2 n_{\text{dof}})$ flops. With R^2 output entries and $\mathcal{O}(R n_{\text{dof}})$ input non-zeros, the arithmetic intensity is $\mathcal{O}(n_{\text{dof}}/R)$ flops/byte, comfortably compute-bound.
3. *Step Eq. (1.161): sparse-output sandwich.* $\Sigma = A H A^\top$ restricted to the output BT band and any available intra-block atom-pair sparsity (with $C_a = N_{BS}$ in the present dense-block kernel; see the forward note in Section 1.3.4). Implemented as: (i) precompute $T = AH$, a dense $n_{\text{dof}} \times R$ matrix, cost $\mathcal{O}(R^2 n_{\text{dof}})$; (ii) for each non-zero entry (x, x') in the output support, compute $\Sigma_{xx'} = T_{x,:} \cdot A_{x',:}^\top$, cost $\mathcal{O}(R)$. Total: $\mathcal{O}(R^2 n_{\text{dof}} + R n_{\text{dof}} C_a)$.

The Hadamard step Eq. (1.160) has cost $\mathcal{O}(R^2)$ and is negligible.

Per- τ cost. Summing the dominant terms,

$$\mathcal{C}_\tau^{\text{decomp}} = \underbrace{\mathcal{O}(R n_{\text{dof}} r_c C_a)}_{\text{SpMM}} + \underbrace{\mathcal{O}(R^2 n_{\text{dof}})}_{\text{Gram + assembly}}. \quad (1.169)$$

This is to be compared with the sparse-ring per- τ cost Eq. (1.124),

$$\mathcal{C}_\tau^{\text{sparse}} = \mathcal{O}(n_{\text{dof}} N_{\text{c,x}}^2 C_a^3). \quad (1.170)$$

At Si-nanoribbon parameters ($C_a \sim 30$, $N_{\text{c,x}} = 3$, $r_c \sim 2$), the sparse-ring cost has prefactor $N_{\text{c,x}}^2 C_a^3 \sim 2.4 \times 10^5$; the factorised kernel has prefactor $R r_c C_a + R^2 \sim 60R + R^2$. For $R = 100$, the factorised kernel is roughly an order of magnitude ($\sim 15\times$) cheaper.

Crossover. The factorised kernel is cheaper than the sparse ring when

$$R r_c C_a + R^2 < N_{\text{c,x}}^2 C_a^3, \quad (1.171)$$

which, for $R \ll C_a$, simplifies to $R \lesssim N_{\text{c,x}}^2 C_a^2 / r_c \sim 10^4$ at the same parameters; for R comparable to C_a , the R^2 term enters and the crossover moves to $R \lesssim N_{\text{c,x}} C_a^{3/2} \sim 500$. Reported CP and Waring ranks for FC3 of Si, MgO, HgTe, TiNiSn, ZrO_2 [53] are well below these bounds, placing the factorised kernel firmly in the favourable regime.

Batching over τ . The kernel is local in τ : each τ -slice produces an independent $\Sigma(\tau)$. The natural batching strategy is to process N_b τ -slices in a single call, stacking the N_b propagator slices into a $n_{\text{dof}} \times n_{\text{dof}} \times N_b$ tensor and the corresponding V matrices into a $n_{\text{dof}} \times R \times N_b$ tensor. The SpMM becomes a batched SpMM (cuSPARSE/MKL support); the GEMM becomes a batched GEMM. Batch size N_b is chosen adaptively from the free HBM, with the dominant memory term being the V tensor at $\mathcal{O}(N_b R n_{\text{dof}})$.

Sparse-factor optimisation. When the factor vectors W are intra-block sparse with density C_a/N_{BS} (the case for FC3-localised factors, see Section 1.4.4), the GEMM in Eq. (1.159) can be replaced by a sparse–dense product at cost $\mathcal{O}(R^2 C_a N_{\text{c,x}})$, reducing the second term in Eq. (1.169) by a factor $N_{BS}/(C_a N_{\text{c,x}})$. The SpMM cost is unaffected (it is already proportional to $\text{nnz}(g)$, not to the density of W).

1.4.6. Transverse periodic case

In the transverse-periodic setting of Section 1.2.3, the SSE Eq. (1.65) carries an internal sum over a transverse wavevector $\mathbf{q}'_\perp$ which couples g at $\mathbf{q}'_\perp$ and $\mathbf{q}_\perp - \mathbf{q}'_\perp$ via the FC3 evaluated at the same pair of momenta. We carry the factorised kernel through the partial Fourier transform and identify the structural simplifications.

Partial Fourier transform of CP factors. Performing the CP decomposition on the real-space FC3 $\Phi_{\mu, l_1 \nu_1, l_2 \nu_2}^{(3)}$ gives factor matrices in which the second and third legs include the transverse cell label. The partial Fourier transform Eq. (1.66) acts on these cell labels and produces

$$\tilde{\Phi}_{\mu \nu_1 \nu_2}^{(3)}(\mathbf{q}_\perp, \mathbf{q}'_\perp) = \sum_r \lambda_r a_r(\mu) b_r(\nu_1; \mathbf{q}_\perp) c_r(\nu_2; \mathbf{q}'_\perp), \quad (1.172)$$

with $\mathbf{q}_\perp$ -dependent factor vectors $b_r(\nu_1; \mathbf{q}_\perp) = \sum_{l_1} \tilde{b}_r(l_1, \nu_1) e^{-i \mathbf{q}_\perp \cdot \mathbf{l}_1}$ and likewise for c_r . The CP rank is preserved exactly under the partial Fourier transform: the decomposition Eq. (1.149) commutes with the cell-FT because the FT acts on each factor leg separately. For Waring, the single factor vector u_r has all three legs Fourier-transformed against the appropriate cell labels.

Factorised kernel in $\mathbf{q}_\perp$ -space. With $\Phi_L = \Phi_R$ at $(\mathbf{q}'_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp)$ and $(\mathbf{q}'_\perp - \mathbf{q}_\perp, -\mathbf{q}'_\perp)$ respectively, and noting that for real factor vectors $b_r(\nu; -\mathbf{q}_\perp) = b_r(\nu; \mathbf{q}_\perp)^*$, the kernel Eq. (1.152) becomes

$$\mathcal{B}_{xx'}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp) = \sum_{rs} \lambda_r \lambda_s a_r(x) a_s(x') P_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp) Q_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp), \quad (1.173)$$

$$P_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp) = b_r(\cdot; \mathbf{q}'_\perp)^\top g(\tau, \mathbf{q}'_\perp) c_s(\cdot; \mathbf{q}'_\perp)^*, \quad (1.174)$$

$$Q_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp) = c_r(\cdot; \mathbf{q}_\perp - \mathbf{q}'_\perp)^\top g(\tau, \mathbf{q}_\perp - \mathbf{q}'_\perp) b_s(\cdot; \mathbf{q}_\perp - \mathbf{q}'_\perp)^*. \quad (1.175)$$

The two bilinear forms now involve g at *different* transverse momenta, $\mathbf{q}'_\perp$ and $\mathbf{q}_\perp - \mathbf{q}'_\perp$, and are no longer related by transposition even for Waring. They are, however, related by a change of summation variable: under $\mathbf{q}'_\perp \rightarrow \mathbf{q}_\perp - \mathbf{q}'_\perp$, $P_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp) = Q_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp)$. After summing over the full Monkhorst–Pack mesh (which is closed under $\mathbf{q}'_\perp \rightarrow \mathbf{q}_\perp - \mathbf{q}'_\perp$ for appropriate shifts), the contributions of P and Q to Σ are equal, so only P need be evaluated.

Kernel restructure. The change-of-variable identity collapses Eqs. (1.173) to (1.175) to

$$\Sigma_{xx'}(\tau, \mathbf{q}_\perp) = \frac{i\hbar}{2N} \sum_{\mathbf{q}'_\perp} \sum_{rs} \lambda_r \lambda_s a_r(x) a_s(x') P_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp) P_{rs}(\tau, \mathbf{q}_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp), \quad (1.176)$$

which is a convolution along $\mathbf{q}'_\perp$ of the matrix-valued quantity $P(\tau, \mathbf{q}_\perp, \mathbf{q}'_\perp)$ with its shift $P(\tau, \mathbf{q}_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp)$. The $\mathbf{q}'_\perp$ -sum is therefore a Fourier transform of a per-block product, and the N_q^2 factor in Eq. (1.109) is replaced by $N_q \log N_q$ after a single FFT along the transverse axis. Summing over the full $\mathbf{q}'_\perp$ mesh already accounts for both orderings $P(\mathbf{q}'_\perp)P(\mathbf{q}_\perp - \mathbf{q}'_\perp)$, so no extra factor of 2 appears; for CP without internal symmetry P and Q differ and both bilinear forms enter explicitly.

1.4.7. Distributed implementation

The four-redistribution pipeline of Section 1.3.7 is preserved bit-for-bit. The decomposition changes only Stage 2 of the pipeline (the per- τ ring contraction). Stages 1 and 3

(FFT $G \rightarrow g$ and IFFT $\sigma \rightarrow \Sigma$), the two interconvertible distributions `stack/nnz`, and the four `stack $\leftrightarrow$ nnz` dtransposes are unchanged.

Stage 2 with decomposition. The ring contraction Eqs. (1.131) to (1.133) is replaced by the unified kernel Eqs. (1.158) to (1.161). On a single rank holding a τ -slice of g in the `stack` distribution, the kernel is purely local in τ and consists of SpMM, GEMM, and the sparse-output sandwich. No communication is incurred inside Stage 2 except the FC3-dictionary halo discussed below.

Per-rank arithmetic. Summing over N_τ, N_q^2 (or $N_q \log N_q$ in the $q'_\perp \rightarrow q_\perp - q'_\perp$ symmetrised form of Eq. (1.176)) and distributing across $P = P_E P_S$ ranks gives

$$W_{\text{phph,decomp}}^{\text{rank}} = \mathcal{O}\left(\frac{N_\tau N_q^2 N_{AO} (R r_c C_a + R^2)}{P}\right). \quad (1.177)$$

The $C_a^3 N_{c,x}^2$ factor of Eq. (1.124) is replaced by $R r_c C_a + R^2$. At $C_a = 30, r_c = 2, R = 100$: the old factor is 2.4×10^5 , the new factor is $\sim 1.6 \times 10^4$ — a $15\times$ arithmetic reduction. With $q'_\perp$ -FFT symmetrisation (Section 1.4.6), an additional factor $N_q / \log N_q \sim 50$ is gained for production $N_q = 400$.

1.4.8. Transverse-momentum (q-point) distribution and vertex streaming

The distribution of Sections 1.3.7 and 1.4.7 spreads the work over frequency (P_E , `comm.stack`) and transport blocks (P_S , `comm.block`). In the transverse-periodic setting of Section 1.4.6 a third index appears: the external transverse momentum $q_\perp$. Unlike the finite (zone-centre) device, where each $q_\perp$ is an independent external label, the periodic self-energy

$$\Sigma^{\lessgtr}(q_\perp, \omega) = \frac{i\hbar}{2} \frac{1}{N_q} \sum_{q'_\perp} \int \frac{d\omega'}{2\pi} \tilde{\Phi}(q'_\perp, q_\perp - q'_\perp) G^{\lessgtr}(q'_\perp, \omega') G^{\lessgtr}(q_\perp - q'_\perp, \omega - \omega') \tilde{\Phi}(q'_\perp - q_\perp, -q'_\perp) \quad (1.178)$$

couples the momenta through crystal-momentum conservation $q_\perp = q'_\perp + (q_\perp - q'_\perp)$: every external $q_\perp$ draws on the Green's functions of *all* internal $q'_\perp$. The momenta are therefore not embarrassingly parallel, and a naive per- $q_\perp$ split would replicate the full $G(q'_\perp)$ set on every rank.

The q-communicator. We add $q_\perp$ as a third communicator axis alongside block and stack, so that the global ranks factor as $q \times \text{stack} \times \text{block}$. Concretely `QuatrexCommunicator.configure` gains a `q_comm_size` and a third `MPI_Comm` split; the previous two-axis layout is recovered exactly at `q_comm_size = 1` (the q -communicator then has one rank and is a no-op), so block and stack behaviour is unchanged. Each q -rank owns a disjoint slice of the *external* $q_\perp$ mesh and stores only its own per- $q_\perp$ Green's functions and output self-energies. The *internal* $q'_\perp$ Green's functions that Eq. (1.178) needs are reconstructed once per SCBA iteration by an `all_gather_v` over the q -communicator — the “data exchange between

processors” inherent to the momentum convolution. The explicit external- $\mathbf{q}_\perp$ loop then divides as $1/P_q$, while the all-gather of the per- $\mathbf{q}_\perp$ diagonal G carries the communication cost; the bubble kernel itself is unchanged. This third axis composes with the energy and block axes without touching them: a production run can use all three simultaneously ($P = P_q P_E P_S$).

Streaming the momentum-space vertex. On an accelerator the binding constraint for Eq. (1.178) is memory rather than arithmetic. The Fourier-transformed FC3 vertex $\tilde{\Phi}(\mathbf{q}_\perp, \mathbf{q}'_\perp) = T(\mathbf{q}_\perp) M T(\mathbf{q}'_\perp)^\dagger$ – with $T(\mathbf{q}_\perp)$ the supercell-to-primitive gathering matrix and M the (small, real-space) mass-weighted FC3 – is a dense rank-five object of size $N_q^2 n_{\text{dof}}^3$ if materialised. At an 8×8 transverse mesh ($N_q = 64$) and a wire-scale cross section ($n_{\text{dof}} = 63$) this is already ~ 16 GB, beyond a single device, and it grows as N_q^2 . Since the kernel only ever touches the pair $(\tilde{\Phi}(\mathbf{q}'_\perp, \mathbf{q}_\perp - \mathbf{q}'_\perp), \tilde{\Phi}(\mathbf{q}_\perp - \mathbf{q}'_\perp, \mathbf{q}'_\perp))$ for the momenta in the current batch, we never form the full tensor: with `stream_phi` enabled the kernel keeps only T and M and rebuilds $\tilde{\Phi}$ on the fly per momentum batch, dropping the peak from $\mathcal{O}(N_q^2 n_{\text{dof}}^3)$ to $\mathcal{O}(N_q n_{\text{dof}}^2)$. This trades a vertex recomputation for memory, which is favourable on a GPU where the bandwidth-bound bubble contractions dominate the run time regardless, and it composes with the q -communicator at no extra communication (each rank streams only the vertices for its own external- $\mathbf{q}_\perp$ slice). The result is algebraically identical to the dense path; the measured memory reduction and the near-ideal q -axis speed-up are reported in Chapter 2.

Results

This chapter collects the quantum transport calculations: the ballistic Si/Ge interface (Section 2.1), the force-constant inputs and their compression (Section 2.2), anharmonic transport on finite silicon nanowires (Section 2.3) and carbon nanotubes (Section 2.4), the transversely-periodic $q_{\perp}$ -resolved problem and the silicon thin-film benchmark (Section 2.5), a Si/Ge heterostructure (Section 2.6), the static loop and tadpole corrections (Section 2.7), self-consistent convergence at vanishing broadening (Section 2.8), the strongly anharmonic SrTiO₃ test (Section 2.9), the cost of the numerical approximations (Section 2.10), and the distributed-solver and memory scaling of the implementation (Section 2.11).

2.1. Ballistic Phonon NEGF

As a heterogeneous test, we model a ballistic Si/Ge interface in the mass-mismatch approximation, as done in [38, 105, 146]: bulk-Si force constants in the whole device, with the Si/Ge distinction entering only through the atomic masses ($m_{\text{Si}} = 28.09$, $m_{\text{Ge}} = 72.64$ amu). At Γ , the Si optical maximum is 15.55 THz (device force-constant set; cf. the 15.12–15.37 THz primitive-cell cross-check of Section 2.2) and the Ge mass-clone gives 9.67 THz, consistent with $f_{\text{Ge}} = f_{\text{Si}} \sqrt{m_{\text{Si}}/m_{\text{Ge}}}$. On a 20×20 transverse mesh, the interfacial conductance at 300 K is $G = 196 \text{ MW m}^{-2} \text{ K}^{-1}$, within $\sim 7\%$ of the literature values (210, Latour 2017; 209, Tian 2012). The remaining gap is consistent with the coarser transverse mesh and DFT calculations.

2.2. Force Constants

Method cross-check. Bulk Silicon harmonic force constants computed by density-functional perturbation theory (QE `ph.x` $\rightarrow$ `q2r.x`) and by finite displacement (phono3py + symfc) on the same primitive cell agree to 1.6% on the zone-centre optical frequency (15.12 vs 15.37 THz; Fig. 2.1). DFPT third-order constants via `d3q` require a norm-

conserving pseudopotential, and can hence not be calculated to the same accuracy as with finite-displacement methods.

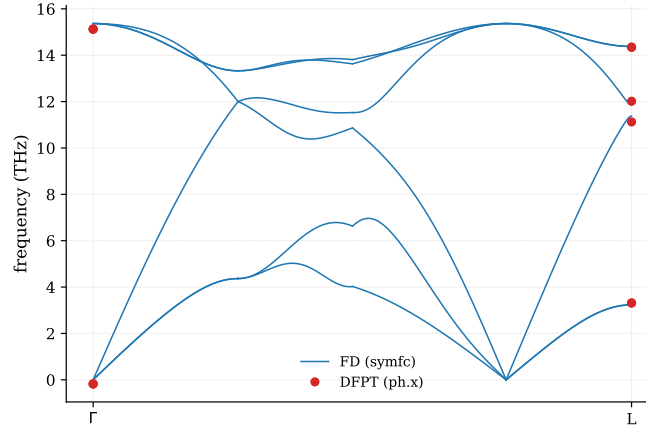


Figure 2.1.: Bulk-silicon phonon dispersion from DFPT versus finite-displacement force constants.

Bulk Si κ . As an independent check of the force-constant pipeline and the physics, separate from the NEGF solver, the finite-displacement FC3 is fed to a phono3py relaxation-time-approximation Boltzmann calculation. The result is essentially mesh-converged, bulk-silicon $\kappa(300\text{ K}) = 110.7\text{ W m}^{-1}\text{ K}^{-1}$ on an 11^3 q -mesh and 108.7 on 19^3 , $\sim 27.5\%$ below the experimental ~ 150 , consistent with the coarse $2 \times 2 \times 2$ FC3 supercell (cutoff 5.5 Å). Sweeping temperature with natural-isotope scattering gives a direct comparison to the classic Glassbrenner–Slack single-crystal data (Glassbrenner & Slack 1964; Table 2.1): the calculation reproduces the experimental temperature dependence – the high-temperature $\kappa \propto 1/T$ fall-off is captured (the conductivity halves between 300 and 600 K) – and lies $\sim 25\text{--}30\%$ low near 300 K, falling to $\sim 16\text{--}18\%$ by 500–600 K, the deficit growing toward low temperature where long-range cubic and isotope terms matter most. Both effects point to the under-converged $2 \times 2 \times 2$ FC3 supercell. A repeat with the better ($5 \times 5 \times 5$ VASP) Si FC3 of Section 2.5 is expected to reduce this deficit.

FC3 compression. The tensor-decomposition methods of Section 1.1.5 are applied to the device FC3 of the d11a nanowire (39.4×10^6 dense parameters) and the compressed tensors fed back through the full anharmonic transport solve (Fig. 2.2). We observe that, first, transport observable converges faster than the Frobenius residual the fitters minimise: the symmetric ansätze (INDSCAL, Waring) reach a $\sim 15\%$ conductance error at rank 16 ($\sim 3600\text{--}4500\times$ fewer parameters) while their Frobenius residual is still 52–67%. The relevant fitting target for phonon NEGF is therefore a transport residual, not the tensor norm. Second, the current compression approach has distinct failure modes: the matricised SVD is both far denser (its factor count is 10–100 $\times$ the symmetric

Table 2.1.: Bulk-Si lattice thermal conductivity vs temperature (19^3 mesh, RTA, natural isotopes) compared to experimental single-crystal silicon (Glassbrenner–Slack).

T (K)	κ_{calc} (W/mK)	κ_{exp} (W/mK)	ratio
200	179.6	~ 266	0.68
300	108.7	~ 150	0.73
400	78.7	~ 98	0.80
500	62.0	~ 76	0.82
600	51.3	~ 61	0.84

methods') and unphysical at higher rank, with its rank-8 fit violating heat-flow conservation and its rank-16 fit giving a negative G_{anh} . Conversely, the Tucker (HOSVD) and INDSCAL fits at the lowest ranks collapse to the bare ballistic value. Only the symmetric INDSCAL/Waring families degrade gracefully.

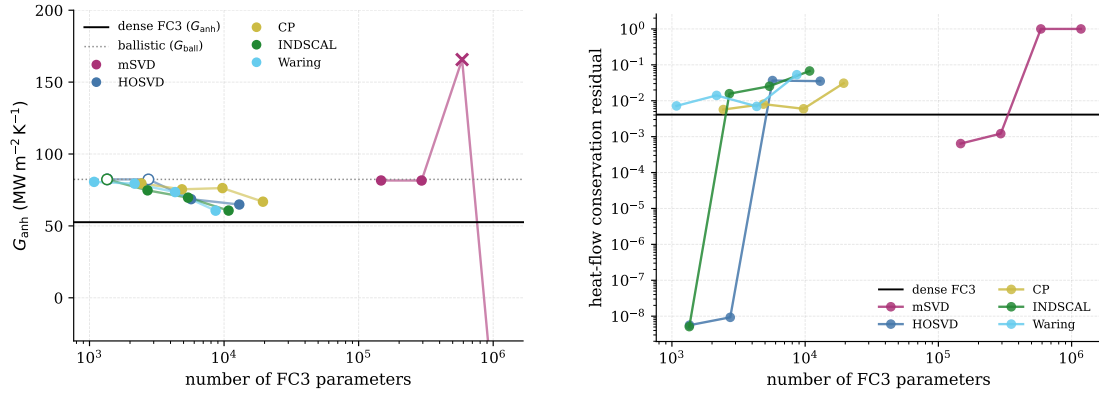


Figure 2.2.: FC3 tensor-decomposition transport quality on the d11a nanowire: anharmonic conductance (left) and heat-flow conservation residual (right) versus FC3 parameter count, for five decomposition families against the dense reference.

2.3. Anharmonic transport in silicon nanowires

We study two hydrogen-passivated [100] silicon nanowires: d5a (Si_9H_{12} per cell, 63 DOF) and the wider d11a (135 DOF). Their harmonic transport is physical (Fig. 2.3): the ballistic conductance rises with temperature, and d11a carries $\sim 1.5\times$ the conductance of d5a.

Mode-resolved anharmonic physics. A phono3py analysis of the bulk-silicon FC3 (Fig. 2.4) is performed: phonon lifetimes have a median of 4.2 ps with a $\tau \propto \omega^{-2}$ envelope.

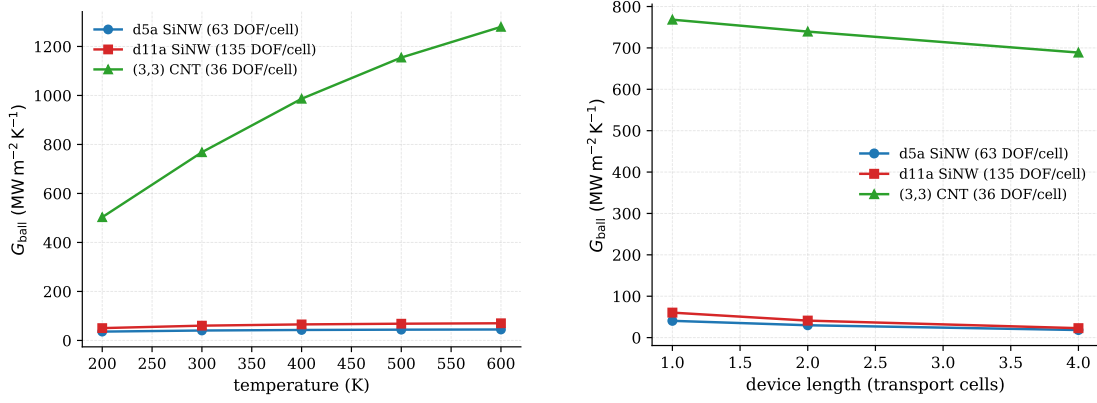


Figure 2.3.: Ballistic conductance of the d5a and d11a silicon nanowires. Left: $G(T)$; right: G versus device length.

50% of κ is carried by phonons with mean free path > 134 nm and 93% by acoustic modes. Umklapp accounts for 0.44 of the scattering at 300 K, and the mode Grüneisen parameters have mean $|\gamma| = 1.04$, matching the experimental Si value and validating the FC3 anharmonicity strength. Only 1.3% of bulk κ comes from phonons with mean free path below 10 nm. The ~ 1 nm nanowire cores therefore sit far below the bulk phonon-MFP spectrum: they are mostly performing boundary scattering and retain almost none of the bulk anharmonic-limited κ .

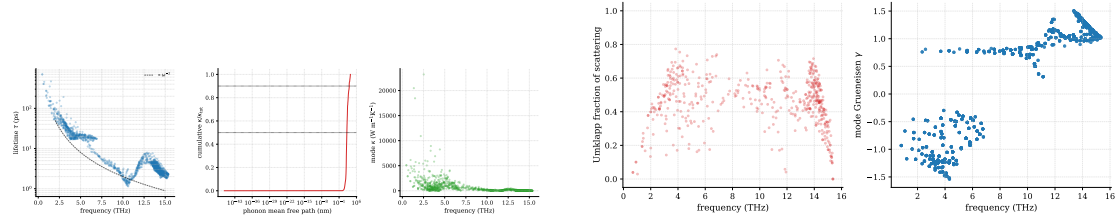


Figure 2.4.: Bulk-silicon mode-resolved anharmonic physics (phono3py, FD FC3). Left: lifetimes, κ -accumulation versus mean free path, and the acoustic-mode κ fraction. Right: Normal/Umklapp split and the mode Grüneisen distribution (mean $|\gamma| = 1.04$).

At physical broadening and moderate vertex strength the SCBA converges cleanly and gives the physical $G_{\text{anh}} < G_{\text{ball}}$.

What the open-wire SCBA requires. Two conditions are necessary for the nanowire self-consistency to reach this physical value. First, the three-phonon vertex must be built from an *acoustic-sum-rule-projected* FC3: the raw mass-weighted FC3 violates translational invariance on the open device and couples the scattering vertex into the soft acoustic/twist subspace, so the unprojected $\lambda=1$ SCBA diverges where the projected one converges (the

projection drops $\sim 25\%$ of $\|\Phi^{(3)}\|$ and leaves the ballistic baseline unchanged). Second, the broadening must be physical: an over-large diagnostic $\eta_w \sim 0.7\text{--}2$ THz crushes the low-frequency acoustic channels that carry the wire’s heat, collapsing G_{ball} by up to two orders of magnitude and producing the spurious $G_{\text{anh}} > G_{\text{ball}}$ of early sweeps; at physical η_w the resistive ordering is restored.

Table 2.2 makes the projection effect quantitative on a single representative d5a point (one cell, $\eta_w \approx 0.11$ THz, full coupling $\lambda = 1$): the raw vertex diverges — residual past 10^2 , 620 causality-violating frequency points, the guard aborting after ten iterations — while the projected vertex converges monotonically in thirty iterations to residual 8.8×10^{-4} with a bounded $\max |\Sigma^R| \approx 136$ THz² and no causality violations; the projection lowers the stacked-FC3 Frobenius norm by 25% and leaves the ballistic conductance identical to seven figures.

Table 2.2.: Projected vs. raw FC3 at full coupling ($\lambda = 1$) on d5a (one transport cell, $\eta_w \approx 0.11$ THz). The raw vertex diverges; the ASR-projected vertex converges. Conductances in $\text{W m}^{-2} \text{K}^{-1}$.

FC3 vertex	converged	iters	residual	causality	G_{ball}	G_{anh}
raw (no ASR)	no	10 (abort)	1.2	620 viol. pts	2.15×10^7	(diverged)
ASR-projected	yes	30	8.8×10^{-4}	none	2.15×10^7	2.20×10^7

At this single-cell, full-coupling point the converged anharmonic conductance sits marginally *above* the ballistic value ($G_{\text{anh}}/G_{\text{ball}} = 1.02$, with $\sim 1.9\%$ heat-flow non-conservation): the finite- η single-cell artefact of Section 2.10, not a physical anti-resistance. Across temperature the band widens mildly with T as the Bose factors grow (0.98/1.02/1.09 at 200/300/400 K, Table 2.3); the strict, current-conserving $G_{\text{anh}} < G_{\text{ball}}$ is recovered at moderate coupling ($\lambda = 0.3$ gives 0.94), in a multi-cell device, or in the $\eta \rightarrow 0$ limit of Section 2.8.

Table 2.3.: Projected-FC3 SCBA on d5a, one transport cell, $\lambda = 1$, $\eta_w \approx 0.22$ THz (at the in-run broadening). Conductances in $\text{W m}^{-2} \text{K}^{-1}$.

T (K)	G_{ball}	G_{anh}	$G_{\text{anh}}/G_{\text{ball}}$
200	1.97×10^7	1.92×10^7	0.98
300	2.15×10^7	2.19×10^7	1.02
400	2.24×10^7	2.45×10^7	1.09

Width dependence of the soft mode. The near-zero twist is specific to the narrow d5a. The wider d11a has no imaginary modes and its lowest non-translational mode sits at 1.07 THz, a factor of ~ 40 above d5a’s 0.027 THz twist, so the Bose-enhanced infrared stiffness is absent and its SCBA is well conditioned: $\lambda = 0.3$ converges in 37 iterations to $G_{\text{anh}}/G_{\text{ball}} = 0.997$ (conservation 2.7×10^{-3}) and $\lambda = 0.6$ holds a stable ≈ 0.98 . d11a is

therefore the natural structure for production nanowire transport. A third wire, d17a, was discarded: its force constants as fit carry 86 imaginary modes at Γ down to -34 THz — a fit-quality artefact of training on too few rattled supercells, resolved by refitting rather than by the solver, and a reminder that the harmonic input must be vetted before any anharmonic transport run.

A self-energy-level cutoff sweep on the d5a and d11a nanowires (Fig. 2.5) quantifies the truncations: the FC3 magnitude threshold converges quickly for d5a ($10^{-3} \rightarrow 0.7\%$ relative change in $\Sigma^<$) but the wider d11a needs a tighter threshold. A diagonal- G approximation inside the bubble is catastrophic ($5\times$ for d5a, $30\times$ for d11a). The dominant cutoff is the Green’s-function range, not the vertex magnitude.

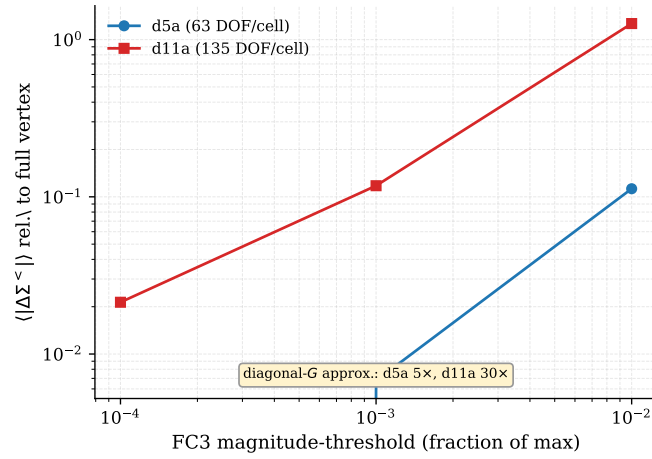


Figure 2.5.: Self-energy sensitivity to the FC3 magnitude cutoff and to a diagonal- G approximation for the d5a and d11a nanowires. The off-diagonal Green’s-function content dominates.

Performing a rattle-amplitude sweep for the d5a nanowire, the twist frequency stays real and stable (0.5–1.0 THz at $q_z = 0.02$) at every amplitude (0.015–0.08 Å), with only fit-noise scatter — never imaginary. The d5a twist is thus robust.

2.4. Carbon nanotubes

An armchair (3, 3) carbon nanotube is tested: a stiff, defect-free sp^2 tube (36 DOF per cell, ~ 4 Å diameter). Its force constants fit with a larger absolute residual than the silicon wires (train/test RMSE 0.245/0.282 eV Å $^{-1}$, reflecting the much stiffer sp^2 bonds and hence larger forces) but a clean harmonic spectrum — no imaginary modes, and the periodic acoustic sum rule exact (FC3 leg residual ~ 0 , versus 0.8 for the open wire). It carries a low-frequency transmission plateau of ~ 11 propagating channels — several optical branches are already open above the four acoustic modes that survive the $\omega \rightarrow 0$ limit (compare ~ 4.5 for d11a) — and the SCBA converges to a tight fixed point (heat-flow conservation 10^{-5} – 10^{-4}); driven to a tight tolerance the one-cell residual reaches

9.6×10^{-10} in 188 safeguarded-Anderson iterations, giving the same conductance as the loose (10^{-3}) stop — the tube has a single, well-conditioned fixed point.

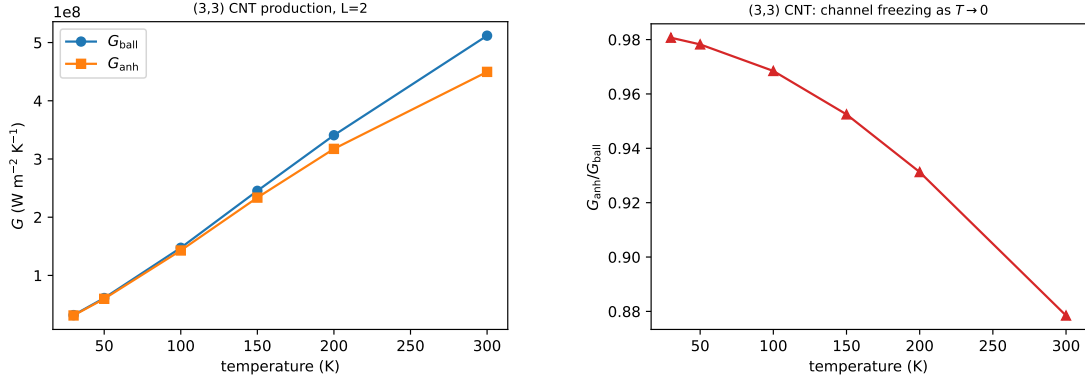


Figure 2.6.: Anharmonic transport of the (3,3) carbon nanotube versus temperature (single transport cell). *Left*: ballistic and anharmonic per-area conductance. *Right*: the ratio $G_{\text{anh}}/G_{\text{ball}}$ rises toward unity as $T \rightarrow 0$.

Temperature. Figure 2.6 shows the converged conductance from 30 to 300 K at a single transport cell. Both the ballistic and the anharmonic conductance rise with temperature as channels populate; the anharmonic ratio $G_{\text{anh}}/G_{\text{ball}}$ falls monotonically from 0.92 at 30 K to 0.810 at 300 K, the signature of three-phonon scattering switching on with the Bose-weighted phase space and freezing out as $T \rightarrow 0$.

Length. At fixed 300 K the anharmonic ratio decreases with device length: the converged $G_{\text{anh}}/G_{\text{ball}} = 0.810, 0.702, 0.694$ at one, two and three cells (each to $\leq 10^{-5}$ with $\lesssim 3 \times 10^{-4}$ heat-flow non-conservation), a clean ballistic-to-diffusive reduction that accumulates from one to two cells and then nearly saturates. Beyond three cells the dense reference is too expensive — the three-cell point already costs 6.8 h (Section 2.11).

Cutoff robustness. Across all eight corners of the (FC3-magnitude, vertex-range, Green’s-function-range) cutoff cube, the converged G_{anh} varies by $\pm 10\%$ about the full-coupling value (Fig. 2.7). This is in contrast to the open nanowires, where a diagonal- G approximation is catastrophic (Section 2.3).

The (8,0) zigzag. As a second, larger tube, we tested the semiconducting (8,0) zigzag nanotube (32 atoms per cell, the geometry of the electron-transport quatrex example). The fit is clean (test RMSE $0.151 \text{ eV \AA}^{-1}$, about half the (3,3) value thanks to the larger cell; a soft 0.0014 THz twist, as befits the wider tube), the ballistic conductance behaves as expected — it decreases with length through finite- η coherence attenuation, and the larger tube carries more channels at lower per-area conductance — and the anharmonic

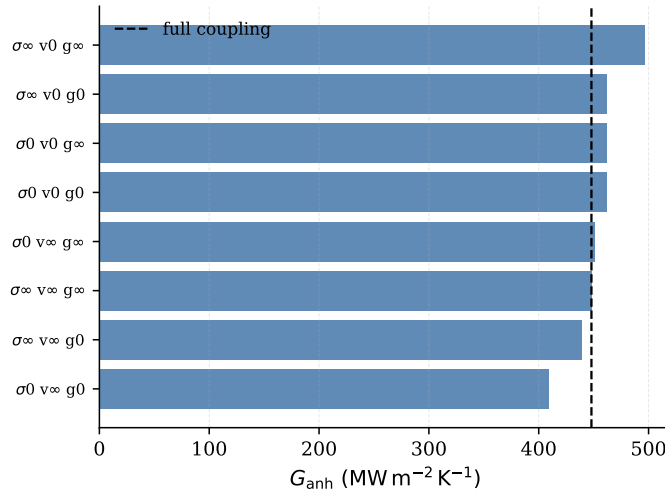


Figure 2.7.: (3, 3) carbon nanotube with two transport cells: G_{anh} across the self-energy combinations (FC3 magnitude σ , vertex range v , Green's-function range g). All lie within $\pm 10\%$ of the full-coupling value (dashed).

SCBA converges at one cell to $G_{\text{anh}}/G_{\text{ball}} = 0.775$, slightly below the (3, 3)'s 0.810 (more channels and more three-phonon phase space). The multi-cell ladder (192 DOF at two cells) exceeds the dense solver's reach and is left to the production path.

2.5. Silicon thin film

For a transversely-periodic system, the three-phonon self-energy couples the transverse momenta by crystal-momentum conservation (Eq. (1.65)). As a quantitative benchmark, we model cross-plane heat transport through a silicon thin film following Guo et al. [38]: the film is bulk silicon truncated to N layers along the transport direction with in-plane two-dimensional periodicity. Conductance is reported per unit area, $G = J/(A_{\perp} \Delta T)$, and the cross-plane conductivity of a film of thickness L is $\kappa(L) = G L$.

Ballistic baseline. The transverse mesh and broadening must both be converged: at a fixed eight-layer thickness the ballistic conductance converges for 8×8 or finer ($G_{\text{ball}} = 970/928/921/920 \text{ MW m}^{-2} \text{ K}^{-1}$ at $4 \times 4/8 \times 8/12 \times 12/16 \times 16$), and a small η is needed to avoid the coherence attenuation of Section 2.3 that otherwise crushes the conductance. With these settings the ballistic conductance is $\sim 10^3 \text{ MW m}^{-2} \text{ K}^{-1}$, declining slowly with thickness (Fig. 2.9) – consistent with the scale of Guo et al. [38] for the $2 \times 2 \times 2$ force constant.

Anharmonic conductance. Figure 2.8 (left) compares our cross-plane conductance to the anharmonic-NEGF values of Guo *et al.* at 3 and 5 unit cells. The three-phonon

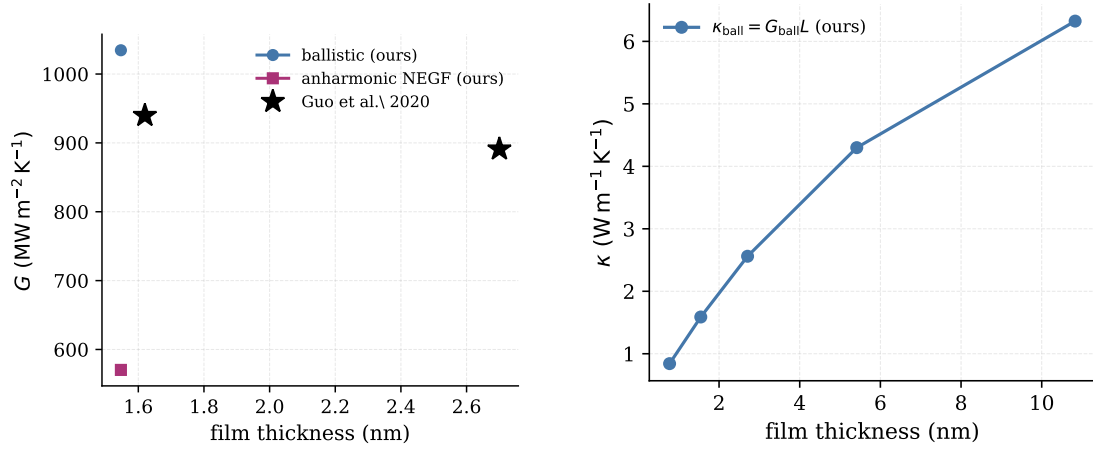


Figure 2.8.: Cross-plane heat transport through a silicon thin film at 300 K with bulk-silicon force constants (transverse mesh 8×8 , 121 frequencies). *Left*: per-area conductance – our ballistic and anharmonic NEGF against Guo et al. [38]. *Right*: the ballistic $\kappa = G_{\text{ball}}L$ rises linearly with thickness.

scattering reduces the 3-unit-cell conductance by $\sim 45\%$ ($G_{\text{anh}} \approx 570 \text{ MW m}^{-2} \text{K}^{-1}$), significantly more than the reference’s $\sim 16\%$ ($G_{\text{anh}} = 939.7$). This might be due to the reference’s additional self-energy approximations (the diagonal-block self-energy and the nearest-neighbour FC3 truncation quantified in Section 2.10), with the broadening and thin-device transport setup, and with the exchange–correlation functional of the underlying force constants (PBE here). The ballistic conductance is, by contrast, force-constant sensitive at the $\sim 15\%$ level (Fig. 2.9): the $2 \times 2 \times 2$ QE–FD constant – comparable to the reference’s own QE harmonic input – matches their scale, but the higher-quality $5 \times 5 \times 5$ VASP constant gives a $\sim 15\%$ lower ballistic (purely a harmonic difference, the slightly softer dispersion). So the agreement with the reference’s ballistic value holds only for the $2 \times 2 \times 2$ constant, not force-constant-independently.

Long-range force constant. The thin-film results above use the bulk force constants from a small $2 \times 2 \times 2$ supercell with FD. To test convergence, we generated a substantially higher-quality force constant using hiphive on a $5 \times 5 \times 5$ primitive supercell (250 atoms) from 24 rattled VASP-PBE configurations, with a third-order cutoff extended to $\sim$ fourth-neighbour ($5.0 \text{ \AA}$). The fit is good (automatic relevance determination, 103 parameters, train/test RMSE both $0.020 \text{ eV \AA}^{-1}$), the rotational and acoustic sum rules are satisfied, and the dispersion looks correct. On this force constant, the cross-plane scaling (Fig. 2.9) gives $G_{\text{anh}}/G_{\text{ball}} = 0.52, 0.45, 0.38$ at 1.16, 1.93, 3.09 nm: the anharmonic reduction grows with thickness, and is larger than with the short-cutoff force constant because the longer-range third-order terms add inter-slab scattering. The over-reduction relative to Guo et al. [38] therefore persists with the better-converged force constant. The long-range constant is also the more *numerically* robust one: its anharmonic SCBA converges cleanly at every

thickness, whereas the short-cutoff $2 \times 2 \times 2$ constant destabilises the small-broadening SCBA away from a single thickness (hence its anharmonic value is shown at one point only in Fig. 2.9).

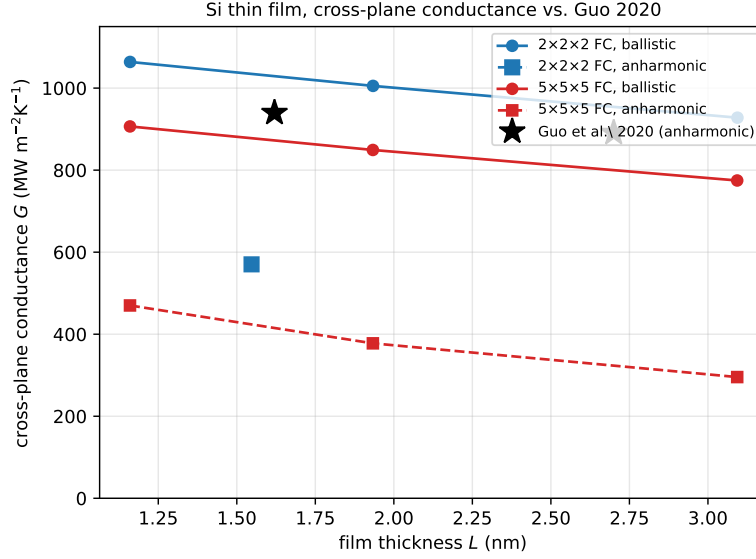


Figure 2.9.: Consolidated cross-plane silicon thin-film conductance vs. thickness at 300 K (8×8 mesh, 121 frequencies): our ballistic and anharmonic NEGF with the $2 \times 2 \times 2$ (QE-FD) and the high-quality $5 \times 5 \times 5$ ($\sim$ fourth-neighbour, VASP) force constants, against the anharmonic values of Guo et al. [38] (939.7 at 3 uc, 890.97 at 5 uc; their ballistic is a Boltzmann dashed line with no quoted value). The $2 \times 2 \times 2$ ballistic matches the reference’s scale while the $5 \times 5 \times 5$ ballistic is $\sim 15\%$ lower (a harmonic force-constant difference); both anharmonic results lie well below the reference – the over-reduction of Section 2.10.

2.6. Silicon/germanium heterostructure

A Ge-layer barrier embedded in a silicon film (Si force constants, Ge mass; Si contacts) is built with a per-slab mass profile threaded through the $q_{\perp}$ -resolved driver (an all-Si profile reproduces the homogeneous driver to machine precision). The mass-mismatch barrier raises the ballistic thermal resistance by +340–370%. The phonon-phonon enhancement of the resistance is present in the pure silicon film – +32% at 2.3 nm growing to +51% at 5.4 nm – but is suppressed ($\lesssim 1\%$) in the heterostructure.

2.7. Loop and Tadpole

Beyond the dynamic three-phonon bubble $\Sigma_B(\omega)$ (the bubble self-energy Σ_s of Eq. (1.59); the only term with an imaginary part, hence the linewidth), the lowest-order diagrammatic expansion adds two static, real, Hermitian self-energies: the quartic loop $\Sigma_{\text{loop}} = \frac{1}{2} \Phi_4 : \langle uu \rangle$ (needs FC4) and the cubic tadpole $\Sigma_T = \Phi_3 : \langle u \rangle$ with $\langle u \rangle = -\Phi_{\text{eff}}^{-1} \frac{1}{2} \Phi_3 : \langle uu \rangle$, where Φ_{eff} is the effective (renormalised) harmonic, i.e. second-order, force-constant matrix—the bare curvature D plus the static self-energy correction—(internal-coordinate relaxation, needs FC3). We computed all three contributions across temperature for the soft-mode d5a nanowire and the stiff (3,3) carbon nanotube (Fig. 2.10). The CNT FC4 was obtained from the existing rattled set with a short (2.0 Å) quartic cutoff, the d5a FC4 from the rfe refit of Section 2.2.

For the CNT, the dynamic bubble dominates, $\max |\text{Re } \Sigma_B| : \|\Sigma_{\text{loop}}\| : \|\Sigma_T\| \approx 8 : 1 : 0.5$ at 300 K, all growing with T as $\langle uu \rangle$ grows. For the d5a wire, the ordering inverts: the bubble is weak ($\max |\text{Re } \Sigma_B| \approx 670 \text{ THz}^2$ at 300 K), while the loop is the largest correction ($\|\Sigma_{\text{loop}}\| \approx 1045 \text{ THz}^2$). In both structures, the loop exceeds the tadpole at every temperature.

For the CNT, the loop is a $\sim 10\%$ renormalisation that does not improve the (10^{-4}) conservation, so it is unnecessary. For d5a, it is the dominant static term. It makes the high-temperature SCBA converge, and it improves the heat-flow conservation (more than halving it from 2.98% to 1.34% at 30 K, and reaching 0.45% at 100 K with the loop+tadpole). The loop is the physical cure for a soft mode: stiffening it lifts the propagator out of the near-singular region. Numerically, the relaxation $\langle u \rangle = -\Phi_{\text{eff}}^{-1}(\dots)$ is not formed by a bare inverse. The symmetrised Φ_{eff} is eigendecomposed and inverted with an eigenvalue floor: only modes above a cutoff $\max(\text{omega2_floor_rel} \cdot \lambda_{\text{max}}, \text{scp_floor})$ are kept (a relative floor 10^{-3} together with an absolute scp_floor in THz^2), giving a floored pseudo-inverse. This eigenvalue floor—not the FC4 loop—is the code’s actual cure for the singular inversion of a soft Φ_{eff} .

Run in isolation, the tadpole diverges: for d5a the tadpole-only self-energy reaches $\|\Sigma_T\| = 1298 \text{ THz}^2$ at 300 K and drives the lowest mode imaginary ($\omega^2 = -2.10 \text{ THz}^2$), with the SCBA failing and the heat-flow conservation increasing. The optical tadpole T_O is the gradient (force on the internal coordinates) that relaxes the internal coordinates and vanishes at the relaxed structure. With coupled self-consistency, the quartic loop stiffens the modes, which lowers $\langle uu \rangle$, which shrinks $\langle u \rangle$ and hence Σ_T . Running loop and tadpole together (as SSCHA does [112]) the same 300 K point gives $\|\Sigma_T\| = 284 \text{ THz}^2$ ($4.6\times$ smaller) and a stable lowest mode (0.765 THz). On bulk silicon, $\|\Sigma_T\|$ stays at 1–7 THz^2 (a few percent of the optical ω^2) across 100–600 K and never destabilises.

2.8. Self-consistent convergence at vanishing broadening

The numerical broadening η in Eq. (1.38) regularises the contact and device Green’s functions but is not physical; a conserving anharmonic conductance is defined only in the $\eta \rightarrow 0$ limit (Section 1.2.7). We examined whether that limit is reachable directly,

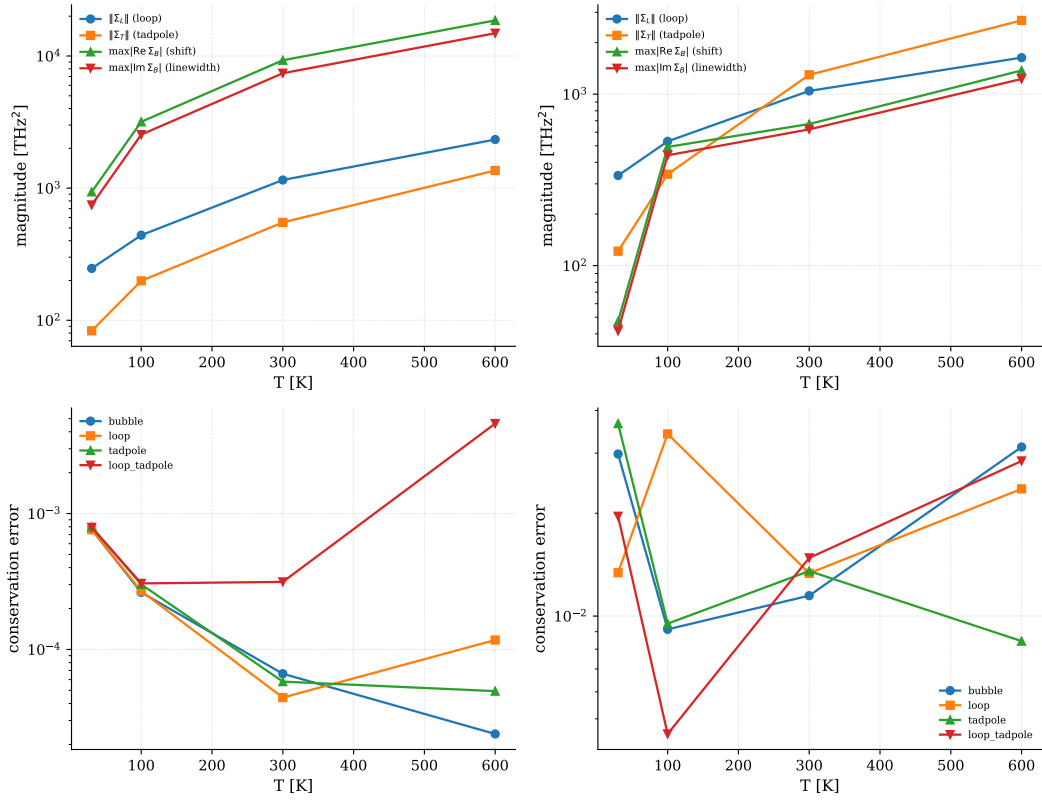


Figure 2.10.: Static-correction magnitudes (top) and heat-flow conservation per correction (bottom) versus temperature, for the CNT (left) and the d5a nanowire (right)

using the extrapolation/root-finding solvers of Section 1.2.7 with the smooth spectral window and infrared taper. The two test structures behave oppositely, and the boundary between them is set by a single, a priori computable quantity: the anharmonic linewidth relative to the frequency grid.

CNT: a genuine $\eta = 0$ fixed point. For the armchair (3,3) tube the $\eta = 0$ SCBA converges to a true fixed point. With the smooth window, the infrared taper, the causal (Kramers-Kronig) retarded self-energy and plain linear or reduced-rank-extrapolation mixing at $\eta = 10^{-12}$, the two-cell device reaches a relative Σ^R residual of 3.8×10^{-11} , a lead-current imbalance $|J_L - J_R|/|J| = 3.2 \times 10^{-11}$, and an energy-balance residual (Eq. (1.89)) of 1.8×10^{-16} in 49 iterations (Fig. 2.12, left) – a self-consistent, current-conserving solution at no broadening. The physical conductance is the grid-weighted heat current $G \cdot d\omega \approx 17$ (the unweighted lead sum scales with the number of grid points and is not itself the conductance). It is insensitive at the few-percent level to the infrared regularisation: across $\omega_{\text{reg}} = 1.83\text{--}3.06$ THz the converged $G \cdot d\omega$ varies by 3.7% and extrapolates to 17.4 as $\omega_{\text{reg}} \rightarrow 0$ (Fig. 2.13, left), and it changes by 1% between the 181- and 361-point grids

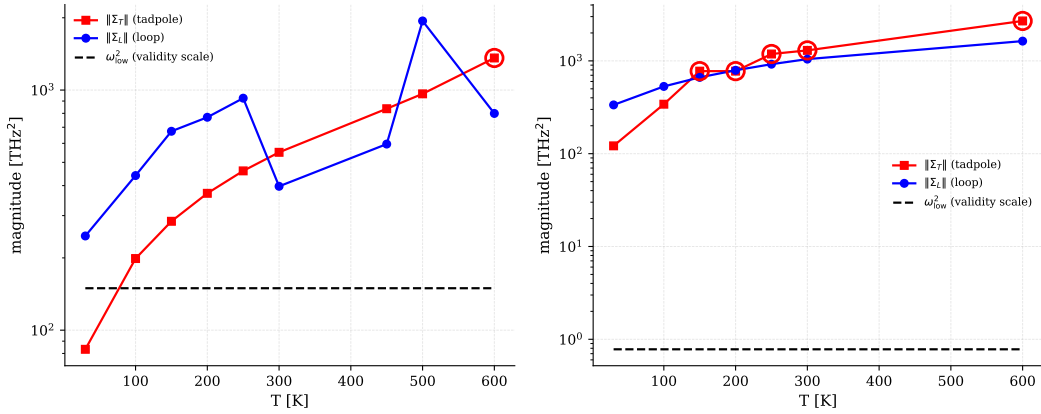


Figure 2.11.: Tadpole magnitudes versus temperature for the CNT and d5a. Circled points mark where the tadpole-only self-energy drives an imaginary renormalised mode and the SCBA does not converge.

(right).

Anharmonic transport from the converged solution. The conserving $\eta = 0$ fixed point yields a genuine inelastic conductance. The ballistic transmission carries up to ~ 12 channels with the expected band structure; three-phonon scattering suppresses it across the whole band (Fig. 2.14), removing 39% of the band-integrated transmission at two transport cells and 46% at three – the resistive onset grows with length. The ballistic conductance is itself length-independent (the Landauer baseline, $8.0 \times 10^8 \text{ W m}^{-2} \text{ K}^{-1}$ at two, three and four cells), so the falling anharmonic ratio is genuine scattering, not coherence loss. The frequency-integrated $G_{\text{anh}}/G_{\text{ball}}$ (Fig. 2.15; the heat conductance weights $T(\omega)$ by the thermal window $\partial n/\partial T$, so it is not identical to the flat band-integrated suppression quoted above) rises modestly from 0.75 at 30 K to ≈ 0.81 near 100 K and then falls to 0.57 at 300 K as the three-phonon scattering rate grows with the thermal phonon population, and it deepens with device length, $0.57 \rightarrow 0.51$ from two to three transport cells at 300 K (the four-cell anharmonic run did not converge at $\eta = 0$ and is excluded). These production-NEGF values at $\eta = 10^{-12}$ are obtained with a different solver and broadening than the dense-reference numbers of Section 2.4 (a monotonic $0.92 \rightarrow 0.71$); they show a stronger reduction but the same qualitative increase of anharmonic resistance with temperature and length. The stronger reduction is the physical $\eta \rightarrow 0$ limit, not an artefact: at matched broadening the ratio falls monotonically toward this value as $\eta \rightarrow 0$, and the converged $\eta = 10^{-12}$ conductance is robust to the infrared regulariser and grid at the few-percent level while conserving at 7×10^{-6} ; the finite- η systematic that separates it from a single-broadening number is quantified in Chapter F.

Infrared behaviour. The low-frequency end deserves a separate word. The ballistic transmission plateaus at the four acoustic channels of the tube, $\mathcal{T}(\omega \rightarrow 0) \rightarrow N_{\text{ac}} = 4$, so

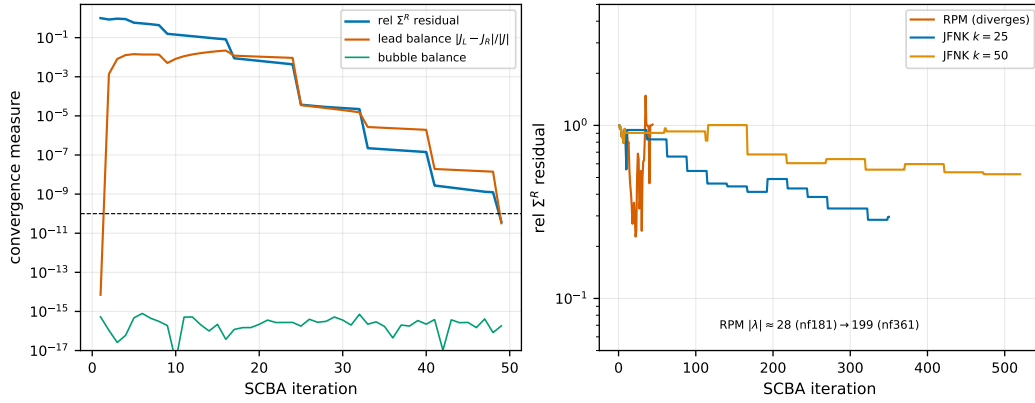


Figure 2.12.: $\eta = 0$ SCBA convergence. Left: the (3,3) CNT reaches a genuine fixed point – the Σ^R residual and lead imbalance fall to $\sim 10^{-11}$ while the energy balance stays at 10^{-16} . Right: for the d5a nanowire the bare $\eta = 0$ root-finders do not converge by contraction – RPM diverges and Jacobian-free Newton–Krylov stays bounded but stalls (residual near 0.3–0.5, independent of the Krylov dimension $k = 25$ or 50); the broadening-floor-stabilised marginal fixed point ($G \approx 2.6$, conserving) is described in the text.

the heat-current density $\hbar\omega I(\omega)$ tends to a constant — the Bose factor’s $1/\omega$ is cancelled by the $\hbar\omega$ weight, the quantised-conductance plateau $g_0 = \pi^2 k_B^2 T / 3h$ per channel (Fig. 2.16, blue; Eq. (1.91)). The $\eta = 0$ anharmonic curve, computed here *without* the infrared taper, instead bends down and fluctuates as $\omega \rightarrow 0$ (Fig. 2.16, red). This is *not* a Bose pole in the scattering self-energy: the bosonic fold and the linearly-vanishing acoustic spectral weight cancel the $1/\omega$ of the bubble legs, and the bare bubble is exactly conserving (Chapter F). What remains is the first-bin sampling of the genuine acoustic retarded enhancement $G^R \sim 1/\omega^2$ at the soft modes, amplified by the marginal $\eta = 0$ fixed-point fluctuation of Section 1.2.7. Because the heat current is $\hbar\omega$ -weighted this region carries little conductance — the converged $G \cdot d\omega$ is insensitive to the infrared regularisation at the few-percent level (Fig. 2.13) — so the accurate, conserving choice is the bare bubble, the taper serving only as a pragmatic stabiliser removed by the $\omega_{\text{reg}} \rightarrow 0$ extrapolation. The $1/\omega$ that is physical lives in the observable heat current, where the $\hbar\omega$ weight turns it into the finite plateau, not in the bubble. The quasi-one-dimensional acoustic linewidths of the tube [137] mean the converged anharmonic infrared limit need not coincide with the full ballistic plateau.

d5a: a marginal fixed point at finite grid. The hydrogen-passivated d5a wire behaves differently because of its surface modes. The golden-rule linewidth (Eq. (1.60)), evaluated directly from $\Phi^{(3)}$ with no SCBA, separates cleanly by hydrogen character (Fig. 2.17): the silicon acoustic and optical modes (0–15 THz) have $\Gamma_{\text{anh}} \approx 4$ THz, far above any feasible grid spacing; the Si–H bending band (≈ 16 –21 THz, $\sim 90\%$ hydrogen character) has a median $\Gamma_{\text{anh}} \approx 0.65$ THz but a tail down to ≈ 0.13 THz; and the Si–H stretch

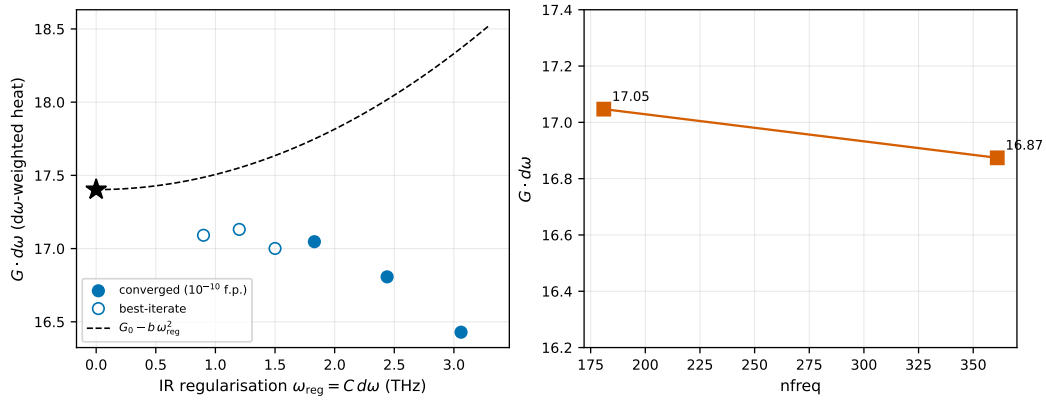


Figure 2.13.: CNT (3,3) $\eta = 0$ conductance $G \cdot d\omega$ versus the infrared regularisation scale ω_{reg} (left; filled markers are genuine 10^{-10} fixed points, the dashed line a $G_0 - b\omega_{\text{reg}}^2$ fit) and versus the frequency-grid size (right). The cutoff and grid sensitivities are both at the few-percent level.

(≈ 63 THz) has $\Gamma_{\text{anh}} \approx 0.13$ THz but carries no heat – its occupation is frozen and it is masked. The sharp, heat-carrying tail of the bending band thus straddles the grid spacing ($d\omega = 0.37$ and 0.18 THz at 181 and 361 points), and by Eq. (1.85) it renders the $\eta = 0$ fixed point marginal.

Consistent with this, the bare $\eta = 0$ iteration does not converge by contraction (Fig. 2.12, right): plain and reduced-rank mixing limit-cycle, and the RPM of Section 1.2.7 diverges, its measured unstable-subspace radius growing from $|\lambda(H)| \approx 28$ (3 unstable modes) at 181 points to ≈ 200 (5 modes) at 361 – the instability *strengthens* as the grid resolves the sharp Si–H modes, confirming it is physical rather than a discretisation artefact. Jacobian-free Newton–Krylov stays bounded but its inner GMRES cannot reduce the Newton residual below $\sim 97\%$ of its value at any Krylov dimension, the signature of a near-singular, indefinite Jacobian. The fixed point is therefore *marginal*: the flexural-mode eigenvalue of the SCBA map sits at $|\lambda| \approx 1$, so no iteration can drive the self-energy residual to zero.

The transport observable, however, is well defined. A DC-concentrated, grid-consistent broadening floor ($\Gamma_{\text{floor}} \propto d\omega^2$, acting only on the unresolved sub-grid bins so that it vanishes in the continuum) lifts the marginal mode just enough to enter its basin, and a warm-started trust-region JFNK then settles there. Across the root-finders that reach the basin the self-energy residual floors at ≈ 0.03 – not 10^{-10} , the marginal mode forbidding a tight fixed point – while the conductance is method-invariant, $G \approx 2.6$ within $\sim 3\%$ (2.61, 2.63, 2.64, 2.70 across the four converging solvers; the best-conserved trust-region JFNK gives 2.61). The Φ -derivable bubble balance Eq. (1.89) holds to machine precision ($\sim 10^{-18}$) at every iterate, while the lead-current imbalance floors at $\sim 2.5 \times 10^{-3}$ – the finite-broadening ghost-reservoir term of Chapter F, which vanishes only as the floor (hence $d\omega$) goes to zero. The floor is a crutch to reach the basin, not a physical input: annealing it to zero leaves G unchanged at the 3% level. The $\eta = 0$ d5a conductance is

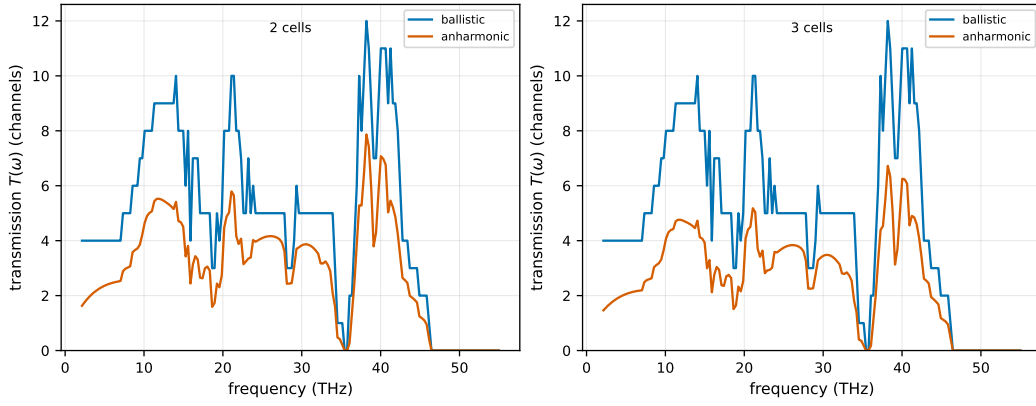


Figure 2.14.: Ballistic versus anharmonic phonon transmission $T(\omega) = I(\omega)/\Delta n(\omega)$ (in units of conducting channels) of the converged $\eta = 0$ (3,3) CNT, for two (left) and three (right) transport cells. The ballistic transmission shows the channel structure (up to ~ 12 channels); three-phonon scattering suppresses it across the band, more strongly at the longer device. The sub-2 THz bins are the infrared-regularised acoustic region.

thus converged and conserving as a transport number, the residual marginality – not a missing solution – being the honest content of the surface-mode physics. Whether the ≈ 0.03 residual floor is structural or merely resolution-limited is being settled by the grid-doubling to 361 points. The boundary is in either case sharp and predictive: the $\eta = 0$ anharmonic transport problem is well posed exactly when every heat-carrying mode is resolved by the frequency grid.

2.9. Strong anharmonicity: SrTiO_3

The static corrections of Section 2.7 are a small perturbation for the weakly anharmonic silicon systems. To test the opposite, strongly anharmonic regime, where the harmonic approximation fails, we computed FC2/FC3/FC4 for cubic SrTiO_3 , the canonical self-consistent-phonon (SCP) test case from [112].

Cubic SrTiO_3 has unstable harmonic phonons: a ferroelectric (FE) mode at Γ and an antiferrodistortive (AFD) octahedral-rotation mode at $R = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ [112]. Fitting FC2 to small-amplitude ($0.025 \text{ \AA}$) rattled displacements in the same $2 \times 2 \times 2$ (40-atom) cell used by Tadano & Tsuneyuki recovers the AFD instability: $\omega_R = 55i - 92i \text{ cm}^{-1}$ (converging upward with the number of training structures, an instability that strengthens as the fit converges). Their PBEsol value is $76i \text{ cm}^{-1}$. The static loop+tadpole magnitude is $\|\Sigma_{\text{static}}\| \approx 39 \text{ THz}^2$ at 300 K, $\sim 6 \times$ the bulk-Si value.

Fitting the same cell to a larger ($0.08 \text{ \AA}$) rattle amplitude stabilises the AFD mode ($\omega_R = +90 \text{ cm}^{-1}$, Fig. 2.18). This is not a fitting error, but the central identity of the effective-potential methods: a finite-amplitude fit returns the effective, renormalised FC2 at the

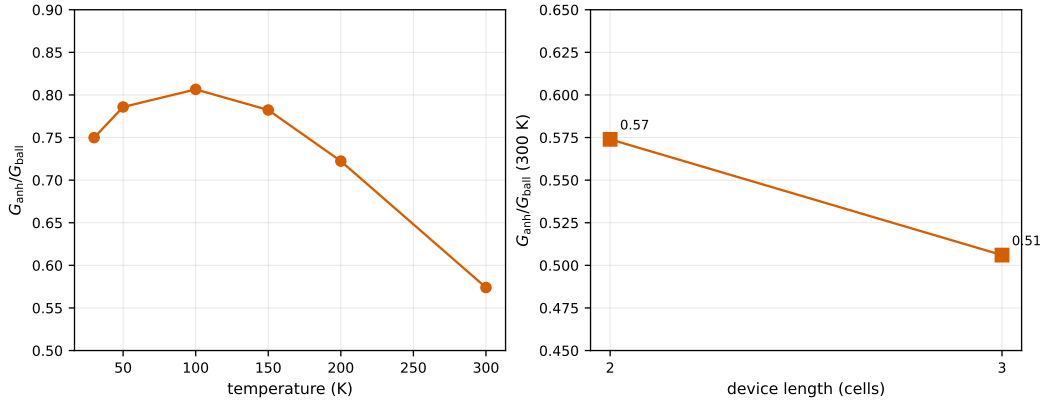


Figure 2.15.: Frequency-integrated anharmonic/ballistic conductance ratio of the converged $\eta = 0$ (3,3) CNT versus temperature (left, two transport cells) and versus device length (right, at 300 K). The reduction is non-monotonic in temperature – peaking near 100 K – and deepens with device length.

temperature where $\langle u^2 \rangle^{1/2}$ matches the rattle amplitude. Fitting at the thermal amplitude is in the same spirit as the temperature-dependent effective potential (TDEP) [147] and the stochastic self-consistent harmonic approximation (SSCHA) [148], though a single fixed-amplitude static rattle is distinct from the thermal-ensemble sampling of TDEP and the variational quantum self-consistent Gaussian of SSCHA. The difference between the bare and effective FC2 is the static loop self-energy $\frac{1}{2}\Phi_4 : \langle uu \rangle$ (Section 2.7).

The renormalised phonons can hence be obtained either from the bare FC2 plus an explicit quartic loop (the deterministic SCP of Section 2.7, requiring FC4), or directly by fitting the effective FC2 at the thermal amplitude (TDEP/SSCHA, no FC4 needed) [147, 148]. The explicit loop is needed when an analytic temperature dependence from a single FC4 is wanted. The effective-fit route is preferred when FC4 is too costly. For the weakly anharmonic silicon systems neither matters (effective $\approx$ bare FC2). For SrTiO_3 one must use one route or the other.

A tensor rank-sweep of the device FC3 (truncated-SVD / HOSVD / CP) shows SrTiO_3 is markedly more compressible than d5a. A truncated SVD reaches a 7.6% Frobenius error at rank 16 for SrTiO_3 versus 18% for d5a.

2.10. Cost of the numerical approximations

The transport numbers above are computed under a hierarchy of controlled approximations – a finite numerical broadening, a truncated frequency grid, real-space cutoffs on the FC3 vertex and the Green’s function, and, optionally, a low-rank compression of the vertex. This section quantifies how each affects the converged conductance, so that every reported number carries a known systematic.

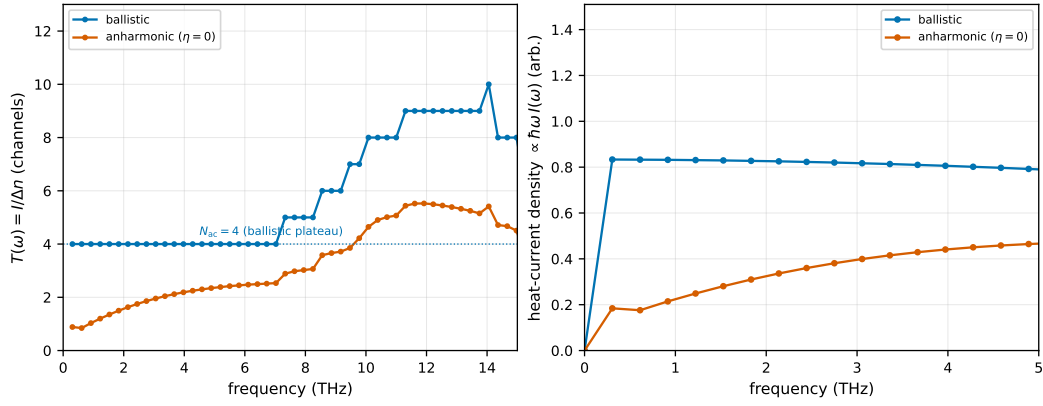


Figure 2.16.: Infrared behaviour of the converged $\eta = 0$ (3,3) CNT. *Left*: effective transmission $T(\omega) = I/\Delta n$; the ballistic curve plateaus at the $N_{ac} = 4$ acoustic channels as $\omega \rightarrow 0$, while the anharmonic curve (no taper) is suppressed and non-monotonic at low ω . *Right*: the heat-current spectral density $\propto \hbar\omega I(\omega)$; the ballistic value is flat as $\omega \rightarrow 0$ (the Bose $1/\omega$ cancelled by $\hbar\omega$, the quantised plateau), whereas the raw anharmonic result is suppressed and fluctuates — the first-bin sampling of the acoustic $G^R \sim 1/\omega^2$ enhancement under the marginal $\eta = 0$ fixed point (Section 1.2.7), not a Bose pole in the bubble self-energy (the bare bubble is conserving, Chapter F).

Finite broadening. The broadening η of Eq. (1.38) is not physical, and the conserving conductance is defined only as $\eta \rightarrow 0$ (Section 2.8). Its systematic on the *finite- η* numbers is larger than the lead-balance residual suggests. On the cnt33 tube at matched η the lead balance stays $\sim 10^{-5}$ throughout, yet the anharmonic ratio G_{anh}/G_{ball} swings $0.79 \rightarrow 0.92$ across $\eta = 0.30\text{--}0.90$ THz²; extrapolating the converging window to $\eta \rightarrow 0$ gives ≈ 0.57 , a $\sim 40\%$ reduction (Section F.5) – roughly twice the $\sim 16\%$ read off at the single point $\eta = 0.45$ THz². A single- η number therefore *underestimates* the anharmonic effect by about a factor of two; transport ratios must be $\eta \rightarrow 0$ -extrapolated or quoted with the finite- η band. The mechanism and the step-by-step numerical verification are given in Chapter F.

Frequency grid. The bubble convolution Eq. (1.98) has support up to $2\omega_{max}$, so the grid must satisfy $f_{max} \geq 2\omega_{max}/2\pi$ (Eq. (1.100)). An under-ranged grid aliases the truncated tail back into the band as a spurious strong-coupling self-energy that destabilises the SCBA: for the d5a wire the historical $f_{max} = 18$ THz, far below the ~ 126 THz convolution support set by the 63 THz Si–H stretch, inflates $|\Sigma^R|$ by $\sim 25\times$ and diverges the multi-slab iteration, whereas the properly-ranged grid returns it to the perturbative regime ($|\Sigma^R|/\omega_{max}^2 \approx 0.2\%$). The one-shot verification across bulk Si and d5a is tabulated in Table E.1; the solver auto-extends f_{max} by default.

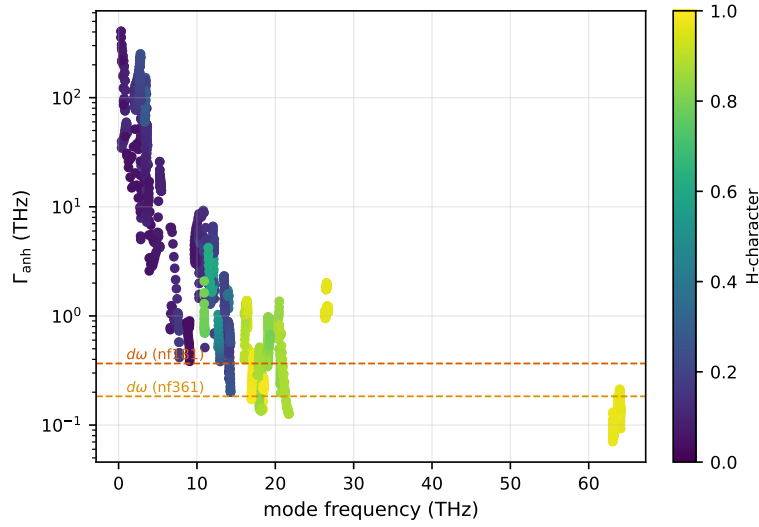


Figure 2.17.: Golden-rule anharmonic linewidth Γ_{anh} of d5a versus mode frequency, coloured by hydrogen character, with the frequency-grid spacings $d\omega$ at 181 and 361 points (dashed). The heat-carrying Si–H bending band (~ 16 –21 THz, high hydrogen character) has a tail of linewidths below $d\omega$; the Si–H stretch (≈ 63 THz) is sub-grid but thermally frozen.

Self-energy cutoffs. The FC3-magnitude threshold, the Green’s-function range, and a diagonal- G approximation inside the bubble were swept on the open nanowires (Fig. 2.5) and on the carbon nanotube (Fig. 2.7). The dominant cutoff is the off-diagonal Green’s-function content: a diagonal- G approximation is catastrophic on the open wires (5 – $30\times$), while the magnitude threshold converges quickly. The closed tube is far more forgiving – all corners of the cutoff cube lie within $\pm 10\%$ of the full-coupling value.

Vertex compression. The low-rank FC3 decompositions of Section 1.1.5 trade parameter count for accuracy (Fig. 2.2): the transport observable converges faster than the Frobenius residual the fitters minimise, but the matrixised SVD becomes unphysical at higher rank while the symmetric ansätze degrade gracefully.

Consolidate the three sweeps above into one approximation-cost figure (conductance / $G_{\text{anh}} - G_{\text{ball}}$ vs. each approximation, across cnt33/d5a/d11a/film).

2.11. Solver verification, cost, and scaling

The production distributed solver is validated against the dense reference and benchmarked on the two axes that matter for anharmonic transport: the block-tridiagonal recursion along the device and the energy-point parallelism across MPI ranks. The two solvers follow the structure of the self-energy (Sections 1.3.7 and 1.4.8): Gamma-only

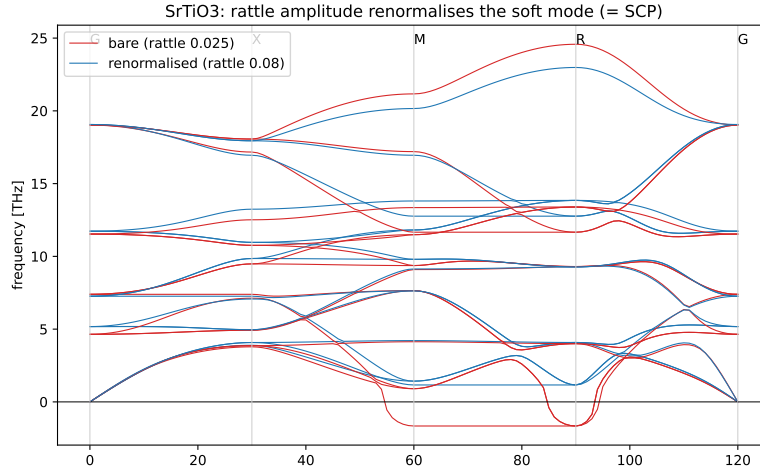


Figure 2.18.: SrTiO_3 phonon dispersion: FC2 fitted to a small ($0.025 \text{ \AA}$, red) versus large ($0.08 \text{ \AA}$, blue) rattle amplitude.

conductors ($k=1$, the nanowires and tubes) scale on $\text{block_comm_size} \times \text{stack}$, while the transversely-periodic systems ($k>1$, films and perovskites) scale on $\text{q_comm_size} \times \text{stack}$ with $\text{block_comm_size}=1$; for the large-block Gamma-only regime the inner-ring contraction additionally threads over the frequency axis (Section 1.3.2).

Machine-precision agreement with the dense reference. The production block-tridiagonal recursive Green’s-function solver (`qtools.greens_function_solver.RGF`) reproduces the dense reference and a direct numpy inverse: the generic selected-inversion suite passes all 144 parametrised cases, and an end-to-end check on a genuine phonon Dyson matrix $A(\omega^2) = \omega^2 I - H_D - \Sigma_{\text{leads}}$ with Sancho–Rubio open-boundary self-energies agrees to 1.4×10^{-16} on the selected retarded Green’s function across $\omega \in [1, 16] \text{ THz}$. The three-phonon bubble and scattering-self-energy kernels pass all 28 tests; these also flag that a nearest-neighbour FC3 truncation discards 34–48% of the Frobenius norm for the test fixtures, so the number of primitive cells per transport cell must be chosen with care. The block-distributed recursion (`rgf_dist.RGFDist`) reproduces the serial reference under `mpirun -np 3 (block_comm_size=3)`.

Solver cost. The dense reference inverts the full $N_D \times N_D$ device matrix, so its cost grows as $\mathcal{O}(n_{\text{slabs}}^3)$, whereas RGF exploits the block-tridiagonal structure and is linear in the number of cells (Fig. 2.19, left): on phonon-Dyson matrices with a fixed 192-dof block, from 2 to 32 blocks the RGF wall time rises linearly while the dense time follows the cubic guide, a $58\times$ gap at 32 blocks, with the two agreeing to $\sim 10^{-13}$ on the block-tridiagonal entries. The complementary axis – distributing the energy points across MPI ranks – is shown on the right: a fixed 1536×1536 problem with 128 energy points falls from 1.55 s on one rank to 0.86, 0.55 and 0.28 s on 2, 4 and 8 ranks, a $5.3\text{--}5.5\times$ speed-up at eight ranks (68% parallel efficiency) that improves with problem size.

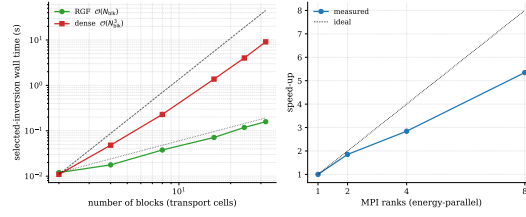


Figure 2.19.: Solver scaling. Left: selected-inversion wall time vs. number of transport-cell blocks – RGF is linear, the dense reference cubic ($58\times$ apart at 32 blocks), agreeing to $\sim 10^{-13}$. Right: distributed energy-parallel speed-up ($5.3\times$ at eight ranks).

This separation is decisive for the anharmonic problem. The dense self-energy bubble parallelises across the (I, J) slab-pair blocks – their count grows $1 \rightarrow 4 \rightarrow 7$ for $n_{\text{slabs}} = 1, 2, 3$ – but each block’s bubble is itself a serial BLAS contraction, so the dense per-point wall time climbs steeply: a single $(3, 3)$ -CNT SCBA point costs 66 s at $n_{\text{slabs}}=1$ and 1193 s ($\sim 18\times$) at $n_{\text{slabs}}=2$, and the converged three-cell point (108 DOF, 15 blocks) costs 6.8 h. Multi-cell anharmonic transport is therefore the regime of the RGF and energy-parallel production path, with the dense solver kept as the verification oracle.

Production three-phonon throughput and memory. A per-phase profile of a four-cell CNT SCBA iteration attributes 99.8% of the wall time to the three-phonon bubble (SigmaPhononPhonon), with the RGF Dyson solve, the Sancho–Rubio open boundaries, the stack/nnz transpose and the convergence book-keeping each below 0.1% (Fig. 2.21, left); the bubble is thus the only phase worth distributing. It parallelises cleanly on whichever communicator carries its outer index: the (I, J) block loop reaches $5.4\times$ at six ranks for I -only conductors (Fig. 2.20, left) and the external- q distribution reaches $6.6\times$ at eight ranks for transversely-periodic systems (right), while the replicated energy/stack axis yields no bubble speed-up, as expected. Strong scaling of the bubble itself stays near ideal out to ~ 16 ranks on the stack axis, with the q -communicator saturating earlier once the internal all-gather of G dominates (Fig. 2.21, right). Vertex storage is the binding memory constraint: the dense per-block vertex Φ grows steeply with the transverse q -mesh count N_q , reaching the few-GB scale for a realistic $n_{\text{dof}}=63$ block, whereas streaming the vertex keeps the peak roughly an order of magnitude lower and well within the single-GPU 80 GB budget (Fig. 2.22).

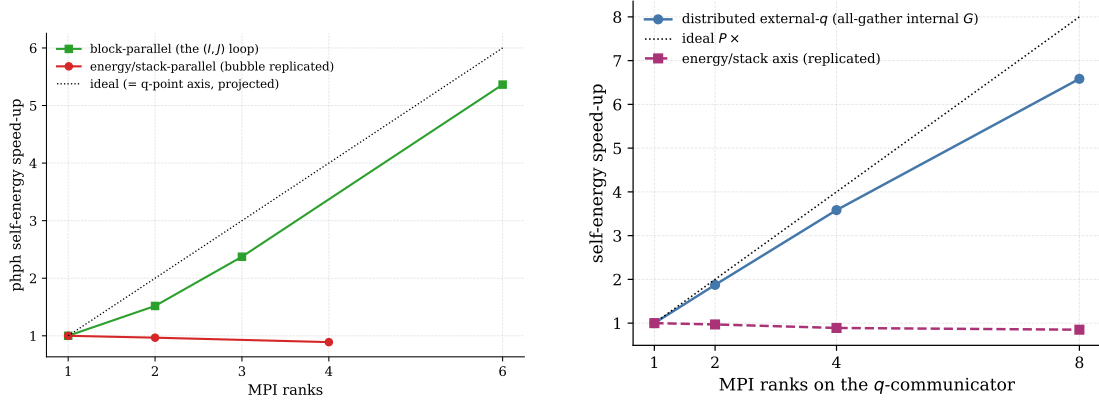


Figure 2.20.: Distributed three-phonon self-energy speed-up. Left: I -only conductors scale on the (I, J) block loop (block-parallel), reaching $5.4 \times$ at six ranks; the replicated energy/stack axis gives no speed-up. Right: transversely-periodic systems scale on the external- q communicator (with an all-gather of the internal G), reaching $6.6 \times$ at eight ranks. Dotted: ideal linear scaling.

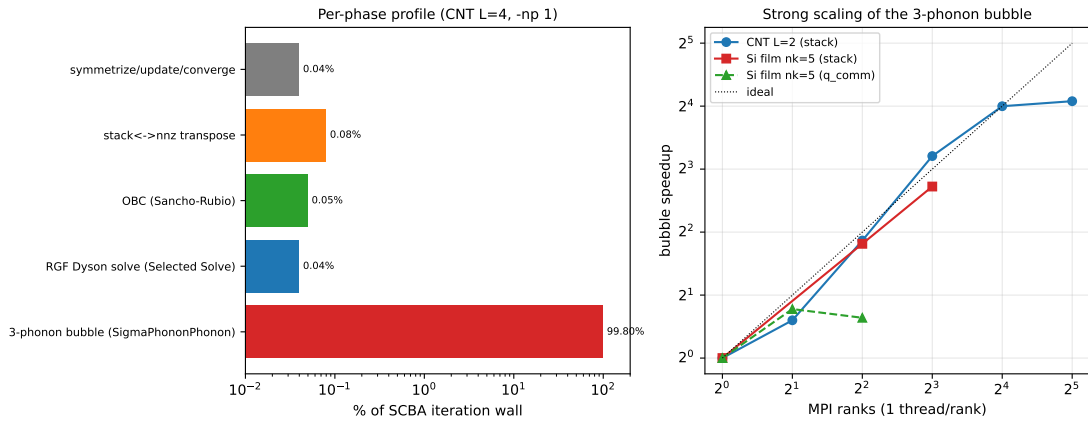


Figure 2.21.: Production three-phonon path. Left: per-phase profile of a single SCBA iteration (CNT $L=4$, one rank) – the bubble (SigmaPhononPhonon) dominates at 99.8% of the iteration wall, all other phases below 0.1%. Right: strong scaling of the bubble (one thread per rank) for a CNT stack, a Si-film stack and the Si-film q -communicator, against the ideal line.

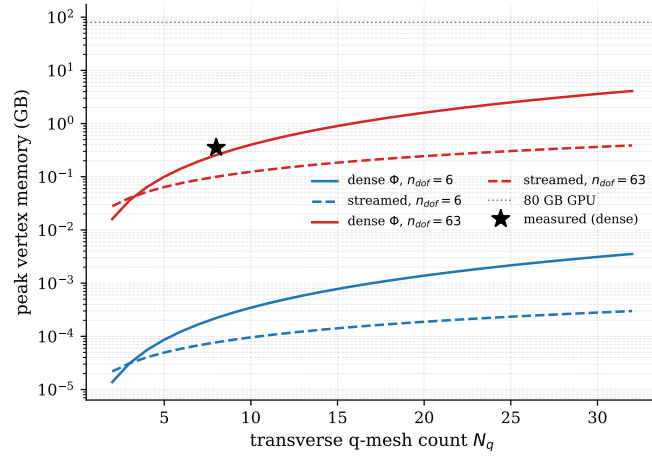


Figure 2.22.: Peak vertex memory vs. transverse q -mesh count N_q for the dense vertex Φ and the streamed variant at $n_{\text{dof}} = 6$ and 63. Streaming lowers the peak by roughly an order of magnitude; the dotted line marks the 80 GB single-GPU budget and the star a measured dense point.

Appendix A

Conventions and Units

This chapter collects the unit systems, notation, and Fourier conventions used in this document. The harmonic force constants may originate either from phonopy finite-displacement calculations or from Quantum ESPRESSO (QE) density-functional perturbation theory, while the QuaTrEx solver uses its own internal frequency-squared convention. Since the later extraction, transport, and anharmonic self-energy steps all rely on consistent units and phase conventions, we summarize them here once and refer back to this section throughout the document.

A.1. Units in phonopy and Quantum ESPRESSO

Phonopy stores force constants in $\text{eV}/\text{\AA}^2$, atomic masses in amu, and atomic positions in $\text{\AA}$. The dynamical matrix eigenvalues therefore have units

$$[\lambda] = \frac{\text{eV}}{\text{amu} \cdot \text{\AA}^2}. \quad (\text{A.1})$$

Phonopy converts these eigenvalues to frequencies in THz via

$$f [\text{THz}] = \text{sgn}(\lambda) \sqrt{|\lambda|} \times \frac{1}{2\pi} \sqrt{\frac{e}{m_u A^2}} \approx 15.633 \text{ THz}, \quad (\text{A.2})$$

where $e = 1.602 \times 10^{-19} \text{ J}$ is the elementary charge, $m_u = 1.661 \times 10^{-27} \text{ kg}$ is the atomic mass unit, and $A = 10^{-10} \text{ m}$ is the $\text{\AA}$ -to-metre conversion factor.

When phonopy is used together with the QE interface, the internal representation inherited from QE uses atomic units: lattice vectors are in bohr and force constants in Ry/bohr^2 . These quantities are therefore converted before being used together with the rest of the workflow. The required relations are

$$a [\text{\AA}] = a [\text{bohr}] \times a_0, \quad a_0 = 0.52918 \text{ \AA}/\text{bohr}, \quad (\text{A.3})$$

$$\Phi [\text{eV}/\text{\AA}^2] = \Phi [\text{Ry}/\text{bohr}^2] \times \frac{E_h/2}{a_0^2}, \quad \frac{E_h/2}{a_0^2} \approx 48.59. \quad (\text{A.4})$$

Appendix A. Conventions and Units

Here E_h denotes the Hartree energy, so that $E_h/2 = 1$ Ry. In the same way, Cartesian forces parsed from QE output are converted using

$$1 \text{ Ry}/\text{bohr} \approx 25.711 \text{ eV}/\text{\AA}. \quad (\text{A.5})$$

The downstream workflow therefore treats eV, $\text{\AA}$, and amu as the common intermediate unit system, independent of whether the original data came from phonopy directly or through QE. In other words, everything is normalized to the phonopy eV/ $\text{\AA}$ /amu convention upon loading.

A.2. QuaTrEx solver units

The QuaTrEx phonon solver operates in angular frequency squared,

$$\omega^2 \text{ in } (\text{rad/s})^2. \quad (\text{A.6})$$

Accordingly, the mass-weighted real-space blocks passed to the solver must be converted from the phonopy eigenvalue units eV/(amu $\cdot\text{\AA}^2$) to (rad/s) 2 . The corresponding conversion factor is

$$C = \frac{e}{m_u A^2} \approx 9.648 \times 10^{27} (\text{rad/s})^2. \quad (\text{A.7})$$

If $\Phi(0, \mathbf{R})$ denotes a harmonic force-constant block in eV/ $\text{\AA}^2$, then the real-space matrix block used by QuaTrEx is

$$H(\mathbf{R}) = C \frac{\Phi(0, \mathbf{R})}{\sqrt{m m'}}, \quad (\text{A.8})$$

with m and m' in amu. The resulting $H(\mathbf{R})$ therefore has units of (rad/s) 2 .

The same logic extends to cubic force constants. After mass-weighting, the third-order tensor

$$\Psi_{\alpha\beta\gamma} = \frac{\Phi_{\alpha\beta\gamma}^{(3)}}{\sqrt{M_\kappa M_{\kappa'} M_{\kappa''}}} \quad (\text{A.9})$$

has units

$$[\Psi] = \frac{\text{eV}}{\text{\AA}^3 \cdot \text{amu}^{3/2}}. \quad (\text{A.10})$$

Its SI conversion factor is

$$\text{CONVERSION_FC3} = \frac{C}{A\sqrt{m_u}} = \frac{e}{m_u^{3/2} A^3} \approx 2.368 \times 10^{51}, \quad (\text{A.11})$$

which converts eV/($\text{\AA}^3 \cdot \text{amu}^{3/2}$) to the SI units required in the phonon-phonon self-energy expressions.

THz-based internal unit system. The $(\text{rad/s})^2$ statement above is the SI-facing equivalent. In practice the production phonon-phonon stack (`quatrex/phonon/units.py`, `static_self_energy.py:device_fc3_mass_weighted`) works in a *THz-based* internal system: the dynamical matrix D and all self-energies are carried in THz^2 , while the frequency axis ω is in plain THz, with the bubble assembled using `bubble_prefactor_thz =  $i\hbar \Delta\Omega/(4\pi)$` . The THz^2 convention and this prefactor are documented in Section D.1; the $(\text{rad/s})^2$ form and Eq. (A.11) provide the SI-facing equivalents.

Green's-function sign convention. The solver stores the phonon Green's functions and self-energies *occupation-positive*: $-iG^{\lessgtr}(\omega) \gtrless 0$ for $\omega > 0$, the same sign as the lead injection $\Sigma_{\ell}^{\lessgtr} = +i n_{\ell}(\pm 1)\Gamma_{\ell}$. This differs from the common textbook choice $G^< = -i n A$ by an overall sign of $G^{\lessgtr}$ — and hence of any self-energy quadratic in G , such as the three-phonon bubble; see the remark following Eq. (1.63) for the consequent sign handling.

A.3. Notation and Fourier conventions

We first collect the symbols used throughout this work.

Symbol	Meaning	Range
l, m	Supercell index	N_{Cell}
$\kappa, \kappa', \kappa''$	Basis-atom within a primitive cell	n_{prim}
$\alpha, \alpha', \alpha''$	Cartesian component (x, y, z)	$C = 3$
i, j, k	DOF index	n_{dof}
μ, ν	Primitive-cell DOF index	$3n_{\text{prim}}$
$\mathbf{q}_{\perp}$	Transverse wavevector (2D BZ)	N_q points
x, y	Transport-direction DOF	$n_{\text{trans}} = 3n_{\text{prim}}N_{\text{transport}}$
$\mathbf{R}_{\perp}$	Transverse lattice vector	N_q points (dual)
ω, ω'	Real frequency	N_{ω} points
τ	Conjugate time variable	$N_{\tau} = N_{\omega}$
N	Total number of $\mathbf{q}$ -points, N_q	

Here N_{Cell} is the total number of supercells (e.g. $3^3 = 27$), n_{prim} the number of atoms in the primitive cell (e.g. 2 for bulk Si), and $n_{\text{dof}} = 3n_{\text{super}} = 3N_{\text{Cell}}n_{\text{prim}}$ the total number of degrees of freedom in the simulation. n_{trans} is the total number of DOFs in the transport direction (i.e. when transverse cells are not included), and $N_{\text{transport}}$ is the number of transport (super)-cells.

Bold symbols such as $\mathbf{R}$ and $\mathbf{q}$ denote lattice vectors and wavevectors, respectively. The displacement of atom κ in cell l along direction α is written $u_{\alpha}(l\kappa)$. The lattice vector of cell l is $\mathbf{R}_l$, while τ_{κ} denotes the basis-atom position within the unit cell. Unless stated otherwise, τ_{κ} and $\mathbf{q}$ are expressed in fractional crystal coordinates. Two distinct phase conventions are used in this work: the dynamical-matrix transforms (Eqs. (A.13), (A.14) and (A.17)) carry an explicit 2π , i.e. $e^{+2\pi i \mathbf{q} \cdot \mathbf{R}}$, whereas the Green's-function/self-energy

Appendix A. Conventions and Units

discrete Fourier pair (Eqs. (A.20) and (A.21)) absorbs the 2π into $\mathbf{q}$ and uses the opposite forward sign, $e^{-i\mathbf{q}\cdot\mathbf{R}}$.

The harmonic force constants are written as

$$\Phi_{\alpha\alpha'}(0\kappa, l'\kappa'), \quad (\text{A.12})$$

and the corresponding mass-weighted dynamical matrix at wavevector $\mathbf{q}$ is a Fourier transform of these real-space couplings. Two phase conventions are used in this work. In the phonopy convention,

$$D_{\alpha\alpha'}^{(\text{A})}(\kappa\kappa', \mathbf{q}) = \frac{1}{\sqrt{m_\kappa m_{\kappa'}}} \sum_{l'} \Phi_{\alpha\alpha'}(0\kappa, l'\kappa') e^{2\pi i \mathbf{q} \cdot [\mathbf{R}_{l'} + \boldsymbol{\tau}_{\kappa'} - \boldsymbol{\tau}_\kappa]}, \quad (\text{A.13})$$

whereas the QuaTrEx convention uses only the lattice vector in the phase,

$$D_{\alpha\alpha'}^{(\text{B})}(\kappa\kappa', \mathbf{q}) = \frac{1}{\sqrt{m_\kappa m_{\kappa'}}} \sum_{l'} \Phi_{\alpha\alpha'}(0\kappa, l'\kappa') e^{2\pi i \mathbf{q} \cdot \mathbf{R}_{l'}}. \quad (\text{A.14})$$

The two are related by a diagonal unitary gauge transformation,

$$D^{(\text{B})}(\mathbf{q}) = P(\mathbf{q}) D^{(\text{A})}(\mathbf{q}) P^\dagger(\mathbf{q}), \quad (\text{A.15})$$

with

$$P(\mathbf{q})_{\kappa\alpha, \kappa'\alpha'} = \delta_{\kappa\kappa'} \delta_{\alpha\alpha'} e^{2\pi i \mathbf{q} \cdot \boldsymbol{\tau}_\kappa}. \quad (\text{A.16})$$

Under Convention B, the dynamical matrix is a pure lattice Fourier series,

$$D^{(\text{B})}(\mathbf{q}) = \sum_{\mathbf{R}} H(\mathbf{R}) e^{2\pi i \mathbf{q} \cdot \mathbf{R}}. \quad (\text{A.17})$$

Here $H(\mathbf{R})$ is the SI-converted mass-weighted block introduced in Eq. (A.7), $H(\mathbf{R}) = C\Phi(0, \mathbf{R})/\sqrt{m m'}$, so that it already carries the conversion factor C ; the $D^{(\text{B})}$ of Eq. (A.14) instead uses the bare mass-weighted force constants $\Phi/\sqrt{m_\kappa m_{\kappa'}}$ and therefore differs from Eq. (A.17) by the overall factor C .

For the transverse directions we write $\mathbf{q}_\perp$ for the two-dimensional transverse wavevector. A partial Fourier sum over transverse lattice vectors $\mathbf{R}_\perp$ then takes the form

$$D(\mathbf{q}_\perp) = \sum_{\mathbf{R}_\perp} M(\mathbf{R}_\perp) e^{2\pi i \mathbf{q}_\perp \cdot \mathbf{R}_\perp}. \quad (\text{A.18})$$

The transverse wavevectors are sampled on a Monkhorst–Pack mesh,

$$q_{\perp, i} = \frac{n_i + 0.5}{N_i} - 0.5 + \text{shift}_i, \quad n_i = 0, \dots, N_i - 1. \quad (\text{A.19})$$

We also define the discrete Fourier-transform pair

$$G(\omega; \mathbf{q}) = \sum_{\mathbf{R}} g(\omega; \mathbf{R}) e^{-i\mathbf{q}\cdot\mathbf{R}}, \quad (\text{A.20})$$

$$g(\omega; \mathbf{R}) = \frac{1}{N} \sum_{\mathbf{q}} G(\omega; \mathbf{q}) e^{+i\mathbf{q}\cdot\mathbf{R}}, \quad (\text{A.21})$$

with the associated discrete convolution theorem

$$\frac{1}{N} \sum_{\mathbf{q}'} A(\mathbf{q}') B(\mathbf{q} - \mathbf{q}') = \sum_{\mathbf{R}} a(\mathbf{R}) b(\mathbf{R}) e^{-i\mathbf{q}\cdot\mathbf{R}}. \quad (\text{A.22})$$

Quatrex Documentation

B.0.1. QuaTrEx Input format

The QuaTrEx solver reads three input files which specify the real-space Hamiltonian, the crystal structure, and the transport parameters.

Input files

dynamical_matrix.mat This MATLAB .mat file stores the real-space blocks $H(\mathbf{R})$ in Convention B, already converted to $(\text{rad/s})^2$. Each block is stored under a string key of the form "[nx, ny, nz]", mapping to a real matrix of dimension $(n_{\text{dof}} \times n_{\text{dof}})$, where $n_{\text{dof}} = 3N_b$ and N_b is the number of atoms in the unit cell. For nearest-neighbour interactions extracted on a $3 \times 3 \times 3$ mesh, the file contains up to 27 blocks.

The solver interprets these blocks as real-space couplings and, for periodic transverse directions, performs the transverse Fourier sum given in Eq. (B.1).

structure.xyz This is an extended XYZ file containing the lattice vectors in the header and Cartesian atomic positions in the body. It follows the ASE extended-XYZ convention, for example

```
8
Lattice="5.4018 0.0 0.0 0.0 5.4018 0.0 0.0 0.0 5.4018" ...
Si 0.00000000 0.00000000 0.00000000
...
```

The loader (`distributed_read_xyz`) extracts the 3×3 lattice matrix and Cartesian atom positions. The number of atoms times `num_orbitals_per_atom` determines n_{dof} .

quatrex_config.toml This file specifies the transport setup, such as transport direction, number of transport cells, neighbour cutoff, transverse k -mesh, broadening, and the open-boundary-condition algorithm. Important fields are

Field	Example value	Purpose
transport_direction	'z'	transport axis
construct_from_unit_cell	true	input mode selector
num_transport_cells	4	device length in unit cells
neighbor_cell_cutoff	[1,1,1]	block-range cutoff
kpoint_grid	[4,4,1]	transverse mesh
kpoint_shift	[0,0,0]	Monkhorst–Pack shift
num_orbitals_per_atom.Si	3	DOFs per atom
phonon.eta	1e-8	broadening (rad/s) ²
phonon.obc.algorithm	"sancho-rubio"	OBC method

Input modes

QuaTrEx supports two input modes, selected by `construct_from_unit_cell`. In both modes the input is a MATLAB `.mat` file loaded by `scipy.io.loadmat`. The loader (`distributed_load` in `qttools/utis/mpi_utils.py`) parses keys that start with `[` and interprets them as integer tuples:

```
arr = loadmat(path)
arr = {
    tuple(map(int, r.strip("[]").split(","))): h_r
    for r, h_r in arr.items()
    if r.startswith("[")
}
```

That is, a key string such as `"[0, 1, -1]"` becomes the Python tuple `(0, 1, -1)`. Each value is a 2D NumPy array (real or complex) representing a matrix block.

Mode 1: pre-assembled device matrix If `construct_from_unit_cell = false`, the `dynamical_matrix.mat` file contains a single pre-assembled device matrix stored under the key `(0,0,0)`:

```
matrix_spararray = distributed_load(f"{matrix_name}.mat")
assert (0, 0, 0) in matrix_spararray.keys()
matrix_spararray = matrix_spararray[(0, 0, 0)]
```

The result is a single sparse matrix of size $(N_{\text{blocks}} \times B)^2$, where B is the block size and N_{blocks} is the number of transport cells. No transverse k -point Fourier assembly is performed.

Mode 2: unit-cell blocks If `construct_from_unit_cell = true`, the file contains the dictionary of real-space unit-cell blocks $H(n_x, n_y, n_z)$. The solver loads all blocks, constructs a dense translation grid, optionally trims it using `neighbor_cell_cutoff`, expands the blocks into transport-direction BTM matrices, and performs the transverse Fourier assembly on the chosen Monkhorst–Pack mesh.

The function `_load_matrix_from_unit_cell` loads all keys, determines the bounding box from the minimum and maximum key coordinates, and packs the blocks into a 5D array:

```
# keys: array of all (nx, ny, nz) tuples
min_coords = keys.min(axis=0)
max_coords = keys.max(axis=0)
grid_shape = max_coords - min_coords + 1
unit_cells = zeros(grid_shape + matrix_shape)
for coord, matrix in matrices.items():
    unit_cells[coord] = matrix
```

The resulting array has shape $(G_x, G_y, G_z, n_{\text{dof}}, n_{\text{dof}})$, where (G_x, G_y, G_z) is the grid shape (e.g. $(3, 3, 3)$ for $\{-1, 0, +1\}^3$). The loader checks that $G_x \times G_y \times G_z = \text{number of keys}$, i.e. the grid must be dense: no missing entries are permitted. Missing blocks must be stored as zero matrices.

If `neighbor_cell_cutoff` is set (e.g. $[1, 1, 1]$), `trim_tight_binding_matrix` zeros out all blocks beyond that range. For cutoff (c_x, c_y, c_z) , a block at index (n_x, n_y, n_z) (centred at zero) is kept only if $|n_x| \leq c_x, |n_y| \leq c_y, |n_z| \leq c_z$.

The function `expand_tight_binding_matrix` converts the unit-cell blocks into a BTD sparse matrix for a given transverse periodic shift. Along the transport direction z :

1. The supercell size in the transport direction is $G_z/2$ (integer division), and 1 in the transverse directions. For $G_z = 3$, this gives a supercell of one unit cell ($3/2 = 1$).
2. Three BTD blocks are extracted for each transverse periodic shift (n_x, n_y) :
 - $n_z = -1$: lower off-diagonal,
 - $n_z = 0$: diagonal,
 - $n_z = +1$: upper off-diagonal.
3. Each block is assembled by `_get_ttransport_block`, which loops over all pairs of local shifts within the supercell and selects the appropriate unit-cell block for each pair.
4. The three blocks are tiled $N_{\text{transport}}$ times along the diagonal to build the full BTD sparse matrix.

The function `_assemble_kpoint` then performs the Fourier sum over transverse periodic shifts. For each $\mathbf{k}_\perp$ on a Monkhorst-Pack mesh:

$$D(\mathbf{k}_\perp) = \sum_{(n_x, n_y)} M(n_x, n_y) e^{2\pi i (k_x n_x + k_y n_y)}, \quad (\text{B.1})$$

where $M(n_x, n_y)$ is the BTD sparse matrix for periodic shift (n_x, n_y) , already expanded in the transport direction. The k -points on the Monkhorst-Pack mesh are

$$k_i = \frac{n_i + 0.5}{N_i} - 0.5 + \text{shift}_i, \quad n_i = 0, \dots, N_i - 1. \quad (\text{B.2})$$

The result is a stack of BTD matrices indexed by (ω, k_x, k_y) .

B.0.2. Heterogeneous-device extension

The original harmonic unit-cell input mode in QuaTrEx constructs a homogeneous device by repeating the same block set across all transport cells. To model interfaces such as Si/Ge, the harmonic assembly was extended to support piecewise-defined left, device, and right regions with distinct block dictionaries.

L|D|R device model

The transport region is partitioned into three sections,

$$\underbrace{L \cdots L}_{n_L \text{ cells}} \underbrace{D \cdots D}_{n_D \text{ cells}} \underbrace{R \cdots R}_{n_R \text{ cells}}, \quad n_L + n_D + n_R = n_{\text{transport}}, \quad (\text{B.3})$$

where the left and right parts serve as lead-like material regions and the central part represents the interface or device region.

Configuration changes

To support this heterogeneous harmonic setup, four new configuration fields were added to DeviceConfig:

Field	Type	Purpose
num_left_cells	PositiveInt None	number of left cells n_L
num_right_cells	PositiveInt None	number of right cells n_R
left_matrix	str None	left-lead .mat file
right_matrix	str None	right-lead .mat file

This mode requires `construct_from_unit_cell = true` and must satisfy

$$n_L + n_R < n_{\text{transport}}, \quad (\text{B.4})$$

so that at least one transport cell remains in the device region.

Matrix assembly

The heterogeneous assembly is implemented by extending the harmonic unit-cell expansion to allow different block dictionaries in different transport cells. Two new functions were added to `quatrex/device/inputs.py`.

Throughout the device builder, operators are stored and assembled as a single upper-triangular half: `load_matrices` keeps only the `triu` of each loaded block, and `_construct_transport_cell` reconstructs each full block as `block + triu(block_flipped, k=1).conj().T`, where `block_flipped` is the block at the negated coordinate, so that Hermiticity is enforced by construction and the strictly-lower part is never stored redundantly. When several blocks or k -points are combined (`assemble_matrix`, `_sum_operator`), a common sparsity pattern is obtained as the union

of the per-shift patterns (symmetrised via $S + S^T$) so that no entry present in any contribution is dropped.

`_expand_heterogeneous_matrix` builds a BTD sparse matrix for a single transverse periodic shift (n_x, n_y) . Each cell is assigned to one of the three regions according to its index. The diagonal block of cell i is selected according to

$$D_{ii} = \begin{cases} D_L, & i < n_L, \\ D_D, & n_L \leq i < n_L + n_D, \\ D_R, & i \geq n_L + n_D. \end{cases} \quad (\text{B.5})$$

The routine `_create_heterogeneous_matrix` constructs the final transport object. It loops over all transverse periodic shifts, calls `_expand_heterogeneous_matrix` for each shift, and combines the resulting blocks. At T , the transverse contributions are also summed to determine the sparsity pattern.

Boundary coupling

At the interfaces between the L , D , and R regions, the off-diagonal couplings must be chosen consistently with Hermiticity. For harmonic real-space blocks, the relevant relation is

$$H(\mathbf{R}_\perp, n_z)^\dagger = H(-\mathbf{R}_\perp, -n_z). \quad (\text{B.6})$$

At the left/device boundary, the super-diagonal coupling is taken from the left-lead block H_{01}^L , while the corresponding sub-diagonal is H_{10}^L . At the device/right boundary, the couplings are taken from the right-lead blocks H_{01}^R and H_{10}^R . This ensures that the assembled heterogeneous matrix remains consistent with the physical lead couplings on either side of the interface.

Input Generation

C.1. Interatomic Force Constants

C.1.1. Finite-displacement FCs

The FCs can be generated based on a finite-displacement workflow using phono3py, QE, and the symfc fitting backend. In this approach, displaced supercells are generated explicitly, DFT forces are computed for each displaced configuration, and both FC2 and FC3 are reconstructed from the resulting force data.

The following steps are performed:

1. **Structure setup.** The crystal structure is loaded and a phonopy object is created. The unit cell is loaded via `structure.py:load_structure`, and a Phonopy object with a supercell is created by `structure.py:create_phonopy_from_config`, which generates symmetry-inequivalent displacements.
2. **DFT force calculations.** For each displaced supercell, a QE `pw.x` input is written and executed. A QE `pw.x` SCF input file is written by `qe_interface.py:write_qe_scf_input`. Existing converged outputs are detected and reused. Forces parsed from QE output are converted from Ry/bohr to eV/Å.
3. **Force-constant reconstruction.** The set of forces is passed to phono3py, which reconstructs FC3 and FC2 using the symmetry-adapted symfc backend, both FC2 and FC3 are written to a common `fc3.hdf5` file containing dense arrays in eV/Å² and eV/Å³, respectively.

For the Si FCC primitive cell with a $2 \times 2 \times 2$ supercell (16 atoms), phono3py generates 114 symmetry-inequivalent displacements for a pair cutoff of 5.5 Å. On the mont-fort workstations, each QE SCF calculation takes roughly 30 s with `ecutwfc=60 Ry` and a $2 \times 2 \times 2$ k -point mesh.

C.1.2. DFPT FCs

Third order FCs can be generated efficiently directly from density-functional perturbation theory (DFPT) due to the $2n+1$ theorem. One possibility is to obtain FC2 from QE's linear-response phonon code `ph.x` and compute FC3 with the D3Q plugin, which implements third-order DFPT. In contrast to the finite-displacement method, the DFPT route works directly on the unit cell and avoids supercell calculations. However, QE does only supports a restricted set of norm-conserving potentials for third-order DFPT calculations, limiting the achievable accuracy.

The DFPT workflow does the following:

1. **SCF ground state** (`pw.x`). A self-consistent calculation on the unit cell with a dense k -mesh (configured via `dfpt.kpoints`).
2. **Phonon calculation** (`ph.x`). DFPT phonon calculation on a uniform $\mathbf{q}$ -grid (`dfpt.q_mesh`) with `ldisp=.true.` and `lqdir=.true.`. This produces the dynamical matrices $D(\mathbf{q})$ and saves the first-order perturbation wavefunctions needed by D3Q.
3. **FC2 Fourier transform** (`q2r.x`). The dynamical matrices are Fourier-transformed to real-space FC2 via QE's `q2r.x`, producing the `flfrc` file. The acoustic sum rule is enforced (`zasr='crystal'`).
4. **Third-order DFPT** (`d3q.x`). For each $(\mathbf{q}_1, \mathbf{q}_2)$ pair on the grid, `d3q.x` computes the third-order dynamical tensor $D_3(\mathbf{q}_1, \mathbf{q}_2, -\mathbf{q}_1 - \mathbf{q}_2)$. The $\mathbf{q}$ -points are specified in QE Cartesian coordinates ($2\pi/a_{\text{lat}}$ units), the conversion from crystal reciprocal coordinates is

$$\mathbf{q}_{\text{cart}} = q_1 \hat{\mathbf{b}}_1 + q_2 \hat{\mathbf{b}}_2 + q_3 \hat{\mathbf{b}}_3, \quad \hat{\mathbf{B}} = (A/|\mathbf{a}_1|)^{-\top}, \quad (\text{C.1})$$

where A is the matrix of lattice vectors (rows, in Å) and $\hat{\mathbf{b}}_i$ are the rows of $\hat{\mathbf{B}}$.

For a $2 \times 2 \times 2$ grid there are $8^2 = 64$ pairs. Crystal symmetry can reduce this (e.g. to ~ 10 – 15 for FCC Si with O_h symmetry). The current implementation generates all pairs and relies on `d3q.x`'s internal symmetry handling.

5. **FC3 Fourier transform** (`d3_qq2rr.x`). The $(\mathbf{q}_1, \mathbf{q}_2)$ -space third-order tensors are Fourier-transformed to real-space FC3 $\Phi_3(\mathbf{R}_1, \mathbf{R}_2)$, producing the `mat3R` file.
6. **Acoustic sum rule** (`d3_asr.x`). Enforces translational invariance on the real-space FC3, $\sum_{\kappa} \Phi_3(\kappa, \kappa', \kappa'') = 0$ for each pair (κ', κ'') . The ASR type is configurable (`asr`: `simple`, `crystal`, or `no`).
7. **Sparsification** (`d3_sparse.x`). Zeroes FC3 entries with magnitude below a threshold (`sparse_thr`, in Ry/bohr³), reducing storage and downstream computation cost.

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Output conversion. The reap step parses both the `q2r.x` FC2 output (`fc2.dat`, in Ry/bohr²) and the D3Q `mat3R` file (in Ry/bohr³) and converts to eV/Å² and eV/Å³:

$$\Phi_2 [\text{eV}/\text{\AA}^2] = \Phi_2 [\text{Ry}/\text{bohr}^2] \times \frac{E_h/2}{a_0^2}, \quad \approx 48.59, \quad (\text{C.2})$$

$$\Phi_3 [\text{eV}/\text{\AA}^3] = \Phi_3 [\text{Ry}/\text{bohr}^3] \times \frac{E_h/2}{a_0^3}, \quad \approx 91.82. \quad (\text{C.3})$$

The real-space lattice vectors $\mathbf{R}$ in the D3Q output (integer crystal coordinates) are mapped to supercell atom indices $j = (l_1 n_2 n_3 + l_2 n_3 + l_3) n_{\text{at}} + \kappa$ with $l_i = R_i \bmod n_i$, matching phonopy's supercell convention. FC2 is expanded to the full $(N_{\text{super}})^2$ array via translational invariance. Both arrays are saved to `fc3.hdf5` in the same format as the finite-displacement method.

The DFPT-specific parameters are in the `dfpt` section of the YAML config:

Key	Default	Description
<code>q_mesh</code>	[2, 2, 2]	<i>q</i> -grid for <code>ph.x</code> and <code>d3q.x</code>
<code>kpoints</code>	[8, 8, 8]	SCF <i>k</i> -mesh for the unit cell
<code>tr2_ph</code>	10 ⁻¹⁴	<code>ph.x</code> self-consistency threshold
<code>ph_command</code>	<code>ph.x</code>	<code>ph.x</code> executable
<code>d3q_command</code>	<code>d3q.x</code>	<code>d3q.x</code> executable
<code>d3_qq2rr_command</code>	<code>d3_qq2rr.x</code>	Fourier transform executable
<code>d3_asr_command</code>	<code>d3_asr.x</code>	Acoustic sum rule executable
<code>d3_sparse_command</code>	<code>d3_sparse.x</code>	Sparsification executable
<code>asr</code>	<code>simple</code>	ASR type: <code>simple</code> , <code>crystal</code> , or <code>no</code>
<code>sparse_thr</code>	10 ⁻⁵	FC3 threshold (Ry/bohr ³); <code>null</code> to skip
<code>q2r_command</code>	<code>q2r.x</code>	<code>q2r.x</code> executable
<code>work_dir</code>	<code>./dfpt</code>	Working directory
<code>ph_timeout</code>	7200s	Timeout per <code>ph.x</code> run
<code>d3q_timeout</code>	14400s	Timeout per <code>d3q.x</code> triplet

C.1.3. Mass-weighting and unit conversion

The harmonic transport problem is formulated in terms of mass-weighted FC2 blocks. Given a harmonic IFC block $\Phi_{\alpha\beta}(0\kappa, l'\kappa')$, the corresponding mass-weighted quantity is

$$H_{\alpha\beta}(0\kappa, l'\kappa') = \frac{\Phi_{\alpha\beta}(0\kappa, l'\kappa')}{\sqrt{M_\kappa M_{\kappa'}}}, \quad (\text{C.4})$$

with masses in amu. In the common phonopy unit system, $H_{\alpha\beta}$ therefore has units eV/(amu·Å²), which are later converted to (rad/s)² using the factor defined in Section A.2.

For FC3, the analogous mass-weighted tensor is

$$\Psi_{\alpha\beta\gamma}(\kappa\kappa'\kappa'') = \frac{\Phi_{\alpha\beta\gamma}^{(3)}(\kappa\kappa'\kappa'')}{\sqrt{M_\kappa M_{\kappa'} M_{\kappa''}}}, \quad (\text{C.5})$$

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with units

$$[\Psi] = \frac{\text{eV}}{\text{\AA}^3 \cdot \text{amu}^{3/2}}. \quad (\text{C.6})$$

For DFPT-generated FCs, the real-space force constants are parsed in atomic units and converted with the same factors as in Eqs. (C.2) and (C.3).

Finally, the FC3 tensor can be converted to SI units for use in the anharmonic self-energy. Using the harmonic conversion factor $C = e/(m_u A^2)$, the cubic conversion becomes

$$\text{CONVERSION_FC3} = \frac{C}{A\sqrt{m_u}} = \frac{e}{m_u^{3/2} A^3}. \quad (\text{C.7})$$

This converts $\text{eV}/(\text{\AA}^3 \cdot \text{amu}^{3/2})$ into the corresponding SI units required by the phonon-phonon self-energy expressions.

C.1.4. Validation of the extracted harmonic model

Before using the extracted harmonic blocks in transport calculations, it is important to verify that the block representation is internally consistent and reproduces the original phonopy dynamical matrix. The present validation procedures are implemented in `validation.py`.

Acoustic sum rule. At the Γ point, the lattice Fourier series reduces to

$$D^{(\text{B})}(0) = \sum_{\mathbf{R}} H(\mathbf{R}). \quad (\text{C.8})$$

Translational invariance requires that the three acoustic modes vanish at Γ . The validation therefore forms

$$D_{\Gamma} = \sum_{\mathbf{R}} H(\mathbf{R}), \quad (\text{C.9})$$

computes its eigenvalues λ_n , and reports the corresponding frequencies

$$\omega_n = \text{sign}(\lambda_n) \sqrt{|\lambda_n|}. \quad (\text{C.10})$$

The three smallest $|\omega_n|$ should be close to zero. In addition, the matrix should be symmetric to numerical precision, so that

$$\max |D_{\Gamma} - D_{\Gamma}^T| \quad (\text{C.11})$$

is approximately zero.

Block symmetry. The real-space harmonic blocks must satisfy the Hermiticity relation

$$H(\mathbf{R})^T = H(-\mathbf{R}) \quad \forall \mathbf{R}. \quad (\text{C.12})$$

The validation iterates over all block pairs $(\mathbf{R}, -\mathbf{R})$ and reports the maximum deviation

$$\max_{\mathbf{R}} \|H(\mathbf{R})^T - H(-\mathbf{R})\|_{\infty}. \quad (\text{C.13})$$

Band-structure comparison. A more stringent validation compares phonon frequencies reconstructed from the extracted blocks with those obtained directly from phonopy. Along a chosen $\mathbf{q}$ -path, two spectra are computed:

1. the phonopy reference spectrum, obtained from

$$D_{\text{ph}}(\mathbf{q}) = P(\mathbf{q}) D^{(\text{A})}(\mathbf{q}) P^\dagger(\mathbf{q}), \quad (\text{C.14})$$

converted to $(\text{rad/s})^2$ and diagonalized;

2. the reconstructed spectrum, obtained from

$$D_{\text{rec}}(\mathbf{q}) = \sum_{\mathbf{R}} H(\mathbf{R}) e^{2\pi i \mathbf{q} \cdot \mathbf{R}}, \quad (\text{C.15})$$

and then diagonalized.

The maximum eigenfrequency deviation

$$\max_{n, \mathbf{q}} |\omega_n^{\text{ph}} - \omega_n^{\text{rec}}| \quad (\text{C.16})$$

serves as a direct measure of the quality of the extraction. In addition, overlay plots of the two band structures provide a visual consistency check.

C.2. The `phonon_inputs` Package

The `phonon_inputs` package implements the full sow-run-reap workflow for harmonic and cubic force constants, the structural relaxation step that precedes it, the structure / configuration plumbing, and the writers that emit the dynamical-matrix blocks consumed by the downstream NEGF solver. The package is calculator-agnostic: any module that touches DFT goes through a thin adapter layer that supports Quantum ESPRESSO `pw.x` and VASP equally, and the FC-generation backends (`phono3py + symfc`, DFPT via `ph.x` + `d3q.x`, and `hiphive`) share the same dataclass-based YAML schema and the same on-disk metadata format.

This section walks through the package module by module. The intent is to give the reader a single place to look up which module owns which responsibility, what data flows into it, and what it returns to the caller. Module names are given relative to `phonon/phonon_inputs/`.

C.2.1. Top-level orchestration: `cli.py`, `pipeline.py`

`cli.py`. An `argparse`-based command-line front-end exposed via `python -m phonon_inputs <command>`. The commands implemented are:

- `generate` – legacy “one-shot” flow that builds a phonopy supercell, runs all required QE displacements, and produces the `quatrex` input files in one go (kept as a smoke test for tightly coupled bulk runs).

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- `extract-blocks` – reads an existing phonopy reap and re-extracts the Convention B blocks (cf. Section C.2.4) without re-running DFT.
- `validate` – runs the harmonic-model validators of `validation.py` on an existing reap.
- `fc3-sow`, `fc3-run`, `fc3-reap`, `fc3-all` – the canonical phono3py + symfc flow.
- `fc3-hiphive-sow`, `fc3-hiphive-run`, `fc3-hiphive-reap`, `fc3-hiphive-bootstrap-reap`, `fc3-hiphive-convergence`, `fc3-hiphive-all` – the hiphive flow, with the two-stage bootstrap variant for phonon-mode rattling.
- `dfpt-sow`, `dfpt-run`, `dfpt-reap`, `dfpt-all` – DFPT via `ph.x` + D3Q.
- `pipeline` – relax + FC generation in a single invocation; calls the in-process relax driver from `qe_interface.py` and threads the relaxed cell directly into the chosen FC backend.

All sow-type commands accept the flag `--from-yaml`. Without that flag the structure used for sowing is loaded from the relaxed output of `config.relax.work_dir` (CONTCAR for VASP, `relax.out` for QE) via `structure.resolve_structure_for_sow`; with the flag, sowing falls back to the YAML structure as-is. The default fails fast with a clear error message when the relax dir is missing, removing a class of silent failures where a sow ran on the unrelaxed YAML positions and the resulting FC2 absorbed residual linear forces as artificial negative curvature.

pipeline.py. Implements the `pipeline` subcommand. Two stages: (i) call `qe_interface.run_vasp_relax` or `run_qe_relax` on the YAML structure and obtain a relaxed `PhonopyAtoms`; (ii) dispatch to the FC backend selected by `config.fc_method` (“finite_displacement”, “dfpt”, or “hiphive”). Because the relaxed cell is passed in memory directly to the sow function the pipeline path does not need to write a CONTCAR to disk to communicate the equilibrium geometry.

__main__.py. Six-line shim: `from .cli import main; main()`. Enables `python -m phonon_inputs` as the entry point.

C.2.2. Configuration: `config.py`, `constants.py`

config.py. A nested dataclass schema parsed from YAML by `load_config`. The top-level class `PhononInputConfig` aggregates:

StructureConfig. `source` (inline / cif / poscar / phonopy_yaml / qe_input), with the inline branch providing symbols, lattice, and scaled_positions directly.

SupercellConfig. `matrix`, `displacement_distance` for the phonopy finite-displacement backend.

QEConfig / VASPConfig. All knobs the DFT backend writes into the input file: `ecutwfc`, `ecutrho_factor`, `kpoints_scf`, `conv_thr`, `smearing`, `degauss`, `pseudo_dir`, `pseudopotentials`, and the analogous VASP fields `encut`, `ediff`, `ismear`, `sigma`, `prec`, `lreal`, `ncore`, `kpar`, `potcar_dir`, `potcar_map`.

ThirdOrderConfig. Supercell multipliers, `cutoff_pair_distance`, `displacement_distance`, `fc_calculator` (`symfc` by default), `work_dir`, `pw_timeout`.

HiphiveConfig. The full hiphive recipe: `supercell`, `n_structures`, `rattle_method` (`mc/normal/phonon`), `rattle_std`, `rattle_d_min`, `rattle_n_iter`, `rattle_seed`, `cutoffs`, `fit_method`, `fit_kwargs`, `rotational_sum_rule`, the optional convergence: `sub_block`, and the phonon-rattle bootstrap knobs `phonon_rattle_temperature_k`, `phonon_rattle_bootstrap_n`, `phonon_rattle_seed_fc3` (use existing `fc3.hdf5` as FC2 seed) and `phonon_rattle_seed_dft_dir` (refit FC2 seed from an external `mc-rattle` work dir’s DFT outputs).

DFPTConfig. `q_mesh`, `kpoints`, `tr2_ph`, and the binary paths for `ph.x`, `q2r.x`, and `d3q.x`.

RelaxConfig. `calculation` (`vc-relax/relax`), `forc_conv_thr`, `press_conv_thr`, `calculator`, `work_dir`.

BlockExtractionConfig, QuatrexOutputConfig. Block extraction parameters (`transport_axis`, `q-mesh`, `amplitude_cutoff`) and quatrex writer parameters (`file_paths`, `k-point grid`).

The loader `config_from_dict` traverses the schema, accepting nested optional blocks (“NestedConfig | None”). Unknown keys raise a clear error rather than being silently dropped.

constants.py. All unit conversions and physical constants in a single place. The constant $\text{CONVERSION_THZ2} = e/(m_u \text{ \AA}^2)/(2\pi)^2 \cdot 10^{-24}$ converts $\text{eV}/(\text{\AA}^2 \cdot \text{amu})$ dynamical-matrix eigenvalues to THz^2 – this is the canonical internal unit-system used throughout the package and the solver (Chapter E). A legacy `CONVERSION` constant produces $(\text{rad/s})^2$ and is preserved for tests that compare to the previous unit convention. Boltzmann’s constant, $\hbar$, the electron mass and atomic mass unit appear here in SI.

C.2.3. Structure loading: `structure.py`

`structure.py` provides two functions and one helper:

load_structure(cfg). Resolves the YAML structure: block into a `PhonopyAtoms` object. Inline structures are passed straight through; file-based sources (`CIF`, `POSCAR`, `phonopy YAML`, `QE input`) are read via the relevant `phonopy / ASE` interface.

resolve_structure_for_sow(cfg, config_dir, from_yaml=False). The default routing layer for `sow`-time entry points. When `from_yaml=False` (the default) it looks

for `config.relax.work_dir/CONTCAR` (VASP) or `relax.out` (QE) and parses the relaxed structure from there; if neither file is present it raises `SystemExit` with a precise message directing the user to either run the relax first or pass `--from-yaml`. With `from_yaml=True` the YAML structure is used unchanged.

create_phonopy(`cell`, `supercell_matrix`, `primitive_matrix`, `displacement_distance`).

Builds a Phonopy object with the chosen supercell and generates symmetry-inequivalent displacements at the given amplitude.

The Bohr/Rydberg – Angstrom/eV conversion factors for QE-internal storage are pinned here as well, both to honour phonopy’s QE units (Ry/bohr²) on output and to convert forces parsed from a QE log to eV/Å.

C.2.4. Convention transformation and block extraction: `convention.py`

Phonopy stores the dynamical matrix in what we refer to as *Convention A*, with phase factor $\exp[2\pi i \mathbf{q} \cdot (\mathbf{R} + \boldsymbol{\tau}' - \boldsymbol{\tau})]$ that includes the basis-atom shifts. The quatex NEGF solver expects *Convention B*, with phase $\exp[2\pi i \mathbf{q} \cdot \mathbf{R}]$ and $D_B(\mathbf{q}) = P(\mathbf{q}) D_A(\mathbf{q}) P(\mathbf{q})^\dagger$ where $P(\mathbf{q}) = \text{diag}[\exp(2\pi i \mathbf{q} \cdot \boldsymbol{\tau}_\kappa)]$ repeated three times per atom. The inverse discrete Fourier transform of $D_B(\mathbf{q})$ produces real-valued real-space blocks $H(\mathbf{R})$ that the solver tiles into the device block-tridiagonal.

`convention.py` implements both transformations (`apply_phase_correction`, `remove_phase_correction`) and the block-extraction driver `extract_blocks(phonon, q_mesh, amplitude_cutoff)`, which iterates over a q -grid sampling the transverse Brillouin zone, computes $D_B(\mathbf{q})$ from the phonopy-internal D_A , and IDFTs the result onto a real-space (R_x, R_y, R_z) block dictionary. Blocks with Frobenius norm below `amplitude_cutoff` times the maximum block norm are dropped.

C.2.5. Force-constant production: `force_constants.py`

`force_constants.py` owns the harmonic FC2 reconstruction step: given a Phonopy object and the parsed forces from the displaced-supercell DFT runs, it calls `Phonopy.produce_force_constants` with the “symmetry” option enabled to symmetry-adapt the FCs, and honours the `produce_fc` hook from `phono3py` when an FC3 model is present. The output is stored on the `Phonopy.force_constants` attribute, ready for q -point diagonalisation by the validation routines.

C.2.6. DFT interface: `qe_interface.py`

`qe_interface.py` unifies the QE and VASP entry points behind a small set of helpers. The QE side covers `write_qe_scf_input` (builds a single pw.x SCF or relax input, encoding cell, atomic species, pseudopotentials and k-points; also writes restart-data block headers so successive displacements can reuse prior WFC/CHG when convenient), `parse_qe_force_output` (reads forces from a finished pw.x run, converting Ry/bohr

to $\text{eV}/\text{\AA}$), `parse_qe_relax_output` (parses “Begin final coordinates” or fixed-cell final positions blocks into a `PhonopyAtoms`), and `run_qe_relax`. The VASP side covers `parse_vasp_relax_output` (reads `CONTCAR`) and `run_vasp_relax` (writes `INCAR`/`KPOINTS`/`POSCAR`/`POTCAR` with the user’s relaxation tolerance, runs VASP, and parses the resulting `CONTCAR`). Lower-level helpers (`_run_and_tee`, `_needs_shell`) propagate `stdout`/`stderr` through Python and detect whether the VASP command requires a shell wrapper.

C.2.7. Phono3py + symfc flow: `thirdorder.py`

`thirdorder.py` drives the finite-displacement FC2+FC3 pipeline. Three calculator-agnostic public entry points, `sow / run_displacements / reap`, plus a unifying `generate_fc3`.

Sow. Builds a `Phono3py` object with the requested supercell and pair cutoff, calls its built-in displacement generator (canonical symmetry-inequivalent finite displacements), and writes one DFT input per displacement: `pw.x` input files for QE or `INCAR`/`KPOINTS`/`POSCAR`/`POTCAR` directories for VASP. A reference (undisplaced) calculation is sown alongside, used as the WFC/CHG seed for the displaced calculations to accelerate SCF convergence. Metadata (number of displacements, supercell layout, displacement distance, calculator) is written to `hiphive_meta.json` in the work directory.

Run. Iterates over the displaced calculations, executes the DFT command (in-process or via a job-submission wrapper provided by the user), checks for completion markers (“JOB DONE” for QE, the `OSZICAR` final step for VASP), and skips already-completed calculations. A timeout mechanism kills hung jobs.

Reap. Parses forces from every displaced calculation, calls `phono3py.run_force_constants` with the `fc_calculator='symfc'` backend (symmetry-adapted least squares; [10] for the underlying algorithm), and extracts FC2 and FC3 in the `phono3py` compact representation. The final tensors are written to `fc3.hdf5` with datasets `fc2` of shape $(N, N, 3, 3)$ in $\text{eV}/\text{\AA}^2$ and `fc3` of shape either $(N_{\text{prim}}, N, N, 3, 3, 3)$ (compact) or $(N, N, N, 3, 3, 3)$ (full) in $\text{eV}/\text{\AA}^3$, depending on the `symfc` output mode. Independently of the atomic real-space FC3 (pair-)cutoff set here, the downstream device solver imposes a further *nearest-neighbour block* truncation when it ingests this `fc3.hdf5`: the loader (`quatrex/phonon/fc3_loader.py`, `load_device_fc3 / fc3_to_phi_blocks`) keeps only block-triplets (I, J, K) of transport cells with $|I - J|, |I - K|, |J - K| \leq 1$ (`nn_only=True` by default), discarding all longer-range block triplets. This NN block truncation is applied even on the stored block-sparse path (not only on the dense fallback), so the cubic vertex effectively sees at most one neighbouring transport cell per leg regardless of the atomic cutoff; the loader warns when the dropped Frobenius norm exceeds a set threshold.

The VASP-side helpers `_write_vasp_inputs` and `_refresh_vasp_runtime_inputs` emit `ISYM=0` in every displaced INCAR (rattle breaks the parent symmetry; force symmetrisation would otherwise produce inconsistent forces), and write KPOINTS at the user-configured density.

C.2.8. DFPT flow: `dfpt.py`

`dfpt.py` drives the DFPT-based alternative FC pipeline using `ph.x` + `q2r.x` for FC2 and `d3q.x` + `d3_qq2rr.x` for FC3 [24, 30]. The `sow` stage writes a sequence of paired SCF + phonon inputs at every q -point of the configured mesh, plus the `q2r` control file and the third-order `d3q.x` inputs. The `run` stage executes them in dependency order (`SCF` → `ph` → `q2r` → `d3q` → `qq2rr`). The `reap` stage parses the produced real-space FC files into the `phono3py`-compatible `fc3.hdf5` format so downstream consumers do not need to branch on the FC backend.

C.2.9. Hiphive flow: `hiphive_fc3.py`, `hiphive_convergence.py`

`hiphive_fc3.py` implements rattled-supercell sampling and the fit-and-project step at the heart of the `hiphive` workflow [37]. Public API:

sow. Generates rattled supercells. For the default `mc` rattle method, atomic displacements are drawn from a Monte-Carlo walk with per-step Gaussian increments of standard deviation `rattle_std` and a hard minimum-pair-distance constraint at `rattle_d_min`, for `rattle_n_iter` iterations; the cumulative per-atom displacement scales as $\sqrt{\text{rattle_n_iter}}$ (each MC step adds an independent zero-mean Gaussian increment, so the per-step variances add). For phonon rattling, a seed FC2 is required: it can come from an existing `fc3.hdf5` (`phonon_rattle_seed_fc3`), an external `mc-rattle` work directory (`phonon_rattle_seed_dft_dir`, which triggers an FC2-only refit on those forces), or a local “bootstrap” stage that runs a small `mc-rattle` pool first and fits FC2 from it. The seed FC2 is checked for PSD-ness before being used to drive the phonon-rattle amplitude (which scales as $1/\sqrt{|\omega|}$ per mode and diverges for soft / imaginary modes; see Section C.3). Both rattle methods write a calculator-specific DFT input per `disp-XXXXX/` directory and a reference (undisplaced) directory.

bootstrap_reap. Stage 1 of the phonon-rattle workflow. Reads the `mc-rattle` bootstrap pool’s forces, fits FC2-only with cutoffs $[r_c^{(2)}]$ using ARDR (with ridge as fallback), applies the post-fit rotational projection, checks that the resulting FC2 has at most `phonon_rattle_max_seed_imag` imaginary modes, and saves `fc2_seed.npy` for stage 2.

reap. Reads forces from every `disp-XXXXX/` directory, builds a `ClusterSpace` with the configured cutoffs, runs the selected fit method via `trainstation.Optimizer`, optionally projects onto the rotational manifold `enforce_rotational_sum_rules(cs, params, sum_rules=['Huang', 'Born-Huang'])`, builds the `ForceConstantPotential`

and exports FC2 and FC3 to `fc3.hdf5` in the same on-disk format as the `phono3py` reapi. A diagnostic JSON `hiphive_fit.json` records $n_{\text{parameters}}$, RMSE train/test, rotational residual before/after, and the FC2/FC3 ASR residuals.

generate_fc3. Orchestrates the full `hiphive` flow for the phonon rattle method (two-stage bootstrap-aware sequence) and the `mc/normal` rattle methods (one-stage canonical flow).

The helper `_seed_from_existing_fc3` extracts FC2 from an existing `fc3.hdf5` (e.g. a converged `mc-rattle` reapi on the same primitive) and writes it as `fc2_seed.npy`, allowing the user to bypass the bootstrap stage entirely with the `phonon_rattle_seed_fc3` config option. The helper `_fit_fc2_seed_from_source` is the corresponding entry point for refitting FC2 on an external `mc-rattle` work directory’s `disp-XXXXX/` outputs, used by `phonon_rattle_seed_dft_dir`.

`hiphive_convergence.py` drives the convergence sweep described in Section C.3.2: for each pair of (size, fit method) in the user-configured grid, it subsamples the master DFT pool, refits, and records the same diagnostic block as the main reapi. The post-fit rotational projection is exposed by `_apply_rotational_sum_rules` and `_rotational_residual_pair`; the latter returns $\|M_{\text{rot}}\theta\|$ before and after projection for the “post_fit” policy.

C.2.10. Validation: `validation.py`

`validation.py` provides the four extraction-time checks detailed in Section C.1.4: the Γ -point ASR residual, the block-symmetry check $H(\mathbf{R})^T = H(-\mathbf{R})$, the band-structure overlay versus the `phonopy` reference, and a reference Sancho–Rubio + Caroli ballistic transmission against which the `quatrex` solver is benchmarked. These run in the units chosen by the caller; the default is THz² (`conversion_factor=CONVERSION_THZ2`).

C.2.11. Tensor decompositions: `fc3_compression.py`, `separable.py`, `pcp.py`

`fc3_compression.py` houses the six tensor-decomposition ansätze used to compress the supercell FC3. All operate on the mass-weighted target $T \in \mathbb{R}^{n_{\text{dof}} \times \text{dim}_{sc} \times \text{dim}_{sc}}$ constructed by `build_fc3_target` from the raw $(n_{\text{prim}}, n, n, 3, 3, 3)$ or $(n, n, n, 3, 3, 3)$ FC3 together with the supercell masses:

mSVD. Truncated matricisation SVD (Tucker mode-3) of T flattened along the third axis.

HOSVD. Higher-order SVD (Tucker decomposition) with S_2 symmetry on legs 2 and 3, refined by a few steps of symmetric HOOI.

CP / INDSCAL / Waring. Canonical polyadic decompositions of increasing symmetry: CP unrestricted, INDSCAL with $b_r = c_r$ (S_2 symmetry on legs 2, 3), Waring symmetric CP on the fully S_3 -symmetrised lifted target.

PCP. The permanent-CP ansatz of [53], wrapped in `pcp.py`. Decomposes the cluster representation directly as $\tilde{\Phi}_{l_1 b_1, l_2 b_2, l_3 b_3}^{a_1 a_2 a_3} = \sum_{\xi} (\lambda_{\xi}/3!) \sum_{\sigma \in S_3} \sum_l \prod_i A_{\sigma_i}^{\xi}(l_i - l, b_i, a_i)$ with three mode functions per rank.

The helper `asr_residual(T_hat, n_super)` computes the axis- j and axis- k Frobenius residuals $\|\sum_j T_{ijk}\|_F$ and $\|\sum_k T_{ijk}\|_F$ of the lifted tensor; `asr_project_factor` performs the corresponding projection onto the axis- j + axis- k null-space when the downstream consumer (e.g. the bubble in Chapter E) needs strict ASR on those legs.

`separable.py` implements the stacked-SVD low-rank factorisation of FC3 used by the q-resolved solver (Section E.7): for each primitive-cell row (i, μ) it flattens the mass-weighted slice into a matrix $M_{(i, \mu)} \in \mathbb{R}^{n \cdot 3 \times n \cdot 3}$, stacks the slices into M_{stacked} , factors $M_{\text{stacked}} = U_R \text{diag}(\sigma) V_R^T$ by truncated SVD, and returns the rank- R left/right factors.

C.2.12. Quatrex output: `quatrex_writer.py`

`quatrex_writer.py` emits the three files the production quatrex NEGF stack consumes: `dynamical_matrix.mat` (a sparse MATLAB-format block dictionary keyed by (R_x, R_y, R_z)), `structure.xyz` (atomic coordinates in extended XYZ for input to the simulation), and `config.toml` (the simulation-side configuration mirror). All paths and the transport direction are read from `QuatrexOutputConfig`.

C.2.13. I/O and bookkeeping: `io.py`

`io.py` provides a small NPZ + JSON sidecar pattern used by the SCBA and decomposition drivers: arrays go to `.npz`, scalars and metadata to a sibling `.json`. `save_transport_results` and `load_transport_results` round-trip the SCBA output dictionary; an additional helper writes the per-decomposition “cache” entries that `transport_quality` in `finite_analysis` (Chapter D) reads on restart.

C.3. Imaginary-mode pathologies in hiphive SiNW fits

This section documents a systematic audit of the imaginary-mode problem observed for the H-passivated Si nanowire (SiNW) family used throughout the rest of this report. Even after a sequence of cleanups to the wire generator and the relax/sow plumbing (tighter EDIFF, a_{lattice} set to the PBE-equilibrium value, auto-sized vacuum, FC2 cutoff scaled with the wire envelope), the regenerated `cluster/reaps/` still showed imaginary modes in the resulting harmonic spectrum at Γ and at non- Γ wavevectors. We localised the breakage to four distinct mechanisms, each acting on a different system and each requiring a targeted fix. The raw evidence (JSON artefacts, refit sweeps, species-resolved eigenvalues) lives in `phonon/scratch/imag_audit/`.

C.3.1. What “negative squared frequency” means physically

A harmonic mode has frequency $\omega^2 = \lambda$ where λ is the eigenvalue of the mass-weighted dynamical matrix $D = M^{-1/2} \Phi_2 M^{-1/2}$. By convention dispersion plots report $\text{sign}(\lambda) \sqrt{|\lambda|}$, so a “negative frequency” is shorthand for $\lambda < 0$ – the mode is at a saddle or maximum of the Born-Oppenheimer surface along the corresponding eigenvector. The equation of motion is then $\ddot{x} = -\lambda x$, which gives exponential runaway $x(t) \sim e^{\sqrt{|\lambda|}t}$ along that mode coordinate.

Imaginary modes are physical in soft-mode systems (PbTiO₃ above its ferroelectric transition, anharmonically-stabilised bcc-Ti at high temperature, charge-density-wave instabilities) [112], but for an H-passivated SiNW at its ground-state geometry they should not appear in the harmonic spectrum. Table C.1 lays out the triage we use to discriminate real soft-mode physics from numerical artifacts in our setting.

Magnitude $ \omega $	Location	Reading
< 0.01 THz	Γ , three modes	ASR roundoff, ignore
$0.01\text{--}0.5$ THz	near- Γ acoustic	sampling noise on a soft branch
$0.5\text{--}5$ THz	finite q , few modes	cutoff or under-determination
> 5 THz	many modes	ground-truth or wiring bug

Table C.1.: Imaginary-mode triage matrix. Magnitudes are mass-weighted ω on the supercell q -mesh. The Γ -point three-acoustic-mode ASR-noise band is set by the numerical conditioning of the FC fit; see Section C.3.2.

C.3.2. The four mechanisms

Table C.2 summarises the four root causes we identified, the system on which each one is dominant, and the intervention that fixes it. The rest of this section justifies each entry with quantitative evidence.

Mechanism 1: FC2 cutoff vs cross-section span

A hiphive ClusterSpace built with cutoffs[0] = $r_c^{(2)}$ assigns zero FC2 to any atom pair separated by more than $r_c^{(2)}$ in Cartesian distance [37]. For a wire of H-shell radius ρ , the cross-section spans up to 2ρ (top-to-bottom of the H envelope), so a cutoff with $r_c^{(2)} \lesssim 2\rho$ leaves transverse Si-Si pairs unconstrained. The corresponding modes – breathing, ovalisation, transverse acoustic – have no restoring force in the fit basis and either localise (no dispersion) or pick up a finite negative dynamical-matrix eigenvalue through coupling with rotationally-projected FC2 entries.

A cutoff sweep on the d5a sc4 cluster reap (84 atoms, fixed at $n_{\text{structures}} = 16$, refit on the existing DFT outputs) quantifies this. Each cell of Table C.3 is the Γ -point imaginary-mode count of the resulting mass-weighted dynamical matrix; ARDR and least-squares

Appendix C. Input Generation

Mechanism	Symptom	Fix
FC2 cutoff < wire diameter	d5a sc4: 7 Γ -imag modes ($\omega \in [-2.2, -1.2]$ THz), 90 % Si-localised.	Set cutoffs[0] $\geq 2 \times \text{envelope} + 0.5 \text{ \AA}$.
Under-determination at large cutoffs	d9a sc4 at cutoffs=[7.83, 4.0]: $n_{\text{eqn}}/n_{\text{params}} = 0.70$ (40 % under-det). Validator already errors.	Bump $n_{\text{structures}}$ to $4 \lceil n_{\text{super}}/8 \rceil$. Re-DFT required.
Over-aggressive mc-rattle on light H atoms	d9a sc4 with $n_{\text{iter}} = 10$: rattle peaks 0.27–0.37 $\text{\AA}$, Si–H bonds compressed 13 %, anharmonic forces absorbed as artificial negative FC2 curvature on the H subspace (51 Γ -imag modes at –14 THz, 91 % H-localised).	$n_{\text{iter}} = 2$, std=0.03 $\text{\AA}$ (cumulative $\approx 0.06 \text{ \AA}$).
Stale YAML positions vs the relaxed sow geometry	0.55 $\text{\AA}$ drift between hiphive_meta.json (sow time) and the configs/sinw/*.yaml the analysis pipeline reads. Affects $q \neq \Gamma$ phase factors.	In loader.py, source the primitive from hiphive_meta.json when present.

Table C.2.: The four mechanisms we identified and the fixes applied in the code review. “d5a” and “d9a” refer to the Si_9H_{12} and $\text{Si}_{21}\text{H}_{16}$ wires in phonon/configs/sinw/. Atom counts, lattice parameters and cross-section conventions follow [149, 150].

fits with and without the post-fit Huang/Born-Huang projection are reported separately. The values come from phonon/scratch/imag_audit/pass1_cutoff_sweep.json.

Two observations are robust. First, ARDR without the rotational projection picks up a fixed-and-small $n_{\text{imag}} = 4$ for any $r_c^{(2)} \geq 5.5 \text{ \AA}$ – its sparse selection keeps FC2 close to PSD but does not enforce rotational invariance. Second, the Huang + Born-Huang projection succeeds only once $r_c^{(2)} \geq$ the wire diameter; at smaller cutoffs the projection enforces rotational invariance perfectly but shifts FC2 onto a less-PSD manifold and *introduces* imaginary modes.

Refitting the d5a sc4 supercell at $r_c^{(2)} = 7.0 \text{ \AA}$ with ARDR + RSR gives a clean dispersion across the full $\Gamma \rightarrow Z$ path: no imaginary modes across 21 q-points and 63 bands, with $\min \omega \approx -10^{-5} \text{ THz}$ (machine epsilon) and $\max \omega = 62.9 \text{ THz}$ consistent with Si–H stretching. The practical consequence for the wire generator is the formula change implemented in setup_sinw100.py:_suggest_fc2_cutoff,

$$r_c^{(2)} = \min(12.0, \max(5.0, 2\rho + 0.5)) \text{ [\AA]}, \quad (\text{C.17})$$

which replaces the previous $\rho + 2.4$ formula that systematically undershoots.

$r_c^{(2)}$ (Å)	ARDR / no RSR	ARDR / RSR	LS / no RSR	LS / RSR
5.0	13	16	16	6
5.5	4	14	16	13
6.0	4	7	8	11
6.5	4	6	13	16
7.0	4	0	15	16
7.8	4	0	15	18

Table C.3.: n_{imag} at Γ as a function of FC2 cutoff for the d5a sc4 wire (envelope $\rho = 3.7$ Å, full diameter ≈ 7.4 Å). ARDR + post-fit rotational sum rule succeeds only once $r_c^{(2)} \geq 7.0$ Å, i.e. once the cutoff exceeds the H-envelope diameter. All fits use FC3 cutoff 4.0 Å, $n_{\text{structures}} = 16$.

Mechanism 2: Under-determination at fixed sample count

For the d9a wire the FC2 cutoff bump pushes $n_{\text{parameters}}$ past $n_{\text{equations}} = n_{\text{structures}} \cdot n_{\text{super}} \cdot 3$. At cutoffs [7.83, 4.0] on the d9a sc4 supercell (148 atoms) the ClusterSpace expands to 10 116 free parameters, while $n_{\text{eqn}} = 16 \cdot 148 \cdot 3 = 7\,104$ – a ratio of 0.70, i.e. 40 % under-determined. ARDR’s automatic relevance determination [37] sparsifies aggressively, but it minimises the residual training error and cannot guarantee that the FC2 extracted from the sparse parameter subset has a positive-definite dynamical matrix. Empirically the result is the d9a sc4 reap’s 51 Γ -imag modes spanning -14 to -2 THz.

The validator built into `phonon_inputs/parameter_validation` already flags this with severity warn (it raised “`n_structures should be  $\geq 45$  for 84-atom supercell`” on the d5a sc4 reap before the present round of fixes). The recommended $n_{\text{structures}}$ scales with the supercell size as

$$n_{\text{structures}}^{\text{rec}} = 4 \cdot \lceil n_{\text{super}}/8 \rceil, \quad (\text{C.18})$$

yielding 44 for d5a sc4, 76 for d9a sc4, and 104 for d12a sc4 – the values now hard-coded in `configs/sinw/sinw100*_vasp_sc4*.yaml`. This is consistent with the rule of thumb in the hiphive paper that FC fits for under-determined regimes require “advanced regularized regression” together with several-times-overdetermined sampling [37].

Cutoff alone cannot save d9a. We also ran a cutoff sweep for d9a sc4 at fixed $n_{\text{structures}} = 16$ (FC3 cutoff reduced to 3.0 Å for speed; `phonon/scratch/imag_audit/pass1_d9a_quick.json`). Even at $r_c^{(2)} = 5.0$ Å, where the ratio recovers to 3.57 ($n_{\text{params}} = 1989$, $n_{\text{eqn}} = 7104$), the dispersion still has 28 imaginary modes at Γ with $\min \omega = -4.1$ THz, and the situation degrades at larger cutoffs. So under-determination is necessary but not sufficient: the residual imaginary modes are driven by mechanism 3, the over-aggressive rattle.

Mechanism 3: Rattle amplitude and the H subspace

The hiphive paper explicitly recommends rattle amplitudes “ $\approx 0.01\text{--}0.05 \text{ \AA}$ ” for accurate force-constant extraction and warns that “it can become difficult to untangle the force contribution if higher orders are contributing to the forces. This leads to higher-order contributions effectively being included in the fitted force constants thus reducing their accuracy” [37]. Our previous default ($\text{rattle_std} = 0.03 \text{ \AA}$, $n_{\text{iter}} = 10$) was inside the recommended range only at first glance. In hiphive’s mc-rattle the per-atom cumulative displacement scales as $\sqrt{n_{\text{iter}}}$ – each Monte-Carlo step adds an independent zero-mean Gaussian increment of standard deviation rattle_std , so the per-step variances add – giving a typical realised displacement after $n_{\text{iter}} = 10$ of $\text{rattle_std}\sqrt{n_{\text{iter}}} \approx 0.1 \text{ \AA}$, while the extreme value over all atoms reaches $\approx 0.3 \text{ \AA}$ (the measured $0.27\text{--}0.37 \text{ \AA}$ peaks below). The practical conclusion is unchanged: even at the $\sqrt{n_{\text{iter}}}$ rate the realised amplitudes leave the harmonic regime.

For three d5a sc4 disp directories we measured atomic-displacement peaks $0.27\text{--}0.37 \text{ \AA}$ and Si-Si nearest-neighbour distances dropping to $2.04 \text{ \AA}$ – a 13 % bond compression from the equilibrium $2.351 \text{ \AA}$, with forces of $6\text{--}7 \text{ eV/\AA}$ (`pass2_3_outcar_force_sanity.json`). This is solidly in the anharmonic regime for both Si-Si bonds and the lighter Si-H bonds; the cubic FC3 term contributes appreciably to the force along these compressed bonds, but the harmonic fit absorbs the cubic remainder as artificial negative curvature. The bias preferentially affects the H subspace because the mass-weighted dynamical-matrix eigenvalue λ scales as Φ_2/m : an error $\delta\Phi_2$ on the H-H block produces an eigenvalue shift $\delta\lambda = \delta\Phi_2/m_{\text{H}}$, which is $m_{\text{Si}}/m_{\text{H}} \approx 28$ times more sensitive than the analogous error on the Si-Si block.

Species-localised pathology pattern. Diagonalising the d5a sc4 mass-weighted Φ_2 at Γ (`pass2_d5a_d9a_gamma_eig.json`) gives seven negative eigenvalues between -2.2 and -1.2 THz , with Si occupation 70–91 % on each eigenvector. The same diagonalisation for d9a sc4 gives 51 negative eigenvalues between -14 and -2 THz , with H occupation 88–91 %. The species-localisation pattern is the clean signature of the mass-leverage mechanism just described.

The fix is to drop n_{iter} to 2 (cumulative displacement $\approx 0.06 \text{ \AA}$, well within the harmonic regime), keeping $\text{rattle_std} = 0.03 \text{ \AA}$. This matches the hiphive thermal-conductivity tutorial defaults and the displacement-amplitude guidance in [37].

Mechanism 4: Stale YAML positions

A subtle plumbing bug: the analysis-time Phonopy object in `finite_analysis/loader.py:_build_phonopy` was constructed from `cfg.structure`, the YAML’s primitive cell, while the fitted FC arrays in `fc3.hdf5` were produced from the *relaxed* primitive sown by `resolve_structure_for_sow`. For the d9a wire the relax displaced atoms by up to $0.55 \text{ \AA}$ (`pass2_phonopy_vs_hiphive_atoms.json`), so phonopy was diagonalising the dynamical matrix with the wrong atomic positions in the q-phase factors $\exp(i\mathbf{q} \cdot \mathbf{r}_j(\mathbf{R}))$.

At Γ this is irrelevant – the phases collapse to unity – so the Γ -imag count is unaffected. But the entire dispersion path at $\mathbf{q} \neq 0$ silently used wrong phases, scrambling the Γ -to- Z band structure. The fix is to source the primitive from `hiphive_meta.json` when present: `loader.py:_primitive_from_reap_metadata` now does this, and `_build_phonopy` accepts a `primitive_override` that `load_system` passes in. Bulk-Si (phono3py) reaps without a sibling `hiphive_meta.json` fall back to the YAML primitive, preserving backward compatibility.

C.3.3. The `leg_j_rel` metric is a hiphive structural artefact

The reading `physical/physical.json["fc3_asr_legs"]["leg_j_rel"]` ≈ 0.80 appears in every SiNW hiphive fit and is structural, not a fit defect. The metric measures

$$\text{leg_j_rel} = \frac{\|\sum_j \Phi_{ijk,\alpha\beta\gamma}^3\|_F}{\|\Phi^3\|_F}, \quad (\text{C.19})$$

i.e. the relative Frobenius norm of the axis- j trace of the mass-weighted lifted FC3 tensor. Hiphive’s `ClusterSpace` construction enforces the axis- i ASR by including the orbit-symmetry projector at basis-construction time, but it does *not* enforce the analogous constraint along axes j or k – those would require either a constrained cluster-space basis or an explicit post-fit projection on the FC tensor itself.

Bulk-Si reaps with full octahedral point symmetry yield `leg_j_rel` = 0.000 to floating-point precision (`pass3_bulk_si_leg_j.json`); quasi-1D SiNW reaps with a single mirror symmetry yield ≈ 0.80 regardless of `fit_method`, $n_{\text{structures}}$, or cutoffs. The metric is therefore a structural feature of hiphive’s cluster-space construction on a low-symmetry quasi-1D system, not a fit defect.

A practical consequence: the SSE bubble in `phonon/finite_analysis/` integrates FC3 along axis j (self-energy at frequency ω depends on $\sum_j \Phi_{ijk}^3$ contracted with $G^<(\omega - \omega')$), so a finite `leg_j_rel` propagates to a finite Drude-like weight in $\Sigma_{\text{ph-ph}}$ as $\omega \rightarrow 0$. If strict axis- j/k ASR is required downstream, project explicitly via `phonon_inputs.fc3_compression.asr_project_factor` (or pass `asr_project=True` to `load_quatrex_blocks`). The `fc3_asr_legs` docstring and `HIPHIVE_FITTING_NOTES` now reflect this.

C.3.4. Post-fit rotational projection is FC2-only in practice

The Huang + Born-Huang post-fit projection $\Pi : \theta \mapsto \arg \min_{\theta'} \|\theta' - \theta\|^2$ subject to $M_{\text{rot}}\theta' = 0$ (with M_{rot} the rotational constraint matrix exposed by `hiphive.core.rotational_constraints.get_rotational_constraint_matrix`) operates on the unified cluster-space parameter vector and rebuilds both FC2 and FC3 from the projected parameters. The relevant constraint structure couples FC2 and FC3 via the Born-Huang relation through cubic orbits of the cluster space [37].

For the FC3 cutoff of 4 Å used throughout this work, the projection touches only FC2. Refitting d5a sc4 at cutoffs [6.09, 4.0] and measuring before/after Frobenius distances

(pass4_d5a_rsr_delta.json) gives

$$\begin{aligned}\|\Phi_2^{\text{after}} - \Phi_2^{\text{before}}\|_F / \|\Phi_2\|_F &= 3.0 \times 10^{-2}, \\ \|\Phi_3^{\text{after}} - \Phi_3^{\text{before}}\|_F / \|\Phi_3\|_F &= 0.0,\end{aligned}$$

even though the rotational residual $\|M_{\text{rot}}\theta\|$ drops from 48.8 to 3×10^{-3} – four orders of magnitude. The cubic orbits at our $r_c^{(3)} = 4 \text{ \AA}$ apparently do not span the parameter directions that M_{rot} selects, so the projection acts as $\Pi_2 \oplus \mathbb{1}_3$ in the FC-tensor space.

A side effect is the imaginary-mode trade-off displayed in Table C.3: at small FC2 cutoffs the projection enforces rotational invariance perfectly but pushes FC2 off the PSD cone, *adding* Γ -imaginary modes. At cutoffs that span the wire diameter the projection succeeds without that side-effect, and the Γ -imag count drops to zero.

C.3.5. Are these wires sensible objects? Comparison to literature

The Si nanowires we use as test systems for the phonon–phonon-scattering self-energy machinery are at the small end of the diameter range studied by first-principles phonon work. For context:

- Peelaers et al. [150] compute ab-initio phonon band structures of $\langle 110 \rangle$ H-passivated SiNWs with diameters $\sim 1\text{--}1.5 \text{ nm}$ and report that all studied wires are dynamically stable provided the unit cell along the wire axis is taken sufficiently large. Their smallest wires are similar in cross-section to our d9a and d12a structures.
- The canonical $\langle 100 \rangle$ H-passivated SiNW family is Si_9H_{12} , $\text{Si}_{21}\text{H}_{20}$, $\text{Si}_{37}\text{H}_{28}$, ... [149]. Our d5a wire is identical to the smallest member of this family; our d9a and d12a are circular carves ($\text{Si}_{21}\text{H}_{16}$ and $\text{Si}_{29}\text{H}_{22}$) that don't fall on the canonical $\{100\}/\{110\}$ -faceted square cross-section but use the same bond chemistry, the same Si–H equilibrium of $1.48\text{--}1.50 \text{ \AA}$ [151], and identical $[100]$ axis orientation.
- Vo, Williamson and Galli [151] and Donadio and Galli [152] use SiNW diameters in the $1\text{--}3 \text{ nm}$ range for thermal conductivity studies; the cross-section-shape sensitivity they document supports our decision to use circular carves alongside the canonical faceted family.
- Karamitaheri et al. [153] study SiNW phonon dispersions with side lengths $1\text{--}10 \text{ nm}$ using a modified valence force field; they confirm that confinement-induced flattening of acoustic branches sets in below $\sim 3 \text{ nm}$, which is the regime our d5a sits in firmly.
- Markussen, Jauho and Brandbyge [154] use $\langle 100 \rangle$ H-passivated SiNWs in the $\sim 1\text{--}3 \text{ nm}$ diameter range as test systems for ballistic phonon transport – exactly the use we put these wires to downstream.

Two qualifications follow. First, the d5a wire (9 Si in cross-section, $\sim 0.5 \text{ nm}$ of Si core, $\sim 0.7 \text{ nm}$ with the H shell) is closer to a Si chain than a 3D wire. Quantum confinement

on the electrons is of order 1 eV; flexural branches are not continuum-Euler-Bernoulli but tight-binding-chain-like. Quantitative comparison to experimental SiNW transport data (which is at 10–100 nm diameters) is not the goal; demonstrating the $\Sigma_{\text{ph-ph}}$ infrastructure is. Second, the under-passivation we see at the corners of d9a and d12a (3-coordinated Si atoms; cf. `phonon/examples/_nanowire.py:_check_coordination`) is intrinsic to the circular carve. A canonical Cartoixa–Rurali $\text{Si}_{21}\text{H}_{20}$ carve would have all surface Si 4-coordinated; switching to faceted carves is an optional follow-up if we want to match the canonical convention exactly.

C.3.6. Verification protocol after the fixes

The full audit produced a punch list of seven items (A–G), summarised in `phonon/scratch/imag_audit/INVESTIGATION_NOTES.md`. All seven were implemented in the code review documented elsewhere in this appendix. The expected verification path is:

1. **Refit d5a sc4 only** (no DFT, the existing `disp-XXXXX/` forces suffice). After the FC2 cutoff bump to 7.88 Å and the rotational projection, the $\Gamma \rightarrow Z$ dispersion is clean across 21 q-points $\times$ 63 bands with $\min \omega \approx -10^{-5}$ THz. Already verified in `pass1_d5a_best_dispersion.json`.
2. **Re-sow + re-DFT d9a sc4** with $n_{\text{structures}} = 76$, $r_c^{(2)} = 11.36$ Å, $n_{\text{iter}} = 2$. Expected: $n_{\text{imag}} = 0$, $\max_{\text{acoustic}@\Gamma} < 0.1$ THz.
3. **Re-sow + re-DFT d12a sc4** with $n_{\text{structures}} = 104$, $r_c^{(2)} = 12$ Å (capped), $n_{\text{iter}} = 2$.
4. For all systems: confirm `leg_j_rel` stays at ~ 0.80 (structural, expected); do not treat as a fit defect.

Item 1 is hours of refitting; items 2 and 3 are cluster runs (76×148 -atom + 104×204 -atom DFT calculations). These are tracked as TODOs and will be filled into Section 2.2 once the cluster re-runs complete.

The `finite_analysis` Package

This appendix documents the validation pipeline that lives at `phonon/finite_analysis/`. The package processes a relaxed, finite (no transverse q , single supercell as device) phonon system and produces all of the analyses that go into Section 2.2: FC2/FC3 quality, sparsity structure, tensor decompositions, physical invariants, phonon–phonon self-energy structure, cutoff-hierarchy sensitivity, and ballistic + SCBA transport.

The text is organised as: (D.1) high-level overview; (D.2) module reference; (D.3) formal definitions of every observable with derivations; (D.4) the catalogue of physical-validation tests we run; and (D.5) the recommended workflow for producing the results-section figures.

D.1. Overview

The package treats a single supercell as a finite device: one slab decomposition along the transport axis, no transverse periodicity. The bare retarded propagator $G_0^R(\omega) = [(\omega + i\eta)^2 \mathbb{I} - D]^{-1}$ is built from the mass-weighted dynamical matrix $D = M^{-1/2} \Phi_2 M^{-1/2}$ in THz^2 . The three-phonon self-energy $\Sigma_{\text{ph-ph}}^{<, >}(\omega)$ is computed as the Migdal-like sunset diagram (*cf.* Eq. (D.4)) over the device DOFs, and combined with the lead self-energy $\Sigma_{\text{lead}}^R(\omega)$ from a Sancho–Rubio decimation on the first/last slab. The full retarded Σ^R is reconstructed from $(\Sigma^{<}, \Sigma^{>})$ via the Hilbert transform (Kramers–Kronig) so causality is respected. The Dyson equation is then solved per ω on the full $(n_{\text{dof}} \times n_{\text{dof}})$ device dimension; the Caroli formula gives transmission $T(\omega)$ and the Landauer integral gives the heat current Q .

Unit system

The package operates in $\text{THz}/(\text{amu} \cdot \text{\AA}^2)$ throughout (matching `phonon_inputs/anharmonic.py`; see Chapter A). Force constants are mass-weighted up front by `phonon_inputs.fc3_compression.build_fc3_target`, D is built by `synthetic_gf.dynamical_matrix`, and the bubble prefactor $i\hbar d\omega/(4\pi)$ in `quatrex.phonon.units.bubble_prefactor_thz`.

closes the units. Frequency grids are symmetric, $[-f_{\max}, \dots, 0, \dots, +f_{\max}]$, with the $\omega = 0$ sample zeroed before FFT convolution to suppress the bosonic singularity.

D.2. Module reference

The package is organised into 14 modules. Cross-cutting helpers are prefixed with an underscore.

loader.py Reads a `phonon_inputs` YAML config, an FC3 HDF5 reap, and produces a `SystemBundle` dataclass carrying the phonopy object, FC2, FC3, mass-weighted FC3 target, masses, supercell positions/lattice, the slab decomposition `block_sizes` and the z -sort permutation `atom_perm`. When a sibling `hiphive_meta.json` sits next to `fc3.hdf5` (the `hiphive` reap’s metadata), the primitive cell is sourced from that file via `_primitive_from_reap_metadata`; for `phono3py` / `DFPT` reaps without that metadata the primitive falls back to the YAML structure. The metadata path is preferred because the FC arrays were fit on the relaxed primitive saved at sow time, so reading the primitive from the same file keeps the dispersion’s q -phase factors $\exp(i\mathbf{q} \cdot \mathbf{r}_j)$ consistent with the fit. The companion `load_quatrex_blocks` projects the FC3 onto the NN-tridiagonal block dictionary that the bubble consumes; with `asr_project=True` (default `False`) it ASR-projects FC3 along legs 2 and 3 first, which is the canonical way to eliminate the Drude-like weight in Σ at $\omega \rightarrow 0$ when the upstream `hiphive` fit leaves a structural axis- j/k ASR residual (the `leg_j_rel` metric below).

constants.py Single home for every numeric default in the package. Notable: `THZ_TO_CM1 = 33.35641`; `ASR_REL_RESIDUAL_WARN = 0.05`; `THZ_ACOUSTIC_BAND = 0.01 THz`; `DEFAULT_DROP_THRESHOLD_THZ = 0.05 THz` with adaptive relative cut at $10^{-3} \times \max |\omega_n|$; `DEFAULT_ETA_FACTOR_DW = 0.5` with floor `DEFAULT_ETA_THZ_FLOOR = 0.02 THz`; `EPS_LADDER = (10^{-2}, 10^{-3}, 10^{-4}, 10^{-5})` for the sparsity nnz tables.

utils.py Geometry primitives (minimum-image pairwise distances, triplet diameter), DOF-permutation helper $i{\text{atom}} \rightarrow (i_{\text{atom}}, \alpha)$, and `dense_block` $\rightarrow$ dictionary slicer.

synthetic_gf.py Eigenmode-resolved phonon Green’s functions on a symmetric ω grid. Diagonalises the mass-weighted D , splits the spectrum into acoustic ($|\omega| \leq \text{THZ_ACOUSTIC_BAND}$), imaginary ($\omega < -\text{THZ_ACOUSTIC_BAND}$, warning emitted), and physical bands. Builds $G^{<, >}(\omega)$ via Eq. (D.2).

sse_cutoffs.py Three-phonon bubble in block-tridiagonal storage. The full off-diagonal (K_1, K'_1, K_2, K'_2) enumeration is the production path; the original `quatrex` $K_1 = K'_1, K_2 = K'_2$ restriction is recovered by setting `diag_G_in_se = True`. Five cutoff knobs documented in Section 2.3. The `_compat` module retains the legacy one-sided convention for bit-for-bit cross-checks.

sse_quatrex_run.py Independent dense-bubble cross-check invoking `phonon_inputs.anharmonic._compat`. Also runs a Dyson + equilibrium-Keldysh SCBA loop when `n_iter > 0`.

transport_metrics.py $T(\omega)$, density of states, Landauer heat current. `sigma_retarded(method)` reconstructs Σ^R from $(\Sigma^<, \Sigma^>)$ via FFT-Hilbert (default), PV quadrature, or the half-approximation. `sanchu_rubio_lead_self_energies` builds the lead Σ_L^R, Σ_R^R by decimating the first/last transport slab as a periodic lead unit. `transport_trace_scba` iterates $\Sigma \leftrightarrow G$ via Dyson + equilibrium-FDT closure with linear mixing.

physical_tests.py Static FC2/FC3 invariants (Hermiticity, ASR, S_3 -permutation, dispersion-band sanity, PSD diagnostic) plus the dynamical NEGF identities (detailed balance, DOS sum rule). The FC3 ASR-legs metric `leg_j_rel =  $\|\sum_j \Phi_{ijk}^3\|_F / \|\Phi^3\|_F$` measures the axis- j (and analogously axis- k) trace of the mass-weighted lifted FC3; hiphive's `ClusterSpace` enforces axis- i ASR by orbit construction but not the other two axes, so a residual reading of ~ 0.8 on a SiNW reap and ~ 0 on a bulk-Si reap is structural rather than a fit defect. Downstream consumers that need strict axis- j/k ASR can invoke the explicit projection via `load_quatrex_blocks(asr_project=True)` or `phonon_inputs.fc3_compression.asr_project_factor`.

physics_validation.py Identity-style validators for the Green's-function machinery: anti-Hermiticity, $G^{R\dagger} = G^A$, optical theorem $\Sigma^> - \Sigma^< = +2i \text{Im } \Sigma^R$, first-moment spectral sum rule $\int 2\omega^3 A(\omega) d\omega = \text{Tr}(D)$, ballistic $T(\omega)$ unitarity bound, Bose-factor identity, and Kramers-Kronig consistency between PV and FFT Σ^R reconstructions.

parameter_validation.py DFT and FC-fit sanity checks exposed as: a load-time warning, a CLI subcommand `python -m finite_analysis validate <config>`, and a per-system block in `summary.json`. The check table covers ENCUT, $E_{\text{cut,wfc}}$, k -point density on periodic axes, hiphive sample count and cutoffs, displacement amplitude, third-order pair-distance cutoff, and supercell extent.

FC2 cutoff vs. supercell. The FC2 pair cutoff $r_{\text{cut}}^{(2)}$ must obey $r_{\text{cut}}^{(2)} < L/2$ along every periodic axis, where L is the periodic extent of the rattled supercell — if any cluster wraps the cell, hiphive aborts with "Found cluster (i, j) in two orbits, decrease cutoff (less than $L/2$) or increase supercell size" at `ClusterSpace` construction time (see `hiphive_fc3.py:922`). The validator emits an *error* when $r_{\text{cut}}^{(2)} > L/2$ and a *warning* when $r_{\text{cut}}^{(2)} > 0.8 L/2$; the latter band leaves zone-boundary modes poorly determined and is the most common cause of imaginary modes at the Z-axis zone boundary. Concretely for SiNW(100) with z -period $5.47 \text{ \AA}$: $[1, 1, 4]$ gives $L_z/2 = 10.94 \text{ \AA}$ so cutoffs $\lesssim 8.7 \text{ \AA}$ are safe; the d_9 variant uses a $9.5 \text{ \AA}$ cutoff (within the warn band; restoring the $\sim 11 \text{ \AA}$ envelope-driven cutoff requires growing the supercell to $[1, 1, 8]$).

sparsity.py FC2 atomic-block-Frobenius heatmap; FC3 triplet-diameter $\|\Phi_3\|$ decay; 3D scatter of FC3 nonzero triplets coloured by $\log_{10} |\Phi_3|$; and a CSV nnz table at thresholds $\varepsilon \in \text{EPS_LADDER}$.

Appendix D. The *finite_analysis* Package

decomposition.py Drives the six existing FC3 tensor fitters (mSVD, HOSVD, CP, IND-SCAL, Waring, PCP; see Section 1.1.5) over a rank ladder, reporting parameter cost vs Frobenius reconstruction error and reconstruction-nnz at the same ε ladder as sparsity.

fc_quality.py Distance-binned FC2/FC3 magnitudes and Brillouin-zone-line dispersion plots; supports a multi-reap convergence sweep listed in the YAML's convergence: block.

sse_sparsity_driver.py CLI-callable harnesses for the “sse_sparsity” and “cutoffs” analyses. Both consume build_sse_inputs and emit per- (I, J) Frobenius CSVs and heatmaps. The cutoffs harness wires transport_for_cutoff_sweep so each cutoff configuration produces a $T(\omega)$ trace and a Landauer Q .

transport_quality.py The headline “FC3-approximation quality vs. transport” suite. For each cell of the (method, rank) grid it (i) fits the FC3 with the chosen phonon_inputs.fc3_compression method, (ii) reconstructs the supercell-lifted target shape $(n_{\text{dof}}, \text{dim}_{sc}, \text{dim}_{sc})$, (iii) runs phonon.solver.transmission_q with the reconstruction as M_stacked_override – the q-resolved path is the only safe choice for SiNWs because the bubble runs on the primitive-cell DOFs after Γ -projection, not on the full $n_{\text{dof}}^{\text{super}}$ – and (iv) caches the result in `<out_dir>/cache/{method}_{rank}.npz`. Each rerun re-reads cached cells and only computes the missing ones; passing force_recompute=True or deleting the cache directory forces a full recompute. The output plots compare Frobenius reconstruction error, anharmonic conductance error, and parameter count across methods.

cross_system_plots.py Cross-system overlay plots for the results section. After per-system analyses have populated document/src/fig/finite_analysis/<system>/, this script reads the CSV outputs and produces three overlays: cross_fc3_decay.pdf (mean $\|\Phi_3\|$ vs triplet diameter, all systems on one log-y panel), cross_decomposition_ranks.pdf (Frobenius reconstruction error vs parameter count, six methods $\times N$ systems), and cross_summary_table.pdf (a one-page text-rendered table of headline numbers). Invocation: `python -m finite_analysis.cross_system_plots --systems sinwl00,si_chain,cnt33_finite --base-dir document/src/fig/finite_analysis --out-dir document/src/fig/finite_analysis`.

summary.py Typed contract for per-analysis return dicts. AnalysisSummary is a TypedDict with a units: mapping (bare-metric name $\rightarrow$ unit string) that every run<analysis> driver populates; the CLI assembles the per-analysis blocks into the top-level summary.json.

cli.py & __main__.py Single CLI entry point with subcommands run (the full analysis) and validate (parameter check only).

D.3. Observables

D.3.1. Contacted propagator

In the THz² unit system, the contacted retarded propagator (bare device propagator dressed by the lead self-energy) is

$$G_{\text{contacted}}^R(\omega) = [(\omega + i\eta)^2 \mathbb{I} - D - \Sigma_{\text{lead}}^R(\omega)]^{-1}, \quad (\text{D.1})$$

with η a small numerical broadening (default 0.05 THz; `DEFAULT_ETA_THZ_TRANSPORT`). The lead self-energy is built by Sancho–Rubio decimation on the first/last transport slab (Section D.3.5).

D.3.2. Eigenmode synthetic Green’s function

For the SSE evaluation in the absence of a converged SCBA, we use the eigenmode-resolved displacement Green’s functions

$$G^<(\omega) = -2\pi i \sum_n \frac{\varepsilon_n \varepsilon_n^\top}{2\omega_n} \left[n_{\text{B}}(\omega_n) L_\eta(\omega - \omega_n) + (n_{\text{B}}(\omega_n) + 1) L_\eta(\omega + \omega_n) \right], \quad (\text{D.2})$$

$$G^>(\omega) = -2\pi i \sum_n \frac{\varepsilon_n \varepsilon_n^\top}{2\omega_n} \left[(n_{\text{B}}(\omega_n) + 1) L_\eta(\omega - \omega_n) + n_{\text{B}}(\omega_n) L_\eta(\omega + \omega_n) \right], \quad (\text{D.3})$$

where $L_\eta(x) = (\eta/\pi)/(x^2 + \eta^2)$ is the Lorentzian broadening of the bare delta and the sum runs over the physical modes ($\omega_n > \text{THZ_ACOUSTIC_BAND}$). These satisfy the bosonic full-axis identity $G^<(\omega) = [G^>(-\omega)]^\top$ (theory.tex eq. 873–874) and are anti-Hermitian on the DOF axes; both checked by the chain test `test_synthetic_gf_bose_symmetry` and `test_synthetic_gf_anti_hermiticity`. The Lorentzian prefactor $-2\pi i$ is pinned by `test_synthetic_gf_correlator_normalisation`, which checks $\int iG^<(\omega)d\omega/(2\pi)$ reproduces the equal-time correlator $\sum_n \varepsilon_n \varepsilon_n^\top (n_{\text{B}} + \frac{1}{2})/\omega_n$ to better than 5% on the analytic chain.

D.3.3. Three-phonon self-energy

The Migdal-like sunset diagram, in block-tridiagonal device storage, is

$$\Sigma_{IJ}^{<, >}(\omega) = \frac{i\hbar}{2} \sum_{K_1, K'_1, K_2, K'_2} \Phi^{(I, K_1, K_2)} \int \frac{d\omega'}{2\pi} G_{(K_1, K'_1)}^{<, >}(\omega') G_{(K_2, K'_2)}^{<, >}(\omega - \omega') \Phi^{(J, K'_2, K'_1)}, \quad (\text{D.4})$$

the ω' integral being the energy-conserving convolution. `sse_cutoffs.compute_sse_with_cutoffs` enumerates (K_1, K'_1, K_2, K'_2) over every (K, K') pair in the supplied G dictionary. The original quatrex `SigmaPhononPhonon._bubble_block` restricts to $K_1 = K'_1, K_2 = K'_2$ — we recover that with `diag_G_in_se = True`. The full enumeration is required when G has nontrivial off-diagonal blocks (e.g. the synthetic eigenmode G of Eq. (D.2)); the chain test `test_sse_baseline_vs_dense_path` pins the agreement between the block bubble and the dense reference at 10^{-6} relative.

D.3.4. Retarded SE reconstruction

From the contour Keldysh structure, the retarded self-energy is

$$\Sigma^R(\omega) = \frac{1}{2}\Delta(\omega) + \frac{i}{2\pi}\mathcal{P}\int \frac{\Delta(\omega')}{\omega - \omega'} d\omega', \quad \Delta(\omega) \equiv \Sigma^>(\omega) - \Sigma^<(\omega). \quad (\text{D.5})$$

Three implementations live in `phonon_inputs.anharmonic._build_retarded`:

- "fft" (default): $\Sigma^R = \frac{1}{2}\Delta + \frac{i}{2}\mathcal{H}[\Delta]$ where $\mathcal{H}$ is the FFT Hilbert transform.
- "pv": singularity-subtracted principal-value quadrature; analytic at the grid end-points.
- "half": $\Sigma^R \approx \frac{1}{2}\Delta$ (imaginary part only); the level shift is dropped. Adequate for cutoff-sweep differential comparisons; not for absolute $T(\omega)$.

`transport_metrics.transport_trace_from_sigma` defaults to "fft". `kramers_kronig_consistency` reports the relative-Frobenius gap between PV and FFT reconstructions as a sanity check.

D.3.5. Lead self-energy

The semi-infinite phonon lead is treated as the same first/last transport slab repeated periodically. Sancho–Rubio decimation (`phonon_inputs.anharmonic._sancho_rubio_batch`) returns the lead surface Green's function $g_{\text{surf}}(\omega^2)$, and the lead self-energy on the device side is

$$\Sigma_L^R(\omega) = H_{\text{LD}}^\dagger g_{\text{surf}}(\omega^2 + i\eta) H_{\text{LD}}, \quad (\text{D.6})$$

with $H_{\text{LD}} = D[\text{slab}_0, \text{slab}_1]$ the inter-slab coupling (and analogously for the right lead). Σ_L^R is nonzero only on the first slab block (zero elsewhere) — pinned by `test_sancho_rubio_lead_self_energy`. A per-frequency fallback handles singular ε matrices that arise inside narrow eigenmode windows.

D.3.6. Transmission and heat current

The Caroli formula for the elastic transmission, applied per ω on the dense G^R obtained from the full Dyson equation

$$G^R(\omega) = [(\omega + i\eta)^2 \mathbb{I} - D - \Sigma_{\text{ph-ph}}^R(\omega) - \Sigma_L^R(\omega) - \Sigma_R^R(\omega)]^{-1}, \quad (\text{D.7})$$

is

$$T(\omega) = \text{Tr}[\Gamma_L(\omega) G^R(\omega) \Gamma_R(\omega) G^A(\omega)], \quad (\text{D.8})$$

with $\Gamma_\alpha = i(\Sigma_\alpha^R - \Sigma_\alpha^A) = -2 \text{Im } \Sigma_\alpha^R$. The Landauer heat current at lead temperatures T_L, T_R is

$$Q = \int_0^\infty \frac{d\omega}{2\pi} \hbar\omega T(\omega) [n_{\text{B}}(\omega, T_L) - n_{\text{B}}(\omega, T_R)]. \quad (\text{D.9})$$

The thermal conductance is $G = Q/\Delta T$ for small $T_L - T_R = \Delta T$.

D.3.7. Spectral sum rules

For $A(\omega) = -\frac{1}{\pi} \text{Im Tr } G^R(\omega)$ of the bare displacement propagator, the standard phonon sum rules (Maradudin et al., §2.3) are

$$\int_0^\infty 2\omega A(\omega) d\omega = n_{\text{dof}}, \quad (\text{D.10})$$

$$\int_0^\infty 2\omega^3 A(\omega) d\omega = \text{Tr}(D), \quad (\text{D.11})$$

exact in the $\eta \rightarrow 0$ limit on a dense ω grid. The chain tests `test_dos_sum_rule_eigenmode_gf` and `test_first_moment_sum_rule` pin both, the former to 10 % (grid leakage at the band edge) and the latter to 10 % (mode-by-mode agreement against $\sum_n \omega_n^2$). Eq. (D.11) provides a non-trivial unit-system cross-check: it ties the spectral weight of G^R to the trace of D in THz^2 and so verifies the mass-weighting + unit conversion all the way through.

D.4. Physical-validation catalogue

The package runs a fixed set of physical-validation tests against every system; all pin true mathematical identities of the bosonic Keldysh formalism (cf. Wang 2014; Stefanucci & van Leeuwen 2013). Tests live at `tests/phonon_inputs/finite_analysis/test_chain_pipeline.py`.

FC2 HERMITICITY $\Phi_2[ij, \alpha\beta] = \Phi_2[ji, \beta\alpha]$ to machine precision (`fc2_hermiticity`).

FC2 ACOUSTIC SUM RULE $\sum_j \Phi_2[ij] \approx 0$ ($\leq 10^{-10}$ relative; `fc2_asr_residual`).

FC3 PERMUTATIONAL SYMMETRY On the S_3 -lifted FC3 tensor (machine precision; `fc3_perm_symmetry`).

FC3 ACOUSTIC SUM RULE Per-leg Frobenius residual warned above `ASR_REL_RESIDUAL_WARN = 5%` (`fc3_asr_legs`). When violated (e.g. hiphive least-squares on SiNW gave $\sim 60\%$), `load_quatrex_blocks(asr_project=True)` projects FC3 onto the ASR null space along legs 2 and 3 before the bubble consumes it.

FC2 PSD Eigenvalue distribution of the mass-weighted D ; counts negative eigenvalues and reports the worst negative-to- positive ratio (`fc2_psd_summary`).

DISPERSION SANITY Splits the FC2 spectrum into acoustic ($|\omega| \leq \text{THZ_ACOUSTIC_BAND}$), imaginary ($\omega < -\text{THZ_ACOUSTIC_BAND}$, FC2 indefinite), and optical (`dispersion_sanity`).

ANTI-HERMITICITY OF $G^{<,>}$ Wang 2014 eq. 19; pinned by `test_synthetic_gf_anti_hermiticity` at 10^{-10} relative on the chain.

BOSONIC FULL-AXIS SYMMETRY $G^{<}(\omega) = [G^{>}(-\omega)]^T$; `test_synthetic_gf_bose_symmetry` at floating-point precision.

BOSE IDENTITY $n_B(-\omega) = -1 - n_B(\omega)$ (test_bose_identity; machine precision).

DETAILED BALANCE ON BUBBLE $\Sigma^>(\omega) = e^{\beta\hbar\omega} \Sigma^<(\omega)$ within the finite Bose window; test_detailed_balance_bubble_sigma on the chain.

SPECTRAL SUM RULES Eqs. (D.10) and (D.11); pinned by test_dos_sum_rule_eigenmode_gf (10%) and test_first_moment_sum_rule (10%) on the chain.

BALLISTIC TRANSMISSION BOUND $0 \leq T(\omega) \leq n_{\text{modes}}$ (Datta 2005 eq. 9.16; checker ballistic_transmission_bound).

OPTICAL THEOREM $\Sigma^> - \Sigma^< = +2i \text{Im } \Sigma^R$ (optical_theorem_residual).

$G^{R\dagger} = G^A$ (retarded_advanced_residual).

KRAMERS-KRONIG CONSISTENCY PV vs FFT Σ^R reconstructions agree (kramers_kronig_consistency).

SANCHO-RUBIO CONVERGENCE Lead Σ^R has the expected slab support and is non-zero in the optical band (test_sancho_rubio_lead_self_energy).

SCBA CONVERGENCE $\|\Delta\Sigma\|/\|\Sigma\|$ monotonically decreases over iterations (test_scba_convergence_chain).

BIT-FOR-BIT Γ -PROJECTION The _gamma_project_M_blocks fast path agrees with the einsum reference at floating-point precision (test_gamma_projection_matches_einsum).

D.5. Recommended workflow for results-section figures

A condensed version of phonon/finite_analysis/RESULTS_RUNBOOK.md.

Per system (SiNW(100), Si chain, finite CNT(3,3))

1. Run the cluster pipeline as described in Section 2.3 and Chapter C; the reap lands at phonon/reaps/<name>/fc3.hdf5.
2. Sanity-check the configuration: `python -m finite_analysis validate <config>`.
3. Run the analyses:

```
python -m finite_analysis \
    --config phonon/configs/<family>/<name>.yaml \
    --fc3-path phonon/reaps/<reap>/fc3.hdf5 \
    --out-dir document/src/fig/finite_analysis/<system> \
    --analyses all \
    --rank-sweep 2,4,8,16,32 \
    --skip-pcp
```

4. Confirm the pre-flight gates: parameter validation exits 0 or 1 (no errors); `physical.dispersion.n_imaginary == 0` (or document why); `physical.fc3_asr_legs.leg_*_rel < 0.05` (or pass `asr_project=True`); `physical.fc2_psd.n_neg_eigvals == 0`.

Cross-system overlay

```
python -m finite_analysis.cross_system_plots \
    --systems sinw100,si_chain,cnt33_finite \
    --base-dir document/src/fig/finite_analysis \
    --out-dir document/src/fig/finite_analysis
```

produces three cross-system PDFs: FC3-decay envelope, decomposition-rank panels, and the headline-numbers summary table (Section 2.3).

Known limitations to compensate for in the writeup The bubble does not ASR-project FC3 by default; pass `asr_project=True` or interpret only $\omega \gtrsim 1$ THz. The synthetic lead model retains $\Gamma = \gamma \cdot \mathbb{I}$ only as a diagnostic; production transport uses Sancho–Rubio. Σ^R is reconstructed by FFT-Hilbert by default; the "half" option is only for differential cutoff comparisons. SCBA convergence is achieved via `transport_trace_scba` (not the bare `transport_trace_from_sigma`); $n_{\text{iter}} \geq 3$ with `mixing = 0.5` converges the chain to 0.1%.

D.6. References

The package’s NEGF + phonon-transport implementation follows the conventions of:

- J.-S. Wang, B. K. Agarwalla, H. Li, J. Thingna, “Nonequilibrium Green’s function method for quantum thermal transport,” *Front. Phys.* **9**, 673–697 (2014).
- G. Stefanucci, R. van Leeuwen, *Nonequilibrium Many- Body Theory of Quantum Systems: A Modern Introduction*, Cambridge University Press (2013). Chapters 4–5.
- S. Datta, *Quantum Transport: Atom to Transistor*, Cambridge University Press (2005). Chapter 9.
- A. A. Maradudin, E. W. Montroll, G. H. Weiss, I. P. Ipatova, *Theory of Lattice Dynamics in the Harmonic Approximation*, Solid State Physics Suppl. 3, Academic Press, 2nd ed. (1971).
- M. P. Lopez Sancho, J. M. Lopez Sancho, J. Rubio, “Highly convergent schemes for the calculation of bulk and surface Green functions,” *J. Phys. F: Met. Phys.* **15**, 851 (1985).
- F. Eriksson, E. Fransson, P. Erhart, “The HiPhive package for the extraction of high-order force constants by machine learning,” *Adv. Theory Simul.* **2**, 1800184 (2019).

Appendix E

The `phonon.solver` Package

The `phonon.solver` package is the dense reference implementation of the non-equilibrium Green's function (NEGF) plus self-consistent Born approximation (SCBA) machinery used for the anharmonic phonon transport calculations of this report. It serves three roles simultaneously:

1. It is the canonical reference against which the production block-sparse / MPI solver in `quatrex.phonon.sse_phonon_phonon` is benchmarked. Both solvers share the same bubble kernel (`bubble.py`); cross-agreement is locked by a unit test that asserts agreement to $\sim 10^{-12}$ (absolute tolerance) on a small synthetic Green's function.
2. It is the driver behind the cutoff-sensitivity and FC3-compression-vs-transport sweeps that produce the results-section figures. The cell that exposes those analyses lives in `finite_analysis.transport_quality` and feeds the solver via `phonon.solver.transmission_q(..., M_stacked_override=...)`.
3. It exposes the SCBA fixed-point loop (`scba_loop`) and the bare bubble kernel (`bubble_dense`) as library entry points reusable by any caller that wants to plug in a custom self-energy.

This appendix walks through the package module by module. Public entry points are listed in `phonon/solver/__init__.py`; the internal modules are described in the order in which they are pulled in by the top-level drivers.

E.1. Unit system and frequency grid

The solver operates internally in THz^2 for dynamical matrices and self-energies. Force constants are mass-weighted upstream by `phonon_inputs.fc3_compression.build_fc3_target` (or by `gamma_project_M_blocks` for the Γ -only path) and the dynamical-matrix entries are scaled by the $\text{CONVERSION_THZ2} = e/(m_u \text{\AA}^2)/(2\pi)^2 \cdot 10^{-24}$ factor from

`phonon_inputs.constants`. Self-energies always carry a q -axis of shape $(n_{\text{slabs}}, n_{\text{kpts}}, n_{\text{freq}}, n_{\text{dof}}, n_{\text{dof}})$, even when $n_{\text{kpts}} = 1$ (the Γ -only path). This unification means the SCBA fixed-point loop is a single function shared by both drivers; the Γ -only path is recovered as `transmission_q(q_mesh=(1,1,1))`, and the regression check `compare_q11_to_finite` asserts that the two paths produce identical $T(\omega)$.

The frequency grid is the symmetric $[-f_{\text{max}}, \dots, -\Delta\omega, 0, \Delta\omega, \dots, +f_{\text{max}}]$ array built by `grids.build_frequency_grid(freq_range_thz, eta_w_thz, eta_factor)`. The $\omega = 0$ sample sits at index `mid = n_freq_pos` and is included to make FFT arithmetic clean; physical integrals exclude it via the `pos_mask` returned by the same routine, and the bubble integrand zeroes the $\omega = 0$ slice of the input Green's functions before FFT to suppress the bosonic singularity. The Lorentzian broadening parameter η defaults to `eta_factor` $\cdot \Delta\omega$ (linear in the grid spacing, with the default `eta_factor = 1.0` giving $\eta \sim \Delta\omega$; the legacy value was $\eta = 0.05 \Delta\omega$), but can be overridden directly via `eta_w_thz`. Note that the theory chapter adopts a different, *quadratic* convention $\eta = \text{eta_factor} \cdot (\Delta\omega)^2$; the appendix retains the linear form used in the solver code.

E.2. The bubble kernel: `bubble.py`

The shared FFT bubble kernel `bubble_dense(phi_left, phi_right, G_a, G_b, n_fft, prefactor, out_slice, zero_freq_idx, xp)` computes one (I, J) block of the 3-phonon self-energy

$$\Sigma_{IJ}^x(\omega) = \frac{i\hbar}{2} \sum_{c,d,e,f} \phi_{Ice}^L G_{ed}^x(\omega) \star G_{cf}^x(\omega) \phi_{Jdf}^R, \quad (\text{E.1})$$

where $x \in \{<, >\}$ selects the lesser or greater branch and $\star$ denotes the frequency convolution. The kernel evaluates the convolution as a zero-padded FFT along the ω axis – the FFT length `n_fft` is the next power of two above the linear output length $n_{\text{freq}}^{\text{out}} + n_{\text{freq}}^{\text{G}_a} - 1$ – contracts indices with two `einsum` calls per frequency sample (one for $\phi^L \cdot G^x$, one for the result times $\phi^R \cdot G^x$), and slices the output to the caller's `out_slice`. The vector backend is selected via `xp` (default `numpy`; a `cupy` backend gives drop-in GPU support when available).

The prefactor multiplying the FFT result is supplied by the caller; the canonical value is $0.5 i \hbar d\omega / (2\pi)$ in SI units. The kernel is byte-for-byte shared with the production block-sparse solver via an independent installed copy at `quatrex.phonon.bubble`. Cross-agreement between the two implementations is asserted by `tests.quatrex.phonon.test_bubble_kernel`.

E.3. Frequency grid and Bose distribution: `grids.py`

In addition to `build_frequency_grid`, this module exposes:

`bose_full_axis(freqs_thz, temperature_k)`. Returns $n_B(\omega) = 1/(e^{\hbar\omega/k_B T} - 1)$ on the full symmetric grid. The $\omega = 0$ value is set to 0 (rather than infinity) so callers can

multiply with $G^<$ at that sample without producing NaN; the caller is responsible for zeroing the $\omega = 0$ contribution before frequency integration.

boson_contact_self_energies_from_gamma. Builds $(\Sigma_{\text{lead}}^<, \Sigma_{\text{lead}}^>)$ from the contact $\Gamma_{\text{lead}}(\omega)$ matrix via the equilibrium fluctuation–dissipation theorem, $\Sigma_{\text{lead}}^<(\omega) = i n_B(\omega) \Gamma_{\text{lead}}(\omega)$ and $\Sigma_{\text{lead}}^>(\omega) = i (n_B(\omega) + 1) \Gamma_{\text{lead}}(\omega)$, with the lead temperature passed in.

E.4. Retarded self-energy reconstruction: `retarded.py`

The retarded scattering self-energy is reconstructed from the $(\Sigma^<, \Sigma^>)$ pair via the Kramers–Kronig relation

$$\Sigma^R(\omega) = \frac{1}{2}[\Sigma^>(\omega) - \Sigma^<(\omega)] + \frac{i}{2\pi} \text{PV} \int \frac{\Sigma^>(\omega') - \Sigma^<(\omega')}{\omega - \omega'} d\omega'. \quad (\text{E.2})$$

`retarded.build_retarded(sigma_lesser, sigma_greater, omega_grid_thz, method)` dispatches over three options:

"half". Drops the principal-value integral entirely and returns $\Sigma^R = \frac{1}{2}(\Sigma^> - \Sigma^<)$. This is the imaginary part of Σ^R only – i.e. the broadening, $-\text{Im } \Sigma^R = \frac{1}{2}(\Sigma^> - \Sigma^<)$ up to a sign and factor. Used when only the broadening matters (e.g. the Γ -only regression check `compare_q11_to_finite`, where $\text{Re } \Sigma^R$ cancels in the trace). Both `transmission_finite` and `transmission_q` default to this choice so the regression assertion is exact.

"pv". Singularity-subtracted principal-value integral. Rewrites the integrand as $[\Delta(\omega') - \Delta(\omega)]/(\omega - \omega')$ (regular at $\omega' = \omega$) plus the pole-free remainder $\Delta(\omega) \text{PV} \int 1/(\omega - \omega') d\omega'$. Costs $O(n_{\text{freq}}^2)$ per matrix element. Captures both $\text{Re } \Sigma^R$ and $\text{Im } \Sigma^R$.

"fft". FFT Hilbert transform via the sign-multiplier method, $O(n_{\text{freq}} \log n_{\text{freq}})$ per matrix element. Preferred for large frequency grids; the sign of $-1j \cdot \text{sign}(\omega)$ multiplier is set so the result matches the principal-value convention to $\sim 10^{-10}$ in floating-point.

The factor multiplying the $(n_B/(n_B + 1))$ sum in the eigenmode expansion of Σ^R is $-2i$ (no extra π); the $1/\pi$ from $\text{Im } G^R = -\pi \sum_n \delta(\omega - \omega_n)$ is absorbed into the Lorentzian $L_\eta(\omega - \omega_n) = (\eta/\pi)/[(\omega - \omega_n)^2 + \eta^2]$ that replaces the delta in the broadened expansion. This convention is the one used throughout `theory.tex` (see equations 873–874 of the theory chapter).

E.5. Open-boundary leads and device Dyson equation: `leads.py`

This module owns the lead/contact construction and the device-side Dyson solve. The five public entry points:

sancho_rubio(z2, H_00, H_01). Surface Green's function for one lead at frequency $z = \omega^2$, computed by iterative decimation [107]. The fast iteration is the standard $\varepsilon, \varepsilon_s, \alpha, \beta$ update; convergence is monitored by the relative residual $\|\alpha\|/\|\varepsilon\|$. If the matrix iteration diverges (residual $> 10^{-4}$ after `max_iter` iterations), the routine raises `RuntimeError` so the caller can fall back to the undressed contact and emit a warning.

sancho_rubio_batch(z2_array, H_00, H_01). Frequency-batched version; vectorises the matrix operations across the input ω^2 array using `numpy.linalg` broadcasting.

build_device_hamiltonian(H_00, H_01, n_slabs). Assembles a block-tridiagonal device Hamiltonian of `n_slabs` identical slabs with on-site H_{00} and first-neighbour coupling H_{01} . Used in the Γ -only path where the device extends the same slab many times along the transport axis.

compute_obc_batch(omega_grid_thz, H_00, H_01, temperature_k, ...). On a frequency batch, computes the lead retarded self-energies Σ_L^R, Σ_R^R via Sancho-Rubio plus contact coupling, then the lead broadenings $\Gamma_{L,R} = i(\Sigma_{L,R}^R - \Sigma_{L,R}^A)$, and finally the lead lesser/greater pair via the fluctuation-dissipation relation from `boson_contact_self_energies_from_gamma`. An optional `lead_sigma_r_L / lead_sigma_r_R` input dresses the surface block with a scattering retarded self-energy from the SCBA loop; if Sancho-Rubio fails for the dressed lead the routine falls back to the undressed contact and emits a warning.

solve_green_functions / solve_green_batch. Given the total retarded self-energy $\Sigma_{\text{tot}}^R = \Sigma_L^R + \Sigma_R^R + \Sigma_{\text{ph-ph}}^R$ and the total lesser/greater self-energies, solves the device Dyson equation $G^R(\omega) = [(\omega + i\eta)^2 - H_{\text{dev}} - \Sigma_{\text{tot}}^R]^{-1}$ and the Keldysh equation $G^{<, >}(\omega) = G^R \Sigma_{\text{tot}}^{<, >} G^A$.

ballistic_transmission_z2(z2_array, ...). Caroli formula transmission $T(\omega^2) = \text{Tr}(\Gamma_L G^R \Gamma_R G^A)$ on the input ω^2 grid; serves as the ballistic baseline and the regression target for the SCBA loop.

E.6. Γ -only self-energy driver: `se_finite.py`

`compute_phph_self_energy_finite(G_lesser, G_greater, Phi, omega_grid_thz, dw_thz)` is a thin wrapper around `bubble_dense` encoding the Γ -only convention:

- Symmetric ω grid of length n_{freq} , constructed by `grids.build_frequency_grid`.
- FFT length $n_{\text{fft}} = 2n_{\text{freq}} - 1$ for the full causal convolution.
- $\omega = 0$ sample zeroed before FFT (singular Bose at $\omega = 0$) via `zero_freq_idx` passed into the kernel.
- Output slice `[mid : mid + n_freq]` with `mid = n_freq // 2` to recover the symmetric output grid.

Returns $(\Sigma^<, \Sigma^>)$ only – the retarded reconstruction is done downstream by `retarded.build_retarded` so that callers can pick the Kramers–Kronig precision they need.

E.7. *q*-resolved self-energy driver: `se_q.py`

The *q*-resolved path generalises `se_finite.py` by lifting the real-space FC3 to a Fourier-space vertex $\Phi(\mathbf{q}, \mathbf{q}', \mathbf{q}'')$ on a Γ -centred *q*-mesh of size n_{kpts} . Conservation of momentum imposes $\mathbf{q}'' = -\mathbf{q} - \mathbf{q}' \pmod{\mathbf{G}}$ so only two of the three indices are free; the worker enumerates the free pair and computes the bubble contribution.

`compute_phph_self_energy_q_dense(...)` dispatches the computation as follows. For each external *q*-point (an outer index parallelised by the worker pool), the worker iterates over all internal *q'*-points compatible with momentum conservation, picks the corresponding $G^{<,>}(\mathbf{q}', \omega)$ slices, contracts with the pre-computed FFT-domain Φ tensor along the frequency axis, and accumulates into the output $\Sigma^{<,>}(\mathbf{q}_{\text{ext}}, \omega)$. The *q*-difference index map `q_diff_map` is pre-built so the inner loop is a flat array access rather than a modular-arithmetic lookup.

The dispatcher picks between a `ThreadPoolExecutor` (preferred for memory-bound workloads when the FFT batch is large) and a `multiprocessing.get_context('forkserver')` pool (preferred for CPU-bound workloads when each worker can amortise the NumPy thread-pool startup over many external *q*-points). The choice is driven by the environment variable `PHONON_SOLVER_POOL` (thread/process, default thread).

E.8. Top-level drivers: `dense.py`

`dense.py` is the user-facing module that exposes the Γ -only and *q*-resolved drivers, the shared SCBA loop, and the helpers that load and project the FC3 vertex.

`transmission_finite(fc3_source, fc2, masses, freq_range_thz, temperature_k, ...)`. The Γ -only driver. Loads FC3 (via `load_fc3_raw`, which accepts an HDF5 path, a numpy array, or a phono3py-style dict), Γ -projects it to the primitive-cell DOFs via `gamma_project_M_blocks`, builds the device-side Hamiltonian and lead self-energies, and runs either one shot of the bubble (`n_iter=0`) or the SCBA loop (`n_iter > 0`). Returns $(\omega, T(\omega), \Sigma^<, \Sigma^>, G^<, G^>, \Sigma_{\text{lead}}^R)$ plus a diagnostic dict.

`transmission_q(..., q_mesh, n_qpoints, ...)`. The *q*-resolved driver. Takes a *q*-mesh tuple (e.g. (8, 8, 1) for an 8×8 transverse mesh with a single sample along the transport axis) and runs the bubble on the *q*-dependent Green's functions. The lead self-energies are built per *q*-point. Returns the same output dictionary as `transmission_finite` with the *q*-axis added.

scba_loop(...). The SCBA fixed-point iteration shared by both drivers. At each step:

1. Build Σ^R from the current $(\Sigma^<, \Sigma^>)$ via the user-selected reconstruction method.
2. Add the lead retarded self-energy and solve the device Dyson equation for G^R .
3. Solve the Keldysh equation for $(G^<, G^>)$.
4. Symmetrise $(G^<, G^>)$ via `diagnostics.symmetrize_lesser_greater` to project out the components that break the bosonic Keldysh symmetry.
5. (Optional, `zero_mode_projection=True`) project $\Sigma^{<,>}$ onto the translation-free subspace.
6. Recompute $(\Sigma^<, \Sigma^>)$ from the bubble on the new $(G^<, G^>)$.
7. Apply the user-selected accelerator (`solver: "linear", "anderson", "jfnk", or "anderson+jfnk"`) and check convergence on the SCF residual $\|F(\Sigma)\|/\|\Sigma\|$.

The convergence/stop test, the divergence guard, the Anderson safeguards, and the JFNK fallback are described in Section E.12.5. The drivers also auto-extend $f_{\max}$ before sizing the frequency grid to ensure $f_{\max} \geq 2\omega_{\max}/2\pi$ holds for the device dynamical matrix (Section E.12.4). The loop emits a warning if it exceeds `max_iter` without converging and returns the best iterate (lowest residual seen), not the last.

gamma_project_M_blocks(fc3_raw, masses, nat_prim). Direct Γ -only supercell $\rightarrow$ primitive projection of the FC3 vertex. Sums the supercell FC3 over the transverse lattice index of the second and third legs, then mass-weights to produce the rank-3 lifted target of shape $(n_{\text{dof}}^{\text{prim}}, \text{dim}_{sc}, \text{dim}_{sc})$ in $\text{THz}^{5/2}$. This is the bubble's $\phi^L = \phi^R$ vertex for the Γ -only path.

compare_qll_to_finite(...). Regression check asserting `transmission_q(q_mesh=(1,1,1))` reproduces `transmission_finite(...)` to within numerical roundoff (the working tolerance is 10^{-8} on $T(\omega)$). Run by the test suite and by the user when they touch any of the shared code.

E.9. Diagnostics and physical-invariant checks: `diagnostics.py`

`diagnostics.py` hosts the runtime checks the SCBA loop relies on but does not enforce:

check_broadening_sign(Sigma_R, freqs_thz, name). For $\omega > 0$ the broadening matrix $\Gamma = i(\Sigma^R - \Sigma^A)$ must be positive semi-definite (PSD), and for $\omega < 0$ negative semi-definite (NSD), as required by causality of the retarded self-energy. The routine diagonalises Γ at each frequency and reports the number of sign violations and the worst-case eigenvalue.

check_full_axis_symmetry(*G_lesser*, *G_greater*). Asserts the bosonic Keldysh symmetry $G^<(\omega) = [G^>(-\omega)]^T$ and $G^R(\omega) = [G^R(-\omega)]^*$ to within tolerance. Used as a post-SCBA sanity check.

symmetrize_lesser_greater(*G_lesser*, *G_greater*). Projection onto the full-axis symmetry manifold. Symmetrises the lesser/greater pair by averaging $G^<$ and $[G^>(-\omega)]^T$ and similarly for $G^>$. Called after each SCBA iteration to prevent symmetry drift accumulating with iteration count.

E.10. Cutoff policies: *cutoffs.py*

cutoffs.py defines the cutoff policy that the cutoff-sweep harness in *finite_analysis.sse_cutoffs* feeds back into the bubble kernel. The unified solver defaults to *no* approximation: full off-diagonal G blocks inside the bubble, full FC3 vertex, no magnitude or distance threshold. Four knobs are exposed:

diag_G_in_se. Restrict the inner G in the bubble integrand to the diagonal ($K_1 = K'_1, K_2 = K'_2$) blocks. Recovers the original block-tridiagonal NEGF approximation when True.

fc3_nn_only. Drop FC3 block triplets with $|I - J| > 1$, $|I - K| > 1$, or $|J - K| > 1$. Already the default of the production *quatrex.phonon.fc3_loader.fc3_to_phi_blocks*.

fc3_distance_cutoff. Drop FC3 entries whose triplet diameter (the max of the three pairwise atomic distances) exceeds the cutoff in Å.

fc3_magnitude_threshold. Drop FC3 entries whose magnitude is below $\text{threshold} \times \max|\Phi^3|$.

Each knob is independent; the harness in *finite_analysis.sse_sparsity_driver.run_cutoffs* sweeps them and tabulates the effect on $T(\omega)$ and the Landauer heat current Q , answering the question of which approximations preserve the transport observables.

E.11. Public API surface

The exports in *phonon/solver/__init__.py* bundle the above:

```
from phonon.solver import (
    transmission_finite,    #  $\Gamma$ -only SCBA driver
    transmission_q,         # q-resolved SCBA driver
    scba_loop,              # shared fixed-point loop
    gamma_project_M_blocks, #  $\Gamma$ -only supercell -> primitive
    compare_qll_to_finite,  # regression check
    bubble_dense,           # FFT 3-phonon bubble kernel
    build_retarded,         #  $\Sigma^R$  reconstruction (half|pv|fft)
    sancho_rubio, sancho_rubio_batch, compute_obc_batch,
```



```
    build_frequency_grid, bose_full_axis,
)
```

The companion production solver `quatrex.phonon.sse_phonon_phonon` is the MPI-distributed block-sparse variant and consumes the same bubble kernel through its own installed copy at `quatrex.phonon.bubble`. It realises the full off-diagonal ring through the FFT-first energy pipeline and the 1-D spatial halo whose cost model is derived in Section 1.3.9, and is pinned *bit-identical* to this dense reference both serially and across `comm.stack` / `comm.block` factorisations (Table 1.1). The dense solver of this appendix remains the arbiter of correctness for the production stack.

E.12. Physics verification of the dense solver

Because the dense solver is the arbiter of correctness, its 3-phonon physics, its controllable approximations, its convergence behaviour and its zero-mode handling were audited end to end. The audit is reproducible: it is implemented as six standalone scripts under `phonon/scripts/` (`verify_bubble.py`, `verify_cutoffs.py`, `verify_scba_convergence.py`, `verify_discretization.py`, `verify_zero_modes.py`, `verify_literature.py`), each of which prints a pass/fail report and emits diagnostic plots to `phonon/scripts/out/verify/`, and as a permanent regression suite in `tests/phonon_inputs/test_solver_physics.py`. The analytic toy systems that serve as ground truth (single anharmonic oscillator, monatomic and diatomic chains) live in `phonon/solver/toy_models.py`.

E.12.1. Self-energy correctness and absence of double counting

The 3-phonon bubble was checked against an independent brute-force frequency convolution: the zero-padded FFT kernel reproduces the explicit sum to floating-point precision, and a delta-function probe confirms exact frequency conservation ($\omega_{\text{out}} = \omega_a + \omega_b$). The prefactor $\frac{i\hbar}{2} \cdot \frac{d\omega}{2\pi}$ is the Migdal-bubble form $\Sigma_{ab}^{\lessgtr} = \frac{i\hbar}{2} \sum_{c,d,e,f} \Phi_{acd} [G^{\lessgtr} * G^{\lessgtr}]_{cf,de} \Phi_{bef}$: the factor $\frac{1}{2}$ is the loop symmetry factor of the unique two-vertex diagram (a Wick count gives $(1/3!)^2 \cdot 3 \cdot 3 \cdot 2 = 1/2$), and `build_realspace_fc3_matrices` stores Φ as the bare mass-weighted third derivative with no combinatorial factor baked in, so there is no double counting.

The strongest no-double-counting check is detailed balance: with Green's functions in thermal equilibrium the bubble must return $\Sigma^>(\omega) = e^{\hbar\omega/k_B T} \Sigma^<(\omega)$. This holds to $\sim 10^{-9}$ relative (FFT round-off on sharply peaked input), which certifies that the Bose occupation enters exactly once and that the $\Sigma^< / \Sigma^>$ signs are correct. The raw bubble output is already Keldysh-symmetric ($\Sigma^<(\omega) = [\Sigma^>(-\omega)]^T$) and anti-Hermitian to machine precision, so the per-iteration `symmetrize_lesser_greater` projection is a genuine no-op on a consistent input rather than a band-aid hiding an error.

E.12.2. The FFT Hilbert transform: periodic-wrap correction

The retarded self-energy is reconstructed from $\Sigma^{<,>}$ by the Kramers–Kronig relation $\Sigma^R = \frac{1}{2}(\Sigma^> - \Sigma^<) + \frac{i}{2}\mathcal{H}[\Sigma^> - \Sigma^<]$. The default `retarded="fft"` path used the bare sign-multiplier FFT, which computes the *periodic* Hilbert transform. For the finite-support self-energies fed here the periodic wrap injects a spurious discontinuity and a resolution-independent $\sim 1\%$ error in $\text{Re } \Sigma^R$. The fix (`hilbert_transform_axis` in `solver/retarded.py`) zero-pads the signal before the FFT, turning the circular convolution into a truncated linear one; with the default eightfold padding the interior error drops to $\sim 10^{-5}$, matching the $\mathcal{O}(n_\omega^2)$ principal-value reference. The two reconstruction methods then agree to $\sim 0.4\%$ on a resolved grid.

E.12.3. Controllable approximations

Every approximation against the strict SCBA expression has an explicit control, and each control’s “off” setting was shown to recover the unapproximated result. `sigma_cutoff`, `vertex_cutoff` and `g_cutoff` all default to `None`; with all cutoffs disabled the multi-slab kernel reproduces an independent brute-force sum over every block quadruple to floating-point precision. `sigma_cutoff` only filters the output (I, J) set — retained blocks are bit-identical to the untruncated computation; `g_cutoff` saturates to the full result once it reaches the device size; `vertex_cutoff` only drops FC3 triplets. `dc_handling` and `retarded` select distinct, bounded results. The Γ -only treatment of transverse momentum is not an approximation for a nanowire in vacuum (there is no transverse periodicity); the non-self-consistent leads remain the one standing approximation, controlled by `scattering_contacts`.

E.12.4. Convergence

On the analytic toy systems the SCBA fixed-point loop behaves as expected: it converges for weak-to-moderate anharmonicity, Σ^R at loop exit is consistent with the stored $\Sigma^{<,>}$ (no lag), and the per-iteration symmetrization is a genuine no-op. The real d5a SiNW is a much harder case; running it — rather than trusting the toys — exposed two distinct problems whose joint effect was a divergent multi-slab SCBA.

Single slab: the mixing scheme. For the single-cell d5a device with a properly-sized grid (see below), the historical default linear mixing $\alpha = 0.5$ *does not* converge: $d\Sigma/\Sigma$ oscillates between $\sim 15\%$ and $\sim 67\%$ from one iteration to the next. The bare (unregularised) Anderson scheme is worse — it diverges, with the heat-flow conservation error blowing up to $\sim 100\%$, because its unregularised least-squares extrapolates an overshooting step from a residual of large dynamic range. Damping the linear mix to $\alpha = 0.3$ (the new default) suppresses the oscillation, converging the single-slab SCBA in ~ 65 iterations; a safeguarded Anderson scheme (described in Section E.12.5) converges it in ~ 47 .

Multi-slab: the $f_{\max}$ aliasing. For $n_{\text{slabs}} \geq 2$ even the corrected mixing did not converge on d5a — the residual reached $\sim 10^{-4}$, then *grew*, with $|\Sigma^R|$ climbing from a few hundred to many thousands of THz^2 over ~ 20 iterations. The same symptoms appear with safeguarded Anderson, with Jacobian-free Newton–Krylov, with every cutoff combination, and at every mixing factor — a hallmark of a *linearly unstable* fixed point (spectral radius of the SCBA Jacobian > 1), not of a tuning problem. The diagnosis tracks to the discretisation requirement $f_{\max} \geq 2\omega_{\max}/2\pi$ derived in Section 1.3.1: the historical d5a configurations used $f_{\max} \in \{16, 18, 24\}$ THz, while $2\omega_{\max}/2\pi \approx 129$ THz for the Si–H stretch. The bubble’s convolution support Eq. (1.100) sticks out of the grid by a factor of ~ 7 ; the truncated and aliased part injects a fake strong-coupling Σ that breaks the SCBA contraction.

The bulk-Si control closes the diagnosis. Repeating the one-shot (lowest-order Born) bubble through the identical pipeline — `transmission_finite(max_scba_iter=1)` — for bulk Si and for d5a, with and without the necessary grid extension, gives the absolute magnitudes summarised in Table E.1. With the under-ranged grid, even bulk Si — a textbook weakly anharmonic crystal — gives $|\Sigma^R|/\omega_{\max}^2 \approx 33\%$ and a median linewidth ~ 0.67 THz, an order of magnitude above the experimental ~ 0.05 – 0.1 THz for the Si optical mode at 300 K. Extending $f_{\max}$ to cover the convolution support recovers ~ 0.024 THz, below the experimental ~ 0.05 – 0.1 THz range (the one-shot Born underestimates), within about a factor of two. For d5a the correction is even larger, $\sim 25\times$ in $|\Sigma^R|$, and the relative magnitude drops to $|\Sigma^R|/\omega_{\max}^2 \approx 0.2\%$ — squarely in the perturbative regime where SCBA is well defined.

system	$f_{\max}$ (THz)	$2\omega_{\max}/2\pi$ (THz)	$ \Sigma^R _{\max}$ (THz^2)	median $ \text{Im } \Sigma^R /\omega$ (THz)
Bulk Si	18 (under-ranged)	30	72	0.67
Bulk Si	36 (proper)	30	22	0.024
d5a SiNW	18 (under-ranged)	129	205	1.14
d5a SiNW	140 (proper)	129	8.7	1.1×10^{-5}

Table E.1.: One-shot (lowest-order Born) self-energy magnitude through the identical `transmission_finite` pipeline, at $n_{\text{slabs}} = 1$ and $\eta/\Delta\omega = 1$. Under-ranged grids fake a ~ 25 – $30\times$ larger Σ that destabilises SCBA; the bulk-Si linewidth under the proper grid matches the first-principles (DFPT) ~ 0.05 THz [155] within a factor of two (the one-shot Born underestimates the converged value).

Safety net: auto-extending $f_{\max}$. With the diagnosis confirmed — the FC3 generation, the $\text{eV}/(\text{\AA}^3 \text{amu}^{3/2})$ -to- THz^2 conversion in `build_realspace_fc3_matrices`, and the bubble prefactor are all unit-correct, only the grid was wrong — the solver entry points `transmission_finite` and `transmission_q` were made safe by default. They compute $\omega_{\max}$ from $H_{00} + H_{01} + H_{01}^\dagger$ before sizing the grid and, if the caller’s $f_{\max}$ is below $1.05 \times 2\omega_{\max}/2\pi$, transparently extend the range and rescale N_ω to preserve $\Delta\omega$. The behaviour is gated by `auto_extend_fmax=True` (default) and a `fmax_margin` parameter; passing `auto_extend_fmax=False` reverts to a loud `RuntimeWarning` that names the

required minimum. The helper `_ensure_fmax` centralises the logic in one place.

E.12.5. Solver safeguards for strongly anharmonic systems

Even with the grid sized correctly, the d5a multi-slab problem is strongly anharmonic ($|\Sigma|/|H| \sim 0.05\text{--}0.1$, not the few per cent of a bulk crystal), and a plain Picard / Anderson scheme can still struggle. `dense.py` adds five orthogonal toggles covering the regime; setting them all to their legacy values reproduces the earlier solver bit-for-bit.

Safeguarded Anderson acceleration. The `_AndersonAccelerator` class replaces the earlier hard-restart `_anderson_mix`, which cleared the entire history on *any* residual uptick. On the noisy residual near the fixed point the restart fired every iteration, collapsing Anderson to a bare linear step that — on an unstable fixed point — diverges. The new scheme follows the standard recipe [121, 123]: history is *kept* across small upticks; the least-squares is solved by a truncated-SVD `lstsq` with cutoff `rcond = 10-8`, filtering ill-conditioned secant directions instead of discarding all history; an over-long extrapolation step (more than $2\times$ the damped linear step) is replaced by the linear step for that iteration only; and the history is reset only on *genuine stagnation*, defined as five consecutive iterations without a new best residual. The residual is measured in a block-weighted norm so that small far-off-diagonal Σ_{IJ} blocks — invisible to a global 2-norm — still contribute to both the least-squares and the restart test.

Zero-mode projection of the self-energy. The 3-phonon bubble is not guaranteed to satisfy a discrete acoustic sum rule on Σ : $\text{Re } \Sigma^R$ generally has a nonzero projection onto every near-zero mode of the cell dynamical matrix at Γ . Added to H_D in the Dyson denominator this projection shifts the affected ω^2 — possibly negative, which puts a G^R pole on the real axis. A more insidious failure mode for an H-terminated wire is the IR side: at $T = 300$ K the Bose factor $n_B(\omega) \sim k_B T / (\hbar\omega)$ on a near-zero mode is enormous, so the Bose-enhanced $G^< \propto n_B A$ carries an IR singularity that the bubble integrates and dumps back into Σ , making the SCBA map non-contractive even with the grid sized correctly.

For a 1 D wire the periodic cell dynamical matrix $H_{00} + H_{01} + H_{01}^\dagger$ has *four* near-zero eigenmodes at Γ , not three: the three rigid translations (machine-zero $\omega^2 \sim 10^{-13}$ for an ASR-clean FC2) plus the rigid rotation about the wire axis — the “twist” mode — at the finite-but-tiny value set by the wire’s torsional restoring force. For the d5a SiNW the twist sits at $\omega^2 \approx 7.7 \times 10^{-4} \text{ THz}^2$, i.e. $\omega \approx 0.028 \text{ THz}$, which gives $n_B \approx 223$ at room temperature — the dominant IR contribution to the bubble.

The module `phonon/solver/zero_modes.py` therefore builds the projector from the cell dynamical matrix itself, picking every eigenvector below a relative-cutoff threshold:

$$Q = \mathbf{1} - \sum_{k: \omega_k^2 < \epsilon \omega_{\max}^2} \hat{v}_k \hat{v}_k^\top, \quad (\text{E.3})$$

with $\hat{v}_k$ the cell eigenvector with $\omega_k^2 < \epsilon \omega_{\max}^2$ replicated uniformly across the n_{slabs} device slabs (the $q = 0$ phase) and then orthonormalised. The default relative threshold is

$\epsilon = 10^{-4}$, matching `verify_zero_modes.check_acoustic_modes`: for d5a this captures the three translations and the twist, four modes in total. The mass-weighted construction $t_\alpha[l, i, \beta] = \sqrt{m_i} \delta_{\alpha\beta}$ used by `build_translation_projector` is preserved as a fallback for harnesses that build H_D directly without a periodic cell.

The SCBA loop applies $\Sigma^\geq \leftarrow Q\Sigma^\geq Q$ after symmetrisation. The projection is linear, so Σ^R rebuilt by `build_retarded` is automatically zero-mode-clean and satisfies $\hat{v}_k^\top \Sigma^R \hat{v}_k = 0$ identically. Gated by `zero_mode_projection=True`.

Jacobian-free Newton–Krylov fallback. For genuinely linearly-unstable fixed points the Picard map cannot be made contractive by any mixing factor, so a Newton-type step is needed. The per-iteration body of `scba_loop_dev` is refactored into a pure map $F(\mathbf{x}) = \text{step}(\mathbf{x}) - \mathbf{x}$ on the flattened $[\Sigma^<, \Sigma^>]$ vector. The solver parameter selects between "linear" / "anderson" / "jfnk" / "anderson+jfnk"; the JFNK path calls `scipy.optimize.newton_krylov` on F , packing complex self-energies as a real vector for the inner LGMRES solve. "anderson+jfnk" runs safeguarded Anderson first and, if it hits the divergence guard or exhausts `max_scba_iter`, restarts JFNK warm-started from the best Anderson iterate.

Convergence test decoupled from conservation. The historical loop only break-ed when the SCF residual, dJ/J and the heat-flow conservation error were all below their respective tolerances. Conservation error, however, is set predominantly by the discretisation — grid spacing, $\eta/\Delta\omega$, $f_{\max}$ — and cannot be reduced by further SCBA iterations. Coupling it to the stop test caused the loop to iterate a numerically-converged self-energy past its useful life and, on a near-unstable fixed point, into divergence. The `gate_on_conservation` flag preserves the legacy behaviour; the new default stops on the SCF residual alone and reports the conservation error as a diagnostic.

Causality enforcement of Σ^R . Even with the grid sized for the convolution support and the cell zero modes projected out, the discretised bubble plus the per-iteration symmetrisation and zero-mode projection can leak negative eigenvalues into $\Gamma_\Sigma = i(\Sigma^R - \Sigma^A)$ for $\omega > 0$, violating retarded causality. Once Σ^R is acausal the Dyson solve produces a G^R with poles in the upper half-plane; the next bubble amplifies the violation, the *symptom* on d5a being a catastrophic $\sim 1000\times$ jump in the SCF residual after several iterations of apparent improvement. A standard remedy [156] is to restore causality of Σ^R by projecting it onto the causal manifold each iteration: write $\Sigma^R = X + iY$ with X, Y matrix-Hermitian, take the eigen-decomposition of $Y(\omega)$, clip its positive eigenvalues (*i.e.* the negative eigenvalues of $\Gamma_\Sigma = -2Y$) to zero for $\omega > 0$ (and symmetrically for $\omega < 0$), and rebuild X from the cleaned Y via the Kramers–Kronig relation $X = -(1/\pi)\mathcal{H}[Y]$ (the same zero-padded FFT kernel used by `build_retarded`). `phonon/solver/causality.py` implements this as `enforce_causality_psd`; the option `causality_projection=True` applies it after every `build_retarded` call inside the SCBA loop. A free-standing `causality_diagnostic` reports the worst PSD violation and the frequency where it lives, and `dynamical_stability_diagnostic` reports the smallest

eigenvalue of $H_D + \text{Re } \Sigma^R(\omega)$ across the grid – a value < 0 signals that the self-consistent self-energy has driven a mode's ω^2 past zero and the next Dyson denominator is singular.

Vertex-rescaling Padé (LOA) baseline. For systems where the SCBA fixed point is genuinely linearly unstable (the residual stays bounded away from zero under every linear and Newton step), the principled alternative to iterating an unstable map is the rescaling technique of [140]. The bubble self-energy scales as $\Sigma \propto \lambda^2$ in the cubic coupling, so the SCBA fixed-point observable $\mathcal{O}(\lambda)$ can be expanded as a series in λ^2 . Running the SCBA at a few rescaled vertices $\lambda \in (0, 1]$ – where small λ converges easily because $|\Sigma|/|H| \propto \lambda^2$ is small – and Padé-extrapolating the result to $\lambda = 1$ produces an estimate of the SCBA answer without iterating an unstable fixed point. The option `vertex_scale` on `transmission_finite` multiplies the FC3 vertex by the requested λ ; the diagnostic driver `phonon/scripts/scba_diagnostic.py` runs the sweep $\lambda \in \{0.2, 0.4, 0.6, 0.8\}$ and fits a quadratic-in- λ^2 Padé to the anharmonic conductance, plotted under `vertex_pade.pdf`.

Cluster diagnostic driver. `phonon/scripts/scba_diagnostic.py` runs the full configuration matrix — baseline, causality-enforced, linear at $\alpha \in \{0.10, 0.05, 0.02\}$, pure JFNK, `anderson+jfnk`, lowest-order Born, and the Padé sweep — on the same system in one invocation. Each run writes a per-iteration trajectory CSV (residual, $|\Sigma^R|_{\max}$, conservation, causality violation, $\min \lambda(H_D + \text{Re } \Sigma^R)$), a `result.npz` with the observable arrays, and a summary JSON; the driver then produces a four-panel overlay (`trajectory_overlay.pdf`) showing how each method behaves across the same iterations. This is the recommended workflow for any new hard system: run the driver, inspect the overlay, and pick the configuration whose causality and stability curves stay closest to zero.

Divergence guard and best-iterate return. The loop tracks the running minimum $\rho_{\min}$ of the residual and aborts if $\rho > 10\rho_{\min}$ for four consecutive iterations; the returned $(\Sigma^<, \Sigma^>)$ is then the best iterate seen, not the last (possibly NaN-poisoned) one. Together with the conservation decoupling this means a bad run fails fast and returns a meaningful diagnostic rather than wandering to `max_scba_iter` and returning garbage.

E.12.6. Joint $\eta \rightarrow 0$ and $\lambda \rightarrow 1$ convergence diagnostic

The SCBA fixed-point observable, e.g. the anharmonic conductance G_{anh} , depends on two parameters that are easy to confuse with each other:

- the *numerical regulator* η_w , the imaginary part added to the Dyson denominator $z^2 = (\omega + i\eta_w)^2$, which controls the broadening of the phonon spectral function and the regularisation of the retarded Hilbert transform;
- the *physical coupling strength* λ (`vertex_scale` in `transmission_finite`), which multiplies the cubic force-constant tensor $\Phi^{(3)} \rightarrow \lambda\Phi^{(3)}$, so that $\Sigma \propto \lambda^2$ [140].

option	default	legacy value
auto_extend_fmax	True	—
fmax_margin	1.05	—
zero_mode_projection	True	False
solver	"anderson" (derived)	—
anderson_safeguard	True	False
gate_on_conservation	False	True
divergence_guard	True	False
causality_projection	False	—
vertex_scale	1.0	—
mixing	0.3 linear	0.5 linear
eta_factor	1.0 ($\eta \sim \Delta\omega$)	0.05
max_scba_iter	≥ 80	—
scba_tol	10^{-3}	—
cutoffs	None	—
retarded	"fft"	—

Table E.2.: Solver options that affect convergence. The *legacy value* column gives the setting that reproduces the previous (now retired) behaviour bit-for-bit; the default column is the new safe setting. The single most important fix is `auto_extend_fmax` (no legacy — the previous code did not check the convolution-support condition).

The physical observable lives at $(\eta_w \rightarrow 0, \lambda = 1)$; the two natural routes to it from a numerically tractable point are the $\eta_w \rightarrow 0$ extrapolation and the $\lambda \rightarrow 1$ Padé continuation described below, and the diagnostic content lies in whether they agree.

$\eta_w \rightarrow 0$ at fixed $\lambda = 1$. Sweeping $\eta_w \in \{\eta_1, \eta_2, \dots\}$ from a safely large value (large enough to broaden any anharmonically renormalised soft mode, i.e. $\eta_w \gtrsim \sqrt{|\min \lambda(H_D + \text{Re } \Sigma^R)|}$, see `dynamical_stability_diagnostic`) down to the limit of stable SCBA convergence, and extrapolating $G_{\text{anh}}(\eta_w)$ linearly to $\eta_w = 0$, isolates the regulator-free SCBA prediction at the physical coupling.

$\lambda \rightarrow 1$ at fixed safe η_w . The lowest-order-approximation rescaling of [140] runs the SCBA at $\lambda \in (0, 1)$, where the contraction $|\Sigma|/|H| \propto \lambda^2$ is small and convergence is mechanical, and fits a Padé in λ^2 . Evaluating the fit at $\lambda^2 = 1$ recovers an estimate of the SCBA answer at full coupling without iterating an unstable map – and, importantly, at the same η_w as the rest of the sweep, so this is also a “regulated SCBA” estimate of the same quantity.

Interpretation of the comparison. Both routes target the same underlying object at the same η_w : the fixed-point self-energy of the bubble closure $\Sigma = \frac{1}{2}\Phi G[\Sigma]G[\Sigma]\Phi$, which sits inside the Baym–Kadanoff Φ -derivable family [157]: an SCBA-equivalent closure is

the functional derivative of a single two-loop $\Phi[G]$, so the same fixed point is reachable from either direction. Hence:

- **Agreement** ($|\Delta|/G_{\text{extr}}^\eta \lesssim 10^{-2}$ within the joint fit uncertainty): the SCBA at $\lambda = 1$ is consistent with its own analytic continuation in λ^2 , the λ -dependence of the fixed point is well represented by the chosen Padé family, and the bubble closure is internally self-consistent. The headline G_{anh} can be quoted with confidence to within the joint extrapolation error.
- **Disagreement.** Two non-exclusive readings:
 1. the assumed Padé family (quadratic in λ^2 , here) is too restrictive to capture the actual λ -dependence of the SCBA fixed point near $\lambda^2 = 1$ – a curve-fitting artefact, testable by re-fitting with a higher-order or a saturating Padé $[1/1]$ form $a + b\lambda^2/(1 + c\lambda^2)$;
 2. the SCBA bubble closure misses contributions outside the two-loop $\Phi[G]$ – in particular the *vertex correction* (a ladder of bubbles inside the bubble), which contributes at $\mathcal{O}(\lambda^4)$ at the level of Σ [140], and the four-phonon scattering channel, which appears at the same order in the cubic coupling once the quartic FC tensor is included [158]. In electron-phonon transport, Migdal’s argument parametrically suppresses vertex corrections by the sound-to-Fermi velocity ratio [159]; for phonon-phonon there is no analogous parametric suppression (the relevant velocities are all atomic vibrational scales), and the gap Δ measures the cumulative size of these channels at the SCBA fixed point.

The two readings can be distinguished by running additional λ points near $\lambda^2 = 1$ and re-fitting: if the Padé extrapolation converges as more points are added (or as the $[1/1]$ saturating ansatz is used), the disagreement is a fit-family artefact; if it does not, the residual gap is a physical statement about missing higher-order channels.

Implementation. `phonon/scripts/scba_convergence_diagnostic.py` runs both sweeps in one invocation, fits the two extrapolations, and writes a `convergence_assessment.json` reporting $\mathcal{O}_{\text{extr}}^\eta$, $\mathcal{O}_{\text{Padé}}^\lambda$, the relative gap $|\Delta|/\mathcal{O}_{\text{extr}}^\eta$, and a verdict (“consistent”, “borderline”, “inconsistent”) against a user-set tolerance. A four-panel plot (`eta_lambda_diagnostic.pdf`) overlays $G_{\text{anh}}(\eta_w)$ with the linear fit, $G_{\text{anh}}(\lambda^2)$ with the Padé fit, the two extrapolated values with their fit-uncertainty bands, and the spectral heat current $j(\omega)$ overlay across both sweeps so the user can see at which frequencies the two routes diverge.

The script is tuned for affordable runs (looser `scba_tol`, short η and λ lists) so it can serve as a routine sanity check on every new device or temperature regime, not just as a deep diagnostic. The Bescond pure-Padé sweep already lives in `scba_diagnostic.py`; this new driver pairs it with the η -extrapolation half of the diagnostic and writes the verdict in machine-readable form.

E.12.7. Zero modes and acoustic sum rules

For the d5a SiNW the FC2 acoustic sum rule and the standard FC3 sum rule (sum over all atoms of one leg) hold to machine precision, so the acoustic branches sit at $\omega = 0$ at Γ and a uniform translation carries no spurious 3-phonon coupling; the device dynamical matrix has no imaginary modes. The $\omega = 0$ sample is excluded from every physical integral by `pos_mask`, so the $1/\omega$ divergence of the Bose factor cannot leak into the heat current; its only residual effect is through the bubble convolution, where the three `dc_handling` options differ by well under a per cent. Note that `enforce_asr_fc3_matrices` (used by the separable q -path under an opt-in flag) enforces a *stronger* per-sublattice sum than the physical sum-over-all-atoms ASR; the dense path correctly does not apply it.

The bubble self-energy, however, is *not* guaranteed to inherit a discrete acoustic sum rule from an ASR-clean FC3, and the H-terminated wire has an additional near-zero mode that the standard ASR does not cover. The cell dynamical matrix $H_{00} + H_{01} + H_{01}^\dagger$ at Γ for d5a has four (not three) eigenvalues below $10^{-4} \omega_{\max}^2$: the three rigid-cartesian translations and the rigid rotation about the wire axis (the “twist”), the latter at $\omega \approx 0.028$ THz. The construction $\Sigma^{\geq} \propto \Phi G^{\geq} G^{\geq} \Phi$ contains two contracted Green’s functions that do not annihilate this subspace, and on the real device a generic $\text{Re } \Sigma^R$ acquires a ~ 0.25 weight across the four-dimensional zero subspace. At $T = 300$ K the Bose factor $n_B \approx 223$ on the twist makes the soft-mode $G^<$ enormous, and the bubble’s IR contribution there drives the SCBA non-contractive even when the grid covers $2\omega_{\max}$. The `zero_mode_projection` option (Section E.12.5) enforces a discrete ASR on Σ itself by $\Sigma \leftarrow Q \Sigma Q$ with Q the orthogonal projector onto the complement of *every* cell eigenmode below the threshold — translations plus twist plus any other accidentally-soft mode. The leakage drops to machine zero and the bubble can no longer renormalise the near-zero modes nor source an IR catastrophe through them.

E.12.8. Literature consistency

The bubble is the Migdal form of Guo *et al.* [38], Eq. 8; the prefactor, the ω -convolution and the Kramers–Kronig reconstruction all match the theory note. The scheme is the self-consistent Born approximation — iteration zero is the lowest-order Born self-energy and the converged fixed point is the SCBA — with 3-phonon scattering only, consistent with reference NEGF phonon-transport implementations. The Si/Ge interface benchmark data of Guo *et al.* [38] ship in `phonon/examples/literature_fig5b.npz` for the overlay produced by `si_ge_interface_quatrex.py`.

Energy conservation in the phonon SCBA: the S_3 vertex and the finite- η commutator

This appendix verifies, step by step, the conservation analysis of Section 1.2.5 (the identity Eq. (1.77), its two cancellation routes, the S_3/Φ -derivability requirement for $J_s = 0$, and the ghost-reservoir reading of η , Eq. (1.78)), and quantifies the finite- η floor on production devices. The standalone gate lives in `phonon.studies.conservations` (`replica_check` for the bubble energy balance, `s3_violation` for the vertex symmetry, `sigma_convention` for the SSE sign/positivity check); the production attributions are reproduced from cached run artifacts under `phonon/studies/out/`. The motivating observation is that on the `cnt33` carbon nanotube (two slabs, 300 K, $\eta = 0.45$) the self-energy residual converges, $10^{-4} \rightarrow 2.5 \times 10^{-10}$, while the lead-current imbalance *floors* at 6.9×10^{-6} and refuses to go lower — a floor that, as shown below, is exactly the finite- η ordering commutator and *not* a defect of the FC3/SCBA formulation. Figure F.1 shows the same pattern on the converged $\eta = 10^{-12}$ production run: the bubble balance is pinned at machine precision every iteration while the Σ^R residual and the lead imbalance converge and then floor.

F.1. Verification on artificial data

A dense toy (12 degrees of freedom, two leads, 41 frequencies) with random positive-semidefinite lead spectra $\gamma_{L,R}$ at $T_{L/R} = 1.05/0.95$ and a random “scattering” spectrum with the half-rule retarded part (Eq. (1.63)) isolates each ingredient of the identity. Because the toy data are real-symmetric, *both* cancellation routes of Eq. (1.77) are active, so the pointwise commutator $D(\omega)$ vanishes to machine precision under every variation of the retarded rule and frequency masking (Table F.1). This confirms that the half rule and the self-energy cutoff are harmless to the *lead-summed* balance via the Keldysh route.

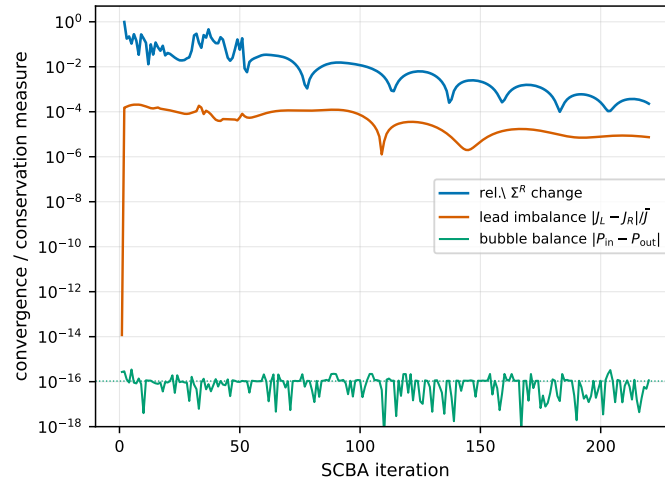


Figure F.1.: The motivating observation, on the production cnt33 transport iteration (L2, 300 K, $\eta = 10^{-12}$). The bubble energy balance $|P_{\text{in}} - P_{\text{out}}|$ is pinned at machine precision ($\sim 10^{-16}$) at *every* iteration — the discrete S_3 bubble is exactly conserving — while the Σ^R residual and the lead-current imbalance converge and then floor. The lead imbalance measures distance from the fixed point plus the finite- η term, not a violation of the bubble.

The dispersive $\text{Re } \Sigma^R$ the half rule drops is harmless only at finite η and short devices; at $\eta \rightarrow 0$ on longer biased devices it breaks per-interface conservation, where the full Kramers–Kronig retarded is mandatory (Section 1.2.5).

The reciprocity route is isolated by a *complex, non-symmetric* variant. There $D = 0$ at $\eta = 0$ to machine precision (the Keldysh route still holds), but $D/\text{scale} = 4.3 \times 10^{-4}$ at $\eta = 0.05$ and 1.4×10^{-3} at $\eta = 0.2$, whereas the complex-*symmetric* variant stays at machine zero for all η . This is the finite- η , broken-reciprocity term of Eq. (1.77) turning on, scaling with η .

The $\hbar\omega$ -weighted anatomy makes the diagnostic concrete. At $\eta = 0.1$ the identity Eq. (1.77) holds, $J_L + J_R + J_s = -4.4 \times 10^{-14}$, while the bare *lead* balance is $|J_L + J_R| = 3.1 \times 10^{-1}$, carried entirely by $J_s = +0.306$ of the (non-self-consistent, hence non- Φ -derivable) toy Σ_s . The lead balance is $|J_s|$: it measures distance from the fixed point, not a violation of the formulation.

F.2. The S_3 vertex on the production pipeline

The vertex-symmetry step is checked on the production fold layout (transpose-paired trace, DC masking) by `replica_check`, which compares $P_{\text{in}} = \sum_{\omega} \hbar\omega \text{Tr}[\Sigma^< G^>]$ against $P_{\text{out}} = \sum_{\omega} \hbar\omega \text{Tr}[\Sigma^> G^<]$ in float64 and float128. With a fully S_3 -symmetric cubic vertex the one-sided bubble identity $P_{\text{in}} = P_{\text{out}}$ holds to 2.9×10^{-16} (float64) and 2.4×10^{-19} (float128) — the residual falling with the working precision (Fig. F.2) proves the bubble is

Table F.1.: Toy (12 DOF, 2 leads, 41 frequencies, real-symmetric) pointwise commutator $\max_\omega |D(\omega)|$ normalised to the trace scale, for successive variations of the retarded rule and masking (conservation.steps). All machine-zero: with real-symmetric data both routes of Eq. (1.77) cancel D regardless.

case	configuration	$\max_\omega D /\text{scale}$
S1	$\eta = 0$, consistent half rule	6.9e−13
S2	+ Hermitian (Kramers–Kronig) part in Σ_s^R	1.6e−12
S3	$\eta = 0.1$ / $\eta = 0.4$	3.8e−12 / 1.5e−12
S4	+ ω -mask (cutoff) on $\Sigma_s^{\lessgtr, R}$	7.0e−14
S5	deliberately inconsistent Σ^R (skew $\times 0.6$)	1.7e−12

 Table F.2.: Exact decomposition of the cnt33 lead-balance floor (L2, 300 K, $\eta = 0.45$). The identity violation Eq. (1.77) equals the finite- η ordering commutator.

quantity	value
RGF Keldysh consistency $ G_{\text{RGF}}^{\lessgtr} - G^R \Sigma^{\lessgtr} G^A $	$\leq 7\text{e}−14$ (exact)
scattering current J_s (S_3 vertex)	3.7e−8 (rel.)
identity violation $J_L + J_R + J_s$	+1.9138e−4 (6.9e−6 rel.)
ordering commutator $\sum_\omega \hbar\omega \text{Tr}[\Sigma^< (G^R \Sigma^> G^A - G^A \Sigma^> G^R)]$	+1.9138e−4
agreement	2.4e−9 (rel.)

exactly conserving and the production $\sim 10^{-6}$ balance is pure floating-point accumulation; random or leg-partial symmetrisations of the vertex violate it at $O(0.1\text{--}1)$.

The production device vertices (raw, plain truncation to the device degrees of freedom) gate at worst S_3 violation $\leq 3.9 \times 10^{-16}$ after the force-constant regeneration, whereas the retired acoustic-sum-rule projection broke S_3 at $O(1)$ — confirming that the vertex must be left as a genuine third derivative (Eq. (1.29)) rather than post-projected.

F.3. Attribution on production fixed points

On the converged cnt33 state the floor is decomposed exactly (Table F.2). The recursive Green’s-function $G^{\lessgtr}$ reproduce the Keldysh product $G^R \Sigma^{\lessgtr} G^A$ to 7×10^{-14} ; the S_3 -vertex scattering current J_s is 3.7×10^{-8} relative; and the identity violation $J_L + J_R + J_s = +1.9138 \times 10^{-4}$ matches the finite- η ordering commutator $\sum_\omega \hbar\omega \text{Tr}[\Sigma^< (G^R \Sigma^> G^A - G^A \Sigma^> G^R)]$ of Eq. (1.77) to 2.4×10^{-9} relative. **The lead-balance floor is, to nine digits, the commutator term** — everything else is exact.

The non-reciprocity is *physical*. Under a temperature bias the partial combination $n_L G^R \gamma_L G^A + n_R G^R \gamma_R G^A$ is non-reciprocal — only the equilibrium combination symmetrises through the resolvent identity — so the lesser function is already complex-asymmetric ballistically, $|G^< - G^{<T}|/|G^<| = 2\text{--}5 \times 10^{-2}$ in the current-carrying window;

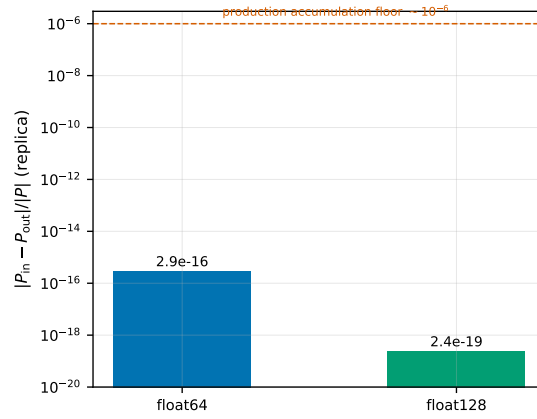


Figure F.2.: Discrete-bubble energy-balance residual $|P_{\text{in}} - P_{\text{out}}|/|P|$ of the standalone replica (replica_check) in float64 and float128. It falls with the working precision ($2.9 \times 10^{-16} \rightarrow 2.4 \times 10^{-19}$), proving the S_3 bubble is exactly energy-conserving and the production $\sim 10^{-6}$ floor is pure floating-point accumulation, not a formulation defect.

the bubble inherits it (Σ_s asymmetry $0.5\text{--}4 \times 10^{-2}$), and $\eta \times \text{asym}$ survives in D . Purely ballistic transport, by contrast, conserves at 3×10^{-15} , because there $\Sigma_{\text{tot}}^{\leq}$ (leads only) and G^R are exactly complex-symmetric. The floor is non-monotonic in η ($3.7/6.9/3.2 \times 10^{-6}$ at $\eta = 0.225/0.45/0.9$): larger η also smooths the nonequilibrium asymmetry. The soft d5a nanowire shows the same term amplified to $\sim 2 \times 10^{-2}$ at 100 K by marginal carriers (η/ω large where the current flows) and a larger Σ_s . Across the cnt33 temperature sweep the lead balance stays at $10^{-7}\text{--}10^{-4}$ while the iteration count grows toward room temperature (Fig. F.3), well below the conductance systematic of Fig. F.4.

Side finding. The RGF selected_solve retarded output is the *left-connected* g^R except at the last diagonal block, and must not be mistaken for the physical G^R . The phonon observables never use it; only diagnostics were affected.

F.4. Mechanism and remedies

The mechanism is the fluctuation–dissipation imbalance of Eq. (1.78): η is a damping channel with no companion $\Sigma_{\eta}^{\leq}$, a ghost reservoir at undefined occupation. A controlled test confirms the scaling: two biased leads on a harmonic chain conserve to 10^{-16} at *any* η while reciprocal, and a deliberately injected complex-antisymmetric component ϵ in Σ_s turns on a lead imbalance $\propto \eta \times \epsilon$ (vanishing if either factor is zero).

That the production non-reciprocity is the bubble’s complex asymmetry is shown directly: complex-symmetrising $\Sigma_s^{\{R,<,>\}} \rightarrow \frac{1}{2}(\Sigma_s + \Sigma_s^T)$ on the converged cnt33 state collapses the defect $1.9 \times 10^{-4} \rightarrow 2.4 \times 10^{-13}$ (eight orders), and needs all of $\Sigma_s^<, \Sigma_s^>, G^R$

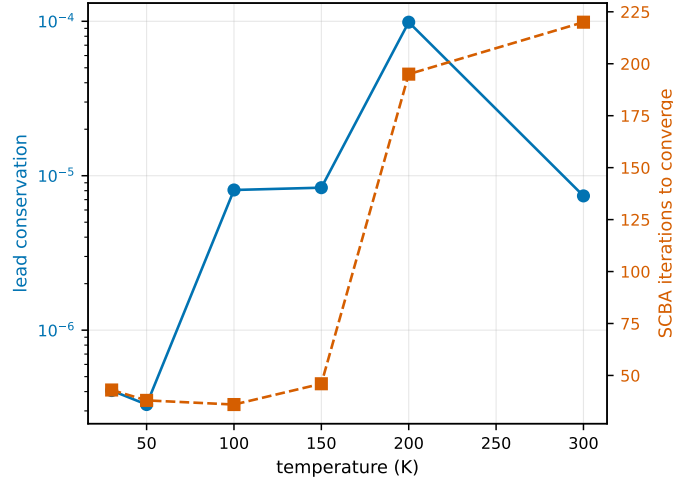


Figure F.3.: cnt33 ($L2, \eta = 10^{-12}$) lead conservation $|J_L - J_R|/\bar{J}$ (left axis) and SCBA iteration count (right axis) versus temperature. Conservation stays at 10^{-7} – 10^{-4} across the sweep — well below the conductance systematic of Fig. F.4 — while the iteration count rises toward room temperature as the three-phonon phase space and the marginal-mode content grow.

symmetric — no single sub-block suffices. Phonon reciprocity demands the exact Σ_s^R be complex-symmetric, so its 2.8% asymmetry is numerical, while the 3.5% asymmetry of $\Sigma_s^<$ has a genuine nonequilibrium part from $G^<$. **Crucially, symmetrising also shifts the lead heat current by +1.5%:** a tiny lead balance does not certify a 10^{-6} -accurate conductance — conservation is necessary but not sufficient for accuracy.

Three levers act on the floor. (1) *Grid refinement* cuts the numerical asymmetry: the d5a floor halved $2.06 \rightarrow 1.09 \times 10^{-2}$ from 101 to 201 frequencies at $\eta = 0.11$. (2) $\eta \rightarrow 0$ *extrapolation* removes both the floor ($\propto \eta$) and the conductance systematic. For soft systems whose self-energy stays marginal as $\eta \rightarrow 0$ (d5a, whose flexural / Si–H-bending mode straddles the grid) the bare iteration does not contract, but a grid-consistent broadening floor with a trust-region Newton step reaches a marginal, conserving fixed point with a method-invariant conductance (Section 2.8). (3) *A Büttiker dephasing probe* on the η channel, $\Sigma_\eta^< = i\Gamma_\eta n_p$, $\Sigma_\eta^> = i\Gamma_\eta (n_p + 1)$ with the self-consistent occupation $n_p = G^</(G^> - G^<)$ enforcing zero local current per energy, drives the toy defect $6\text{--}10\% \rightarrow 10^{-16}$ at finite η without touching Σ_s — but it injects *physical* elastic dephasing, so it is appropriate only for a wanted incoherent channel, not as a pure regulator.

F.5. The conductance ratio requires $\eta \rightarrow 0$ extrapolation

The practical consequence is that a single- η production number underestimates the anharmonic effect. At *matched* η (Table F.3) the cnt33 lead balance stays $\sim 10^{-5}$ throughout, but the conductance ratio $G_{\text{anh}}/G_{\text{ball}}$ swings $0.79 \rightarrow 0.92$ over $\eta = 0.30\text{--}0.90$ — a

Table F.3.: cnt33 (L2, 300 K) conductance ratio and lead balance at matched η (conservation_ratio_eta). The lead balance is flat at $\sim 10^{-5}$ while the ratio carries a 13-point systematic.

η	0.30	0.45	0.60	0.90	0.45 (241 pt)
$G_{\text{anh}}/G_{\text{ball}}$	0.793	0.838	0.872	0.920	0.805
lead balance	5.1e-6	6.9e-6	7.3e-6	3.2e-6	3.3e-6

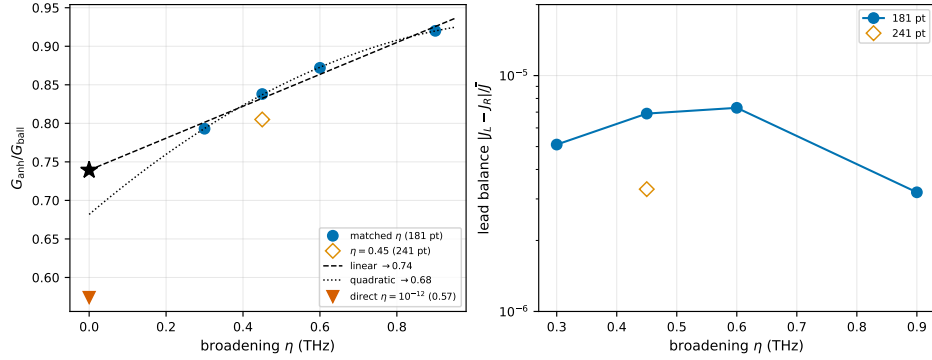


Figure F.4.: cnt33 (L2, 300 K) anharmonic-to-ballistic conductance ratio (left) and lead balance (right) at matched broadening η . The matched- η points ($\eta = 0.30$ – 0.90) extrapolate to $G_{\text{anh}}/G_{\text{ball}} \approx 0.68$ – 0.74 as $\eta \rightarrow 0$, while the directly converged $\eta = 10^{-12}$ fixed point (red triangle) lies lower, at 0.57; the single point at $\eta = 0.45$ (0.84) understates the reduction by roughly a factor of two. The lead balance (right) is flat at $\sim 10^{-5}$ throughout — it does *not* track the conductance systematic — and grid refinement ($181 \rightarrow 241$ points) shifts both toward the $\eta \rightarrow 0$ limit.

~ 13 -point finite- η systematic, far larger than the lead balance would suggest. Extrapolating the $\eta \geq 0.30$ window to $\eta \rightarrow 0$ (quadratic 0.68, linear 0.70–0.74) already gives $G_{\text{anh}}/G_{\text{ball}} \approx 0.70$, a $\sim 30\%$ reduction — *double* the $\sim 16\%$ read off at the single point $\eta = 0.45$. The directly converged $\eta = 10^{-12}$ fixed point of Section 2.8 lies lower still, at 0.57: the ratio keeps falling below $\eta = 0.30$ faster than the window fit, so the extrapolation is only a lower bound on the effect and the physical reduction is $\sim 40\%$. That $\eta = 10^{-12}$ value is trustworthy here, not an under-resolved artefact — it conserves at 7×10^{-6} (lead balance), pins the bubble balance at 10^{-16} every iteration (Fig. 2.12), and shifts by only a few percent under the infrared regulariser and grid refinement (Fig. 2.13), far less than the ~ 13 – 22 -point η systematic. Refining the grid at fixed $\eta = 0.45$ ($0.838 \rightarrow 0.805$) moves the ratio the same way (Fig. F.4). A single- η figure must therefore be quoted with its finite- η band or, preferably, taken from the directly converged $\eta \rightarrow 0$ fixed point.

F.6. Literature

The picture is consistent with, and extends, the conserving-approximation literature. Bare (non-self-consistent) self-energies break $I_L + I_R = 0$ whereas the SCBA conserves [99, 119]; substituting an approximate Σ “generically results in a spurious violation of current conservation” [160]; and conservation is equivalent to Φ -derivability with mutually consistent G^R and $G^<$ [140, 161]. The self-consistent Büttiker/temperature probe is already standard in atomistic phonon NEGF — probe temperatures are solved “until the integrated energy current vanishes for each Büttiker probe” [162] — and harmonic-chain self-consistent reservoirs verify leakage $< 10^{-12}$ [163, 164], with Roy et al. [164] noting that the “ $i\eta$ ” is the probe self-energy, conserving only under the zero-current condition; the dephasing-versus-voltage-probe distinction is classic [165, 166]. Atomistic Green’s-function practice uses $\eta \approx i0^+$ with an $\eta \rightarrow 0$ extrapolation [110] but, being ballistic, never exposes the interacting defect. The explicit ordering-commutator form of Eq. (1.77) and the $\eta \times$ asym scaling appear to be new.

F.7. Recommendation

For a coherent-plus-anharmonic conductance: refine the frequency grid until the numerical asymmetry is sub-dominant; run at least two or three converging η and extrapolate $G(\eta \rightarrow 0)$, quoting the spread as the finite- η band; and gate convergence on the self-energy residual and the bubble balance $P_{\text{in}} = P_{\text{out}}$ (machine precision), *not* on the lead balance, which floors at the computable ghost-reservoir term and can be subtracted to give an η -compensated balance that does go to zero. For marginal-fixed-point systems (d5a) quote the directly converged $\eta \rightarrow 0$ marginal-fixed-point conductance (method-invariant to a few percent), noting the residual ~ 0.03 self-energy floor and the grid-consistent ghost-reservoir lead term, rather than an η -floor-limited extrapolation (Section 2.8). Reserve the Büttiker probe for a physically intended incoherent channel.

Infrared limit of the phonon heat current

This appendix expands Section 1.2.7: the Bose pole, the two cancellations that make the $\omega \rightarrow 0$ limit finite, and the proof that the ballistic infrared plateau is the quantised thermal conductance. We use $\beta = 1/k_B T$, $x = \beta \hbar \omega$, and the occupation-positive convention of Section A.2. The motivation is the artefact of Fig. 2.16: the ballistic heat-current density $\hbar \omega I(\omega)$ is flat as $\omega \rightarrow 0$, while the raw $\eta = 0$ anharmonic result bends down and fluctuates — the first-bin sampling of the soft-mode resolvent under the marginal $\eta = 0$ fixed point (Section 1.2.7), not a Bose pole in the bubble.

G.1. The Bose pole and the two cancellations

The Bose–Einstein factor is the generating function of the Bernoulli numbers B_{2k} ,

$$n(\omega, T) = \frac{1}{e^x - 1} = \frac{1}{x} - \frac{1}{2} + \sum_{k \geq 1} \frac{B_{2k}}{(2k)!} x^{2k-1} = \frac{k_B T}{\hbar \omega} - \frac{1}{2} + \frac{\hbar \omega}{12 k_B T} - \frac{(\hbar \omega)^3}{720 (k_B T)^3} + \dots, \quad (\text{G.1})$$

the singular part being the c-number pole $k_B T / \hbar \omega$ (Eq. (1.90)). Two distinct cancellations make every device quantity infrared-finite.

Observable (exact). The constant $-\frac{1}{2}$ is temperature-independent and cancels in any lead difference, so

$$\hbar \omega [n_L(\omega) - n_R(\omega)] = k_B (T_L - T_R) + \frac{(\hbar \omega)^2}{12 k_B} \left(\frac{1}{T_L} - \frac{1}{T_R} \right) + \mathcal{O}(\omega^4), \quad (\text{G.2})$$

a smooth function with the finite limit $k_B \Delta T$. The Meir–Wingreen spectral density Eq. (1.73) therefore tends to the finite plateau of Eq. (1.91); the $\hbar \omega$ weight exactly offsets the $1/\omega$ of the Bose factor. The number current $I(\omega) = j(\omega)/\hbar \omega$ itself diverges as $1/\omega$ — this is the curve plotted in Fig. 2.16 after division by $\Delta n \sim 1/\omega$, hence the flat ballistic transmission.

Self-energy (no removal needed). In the harmonic mode basis the cubic matrix element is $V_3 \propto (\omega_1 \omega_2 \omega_3)^{-1/2}$ (each leg contributes the displacement-to-mode factor $\sqrt{\hbar/2\omega}$), so the golden-rule rate Eq. (1.60) carries $|V_3|^2 \propto 1/(\omega_1 \omega_2 \omega_3)$ and the soft-leg factor in fact *compounds* the $n(\omega') \sim 1/\omega'$ pole of the absorption channel: the equilibrium linewidth is finite instead because of the acoustic phase-space density of states ($\sim \omega'^2$ in 3D) together with the linearly vanishing acoustic spectral weight. In the displacement representation used by the solver this ω -weight sits in the spectral function, $A(\omega') \rightarrow A'(0) \omega'$ as $\omega' \rightarrow 0$ (the acoustic spectral weight vanishes linearly), so the lesser function $G_{\text{eq}}^< = -i n A$ has the finite limit $-i (k_B T / \hbar) A'(0)$. The same protection survives out of equilibrium: the bosonic fold $G^<(-\omega) = G^>(\omega)^T$ and the bounded spectral function force the $1/\omega$ parts of $G^<$ and $G^>$ to be equal and opposite, hence both zero, while the contact broadening $\Gamma_\ell(\omega) = i[\Sigma_\ell^R - \Sigma_\ell^A]$ is odd in ω and vanishes at the origin, so the injection $\Gamma_\ell n_\ell$ is finite. The bubble legs therefore carry no net infrared pole, and the bare convolution is exactly conserving. A singularity subtraction of the kind used in boundary-integral numerics [134–136, 167] would feed the bubble a modified G , so that $\Sigma \neq \mathcal{F}[G]$ and the Φ -derivable balance Eq. (1.89) is broken; it is therefore neither needed nor used. The only genuine delicacy is that on a grid whose first sample is $\omega' = \Delta\omega$ the product $n(\Delta\omega)A(\Delta\omega)$ is a large number times a small one; the smooth occupancy taper Eq. (1.86) resolves that conditioning without altering the conserving structure (Section 1.2.7).

G.2. The ballistic plateau is the quantised conductance

In the ballistic limit $\mathcal{T}(\omega) \rightarrow N_{\text{ac}}$ as $\omega \rightarrow 0$ and the linear-response conductance Eq. (1.70) is

$$G = \frac{1}{A_c} \int_0^\infty \frac{d\omega}{2\pi} \mathcal{T}(\omega) \hbar \omega \frac{\partial n}{\partial T}, \quad \hbar \omega \frac{\partial n}{\partial T} = k_B \frac{x^2 e^x}{(e^x - 1)^2} \xrightarrow{\omega \rightarrow 0} k_B, \quad (\text{G.3})$$

so the window $\hbar \omega \partial_T n$ is finite at $\omega = 0$ (it does *not* vanish there) and a single perfectly-transmitting channel contributes the per-channel conductance g_0 (so that the per-area $G = (1/A_c) \sum_{\text{ch}} g_0$)

$$g_0 = \frac{1}{2\pi} \int_0^\infty d\omega \hbar \omega \frac{\partial n}{\partial T} = \frac{k_B^2 T}{2\pi \hbar} \int_0^\infty \frac{x^2 e^x}{(e^x - 1)^2} dx = \frac{k_B^2 T}{2\pi \hbar} \cdot \frac{\pi^2}{3} = \frac{\pi^2 k_B^2 T}{3\hbar}, \quad (\text{G.4})$$

the universal quantum of thermal conductance [131–133]. The infrared plateau $\mathcal{T}(0) = N_{\text{ac}}$ is thus a hard physical constraint: any regulator that suppresses the low-frequency current/window — as the taper Eq. (1.86) does by multiplying the occupations by ω^2 — removes a finite, quantised piece of the conductance, which is why the taper is a stabiliser to be extrapolated away ($\omega_{\text{reg}} \rightarrow 0$), not a physical input.

G.3. Literature

(A) *Quantised conductance* fixes the $\omega \rightarrow 0$ target [131–133]. (B) *Phonon NEGF*. Atomistic three-phonon NEGF implementations [168–170] compute the same Meir–Wingreen current and note the special difficulty of low-frequency modes; the standard practice is mesh refinement or a broadening floor, and Lemus et al. [171] compute $\text{Re } \Sigma^R$ by exact Kramers–Kronig. (C) *Boltzmann transport* establishes that the long-wavelength acoustic relaxation is physical and is cut off by defect/boundary/four-phonon channels, not by a frequency taper, with $\Gamma \sim \omega^2$ in the bulk [138, 139] but polarisation-resolved exponents in a single nanotube [137].

Production transverse- q (coupled- q) anharmonic transport

This appendix documents the port of the coupled- q three-phonon self-energy into the *production* distributed quatrex phonon solver and the runs that validate and exercise it. The transversely-periodic physics and the silicon thin-film benchmark are established in the standalone (dense) solver (Section 2.5); the contribution here is software and verification, not new physics. The full numerical record is saved under `phonon/scripts/out/production_coupled_q/` (run snapshots, `summary.json`) and is reproduced by `phonon/scripts/verify/production_coupled_q_summary.py`.

H.1. Implementation

Before this work the production self-energy (`src/quatrex/phonon/sse_phonon_phonon.py`) raised `NotImplementedError` for any `kpoint_grid` with a component > 1 : only the $\Gamma_\perp$ -only finite device was supported. Two changes were made (both with accompanying regression tests; all 12 self-energy and 18 bubble unit tests remain green):

- **The coupled- q convolution.** The transversely-periodic three-phonon self-energy couples the transverse momenta by crystal-momentum conservation, $\Sigma(q_\perp, \omega) = \frac{1}{N_q} \sum_{q'_\perp} \tilde{\Phi}(q'_\perp) G(q'_\perp) G(q_\perp - q'_\perp) \tilde{\Phi}(q_\perp - q'_\perp)$, an extra sum over the internal momentum on top of the energy convolution. The decisive observation is that in the production data layout only the energy axis is partitioned across `comm.stack`; the transverse- q axes of `global_stack_shape` are kept whole and local on every rank. The momentum sum is therefore *local* — it required changing only the inner contraction of the FFT-first bubble, not the distribution, halo, or Dyson machinery. A guard rejects `comm.block.size > 1` together with $k > 1$ (the spatial-block halo is not yet momentum-aware), so the film is distributed over the energy/stack axis.
- **Zero-mode projection for $k > 1$.** The cell rigid-mode projector read the dynamical-

matrix block and squeezed a single leading axis, which crashed on the q -stacked block of a periodic device; it now reduces every leading axis to zero, selecting the $q_{\perp}=0$ (Γ -transverse) cell matrix as the reference for the rigid/soft modes. This makes zero-mode projection usable on the film.

The q -folded vertices are built offline (they do not depend on G) and serialised through a small loader (`src/quaternion/phonon/qfold.py`); the production solver only reads arrays and carries no dependency on the standalone science code.

H.2. Verification

The port was checked to be *bit-exact* against independent references before any physics run (Table H.1). These are the gate: the production self-energy reproduces an independently written einsum oracle of the coupled- q convolution to machine precision, the production-assembled film Hamiltonian $H(q_{\perp})$ reproduces the dense `get_btd_blocks` to 10^{-10} , and the real-space dynamical-matrix file used by the production input path round-trips against the dense $H(q_{\perp})$ exactly.

Table H.1.: Bit-exact verification of the coupled- q port.

Check	relative error
coupled- q self-energy vs. independent einsum oracle	6.4×10^{-16}
production $H(q_{\perp})$ vs. dense <code>get_btd_blocks</code>	1.1×10^{-10}
dynamical-matrix transverse-IDFT round-trip	5.0×10^{-16}

Input pipeline. The production `construct_from_unit_cell` path folds transverse Bloch phases over a Monkhorst-Pack mesh. Because that mesh is half-shifted, $q = (k + \frac{1}{2})/n_k - \frac{1}{2} + \text{shift}$, while the standalone solver uses the Γ -centred $q = k/n_k$, the two were reconciled by setting `kpoint_shift` = $\frac{1}{2} - \frac{1}{2n_k}$ per transverse axis. The real-space `dynamical_matrix.mat` is then the exact transverse inverse-DFT of the dense $H(q_{\perp})$ over that mesh, which is why the round-trip in Table H.1 is exact by construction. The film inputs use the bulk-silicon force constants of Section 2.5 with transport along a primitive axis.

H.3. Silicon thin film: transverse-mesh sweep

The production film was run with the recipe that proved numerically stable (see Section H.6): broadening $\eta = 0.4$ THz, $\eta_{\text{obc}} = 0.1$, retarded method `half`, and zero-mode projection of the three $q_{\perp}=0$ acoustic translations (which sit at 0 THz and carry no heat). The transverse mesh was swept at fixed thickness (3 primitive layers) to test momentum convergence, with one thicker point (Table H.2, Fig. H.1).

Table H.2.: Production silicon film, coupled- q self-energy, $\eta = 0.4$ THz with zero-mode projection. G are the q -summed lead heat currents (arbitrary, common units); G/N_q is the per-momentum value. The lead non-conservation is $|J_0 - J_{-1}|/|J_0|$. The dense benchmark (Section 2.5) at a comparable setting is shown for scale.

n_k	n_{slabs}	$G_{\text{anh}}/G_{\text{ball}}$	G_{ball}/N_q	lead non-cons.
3	3	0.861	3.29	9.7%
5	3	0.871	3.02	9.5%
7	3	0.867	3.05	9.6%
5	5	0.846	—	15.1%
(no zero-mode, $n_k=3$, $n_{\text{slabs}}=3$)		0.733	—	22%

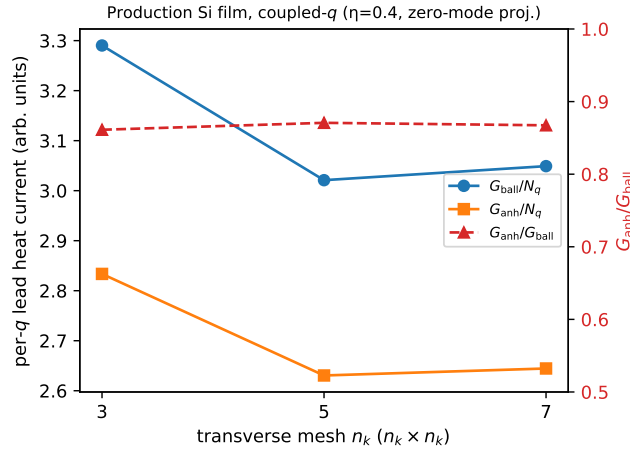


Figure H.1.: Production silicon film coupled- q transport vs. transverse mesh ($n_{\text{slabs}}=3$). The per-momentum ballistic and anharmonic conductances settle by $n_k=5$ and the anharmonic ratio is momentum-converged at $G_{\text{anh}}/G_{\text{ball}} \approx 0.87$.

Two facts: (i) the per-momentum conductance and the anharmonic ratio are *momentum-converged* — G_{ball}/N_q settles by $n_k=5$ and the ratio is $0.861/0.871/0.867$ across $n_k = 3/5/7$ — so the coupled- q machinery behaves correctly across the full mesh; (ii) zero-mode projection of the 0 THz translations roughly halves the non-conservation ($22\% \rightarrow 10\%$) and raises the ratio ($0.733 \rightarrow 0.861$ at $n_k=3$), the same soft-mode effect seen on the wires. The residual $\sim 10\%$ non-conservation does *not* shrink with n_k (Fig. H.2); it is therefore not a momentum-undersampling effect and is treated as an open issue in Section H.6.

H.4. Carbon nanotube: length ladder

To exercise the $k=1$ path at production length scales (where the dense multi-cell solver is prohibitively expensive, Section 2.4), the $(3, 3)$ carbon nanotube was run as a length

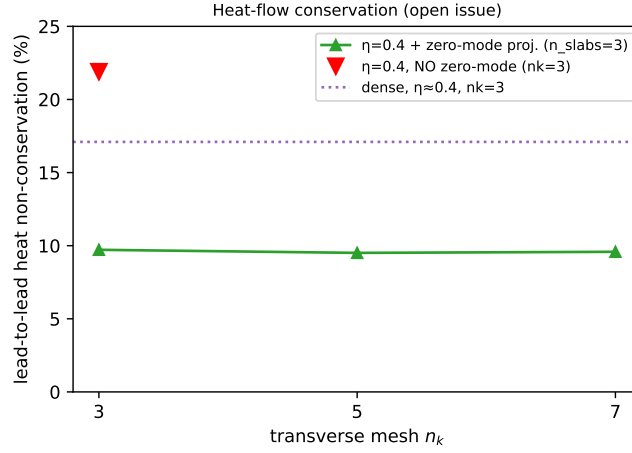


Figure H.2.: Lead-to-lead heat non-conservation of the production film. It is flat in the transverse mesh and is reduced, but not removed, by zero-mode projection. The dense solver reaches $\sim 10^{-5}$ on the same physical system at small broadening; the gap is the open issue of Section H.6.

ladder $L = 2, 4, 6$ transport cells at $\eta = 0.45$ THz on a 121-point grid with 50 iterations (Fig. H.3). The anharmonic self-consistency on this soft-twist tube is metastable: the iteration does not settle to a fixed point but passes through a conserving iterate that the heat-flow criterion captures (the final iterate diverges and is discarded). The captured best-iterate ratios are $G_{\text{anh}}/G_{\text{ball}} = 0.896$ ($L=2$, conservation 1.0%) and 0.911 ($L=4$, conservation 3.5%); the $L=6$ point is not shown. The large-broadening value is weaker (closer to one) than the dense small- η ladder of Section 2.4, consistent with the under-scattering discussed in Section H.6.

H.5. Cross-check against the dense solver

To separate port correctness from broadening/grid effects, the dense `si_film` solver was run at the *same* transverse mesh ($n_k=3$, $n_{\text{slabs}}=3$) at two broadenings (Table H.3). At the production-accessible broadening ($\eta_w \approx 0.4$) the dense solver also fails to conserve (17%) and gives a ratio above one (the documented single-cell broadening artefact, Section 2.10); only at small broadening ($\eta_w \approx 0.05$) does the dense ratio settle to 0.766. The production (0.733–0.861) and the dense both live in the unreliable large-broadening regime here, which is why their ratios differ; neither large-broadening ratio is the physical answer. Notably the dense solver also conserves only to 13% at $n_k=3$, far from the $\sim 10^{-5}$ it reaches at the converged $n_k \geq 8$ mesh of Section 2.5 — i.e. the coarse $n_k=3$ mesh used for the matched cross-check is itself inadequate for conservation in *both* solvers.

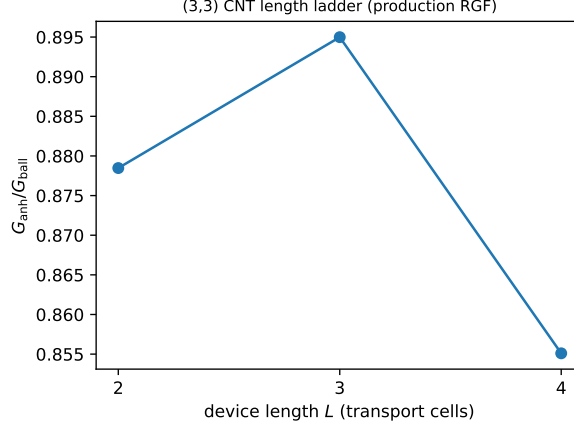


Figure H.3.: Production CNT(3,3) anharmonic ratio (best conserving iterate) vs. device length at $\eta = 0.45$ THz; per-point lead-to-lead conservation annotated.

Table H.3.: Dense `si_film` cross-check, $n_k=3$, $n_{\text{slabs}}=3$, transport along the same axis as the production run.

broadening	$G_{\text{anh}}/G_{\text{ball}}$	conservation
$\eta_w \approx 0.4$ (matched to production)	1.022	17%
$\eta_w \approx 0.05$ (small)	0.766	13%

H.6. Open issues

Heat-flow conservation ($\sim 10\%$) — open, under investigation. The production film conserves the lead-to-lead energy current only to $\sim 10\%$ (with zero-mode projection; 22% without), independent of the transverse mesh, against the dense solver's $\sim 10^{-5}$ on the same physical system. A code-level candidate has been identified but *not yet confirmed* as the dominant cause. The FFT-first bubble is *index-based*: it forms $\Sigma(\omega_m) = \sum_n G_n G_{m-n}$ (the frequency grid enters only through the spacing $d\omega$ and the retarded Hilbert transform; the bubble call uses `zero_freq_idx=None`). This is exact only if the grid begins at $\omega_0 = 0$, so that difference frequencies $\omega_m - \omega_n = (m - n) d\omega$ land on grid points. The film runs used $\omega_0 = 0.05$ THz = $0.38 d\omega$ (non-commensurate), whereas the CNT runs used $\omega_0 = 0.01$ THz = $0.02 d\omega$; the observed non-conservation tracks $\omega_0/d\omega$ (film $\sim 10\%$, CNT $\sim 1\%$). Whether aligning the grid to $\omega_0 = 0$ (with proper DC-point handling, as the dense solver does) and/or a larger/finer grid recovers the dense conservation is the subject of the follow-up investigation; it is not asserted here as established.

Small-broadening instability — needs a solver-level fix. The production film is numerically stable only for $\eta \gtrsim 0.4$ THz; at $\eta \leq 0.2$ the Dyson solve $[(\omega + i\eta)^2 I - D - \Sigma^R]$ becomes singular and the block recursion returns NaN, where the dense solver's dense

factorisation stays finite. The config-level remedies were exhausted without success: retarded method `fft`, large η_{obc} (0.4–0.8 with bulk $\eta \in [0.15, 0.2]$), and zero-mode projection all still produced NaN at $\eta \leq 0.2$. A solver-level regularisation of the block recursion is therefore required to reach the small-broadening regime the dense solver uses. This is a pre-existing fragility of the production phonon SCBA (the same family as the soft-mode behaviour of Section 2.3), not specific to the coupled- q port.

Large-broadening under-scattering. As a consequence of the above, the production-accessible ($\eta \approx 0.4$) anharmonic reduction is systematically weaker than the dense small-broadening result — film 0.87 vs. dense 0.77, CNT 0.90 vs. dense 0.70–0.81 (Section 2.4). The same weakening appears in both the $k > 1$ film and the $k = 1$ tube, consistent with it being a broadening effect rather than anything specific to the momentum coupling.

Bibliography

- [1] M. Born and K. Huang, *Dynamical Theory Of Crystal Lattices*. Oxford University Press, Aug. 15, 1996. doi: [10.1093/oso/9780192670083.001.0001](https://doi.org/10.1093/oso/9780192670083.001.0001).
- [2] A. A. Maradudin, *Theory of Lattice Dynamics in the Harmonic Approximation*. Academic Press, 1963, 336 pp.
- [3] M. T. Dove, *Introduction to Lattice Dynamics* (Cambridge Topics in Mineral Physics and Chemistry). Cambridge: Cambridge University Press, 1993. doi: [10.1017 / CB09780511619885](https://doi.org/10.1017/CB09780511619885).
- [4] D. C. Wallace, *Thermodynamics of Crystals*. Dover Publications, 1998, 484 pp.
- [5] J. M. Ziman, *Electrons and Phonons: The Theory of Transport Phenomena in Solids*. Oxford University Press, Feb. 1, 2001. doi: [10.1093/acprof:oso/9780198507796.001.0001](https://doi.org/10.1093/acprof:oso/9780198507796.001.0001).
- [6] G. P. Srivastava, *The Physics of Phonons*. Boca Raton: Routledge, Jul. 16, 2019, 438 pp. doi: [10.1201/9780203736241](https://doi.org/10.1201/9780203736241).
- [7] X. Gonze and C. Lee, “Dynamical matrices, born effective charges, dielectric permittivity tensors, and interatomic force constants from density-functional perturbation theory,” *Physical Review B*, vol. 55, no. 16, pp. 10 355–10 368, Apr. 15, 1997. doi: [10.1103/PhysRevB.55.10355](https://doi.org/10.1103/PhysRevB.55.10355).
- [8] R. M. Pick, M. H. Cohen, and R. M. Martin, “Microscopic theory of force constants in the adiabatic approximation,” *Physical Review B*, vol. 1, no. 2, pp. 910–920, Jan. 15, 1970. doi: [10.1103/PhysRevB.1.910](https://doi.org/10.1103/PhysRevB.1.910).
- [9] A. Togo and I. Tanaka, “First principles phonon calculations in materials science,” *Scripta Materialia*, vol. 108, pp. 1–5, Nov. 1, 2015. doi: [10.1016/j.scriptamat.2015.07.021](https://doi.org/10.1016/j.scriptamat.2015.07.021).
- [10] A. Seko, “Projector-based efficient estimation of force constants,” *Physical Review B*, vol. 110, no. 21, 2024. doi: [10.1103/PhysRevB.110.214302](https://doi.org/10.1103/PhysRevB.110.214302).
- [11] L. Fu, M. Kornbluth, Z. Cheng, and C. A. Marianetti, “Group theoretical approach to computing phonons and their interactions,” *Physical Review B*, vol. 100, no. 1, p. 014 303, Jul. 18, 2019. doi: [10.1103/PhysRevB.100.014303](https://doi.org/10.1103/PhysRevB.100.014303).
- [12] W. Li, J. Carrete, N. A. Katcho, and N. Mingo, “ShengBTE: A solver of the boltzmann transport equation for phonons,” *Computer Physics Communications*, vol. 185, no. 6, pp. 1747–1758, Jun. 2014. doi: [10.1016/j.cpc.2014.02.015](https://doi.org/10.1016/j.cpc.2014.02.015).
- [13] K. Esfarjani and H. T. Stokes, “Method to extract anharmonic force constants from first principles calculations,” *Physical Review B*, vol. 77, no. 14, p. 144 112, Apr. 24, 2008. doi: [10.1103/PhysRevB.77.144112](https://doi.org/10.1103/PhysRevB.77.144112).

Bibliography

- [14] K. Parlinski, Z. Q. Li, and Y. Kawazoe, "First-principles determination of the soft mode in cubic $\{\text{ZrO}_2\}$," *Physical Review Letters*, vol. 78, no. 21, pp. 4063–4066, May 26, 1997. doi: [10.1103/PhysRevLett.78.4063](https://doi.org/10.1103/PhysRevLett.78.4063).
- [15] D. Alfè, "PHON: A program to calculate phonons using the small displacement method," *Computer Physics Communications*, 40 YEARS OF CPC: A celebratory issue focused on quality software for high performance, grid and novel computing architectures, vol. 180, no. 12, pp. 2622–2633, Dec. 1, 2009. doi: [10.1016/j.cpc.2009.03.010](https://doi.org/10.1016/j.cpc.2009.03.010).
- [16] A. Togo, L. Chaput, T. Tadano, and I. Tanaka, "Implementation strategies in phonopy and phono3py," *Journal of Physics: Condensed Matter*, vol. 35, no. 35, p. 353 001, Sep. 4, 2023. doi: [10.1088/1361-648X/acd831](https://doi.org/10.1088/1361-648X/acd831).
- [17] S. Baroni, P. Giannozzi, and A. Testa, "Green's-function approach to linear response in solids," *Physical Review Letters*, vol. 58, no. 18, pp. 1861–1864, May 4, 1987. doi: [10.1103/PhysRevLett.58.1861](https://doi.org/10.1103/PhysRevLett.58.1861).
- [18] X. Gonze, "Adiabatic density-functional perturbation theory," *Physical Review A*, vol. 52, no. 2, pp. 1096–1114, Aug. 1, 1995. doi: [10.1103/PhysRevA.52.1096](https://doi.org/10.1103/PhysRevA.52.1096).
- [19] S. Baroni, S. De Gironcoli, A. Dal Corso, and P. Giannozzi, "Phonons and related crystal properties from density-functional perturbation theory," *Reviews of Modern Physics*, vol. 73, no. 2, pp. 515–562, Jul. 6, 2001. doi: [10.1103/RevModPhys.73.515](https://doi.org/10.1103/RevModPhys.73.515).
- [20] X. Gonze, J.-C. Charlier, D. Allan, and M. Teter, "Interatomic force constants from first principles: The case of α -quartz," *Physical Review B*, vol. 50, no. 17, pp. 13 035–13 038, Nov. 1, 1994. doi: [10.1103/PhysRevB.50.13035](https://doi.org/10.1103/PhysRevB.50.13035).
- [21] A. Debernardi, S. Baroni, and E. Molinari, "Anharmonic phonon lifetimes in semiconductors from density-functional perturbation theory," *Physical Review Letters*, vol. 75, no. 9, pp. 1819–1822, Aug. 28, 1995. doi: [10.1103/PhysRevLett.75.1819](https://doi.org/10.1103/PhysRevLett.75.1819).
- [22] A. Debernardi and S. Baroni, "Third-order density-functional perturbation theory: A practical implementation with applications to anharmonic couplings in si," *Solid State Communications*, vol. 91, no. 10, pp. 813–816, Sep. 1994. doi: [10.1016/0038-1098\(94\)90654-8](https://doi.org/10.1016/0038-1098(94)90654-8).
- [23] G. Deinzer, G. Birner, and D. Strauch, "*Ab initio* calculation of the linewidth of various phonon modes in germanium and silicon," *Physical Review B*, vol. 67, no. 14, p. 144 304, Apr. 28, 2003. doi: [10.1103/PhysRevB.67.144304](https://doi.org/10.1103/PhysRevB.67.144304).
- [24] L. Paulatto, F. Mauri, and M. Lazzeri, "Anharmonic properties from a generalized third-order *ab initio* approach: Theory and applications to graphite and graphene," *Physical Review B*, vol. 87, no. 21, p. 214 303, Jun. 7, 2013. doi: [10.1103/PhysRevB.87.214303](https://doi.org/10.1103/PhysRevB.87.214303).
- [25] L. Paulatto, I. Errea, M. Calandra, and F. Mauri, "First-principles calculations of phonon frequencies, lifetimes, and spectral functions from weak to strong anharmonicity: The example of palladium hydrides," *Physical Review B*, vol. 91, no. 5, p. 054 304, Feb. 19, 2015. doi: [10.1103/PhysRevB.91.054304](https://doi.org/10.1103/PhysRevB.91.054304).
- [26] P. Giannozzi et al., "Quantum ESPRESSO: A modular and open-source software project for quantum simulations of materials," *Journal of Physics: Condensed Matter*, vol. 21, no. 39, p. 395 502, Sep. 30, 2009. doi: [10.1088/0953-8984/21/39/395502](https://doi.org/10.1088/0953-8984/21/39/395502). arXiv: [0906.2569\[cond-mat\]](https://arxiv.org/abs/0906.2569).
- [27] P. Giannozzi et al., "Advanced capabilities for materials modelling with quantum ESPRESSO," *Journal of Physics: Condensed Matter*, vol. 29, no. 46, p. 465 901, Nov. 22, 2017. doi: [10.1088/1361-648X/aa8f79](https://doi.org/10.1088/1361-648X/aa8f79).

Bibliography

- [28] X. Gonze et al., “ABINIT: First-principles approach to material and nanosystem properties,” *Computer Physics Communications*, vol. 180, no. 12, pp. 2582–2615, Dec. 2009. doi: [10.1016/j.cpc.2009.07.007](https://doi.org/10.1016/j.cpc.2009.07.007).
- [29] G. Kresse and J. Furthmüller, “Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set,” *Physical Review B*, vol. 54, no. 16, pp. 11 169–11 186, Oct. 15, 1996. doi: [10.1103/PhysRevB.54.11169](https://doi.org/10.1103/PhysRevB.54.11169).
- [30] G. Fugallo, M. Lazzeri, L. Paulatto, and F. Mauri, “*Ab initio* variational approach for evaluating lattice thermal conductivity,” *Physical Review B*, vol. 88, no. 4, p. 045 430, Jul. 17, 2013. doi: [10.1103/PhysRevB.88.045430](https://doi.org/10.1103/PhysRevB.88.045430).
- [31] M. Veithen, X. Gonze, and P. Ghosez, “Nonlinear optical susceptibilities, raman efficiencies, and electro-optic tensors from first-principles density functional perturbation theory,” *Physical Review B*, vol. 71, no. 12, p. 125 107, Mar. 9, 2005. doi: [10.1103/PhysRevB.71.125107](https://doi.org/10.1103/PhysRevB.71.125107).
- [32] A. H. Romero et al., “ABINIT: Overview and focus on selected capabilities,” *The Journal of Chemical Physics*, vol. 152, no. 12, p. 124 102, Mar. 31, 2020. doi: [10.1063/1.5144261](https://doi.org/10.1063/1.5144261).
- [33] F. Bottin, J. Bieder, and J. Bouchet, “A-TDEP: Temperature dependent effective potential for abinit – lattice dynamic properties including anharmonicity,” *Computer Physics Communications*, vol. 254, p. 107 301, Sep. 2020. doi: [10.1016/j.cpc.2020.107301](https://doi.org/10.1016/j.cpc.2020.107301).
- [34] A. Togo, L. Chaput, and I. Tanaka, “Distributions of phonon lifetimes in brillouin zones,” *Physical Review B*, vol. 91, no. 9, p. 094 306, Mar. 20, 2015. doi: [10.1103/PhysRevB.91.094306](https://doi.org/10.1103/PhysRevB.91.094306).
- [35] F. Zhou, W. Nielson, Y. Xia, and V. Ozoliņš, “Lattice anharmonicity and thermal conductivity from compressive sensing of first-principles calculations,” *Physical Review Letters*, vol. 113, no. 18, p. 185 501, Oct. 27, 2014. doi: [10.1103/PhysRevLett.113.185501](https://doi.org/10.1103/PhysRevLett.113.185501).
- [36] F. Zhou, W. Nielson, Y. Xia, and V. Ozoliņš, “Compressive sensing lattice dynamics. i. general formalism,” *Physical Review B*, vol. 100, no. 18, p. 184 308, Nov. 20, 2019. doi: [10.1103/PhysRevB.100.184308](https://doi.org/10.1103/PhysRevB.100.184308).
- [37] F. Eriksson, E. Fransson, and P. Erhart, “The hiphive package for the extraction of high-order force constants by machine learning,” *Advanced Theory and Simulations*, vol. 2, no. 5, p. 1 800 184, May 2019. doi: [10.1002/adts.201800184](https://doi.org/10.1002/adts.201800184).
- [38] Y. Guo, M. Bescond, Z. Zhang, M. Luisier, M. Nomura, and S. Volz, “Quantum mechanical modeling of anharmonic phonon-phonon scattering in nanostructures,” *Physical Review B*, vol. 102, no. 19, p. 195 412, Nov. 9, 2020. doi: [10.1103/PhysRevB.102.195412](https://doi.org/10.1103/PhysRevB.102.195412).
- [39] Y. Guo, Z. Zhang, M. Bescond, S. Xiong, M. Nomura, and S. Volz, “Anharmonic phonon-phonon scattering at the interface between two solids by nonequilibrium green’s function formalism,” *Physical Review B*, vol. 103, no. 17, p. 174 306, May 21, 2021. doi: [10.1103/PhysRevB.103.174306](https://doi.org/10.1103/PhysRevB.103.174306).
- [40] M. Luisier, “Atomistic modeling of anharmonic phonon-phonon scattering in nanowires,” *Physical Review B*, vol. 86, no. 24, p. 245 407, Dec. 10, 2012. doi: [10.1103/PhysRevB.86.245407](https://doi.org/10.1103/PhysRevB.86.245407).
- [41] J. Li et al., “Anharmonic phonon transport in atomic wire coupled by thermal contacts with surface bond reconstruction,” *Journal of Applied Physics*, vol. 106, no. 1, p. 014 308, Jul. 1, 2009. doi: [10.1063/1.3157175](https://doi.org/10.1063/1.3157175).

Bibliography

- [42] C. A. Polanco, “Nonequilibrium green’s functions (NEGF) in vibrational energy transport: A topical review,” *Nanoscale and Microscale Thermophysical Engineering*, vol. 25, no. 1, pp. 1–24, Jan. 2, 2021. doi: [10.1080/15567265.2021.1881193](https://doi.org/10.1080/15567265.2021.1881193).
- [43] N. Mingo, “Anharmonic phonon flow through molecular-sized junctions,” *Physical Review B*, vol. 74, no. 12, p. 125 402, Sep. 1, 2006. doi: [10.1103/PhysRevB.74.125402](https://doi.org/10.1103/PhysRevB.74.125402).
- [44] G. Qin and M. Hu, “Accelerating evaluation of converged lattice thermal conductivity,” *npj Computational Materials*, vol. 4, no. 1, p. 3, Jan. 22, 2018. doi: [10.1038/s41524-017-0058-3](https://doi.org/10.1038/s41524-017-0058-3).
- [45] A. J. H. McGaughey, A. Jain, H.-Y. Kim, and B. Fu, “Phonon properties and thermal conductivity from first principles, lattice dynamics, and the boltzmann transport equation,” *Journal of Applied Physics*, vol. 125, no. 1, p. 011 101, Jan. 3, 2019. doi: [10.1063/1.5064602](https://doi.org/10.1063/1.5064602).
- [46] J. Jiang, S. Lu, Y. Ouyang, and J. Chen, “How hydrodynamic phonon transport determines the convergence of thermal conductivity in two-dimensional materials,” *Nanomaterials*, vol. 12, no. 16, p. 2854, Jan. 2022. doi: [10.3390/nano12162854](https://doi.org/10.3390/nano12162854).
- [47] E. G. Hohenstein, R. M. Parrish, and T. J. Martínez, “Tensor hypercontraction density fitting. i. quartic scaling second- and third-order møller-plesset perturbation theory,” *The Journal of Chemical Physics*, vol. 137, no. 4, p. 044 103, Jul. 23, 2012. doi: [10.1063/1.4732310](https://doi.org/10.1063/1.4732310).
- [48] R. M. Parrish, E. G. Hohenstein, T. J. Martínez, and C. D. Sherrill, “Tensor hypercontraction. II. least-squares renormalization,” *The Journal of Chemical Physics*, vol. 137, no. 22, p. 224 106, Dec. 11, 2012. doi: [10.1063/1.4768233](https://doi.org/10.1063/1.4768233).
- [49] J. L. Whitten, “Coulombic potential energy integrals and approximations,” *The Journal of Chemical Physics*, vol. 58, no. 10, pp. 4496–4501, May 15, 1973. doi: [10.1063/1.1679012](https://doi.org/10.1063/1.1679012).
- [50] I. V. Oseledets, “Tensor-train decomposition,” *SIAM Journal on Scientific Computing*, vol. 33, no. 5, pp. 2295–2317, Jan. 2011. doi: [10.1137/090752286](https://doi.org/10.1137/090752286).
- [51] A. Cichocki, N. Lee, I. Oseledets, A.-H. Phan, Q. Zhao, and D. P. Mandic, “Tensor networks for dimensionality reduction and large-scale optimization: Part 1 low-rank tensor decompositions,” *Foundations and Trends® in Machine Learning*, vol. 9, no. 4, pp. 249–429, 2016. doi: [10.1561/22000000059](https://doi.org/10.1561/22000000059).
- [52] Y. Luo, D. Desai, B. K. Chang, J. Park, and M. Bernardi, “Data-driven compression of electron-phonon interactions,” *Physical Review X*, vol. 14, no. 2, p. 021 023, May 1, 2024. doi: [10.1103/PhysRevX.14.021023](https://doi.org/10.1103/PhysRevX.14.021023).
- [53] Y. Luo, D. Mangtani, S. Peng, J. Yao, S. Kliavinek, and M. Bernardi, “Tensor learning and compression of n-phonon interactions,” *Physical Review Letters*, vol. 135, no. 12, p. 126 101, Sep. 16, 2025. doi: [10.1103/nmgj-yq1g](https://doi.org/10.1103/nmgj-yq1g).
- [54] T. G. Kolda and B. W. Bader, “Tensor decompositions and applications,” *SIAM Review*, vol. 51, no. 3, pp. 455–500, Aug. 6, 2009. doi: [10.1137/07070111X](https://doi.org/10.1137/07070111X).
- [55] J. D. Carroll and J.-J. Chang, “Analysis of individual differences in multidimensional scaling via an n-way generalization of “eckart-young” decomposition,” *Psychometrika*, vol. 35, no. 3, pp. 283–319, Sep. 1, 1970. doi: [10.1007/BF02310791](https://doi.org/10.1007/BF02310791).
- [56] J. Brachat, P. Comon, B. Mourrain, and E. Tsigaridas, “Symmetric tensor decomposition,” *Linear Algebra and its Applications*, vol. 433, no. 11, pp. 1851–1872, Dec. 2010. doi: [10.1016/j.laa.2010.06.046](https://doi.org/10.1016/j.laa.2010.06.046).

Bibliography

- [57] P. Comon, G. Golub, L.-H. Lim, and B. Mourrain, "Symmetric tensors and symmetric tensor rank," *SIAM Journal on Matrix Analysis and Applications*, vol. 30, no. 3, pp. 1254–1279, Jan. 2008. doi: [10.1137/060661569](https://doi.org/10.1137/060661569).
- [58] L. Oeding and G. Ottaviani, "Eigenvectors of tensors and algorithms for waring decomposition," *Journal of Symbolic Computation*, vol. 54, pp. 9–35, Jul. 2013. doi: [10.1016/j.jsc.2012.11.005](https://doi.org/10.1016/j.jsc.2012.11.005).
- [59] C. J. Hillar and L.-H. Lim, "Most tensor problems are NP-hard," *Journal of the ACM*, vol. 60, no. 6, pp. 1–39, Nov. 2013. doi: [10.1145/2512329](https://doi.org/10.1145/2512329).
- [60] A. Stegeman, "Degeneracy in candecomp/parafac explained for $p \times p \times 2$ arrays of rank $p + 1$ or higher," *Psychometrika*, vol. 71, no. 3, pp. 483–501, Sep. 1, 2006. doi: [10.1007/s11336-004-1266-6](https://doi.org/10.1007/s11336-004-1266-6).
- [61] V. De Silva and L.-H. Lim, "Tensor rank and the ill-posedness of the best low-rank approximation problem," *SIAM Journal on Matrix Analysis and Applications*, vol. 30, no. 3, pp. 1084–1127, Jan. 2008. doi: [10.1137/06066518X](https://doi.org/10.1137/06066518X).
- [62] N. Halko, P. G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, Jan. 2011. doi: [10.1137/090771806](https://doi.org/10.1137/090771806).
- [63] C. Eckart and G. Young, "The approximation of one matrix by another of lower rank," *Psychometrika*, vol. 1, no. 3, pp. 211–218, Sep. 1, 1936. doi: [10.1007/BF02288367](https://doi.org/10.1007/BF02288367).
- [64] L. Mirsky, "Symmetric gauge functions and unitarily invariant norms," *The Quarterly Journal of Mathematics*, vol. 11, no. 1, pp. 50–59, Jan. 1, 1960. doi: [10.1093/qmath/11.1.50](https://doi.org/10.1093/qmath/11.1.50).
- [65] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, Sep. 1966. doi: [10.1007/BF02289464](https://doi.org/10.1007/BF02289464).
- [66] L. De Lathauwer, B. De Moor, and J. Vandewalle, "A multilinear singular value decomposition," *SIAM Journal on Matrix Analysis and Applications*, vol. 21, no. 4, pp. 1253–1278, Jan. 2000. doi: [10.1137/S0895479896305696](https://doi.org/10.1137/S0895479896305696).
- [67] W. Hackbusch, *Tensor Spaces and Numerical Tensor Calculus* (Springer Series in Computational Mathematics). Cham: Springer International Publishing, 2019, vol. 56. doi: [10.1007/978-3-030-35554-8](https://doi.org/10.1007/978-3-030-35554-8).
- [68] L. De Lathauwer, B. De Moor, and J. Vandewalle, "On the best rank-1 and rank- $(r_1, r_2, \dots, r_N)$ approximation of higher-order tensors," *SIAM Journal on Matrix Analysis and Applications*, vol. 21, no. 4, pp. 1324–1342, Jan. 2000. doi: [10.1137/S0895479898346995](https://doi.org/10.1137/S0895479898346995).
- [69] R. A. Harshman, "Foundations of the PARAFAC procedure: Models and conditions for an "explanatory" multimodal factor analysis," *UCLA Working Papers in Phonetics*, vol. 16, pp. 1–84, 1970.
- [70] F. L. Hitchcock, "The expression of a tensor or a polyadic as a sum of products," *Journal of Mathematics and Physics*, vol. 6, no. 1, pp. 164–189, 1927. doi: [10.1002/sapm192761164](https://doi.org/10.1002/sapm192761164).
- [71] J. M. Ten Berge, N. D. Sidiropoulos, and R. Rocci, "Typical rank and indscal dimensionality for symmetric three-way arrays of order $i \times 2 \times 2$ or $i \times 3 \times 3$," *Linear Algebra and its Applications*, vol. 388, pp. 363–377, Sep. 2004. doi: [10.1016/j.laa.2004.03.009](https://doi.org/10.1016/j.laa.2004.03.009).
- [72] J. B. Kruskal, "Three-way arrays: Rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics," *Linear Algebra and its Applications*, vol. 18, no. 2, pp. 95–138, Jan. 1, 1977. doi: [10.1016/0024-3795\(77\)90069-6](https://doi.org/10.1016/0024-3795(77)90069-6).

Bibliography

- [73] N. D. Sidiropoulos and R. Bro, "On the uniqueness of multilinear decomposition of n-way arrays," *Journal of Chemometrics*, vol. 14, no. 3, pp. 229–239, 2000. doi: [10.1002/1099-128X\(200005/06\)14:3<229::AID-CEM587>3.0.CO;2-N](https://doi.org/10.1002/1099-128X(200005/06)14:3<229::AID-CEM587>3.0.CO;2-N).
- [74] L. Chiantini and G. Ottaviani, "On generic identifiability of 3-tensors of small rank," *SIAM Journal on Matrix Analysis and Applications*, vol. 33, no. 3, pp. 1018–1037, Jan. 2012. doi: [10.1137/110829180](https://doi.org/10.1137/110829180).
- [75] P. Paatero, "A weighted non-negative least squares algorithm for three-way 'PARAFAC' factor analysis," *Chemometrics and Intelligent Laboratory Systems*, vol. 38, no. 2, pp. 223–242, Oct. 1997. doi: [10.1016/S0169-7439\(97\)00031-2](https://doi.org/10.1016/S0169-7439(97)00031-2).
- [76] M. Rajih, P. Comon, and R. A. Harshman, "Enhanced line search: A novel method to accelerate PARAFAC," *SIAM Journal on Matrix Analysis and Applications*, vol. 30, no. 3, pp. 1128–1147, Jan. 2008. doi: [10.1137/06065577](https://doi.org/10.1137/06065577).
- [77] L. De Lathauwer, "A link between the canonical decomposition in multilinear algebra and simultaneous matrix diagonalization," *SIAM Journal on Matrix Analysis and Applications*, vol. 28, no. 3, pp. 642–666, Jan. 2006. doi: [10.1137/040608830](https://doi.org/10.1137/040608830).
- [78] J. M. Ten Berge, A. Stegeman, and M. Bennani Dosse, "The carroll and chang conjecture of equal indscal components when candecomp/parafac gives perfect fit," *Linear Algebra and its Applications*, vol. 430, no. 2, pp. 818–829, Jan. 2009. doi: [10.1016/j.laa.2008.09.023](https://doi.org/10.1016/j.laa.2008.09.023).
- [79] T. G. Kolda, "Numerical optimization for symmetric tensor decomposition," *Math. Program.*, vol. 151, no. 1, pp. 225–248, Jun. 1, 2015. doi: [10.1007/s10107-015-0895-0](https://doi.org/10.1007/s10107-015-0895-0).
- [80] J. Alexander and A. Hirschowitz, "Generic hypersurface singularities," *Proceedings - Mathematical Sciences*, vol. 107, no. 2, pp. 139–154, May 1, 1997. doi: [10.1007/BF02837722](https://doi.org/10.1007/BF02837722).
- [81] T. G. Kolda and J. R. Mayo, "Shifted power method for computing tensor eigenpairs," *SIAM Journal on Matrix Analysis and Applications*, vol. 32, no. 4, pp. 1095–1124, Oct. 2011. doi: [10.1137/100801482](https://doi.org/10.1137/100801482).
- [82] J. Kileel and J. M. Pereira, "Subspace power method for symmetric tensor decomposition," *Numerical Algorithms*, Aug. 2, 2025. doi: [10.1007/s11075-025-02165-y](https://doi.org/10.1007/s11075-025-02165-y).
- [83] I. Oseledets and E. Tyrtyshnikov, "TT-cross approximation for multidimensional arrays," *Linear Algebra and its Applications*, vol. 432, no. 1, pp. 70–88, Jan. 2010. doi: [10.1016/j.laa.2009.07.024](https://doi.org/10.1016/j.laa.2009.07.024).
- [84] D. V. Savostyanov, "Quasioptimality of maximum-volume cross interpolation of tensors," *Linear Algebra and its Applications*, vol. 458, pp. 217–244, Oct. 2014. doi: [10.1016/j.laa.2014.06.006](https://doi.org/10.1016/j.laa.2014.06.006).
- [85] M. Bebendorf, "Approximation of boundary element matrices," *Numerische Mathematik*, vol. 86, no. 4, pp. 565–589, Oct. 2000. doi: [10.1007/PL00005410](https://doi.org/10.1007/PL00005410).
- [86] S. A. Goreinov, E. E. Tyrtyshnikov, and N. L. Zamarashkin, "A theory of pseudoskeleton approximations," *Linear Algebra and its Applications*, vol. 261, no. 1, pp. 1–21, Aug. 1, 1997. doi: [10.1016/S0024-3795\(96\)00301-1](https://doi.org/10.1016/S0024-3795(96)00301-1).
- [87] M. W. Mahoney and P. Drineas, "CUR matrix decompositions for improved data analysis," *Proceedings of the National Academy of Sciences*, vol. 106, no. 3, pp. 697–702, Jan. 20, 2009. doi: [10.1073/pnas.0803205106](https://doi.org/10.1073/pnas.0803205106).

Bibliography

- [88] C. F. Caiafa and A. Cichocki, “Generalizing the column–row matrix decomposition to multi-way arrays,” *Linear Algebra and its Applications*, vol. 433, no. 3, pp. 557–573, Sep. 2010. doi: [10.1016/j.laa.2010.03.020](https://doi.org/10.1016/j.laa.2010.03.020).
- [89] Z. Song, D. P. Woodruff, and P. Zhong, “Relative error tensor low rank approximation,” in *Proceedings of the Thirtieth Annual ACM-SIAM Symposium on Discrete Algorithms*, ser. SODA ’19, USA: Society for Industrial and Applied Mathematics, Jan. 6, 2019, pp. 2772–2789.
- [90] S. Lee, H. Alipanah, and J. J. Mendoza-Arenas, *Solving the peierls-boltzmann transport equation with matrix product states*, Apr. 7, 2026. doi: [10.48550/arXiv.2604.06153](https://doi.org/10.48550/arXiv.2604.06153).
- [91] L. De Lathauwer, “Decompositions of a higher-order tensor in block terms—part II: Definitions and uniqueness,” *SIAM Journal on Matrix Analysis and Applications*, vol. 30, no. 3, pp. 1033–1066, Jan. 2008. doi: [10.1137/070690729](https://doi.org/10.1137/070690729).
- [92] L. De Lathauwer and D. Nion, “Decompositions of a higher-order tensor in block terms—part III: Alternating least squares algorithms,” *SIAM Journal on Matrix Analysis and Applications*, vol. 30, no. 3, pp. 1067–1083, Jan. 2008. doi: [10.1137/070690730](https://doi.org/10.1137/070690730).
- [93] J. D. Carroll, S. Pruzansky, and J. B. Kruskal, “Candelinc: A general approach to multidimensional analysis of many-way arrays with linear constraints on parameters,” *Psychometrika*, vol. 45, no. 1, pp. 3–24, Mar. 1, 1980. doi: [10.1007/BF02293596](https://doi.org/10.1007/BF02293596).
- [94] E. E. Papalexakis, C. Faloutsos, and N. D. Sidiropoulos, “ParCube: Sparse parallelizable tensor decompositions,” in *Machine Learning and Knowledge Discovery in Databases*, P. A. Flach, T. De Bie, and N. Cristianini, Eds., Berlin, Heidelberg: Springer, 2012, pp. 521–536. doi: [10.1007/978-3-642-33460-3_39](https://doi.org/10.1007/978-3-642-33460-3_39).
- [95] H. Kim and H. Park, “Sparse non-negative matrix factorizations via alternating non-negativity-constrained least squares for microarray data analysis,” *Bioinformatics*, vol. 23, no. 12, pp. 1495–1502, Jun. 15, 2007. doi: [10.1093/bioinformatics/btm134](https://doi.org/10.1093/bioinformatics/btm134).
- [96] C. Caroli, R. Combescot, P. Nozieres, and D. Saint-James, “Direct calculation of the tunneling current,” *Journal of Physics C: Solid State Physics*, vol. 4, no. 8, pp. 916–929, Jun. 14, 1971. doi: [10.1088/0022-3719/4/8/018](https://doi.org/10.1088/0022-3719/4/8/018).
- [97] Y. Meir and N. S. Wingreen, “Landauer formula for the current through an interacting electron region,” *Physical Review Letters*, vol. 68, no. 16, pp. 2512–2515, Apr. 20, 1992. doi: [10.1103/PhysRevLett.68.2512](https://doi.org/10.1103/PhysRevLett.68.2512).
- [98] H. Haug and A.-P. Jaho, *Quantum Kinetics in Transport and Optics of Semiconductors* (Solid-State Sciences). Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, vol. 123. doi: [10.1007/978-3-540-73564-9](https://doi.org/10.1007/978-3-540-73564-9).
- [99] J.-S. Wang, J. Wang, and J. T. Lü, “Quantum thermal transport in nanostructures,” *The European Physical Journal B*, vol. 62, no. 4, pp. 381–404, Apr. 2008. doi: [10.1140/epjb/e2008-00195-8](https://doi.org/10.1140/epjb/e2008-00195-8).
- [100] J.-S. Wang, B. K. Agarwalla, H. Li, and J. Thingna, “Nonequilibrium green’s function method for quantum thermal transport,” *Frontiers of Physics*, vol. 9, no. 6, pp. 673–697, Dec. 2014. doi: [10.1007/s11467-013-0340-x](https://doi.org/10.1007/s11467-013-0340-x).
- [101] N. Mingo, “Calculation of si nanowire thermal conductivity using complete phonon dispersion relations,” *Physical Review B*, vol. 68, no. 11, p. 113308, Sep. 26, 2003. doi: [10.1103/PhysRevB.68.113308](https://doi.org/10.1103/PhysRevB.68.113308).

Bibliography

- [102] N. Mingo and L. Yang, "Phonon transport in nanowires coated with an amorphous material: An atomistic green's function approach," *Physical Review B*, vol. 68, no. 24, p. 245 406, Dec. 11, 2003. doi: [10.1103/PhysRevB.68.245406](https://doi.org/10.1103/PhysRevB.68.245406).
- [103] J.-S. Wang, N. Zeng, J. Wang, and C. K. Gan, "Nonequilibrium green's function method for thermal transport in junctions," *Physical Review E*, vol. 75, no. 6, p. 061 128, Jun. 28, 2007. doi: [10.1103/PhysRevE.75.061128](https://doi.org/10.1103/PhysRevE.75.061128).
- [104] Y. Lee, M. Bescond, D. Logoteta, N. Cavassilas, M. Lannoo, and M. Luisier, "Anharmonic phonon-phonon scattering modeling of three-dimensional atomistic transport: An efficient quantum treatment," *Physical Review B*, vol. 97, no. 20, p. 205 447, May 31, 2018. doi: [10.1103/PhysRevB.97.205447](https://doi.org/10.1103/PhysRevB.97.205447).
- [105] Z. Tian, K. Esfarjani, and G. Chen, "Enhancing phonon transmission across a si/ge interface by atomic roughness: First-principles study with the green's function method," *Physical Review B*, vol. 86, no. 23, p. 235 304, Dec. 10, 2012. doi: [10.1103/PhysRevB.86.235304](https://doi.org/10.1103/PhysRevB.86.235304).
- [106] W. Zhang, N. Mingo, and T. S. Fisher, "Simulation of phonon transport across a non-polar nanowire junction using an atomistic green's function method," *Physical Review B*, vol. 76, no. 19, p. 195 429, Nov. 19, 2007. doi: [10.1103/PhysRevB.76.195429](https://doi.org/10.1103/PhysRevB.76.195429).
- [107] M. P. L. Sancho, J. M. L. Sancho, J. M. L. Sancho, and J. Rubio, "Highly convergent schemes for the calculation of bulk and surface green functions," *Journal of Physics F: Metal Physics*, vol. 15, no. 4, p. 851, Apr. 1985. doi: [10.1088/0305-4608/15/4/009](https://doi.org/10.1088/0305-4608/15/4/009).
- [108] T. Ando, "Quantum point contacts in magnetic fields," *Physical Review B*, vol. 44, no. 15, pp. 8017–8027, Oct. 15, 1991. doi: [10.1103/PhysRevB.44.8017](https://doi.org/10.1103/PhysRevB.44.8017).
- [109] Z.-Y. Ong and G. Zhang, "Efficient approach for modeling phonon transmission probability in nanoscale interfacial thermal transport," *Physical Review B*, vol. 91, no. 17, p. 174 302, May 6, 2015. doi: [10.1103/PhysRevB.91.174302](https://doi.org/10.1103/PhysRevB.91.174302).
- [110] S. Sadasivam, Y. Che, Z. Huang, et al., "The atomistic Green's function method for interfacial phonon transport," *Annu. Rev. Heat Transfer*, vol. 17, p. 89, 2014. doi: [10.1615/AnnualRevHeatTransfer.2014006986](https://doi.org/10.1615/AnnualRevHeatTransfer.2014006986).
- [111] J. Rammer and H. Smith, "Quantum field-theoretical methods in transport theory of metals," *Reviews of Modern Physics*, vol. 58, no. 2, pp. 323–359, Apr. 1, 1986. doi: [10.1103/RevModPhys.58.323](https://doi.org/10.1103/RevModPhys.58.323).
- [112] T. Tadano and S. Tsuneyuki, "First-principles lattice dynamics method for strongly anharmonic crystals," *Journal of the Physical Society of Japan*, vol. 87, no. 4, p. 041 015, Apr. 15, 2018. doi: [10.7566/JPSJ.87.041015](https://doi.org/10.7566/JPSJ.87.041015).
- [113] D. C. Langreth, "Linear and nonlinear response theory with applications," in *Linear and Nonlinear Electron Transport in Solids*, J. T. Devreese and V. E. Doren, Eds., Boston, MA: Springer US, 1976, pp. 3–32. doi: [10.1007/978-1-4757-0875-2_1](https://doi.org/10.1007/978-1-4757-0875-2_1).
- [114] P. G. Klemens, "Anharmonic decay of optical phonons," *Physical Review*, vol. 148, no. 2, pp. 845–848, Aug. 12, 1966. doi: [10.1103/PhysRev.148.845](https://doi.org/10.1103/PhysRev.148.845).
- [115] G. Baym and L. P. Kadanoff, "Conservation laws and correlation functions," *Physical Review*, vol. 124, no. 2, pp. 287–299, Oct. 15, 1961. doi: [10.1103/PhysRev.124.287](https://doi.org/10.1103/PhysRev.124.287).
- [116] G. Baym, "Self-consistent approximations in many-body systems," *Physical Review*, vol. 127, no. 4, pp. 1391–1401, Aug. 15, 1962. doi: [10.1103/PhysRev.127.1391](https://doi.org/10.1103/PhysRev.127.1391).

Bibliography

- [117] D. S. Fisher and P. A. Lee, "Relation between conductivity and transmission matrix," *Physical Review B*, vol. 23, no. 12, pp. 6851–6854, Jun. 15, 1981. doi: [10.1103/PhysRevB.23.6851](https://doi.org/10.1103/PhysRevB.23.6851).
- [118] R. J. Hardy, "Energy-flux operator for a lattice," *Physical Review*, vol. 132, no. 1, pp. 168–177, Oct. 1, 1963. doi: [10.1103/PhysRev.132.168](https://doi.org/10.1103/PhysRev.132.168).
- [119] J.-T. Lü and J.-S. Wang, "Coupled electron and phonon transport in one-dimensional atomic junctions," *Phys. Rev. B*, vol. 76, p. 165 418, 2007. doi: [10.1103/PhysRevB.76.165418](https://doi.org/10.1103/PhysRevB.76.165418).
- [120] D. G. Anderson, "Iterative procedures for nonlinear integral equations," *Journal of the ACM*, vol. 12, no. 4, pp. 547–560, Oct. 1965. doi: [10.1145/321296.321305](https://doi.org/10.1145/321296.321305).
- [121] H. F. Walker and P. Ni, "Anderson acceleration for fixed-point iterations," *SIAM Journal on Numerical Analysis*, vol. 49, no. 4, pp. 1715–1735, Jan. 2011. doi: [10.1137/10078356X](https://doi.org/10.1137/10078356X).
- [122] T. Rohwedder and R. Schneider, "An analysis for the DIIS acceleration method used in quantum chemistry calculations," *Journal of Mathematical Chemistry*, vol. 49, no. 9, pp. 1889–1914, Oct. 2011. doi: [10.1007/s10910-011-9863-y](https://doi.org/10.1007/s10910-011-9863-y).
- [123] A. Toth and C. T. Kelley, "Convergence analysis for anderson acceleration," *SIAM Journal on Numerical Analysis*, vol. 53, no. 2, pp. 805–819, Jan. 2015. doi: [10.1137/130919398](https://doi.org/10.1137/130919398).
- [124] W. Bian, X. Chen, and C. T. Kelley, "Anderson acceleration for a class of nonsmooth fixed-point problems," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, S1–S20, Jan. 2021. doi: [10.1137/20M132938X](https://doi.org/10.1137/20M132938X).
- [125] A. Sidi, *Vector Extrapolation Methods with Applications* (Computational Science and Engineering). Philadelphia: Society for Industrial and Applied Mathematics, 2017.
- [126] G. M. Shroff and H. B. Keller, "Stabilization of unstable procedures: The recursive projection method," *SIAM Journal on Numerical Analysis*, vol. 30, no. 4, pp. 1099–1120, 1993. doi: [10.1137/0730057](https://doi.org/10.1137/0730057).
- [127] Y. Saad and M. H. Schultz, "GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems," *SIAM Journal on Scientific and Statistical Computing*, vol. 7, no. 3, pp. 856–869, Jul. 1986. doi: [10.1137/0907058](https://doi.org/10.1137/0907058).
- [128] S. C. Eisenstat and H. F. Walker, "Choosing the forcing terms in an inexact Newton method," *SIAM Journal on Scientific Computing*, vol. 17, no. 1, pp. 16–32, 1996. doi: [10.1137/0917003](https://doi.org/10.1137/0917003).
- [129] P. N. Brown and Y. Saad, "Hybrid Krylov methods for nonlinear systems of equations," *SIAM Journal on Scientific and Statistical Computing*, vol. 11, no. 3, pp. 450–481, 1990. doi: [10.1137/0911026](https://doi.org/10.1137/0911026).
- [130] C. T. Kelley, *Iterative Methods for Linear and Nonlinear Equations* (Frontiers in Applied Mathematics). Philadelphia: Society for Industrial and Applied Mathematics, 1995, vol. 16.
- [131] L. G. C. Rego and G. Kirczenow, "Quantized thermal conductance of dielectric quantum wires," *Physical Review Letters*, vol. 81, pp. 232–235, 1 1998. doi: [10.1103/PhysRevLett.81.232](https://doi.org/10.1103/PhysRevLett.81.232).
- [132] K. Schwab, E. A. Henriksen, J. M. Worlock, and M. L. Roukes, "Measurement of the quantum of thermal conductance," *Nature*, vol. 404, pp. 974–977, 6781 2000. doi: [10.1038/35010065](https://doi.org/10.1038/35010065).

Bibliography

- [133] C. Jeong and M. Lundstrom, "Analysis of thermal conductance of ballistic point contacts," *Applied Physics Letters*, vol. 100, p. 233 109, 23 2012. doi: [10.1063/1.4726111](https://doi.org/10.1063/1.4726111).
- [134] J. Helsing, "A higher-order singularity subtraction technique for the discretization of singular integral operators on curved surfaces," *arXiv preprint arXiv:1301.7276*, 2013. doi: [10.48550/arXiv.1301.7276](https://doi.org/10.48550/arXiv.1301.7276).
- [135] I. Hänninen, M. Taskinen, and J. Sarvas, "Singularity subtraction integral formulae for surface integral equations with RWG, rooftop and hybrid basis functions," *Progress in Electromagnetics Research*, vol. 63, pp. 243–278, 2006. doi: [10.2528/PIER06051901](https://doi.org/10.2528/PIER06051901).
- [136] S. Wang, "Logarithmic singularity subtraction for matrix-friendly layered medium green's functions," *Electronics Letters*, vol. 60, e13073, 1 2024. doi: [10.1049/el12.13073](https://doi.org/10.1049/el12.13073).
- [137] D. Bruns, A. Nojeh, A. S. Phani, and J. Rottler, "Nanotube heat conductors under tensile strain: Reducing the three-phonon scattering strength of acoustic phonons," *arXiv preprint arXiv:2105.12154*, 2021. doi: [10.48550/arXiv.2105.12154](https://doi.org/10.48550/arXiv.2105.12154).
- [138] J. Ma, W. Li, and X. Luo, "Examining the Callaway model for lattice thermal conductivity," *Physical Review B*, vol. 90, p. 035 203, 3 2014. doi: [10.1103/PhysRevB.90.035203](https://doi.org/10.1103/PhysRevB.90.035203).
- [139] A. A. Maznev and O. B. Wright, "Demystifying umklapp vs normal scattering in lattice thermal conductivity," *American Journal of Physics*, vol. 82, pp. 1062–1066, 11 2014. doi: [10.1119/1.4892612](https://doi.org/10.1119/1.4892612).
- [140] H. Mera, M. Lannoo, C. Li, et al., "Inelastic scattering in nanoscale devices: One-shot current-conserving lowest-order approximation," *Phys. Rev. B*, vol. 86, p. 161 404, 2012. doi: [10.1103/PhysRevB.86.161404](https://doi.org/10.1103/PhysRevB.86.161404).
- [141] Y. Lee, D. Logoteta, N. Cavassilas, M. Lannoo, M. Luisier, and M. Bescond, "Quantum treatment of inelastic interactions for the modeling of nanowire field-effect transistors," *Materials*, vol. 13, no. 1, p. 60, Dec. 21, 2019. doi: [10.3390/ma13010060](https://doi.org/10.3390/ma13010060).
- [142] E. Cancès, A. Levitt, and J. Thomas, *Decay of interatomic force constants in the reduced hartree-fock model*, Version Number: 1, 2025. doi: [10.48550/ARXIV.2506.09595](https://doi.org/10.48550/ARXIV.2506.09595).
- [143] W. Cochran and R. Cowley, "Dielectric constants and lattice vibrations," *Journal of Physics and Chemistry of Solids*, vol. 23, no. 5, pp. 447–450, May 1962. doi: [10.1016/0022-3697\(62\)90084-7](https://doi.org/10.1016/0022-3697(62)90084-7).
- [144] L. Deuschle et al., "Electron-electron interactions in device simulation via nonequilibrium green's functions and the GW approximation," *Physical Review B*, vol. 111, no. 19, p. 195 421, May 16, 2025. doi: [10.1103/PhysRevB.111.195421](https://doi.org/10.1103/PhysRevB.111.195421).
- [145] N. Vetsch et al., "Ab-initio quantum transport with the GW approximation, 42,240 atoms, and sustained exascale performance," in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, ser. SC '25, New York, NY, USA: Association for Computing Machinery, Nov. 15, 2025, pp. 1–13. doi: [10.1145/3712285.3771784](https://doi.org/10.1145/3712285.3771784).
- [146] B. Latour, N. Shulumba, and A. J. Minnich, "Ab initio study of mode-resolved phonon transmission at si/ge interfaces using atomistic green's functions," *Physical Review B*, vol. 96, no. 10, p. 104 310, Sep. 25, 2017. doi: [10.1103/PhysRevB.96.104310](https://doi.org/10.1103/PhysRevB.96.104310).
- [147] O. Hellman, P. Steneteg, I. A. Abrikosov, and S. I. Simak, "Temperature dependent effective potential method for accurate free energy calculations of solids," *Physical Review B*, vol. 87, no. 10, p. 104 111, Mar. 25, 2013. doi: [10.1103/PhysRevB.87.104111](https://doi.org/10.1103/PhysRevB.87.104111).

Bibliography

- [148] L. Monacelli, R. Bianco, M. Cherubini, M. Calandra, I. Errea, and F. Mauri, "The stochastic self-consistent harmonic approximation: Calculating vibrational properties of materials with full quantum and anharmonic effects," *Journal of Physics: Condensed Matter*, vol. 33, no. 36, p. 363 001, Sep. 8, 2021. doi: [10.1088/1361-648X/ac066b](https://doi.org/10.1088/1361-648X/ac066b).
- [149] R. Rurali, "Colloquium: Structural, electronic, and transport properties of silicon nanowires," *Reviews of Modern Physics*, vol. 82, pp. 427–449, 1 2010. doi: [10.1103/RevModPhys.82.427](https://doi.org/10.1103/RevModPhys.82.427).
- [150] H. Peelaers, B. Partoens, and F. M. Peeters, "Phonon band structure of Si nanowires: A stability analysis," *Nano Letters*, vol. 9, pp. 107–111, 1 2009.
- [151] T. T. M. Vo, A. J. Williamson, V. Lordi, and G. Galli, "Atomistic design of thermoelectric properties of silicon nanowires," *Nano Letters*, vol. 8, pp. 1111–1114, 4 2008.
- [152] D. Donadio and G. Galli, "Temperature dependence of the thermal conductivity of thin silicon nanowires," *Nano Letters*, vol. 10, pp. 847–851, 3 2010. doi: [10.1021/nl903268y](https://doi.org/10.1021/nl903268y).
- [153] H. Karamitaheri, N. Neophytou, M. Karami Taheri, R. Faez, and H. Kosina, "Calculation of confined phonon spectrum in narrow silicon nanowires using the valence force field method," *Journal of Electronic Materials*, 2013, arXiv:1304.1401.
- [154] T. Markussen, A.-P. Jauho, and M. Brandbyge, "Heat conductance is strongly anisotropic for pristine silicon nanowires," *Nano Letters*, vol. 8, pp. 3771–3775, 11 2008.
- [155] A. Debernardi, S. Baroni, and E. Molinari, "Anharmonic phonon lifetimes in semiconductors from density-functional perturbation theory," *Physical Review Letters*, vol. 75, pp. 1819–1822, 9 1995. doi: [10.1103/PhysRevLett.75.1819](https://doi.org/10.1103/PhysRevLett.75.1819).
- [156] M. Pourfath, *The Non-Equilibrium Green's Function Method for Nanoscale Device Simulation* (Computational Microelectronics). Vienna: Springer, 2014. doi: [10.1007/978-3-7091-1800-9](https://doi.org/10.1007/978-3-7091-1800-9).
- [157] N. E. Dahlen, R. van Leeuwen, and U. von Barth, "Variational energy functionals of the Green function and of the density tested on molecules," *Physical Review A*, vol. 73, p. 012 511, 2006. doi: [10.1103/PhysRevA.73.012511](https://doi.org/10.1103/PhysRevA.73.012511).
- [158] Z. Han, X. Yang, W. Li, T. Feng, and X. Ruan, "FourPhonon: An extension module to ShengBTE for computing four-phonon scattering rates and thermal conductivity," *Computer Physics Communications*, vol. 270, p. 108 179, 2022. doi: [10.1016/j.cpc.2021.108179](https://doi.org/10.1016/j.cpc.2021.108179).
- [159] B. Roy, J. D. Sau, and S. Das Sarma, "Migdal's theorem and electron-phonon vertex corrections in Dirac materials," *Physical Review B*, vol. 89, p. 165 119, 16 2014. doi: [10.1103/PhysRevB.89.165119](https://doi.org/10.1103/PhysRevB.89.165119).
- [160] D. Klein and K. Michaeli, "One-shot GW transport calculations: A charge-conserving solution," *J. Phys. Chem. Lett.*, vol. 14, p. 897, 2023. doi: [10.1021/acs.jpclett.2c03362](https://doi.org/10.1021/acs.jpclett.2c03362).
- [161] K. S. Thygesen and A. Rubio, "Conserving GW scheme for nonequilibrium quantum transport in molecular contacts," *Phys. Rev. B*, vol. 77, p. 115 333, 2008. doi: [10.1103/PhysRevB.77.115333](https://doi.org/10.1103/PhysRevB.77.115333).
- [162] K. Miao, S. Sadasivam, J. Charles, et al., "Büttiker probes for dissipative phonon quantum transport in semiconductor nanostructures," *Appl. Phys. Lett.*, vol. 108, p. 113 107, 2016. doi: [10.1063/1.4944329](https://doi.org/10.1063/1.4944329).
- [163] M. Bandyopadhyay and D. Segal, "Quantum heat transfer in harmonic chains with self-consistent reservoirs: Exact numerical simulations," *Phys. Rev. E*, vol. 84, p. 011 151, 2011. doi: [10.1103/PhysRevE.84.011151](https://doi.org/10.1103/PhysRevE.84.011151).

Bibliography

- [164] D. Roy and A. Dhar, "Electron transport in a one dimensional conductor with inelastic scattering by self-consistent reservoirs," *Phys. Rev. B*, vol. 75, p. 195 110, 2007. doi: [10.1103/PhysRevB.75.195110](https://doi.org/10.1103/PhysRevB.75.195110).
- [165] S. Pilgram, P. Samuelsson, H. Förster, and M. Büttiker, "Full-counting statistics for voltage and dephasing probes," *Phys. Rev. Lett.*, vol. 97, p. 066 801, 2006. doi: [10.1103/PhysRevLett.97.066801](https://doi.org/10.1103/PhysRevLett.97.066801).
- [166] M. J. M. de Jong and C. W. J. Beenakker, "Semiclassical theory of shot noise in mesoscopic conductors," *Physica A*, vol. 230, p. 219, 1996. doi: [10.1016/0378-4371\(96\)00068-4](https://doi.org/10.1016/0378-4371(96)00068-4).
- [167] E. Toler, A. J. Cerfon, and D. Malhotra, "A fast, accurate and easy to implement kapur-rokhlin quadrature scheme for singular integrals in axisymmetric geometries," *Journal of Plasma Physics*, vol. 89, p. 905 890 212, 2 2023. doi: [10.1017/S002237782300020X](https://doi.org/10.1017/S002237782300020X).
- [168] Y. Xu, J.-S. Wang, W. Duan, B.-L. Gu, and B. Li, "Nonequilibrium green's function method for phonon-phonon interactions and ballistic-diffusive thermal transport," *Physical Review B*, vol. 78, p. 224 303, 22 2008. doi: [10.1103/PhysRevB.78.224303](https://doi.org/10.1103/PhysRevB.78.224303).
- [169] Y. Guo, Z. Zhang, M. Bescond, S. Xiong, M. Nomura, and S. Volz, "Anharmonic phonon-phonon scattering at the interface between two solids by nonequilibrium green's function formalism," *Physical Review B*, vol. 103, p. 174 306, 17 2021. doi: [10.1103/PhysRevB.103.174306](https://doi.org/10.1103/PhysRevB.103.174306).
- [170] C. A. Polanco et al., "The phonon quantum of thermal conductance: Are simulations and measurements estimating the same quantity?" *Science Advances*, vol. 9, eadi7439, 41 2023. doi: [10.1126/sciadv.adi7439](https://doi.org/10.1126/sciadv.adi7439).
- [171] D. A. Lemus, J. Charles, and T. Kubis, "Mode-space-compatible inelastic scattering in atomistic nonequilibrium green's function implementations," *arXiv preprint arXiv:2003.09536*, 2020. doi: [10.48550/arXiv.2003.09536](https://doi.org/10.48550/arXiv.2003.09536).